

Introduction to Statistics: A Simulation-First Approach

An Open Educational Resource

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Welcome

This is an open educational resource (OER) for introductory statistics, built on a **simulation-first** pedagogy. Students develop statistical reasoning through hands-on simulation before encountering formal mathematical theory.

Why simulation first?

Traditional introductory statistics courses begin with probability rules and distribution theory, then use those tools to build confidence intervals and hypothesis tests. Students memorize formulas without understanding *why* they work.

This textbook reverses that order:

1. **Explore data** — See patterns, describe distributions, visualize relationships.
2. **Simulate inference** — Use resampling (bootstrap, permutation) to understand *what inference is* and *why it works*.
3. **Approximate with theory** — Learn that the normal and t -distributions are mathematical shortcuts for what simulation does directly.
4. **Formalize probability** — Study probability rules and random variables *after* students already have deep intuition about randomness.
5. **Model relationships** — Apply all inference tools to regression and categorical data.

By the time students encounter the Central Limit Theorem, they have already *built* sampling distributions hundreds of times. The theorem becomes an elegant confirmation of what they've seen, not an abstract declaration to accept on faith.

How to use this book

Throughout this text, **StatBench** activities invite you to explore statistical concepts interactively. Look for boxes like this one. Each activity links to a browser-based simulation tool — no software installation required.

[Launch StatBench](#) to explore all available tools.

Guided Practice exercises appear throughout each chapter with solutions in footnotes — try them before peeking.

Worked Examples walk through complete analyses step by step.

Definitions highlight key statistical vocabulary.

About this book

This textbook remixes material from two outstanding open-source statistics textbooks:

- [Introduction to Modern Statistics](#) (2nd edition) by Mine Çetinkaya-Rundel and Johanna Hardin
- [OpenIntro Statistics](#) by David Diez, Mine Çetinkaya-Rundel, and Christopher Barr

Both are licensed under [Creative Commons BY-SA 3.0](#), and this remix is shared under the same license.

Interactive simulations are powered by [StatBench](#), an open-source statistics visualization toolkit developed at the University of Wisconsin-La Crosse.

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Chapter 1

Preface

For instructors

This textbook is designed around the **simulation-based inference** (SBI) approach popularized by Lock, Lock, Lock, Lock, and Lock in *Statistics: Unlocking the Power of Data*. The pedagogical sequence is:

EDA → Simulation inference → Theoretical inference → Probability → Modeling

The book contains more material than most one-semester courses will cover. The 28 chapters are organized into six parts, and instructors should select the chapters that fit their curriculum. A typical 15-week course might cover:

- **Part I: Explore** (Chapters 1–5): Weeks 1–3
- **Part II: Simulate** (Chapters 6–9): Weeks 4–6
- **Part III: Calculate** (Chapters 10–13): Weeks 7–9
- **Part IV: Relationships** (Chapters 14–20): Weeks 10–13
- **Part V: Foundations** (Chapters 21–25): Select chapters as needed
- **Part VI: Design** (Chapters 26–28): As time permits

Parts V (probability foundations) and VI (power/sample size) are included for courses that need them but can be omitted without loss of continuity.

StatBench integration

[StatBench](#) is a free, browser-based statistics toolkit that supports the simulation-first approach. The text features StatBench throughout but is written to be self-contained — instructors who prefer StatKey, R’s `infer` package, or other tools can substitute freely.

StatBench activities in this text use URL parameters for reproducibility. For example:

`https://datascienceuwl.github.io/statbench/simulate/bootstrap-mean/?seed=42&n=30`

The seed parameter ensures every student who uses the same link gets identical results, enabling graded simulation-based assignments.

Accessibility

This textbook is designed to meet [WCAG 2.1 AA](#) accessibility standards:

- **Text:** Atkinson Hyperlegible font, designed for readers with low vision
- **Math:** MathJax rendering with screen reader support
- **Color:** All color choices meet 4.5:1 contrast ratio; information is never conveyed by color alone
- **Structure:** Semantic HTML with proper heading hierarchy
- **Images:** All figures include descriptive alt text
- **Navigation:** Full keyboard accessibility with visible focus indicators

If you encounter accessibility barriers, please file an issue on the book's GitHub repository.

Acknowledgments

This book would not exist without the extraordinary generosity of the OpenIntro project. Mine Çetinkaya-Rundel, Johanna Hardin, David Diez, and Christopher Barr have created and maintained open-source statistics textbooks that have reached millions of students worldwide. Their commitment to open education made this remix possible.

The simulation-first pedagogy is inspired by the Lock family's pioneering work in *Statistics: Unlocking the Power of Data*.

StatBench was developed at the University of Wisconsin-La Crosse with support from the Department of Mathematics and Statistics.

PART I

EXPLORE

Before we can draw conclusions from data, we need to understand what data *is*, where it comes from, and what it looks like. We develop the vocabulary and visual tools for exploring data — identifying variable types, visualizing distributions, and describing relationships. These skills form the foundation for everything that follows.

Chapter 2

Introduction to Data

Scientists seek to answer questions using rigorous methods and careful observations. These observations — collected from the likes of field notes, surveys, and experiments — form the backbone of a statistical investigation and are called **data**. Statistics is the study of how best to collect, analyze, and draw conclusions from data. In this first chapter, we focus on both the properties of data and on the collection of data. We will introduce the building blocks of statistical thinking: observations, variables, data frames, and the types of questions that statistics can help us answer.

2.1 Case study: Using stents to prevent strokes

In this section we introduce a classic challenge in statistics: evaluating the efficacy of a medical treatment. The terms and ideas introduced here will be revisited throughout the course. For now, the goal is simply to get a sense of the role statistics can play in practice.

A stent is a small mesh tube that is placed inside a narrow or weak artery to assist in patient recovery after cardiac events and reduce the risk of an additional heart attack or death. Many doctors have hoped that there would be similar benefits for patients at risk of stroke. This leads to the principal question the researchers hoped to answer:

Does the use of stents reduce the risk of stroke?

The researchers who asked this question conducted an experiment with 451 at-risk patients.¹ Each volunteer patient was randomly assigned to one of two groups:

- **Treatment group.** Patients in the treatment group received a stent and medical management. The medical management included medications, management of risk factors, and help in lifestyle modification.
- **Control group.** Patients in the control group received the same medical management as the treatment group, but they did not receive stents.

Researchers randomly assigned 224 patients to the treatment group and 227 to the control group. In this study, the control group provides a reference point against which we can measure the medical impact of stents in the treatment group.

¹Chimowitz MI, Lynn MJ, Derdeyn CP, et al. 2011. Stenting versus Aggressive Medical Therapy for Intracranial Arterial Stenosis. *New England Journal of Medicine* 365:993–1003.

2.1.1 Examining the data

Researchers studied the effect of stents at two time points: 30 days after enrollment and 365 days after enrollment. Patient outcomes were recorded as “stroke” or “no event,” representing whether the patient had a stroke during that time period.

It would be difficult to answer the research question by looking at individual patient records one at a time. Instead, we organize the results into a summary table that lets us quickly compare the two groups.

Table 2.1: Descriptive statistics for the stent study.

	30 days: Stroke	30 days: No event	365 days: Stroke	365 days: No event
Treatment	33	191	45	179
Control	13	214	28	199
Total	46	405	73	378

Of the 224 patients in the treatment group, 45 had a stroke by the end of the first year. Using these two numbers, compute the proportion of patients in the treatment group who had a stroke by the end of their first year. (Note: answers to all Guided Practice exercises are provided in footnotes!)

Show answer

The proportion of the 224 patients who had a stroke within 365 days: $45/224 = 0.20 = 20\%$.

2.1.2 Drawing conclusions from the data

We can compute **summary statistics** from Table 2.1 to give us a better idea of how the impact of the stent treatment differed between the two groups.

A **summary statistic** is a single number that summarizes data from a sample. Summary statistics condense a large amount of information into a form that is easy to digest and compare.

For instance, the primary results of the study after one year could be described by two summary statistics: the proportion of people who had a stroke in the treatment and control groups.

- Proportion who had a stroke in the treatment (stent) group: $45/224 = 0.20 = 20\%$.
- Proportion who had a stroke in the control group: $28/227 = 0.12 = 12\%$.

These two summary statistics are useful in looking for differences between the groups, and we are in for a surprise: an additional 8% of patients in the treatment group had a stroke! This is important for two reasons. First, it is contrary to what doctors expected, which was that stents would *reduce* the rate of strokes. Second, it leads to a statistical question: do the data show a “real” difference between the groups?

This second question is subtle. Suppose you flip a coin 100 times. While the chance a coin lands heads in any given coin flip is 50%, we probably will not observe exactly 50 heads. This type of variation is part of almost any type of data generating process. It is possible that the 8% difference in the stent study is due to this natural variation. However, the larger the difference we observe (for a particular sample size), the less believable it is that the difference is due to chance. So what we are really asking is the following: if in fact stents have no effect, how unlikely is it that we would observe such a large difference?

While we do not yet have the statistical tools to fully address this question on our own, we can state the conclusion of the published analysis: there was compelling evidence of harm by stents in this study of stroke patients.

Be careful about overgeneralizing. Do not generalize the results of this study to all patients and all stents. This study looked at patients with very specific characteristics who volunteered to be part of the study and who may not be representative of all stroke patients. In addition, there are many types of stents, and this study considered only the self-expanding Wingspan stent (Boston Scientific). However, the study does leave us with an important lesson: we should keep our eyes open for surprises.

Does this study prove that stents *cause* strokes? What features of the study design might strengthen or weaken this conclusion?

Show answer

Because patients were *randomly assigned* to treatment and control groups, this is a randomized experiment. Random assignment helps ensure the groups are similar in all respects except the treatment itself. This strengthens the case for a causal conclusion. However, we should still ask: could the difference (20% vs. 12%) have arisen by chance alone? We will develop tools to answer this question rigorously later in the course.

This case study highlights a key principle that will guide much of our work in this course: **a single study's results can be surprising, and statistics gives us the tools to evaluate whether an observed result is real or could be due to random chance.** We will return to this idea again and again as we develop the tools of statistical inference.

2.2 Data basics

Effective presentation and description of data is a first step in most analyses. This section introduces one structure for organizing data as well as some terminology that will be used throughout the course.

2.2.1 Observations, variables, and data matrices

An **observation** (also called a **case** or **unit of observation**) is a single entity in a dataset — one person, one county, one transaction. A **variable** is a characteristic that is measured or recorded for each observation. A **data frame** (or **data matrix**) organizes observations as rows and variables as columns. Each cell holds a single value.

Consider data for 50 randomly sampled loans offered through a peer-to-peer lending company. Each row in the table represents a single loan — that is the observation, or unit of observation. The columns represent characteristics of each loan, and each column is a variable. For example, the first row might represent a loan of \$22,000 with an interest rate of 10.90%, where the borrower is based in New Jersey (NJ) and has an income of \$59,000.

Table 2.2: Six observations from a loan dataset.

	loan_amount	interest_rate	term	grade	state	total_income	homeownership
1	22,000	10.90	60	B	NJ	59,000	rent
2	6,000	9.92	36	B	CA	60,000	rent
3	25,000	26.30	36	E	SC	75,000	mortgage
4	6,000	9.92	36	B	CA	75,000	rent
5	25,000	9.43	60	B	OH	254,000	mortgage
6	6,400	9.92	36	B	IN	67,000	mortgage

In practice, it is especially important to ask clarifying questions to ensure important aspects of the data are understood. For instance, it is always important to be sure we know what each variable means and its units of measurement. Table 2.3 describes each variable in the loan dataset.

Table 2.3: Variables and their descriptions for the loan dataset.

Variable	Description
loan_amount	Amount of the loan received, in US dollars.
interest_rate	Interest rate on the loan, in an annual percentage.
term	The length of the loan, which is always set as a whole number of months.
grade	Loan grade, which takes values A through G and represents the quality of the loan and its likelihood of being repaid.
state	US state where the borrower resides.
total_income	Borrower's total income, including any second income, in US dollars.
homeownership	Indicates whether the person owns, owns but has a mortgage, or rents.

What is the grade of the first loan in Table 2.2? And what is the homeownership status of the borrower for that first loan?

Show answer

The loan's grade is B, and the borrower rents their residence.

2.2.2 Tidy data

A data frame where each row is a unique case (observational unit), each column is a variable, and each cell is a single value is commonly referred to as **tidy data**.

Tidy data is a standard way of mapping the meaning of a dataset to its structure. A dataset is tidy when:

1. Each row represents a single observation.
2. Each column represents a single variable.
3. Each cell contains a single value.

When recording data, use a tidy data frame unless you have a very good reason to use a different structure. This structure allows new cases to be added as rows or new variables as new columns and facilitates visualization, summarization, and other statistical analyses.

The grades for assignments, quizzes, and exams in a course are often recorded in a gradebook that takes the form of a data frame. How might you organize a course's grade data using a data frame? Describe the observational units and variables.

Show answer

There are multiple strategies that can be followed. One common strategy is to have each student represented by a row, and then add a column for each assignment, quiz, or exam. Under this setup, it is easy to review a single line to understand the grade history of a student. There should also be columns to include student information, such as one column to list student names.

2.2.3 A larger dataset: US counties

We consider data for 3,142 counties in the United States, which includes the name of each county, the state where it resides, its population in 2017, the population change from 2010 to 2017, poverty rate, and several additional characteristics.

How might these data be organized in a data frame? What are the observations and variables?

Show answer

Each county may be viewed as a case, and there are multiple pieces of information recorded for each case. A table with 3,142 rows and 14 columns could hold these data, where each row represents a county and each column represents a particular piece of information.

Table 2.4: Six observations and six variables from a US county dataset.

name	state	pop2017	pop_change	unemployment_rate	median_edu
Autauga County	Alabama	55,504	1.48	3.86	some_college
Baldwin County	Alabama	212,628	9.19	3.83	some_college
Barbour County	Alabama	25,270	-6.22	6.44	hs_diploma
Bibb County	Alabama	22,668	0.73	4.39	hs_diploma
Blount County	Alabama	58,013	0.68	4.02	hs_diploma
Bullock County	Alabama	10,309	-2.28	7.72	hs_diploma

Table 2.5: Variables and their descriptions for the county dataset.

Variable	Description
name	Name of county.
state	Name of state.
pop2000	Population in 2000.
pop2010	Population in 2010.
pop2017	Population in 2017.
pop_change	Population change from 2010 to 2017 (in percent).
poverty	Percent of population in poverty in 2017.
homeownership	Homeownership rate, 2006–2010.
multi_unit	Percent of housing units that are in multi-unit structures, 2006–2010.
unemployment_rate	Unemployment rate in 2017.
metro	Whether the county contains a metropolitan area (yes or no).
median_edu	Median education level (2013–2017): below_hs, hs_diploma, some_college, or bachelors.
per_capita_income	Per capita (per person) income (2013–2017).
median_hh_income	Median household income.
smoking_ban	Type of county-level smoking ban in place in 2010: none, partial, or comprehensive.

2.3 Types of variables

Examine the variables `unemployment_rate`, `pop2017`, `state`, and `median_edu` in the county dataset. Each of these variables is inherently different from the other three, yet some share certain characteristics. Understanding these distinctions is fundamental to choosing the right analysis tools.

2.3.1 Numerical variables

A **numerical variable** (also called a **quantitative variable**) takes on values where arithmetic operations like adding, subtracting, or averaging make sense.

Consider `unemployment_rate`, which can take a wide range of numerical values, and it is sensible to add, subtract, or take averages with those values. On the other hand, we would not classify a variable reporting telephone area codes as numerical since the average, sum, and difference of area codes does not have any clear meaning.

Numerical variables can be further subdivided:

- **Continuous variables** can take any value within a range, including decimals. The `unemployment_rate` is continuous — it could be 3.86%, 4.127%, or any value in between.
- **Discrete variables** can only take counting values (typically whole numbers: 0, 1, 2, ...). The variable `pop2017` (population count) is discrete — you cannot have 55,504.7 people.

The key test for numerical vs. categorical. Ask yourself: does it make sense to compute the average of this variable? If you can meaningfully average the values, it is numerical. If averaging does not make sense (as with zip codes or phone numbers), the variable is categorical, even if it happens to be recorded using digits.

2.3.2 Categorical variables

A **categorical variable** (also called a **qualitative variable**) takes on values that are category names or labels. The possible values of a categorical variable are called its **levels**.

The variable `state` can take up to 51 values (after accounting for Washington, DC): Alabama, Alaska, ..., and Wyoming. Because the responses themselves are categories, `state` is a categorical variable, and the possible values (states) are the variable's levels.

Categorical variables can be further subdivided:

- **Nominal variables** have categories with no natural ordering. The variable `state` is nominal — there is no inherent reason to rank Alabama above Alaska or Wyoming.
- **Ordinal variables** have categories with a meaningful order. The variable `median_edu` takes values `below_hs`, `hs_diploma`, `some_college`, or `bachelors`. These categories have a natural ordering (less education to more education), making `median_edu` ordinal.

In most of this course, we will treat categorical variables as nominal (unordered). The distinction between nominal and ordinal becomes more important in advanced courses when special methods are used to take advantage of the ordering.

2.3.3 Variable type hierarchy

The breakdown of variable types is summarized in the diagram below.

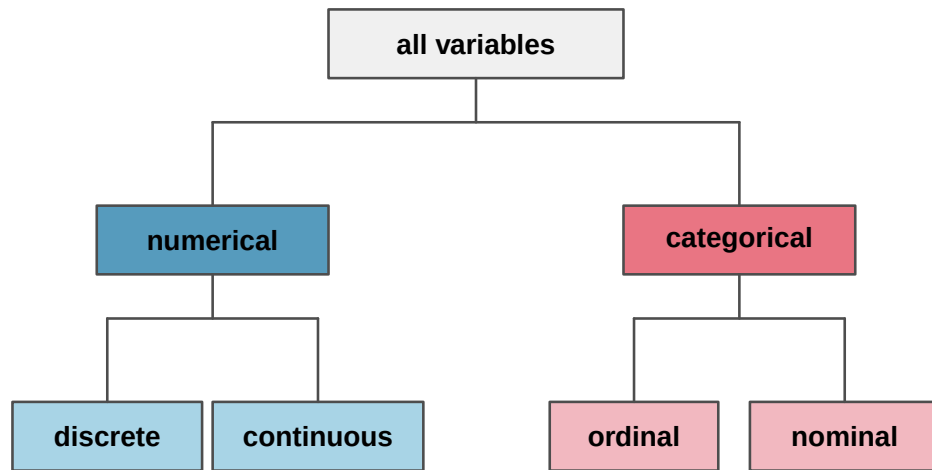


Figure 2.1: Breakdown of variables into their respective types. All variables are either numerical or categorical. Numerical variables are further classified as discrete or continuous. Categorical variables are further classified as ordinal or nominal.

Data were collected about students in a statistics course. Three variables were recorded for each student: number of siblings, student height, and whether the student had previously taken a statistics course. Classify each of the variables as continuous numerical, discrete numerical, or categorical.

The number of siblings and student height represent numerical variables. Because the number of siblings is a count, it is discrete. Height varies continuously, so it is a continuous numerical variable. The last variable classifies students into two categories — those who have and those who have not taken a statistics course — which makes this variable categorical.

An experiment is evaluating the effectiveness of a new drug in treating migraines. A group variable is used to indicate the experiment group for each patient: treatment or control. The `num_migraines` variable represents the number of migraines the patient experienced during a 3-month period. Classify each variable as either numerical or categorical.

Show answer

The group variable can take just one of two group names, making it categorical. The `num_migraines` variable describes a count of the number of migraines, which is an outcome where basic arithmetic is sensible, which means this is a numerical outcome; more specifically, since it represents a count, `num_migraines` is a discrete numerical variable.

Classify each variable as continuous numerical, discrete numerical, nominal categorical, or ordinal categorical:

- Number of text messages sent per day
- Favorite type of music
- Temperature in Fahrenheit
- Satisfaction rating (1 = very unsatisfied, ..., 5 = very satisfied)

Show answer

- a. Discrete numerical — it is a count.
- b. Nominal categorical — the categories (rock, jazz, classical, etc.) have no natural ordering.
- c. Continuous numerical — temperature can take any value in a range.
- d. Ordinal categorical — the numbers have a meaningful order, but the intervals between them may not be equal, and averaging satisfaction ratings is debatable.

2.4 Relationships between variables

Many analyses are motivated by a researcher looking for a relationship between two or more variables. A social scientist might like to answer questions such as:

Does a higher-than-average increase in county population tend to correspond to counties with higher or lower median household incomes?

If homeownership in one county is lower than the national average, will the percent of housing units that are in multi-unit structures in that county tend to be above or below the national average?

How much can the median education level explain the median household income for counties in the US?

To answer these questions, data must be collected, such as the county dataset shown in Table 2.4. Examining summary statistics can provide numerical insights about the specifics of each of these questions. Alternatively, graphs can be used to visually explore the data, potentially providing more insight than a summary statistic.

2.4.1 Scatterplots

A **scatterplot** is a graph that displays the relationship between two numerical variables. Each observation is represented by a point, with one variable on the horizontal axis and the other on the vertical axis.

The figure below displays the relationship between homeownership and `multi_unit` (the percent of housing units that are in multi-unit structures such as apartments and condos) for US counties. Each point on the plot represents a single county. For instance, one highlighted point corresponds to Chatham County, Georgia, which has 39.4% of housing units in multi-unit structures and a homeownership rate of 31.3%.

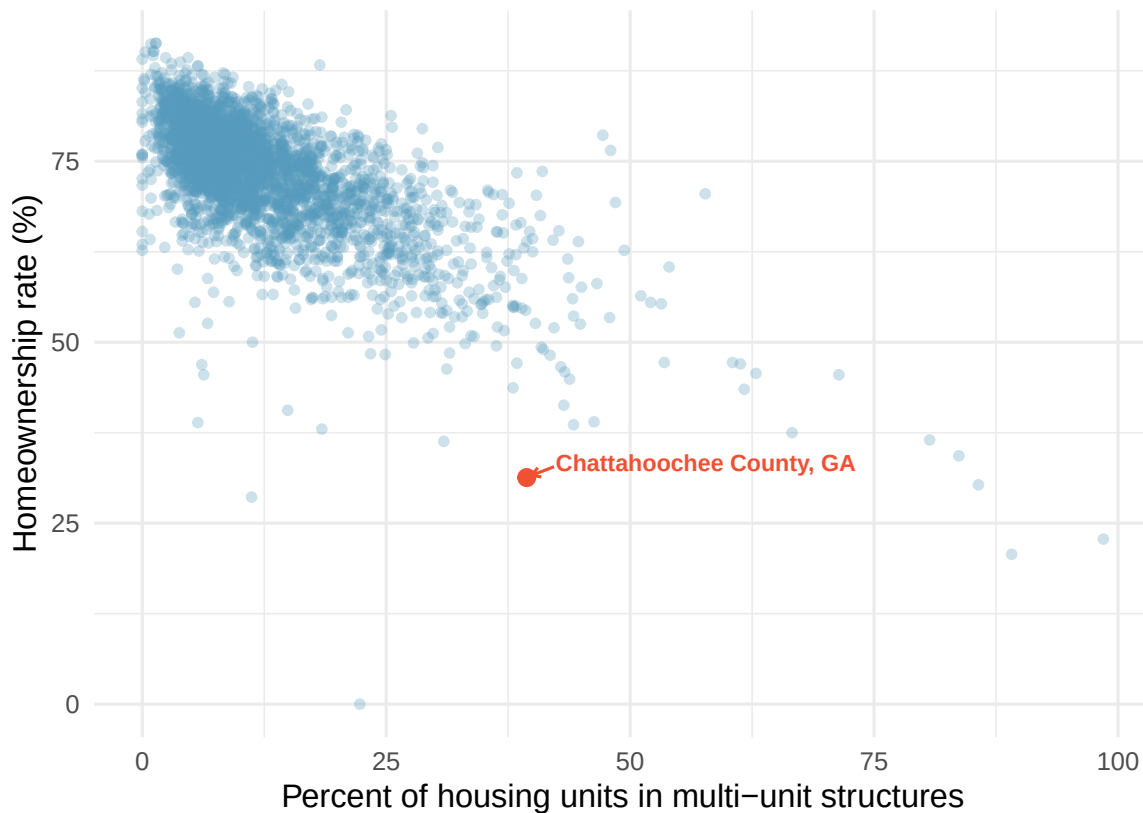


Figure 2.2: A scatterplot of homeownership rate versus the percent of housing units in multi-unit structures for US counties. There is a clear negative trend: as the multi-unit rate increases, homeownership tends to decrease. Chattahoochee County, Georgia is highlighted.

The scatterplot suggests a relationship between the two variables: counties with a higher rate of housing units in multi-unit structures tend to have lower homeownership rates. We might brainstorm as to why this relationship exists and investigate each idea to determine which are the most reasonable explanations.

2.4.2 Associated and independent variables

When two variables show some connection or pattern with one another, they are called **associated** variables (or **related** variables). If two variables are not associated — there is no evident relationship between them — they are said to be **independent**.

Because there is a downward trend in the homeownership vs. multi-unit scatterplot — counties with more multi-unit housing are associated with lower homeownership — these variables are said to be **negatively associated**.

A **positive association** means that as one variable increases, the other tends to increase as well. A **negative association** means that as one variable increases, the other tends to decrease.

Consider the relationship between the percent change in population from 2010 to 2017 and median household income for counties, as shown in the scatterplot below. Are these variables associated?

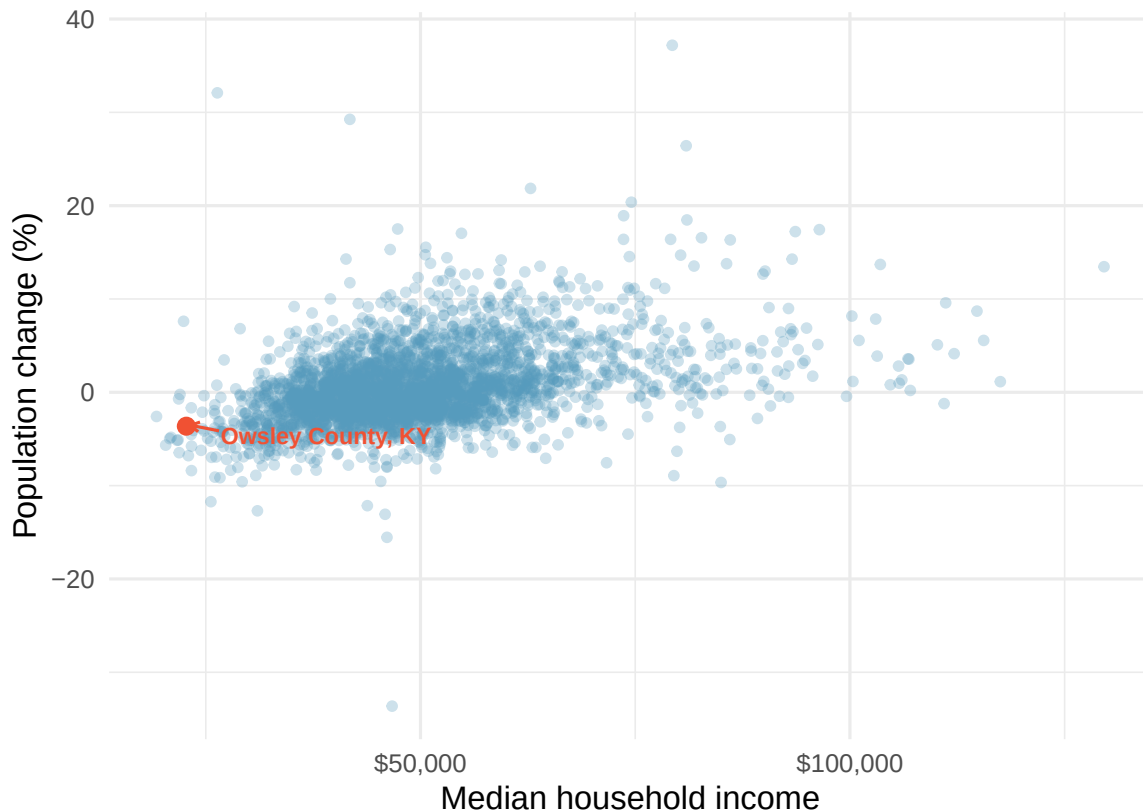


Figure 2.3: A scatterplot showing population change against median household income for US counties. There is a positive trend: counties with higher median household income tend to have higher population growth. Owsley County, Kentucky is highlighted.

The larger the median household income for a county, the higher the population growth observed for the county. While it is not true that every county with a higher median household income has higher population growth, the trend in the plot is evident. Since there is some relationship between the variables, they are associated. Because both variables tend to increase together, this is a positive association.

Associated or independent, not both. A pair of variables are either related in some way (associated) or not (independent). No pair of variables is both associated and independent.

Examine the variables in the loan dataset, which are described in Table 2.3. Create two questions about possible relationships between variables in the loan data that are of interest to you.

Show answer

Two example questions: (1) What is the relationship between loan amount and total income? (2) If someone's income is above the average, will their interest rate tend to be above or below the average?

2.5 Explanatory and response variables

When we ask questions about the relationship between two variables, we sometimes also want to determine if the change in one variable causes a change in the other. Consider the following rephrasing of an earlier question about the county dataset:

If there is an increase in the median household income in a county, does this *drive* an increase in its population?

In this question, we are asking whether one variable affects another.

When we suspect one variable might causally affect another, we label the first variable the **explanatory variable** and the second the **response variable**.

explanatory variable $\rightarrow$ *might affect* $\rightarrow$ response variable

We also use the terms explanatory and response to describe variables where the response might be *predicted* using the explanatory variable, even if there is no causal relationship.

For many pairs of variables, there is no hypothesized relationship, and these labels would not be applied to either variable.

In the stent study from Section 2.1, identify the explanatory and response variables.

The explanatory variable is the treatment group assignment (stent or control), because this is the factor that the researchers manipulated. The response variable is whether the patient had a stroke, because this is the outcome the researchers measured to evaluate the treatment's effect.

Labeling one variable as “explanatory” does not guarantee that it *causes* changes in the response. A formal evaluation of whether one variable causes changes in another requires an experiment. We discuss this distinction further in Section 2.6.

Some disciplines use the terms **independent variable** and **dependent variable** instead of explanatory and response. We avoid this language because the word “independent” has a different technical meaning in statistics — it refers to variables that are not associated. Using “independent” in two different ways can cause confusion.

A researcher is studying whether hours spent studying for an exam are related to exam scores. Identify the explanatory and response variables.

Show answer

The explanatory variable is hours spent studying, because this is the factor that might influence the outcome. The response variable is exam score, because this is the outcome being measured.

2.6 Observational studies and experiments

There are two primary types of data collection: experiments and observational studies. Understanding the difference is critical, because the type of study determines what conclusions we can draw.

2.6.1 Experiments

An **experiment** is a study in which the researcher actively imposes treatments on the subjects in order to observe their responses. When subjects are randomly assigned to treatment groups, the study is called a **randomized experiment**.

For instance, we may suspect that drinking a high-calorie energy drink will improve performance in a race. To check if there really is a causal relationship between the explanatory variable (whether the runner drank an energy drink or not) and the response variable (the race time), researchers identify a sample of individuals and split them into groups. The individuals in each group are *assigned* a treatment. Random assignment organizes the participants into groups that are roughly equal on all

aspects, thus allowing us to control for any other variables that might affect the outcome (such as fitness level, racing experience, or age).

For example, each runner in the experiment could be randomly assigned — perhaps by flipping a coin — into one of two groups: the first group receives a **placebo** (a fake treatment, in this case a no-calorie drink that looks identical to the energy drink) and the second group receives the high-calorie energy drink.

A **placebo** is a fake treatment that is made to look like the real treatment. Placebos are used so that subjects in the control group have the same experience as those in the treatment group (except for the actual treatment), which helps researchers isolate the effect of the treatment itself.

See the stent case study in Section 2.1 for another example of an experiment, though that study did not employ a placebo.

2.6.2 Observational studies

An **observational study** is a study in which the researcher collects data without interfering with how the data arise. In an observational study, the researcher merely observes what happens naturally.

Researchers perform observational studies when they collect data via surveys, review medical or company records, or follow a **cohort** (a group of similar individuals) over time to form hypotheses about why certain outcomes might occur. In each of these situations, researchers merely observe the data that arise — they do not assign treatments to subjects.

A researcher wants to know if smoking causes lung cancer. She follows 10,000 adults over 20 years, recording whether each person smokes and whether they develop lung cancer. Is this an experiment or an observational study?

This is an observational study. The researcher did not assign anyone to smoke or not smoke — she simply observed the participants' behavior and health outcomes. While the study might reveal an association between smoking and lung cancer, it cannot by itself prove causation, because there may be other differences between smokers and non-smokers (such as diet, exercise habits, or genetics) that contribute to the outcome.

2.6.3 The key distinction: association vs. causation

In general, observational studies can provide evidence of a naturally occurring association between variables, but they cannot by themselves show a causal connection. This is because in observational studies, researchers cannot control for all the other variables that might be influencing the outcome.

Association does not imply causation.

In general, association does not imply causation. An advantage of a randomized experiment is that it is easier to establish causal relationships, because random assignment tends to balance out the effects of other variables across treatment groups. Establishing causation from observational studies requires advanced statistical methods that are beyond the scope of this course. We revisit these ideas when we discuss study design in Chapter 2.

Determine whether each of the following is an experiment or an observational study:

- A pharmaceutical company randomly assigns 200 patients to receive either a new drug or a placebo and measures their cholesterol levels after 6 months.
- A university surveys its graduates to determine whether those who participated in study-abroad programs earn higher salaries five years after graduation.

- c. A teacher assigns students in her morning class to use a new learning app and students in her afternoon class to study using traditional methods, then compares exam scores.

-
- a. This is an experiment — the company randomly assigns patients to treatments.
 b. This is an observational study — the university simply collects data on students who chose (or chose not) to study abroad. The students were not randomly assigned.
 c. This is tricky. The teacher assigns groups, so it might seem like an experiment. However, because students were not randomly assigned (they ended up in morning vs. afternoon classes for their own reasons), this is better described as a quasi-experiment or an observational study. Differences in outcomes could be due to the app, or they could be due to systematic differences between the types of students who take morning vs. afternoon classes.

2.6.4 Putting it all together

The table below summarizes the key differences between experiments and observational studies.

	Experiment	Observational Study
Researcher's role	Assigns treatments	Observes what happens naturally
Random assignment?	Yes (in a randomized experiment)	No
Can establish causation?	Yes	Generally no
Example	Randomly assign patients to drug vs. placebo	Survey people about their exercise habits and health

2.7 Chapter review

2.7.1 Summary

This chapter introduced you to the world of data. Here are the key ideas:

- **Data** are observations collected about the world around us. **Statistics** is the science of collecting, analyzing, and drawing conclusions from data.
- A **data frame** organizes data so that each row represents an observation and each column represents a variable. Data organized this way is called **tidy data**.
- Variables come in two fundamental types: **numerical** (continuous or discrete) and **categorical** (nominal or ordinal).
- When two variables show a pattern together, they are **associated**; if not, they are **independent**. Associations can be **positive** (both increase together) or **negative** (one increases while the other decreases).
- The **explanatory variable** is the variable we think might influence the **response variable**.
- **Experiments** randomly assign subjects to treatments and can establish causation. **Observational studies** merely observe what happens and can identify associations, but generally cannot establish causation.
- **Association does not imply causation** — this is one of the most important principles in statistics.

Many of the ideas from this chapter will be revisited as we move on to doing end-to-end data analyses. In the next chapter, you will learn about how we can design studies to collect the data we need to make conclusions with the desired scope of inference.

2.7.2 Key terms

The following terms were introduced in this chapter. You should be able to define and give an example of each.

Term	Term	Term
associated	data	data frame
case	categorical variable	cohort
continuous	dependent variable	discrete
experiment	explanatory variable	independent
level	negative association	nominal
numerical variable	observation	observational study
ordinal	placebo	positive association
randomized experiment	response variable	summary statistic
tidy data	unit of observation	variable

2.7.3 Exercises

Exercises for this chapter are under development.

Chapter 3

Data Collection and Study Design

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Before digging into the details of working with data, we pause to think about how data come to be. If data are to be used to draw broad conclusions, then it is important to understand who or what the data represent. One important aspect is **sampling** — knowing how the observational units were selected from a larger group allows us to generalize back to the population from which the data were drawn. Additionally, by understanding the structure of the study, we can separate causal relationships from mere associations. A good question to ask before analyzing any dataset is: *How were these observations collected?* You will learn a lot about the data by understanding its source.

3.1 Populations and samples

The first step in conducting research is to identify the question to be investigated. A clearly stated research question helps identify what subjects or cases should be studied and what variables are important. It is also important to consider *how* data are collected so that they are reliable and help achieve the research goals.

Consider the following three research questions:

1. What is the average mercury content in swordfish in the Atlantic Ocean?
2. Over the last five years, what is the average time to complete a degree for UW–La Crosse undergrads?
3. Does a new drug reduce the number of deaths in patients with severe heart disease?

The **population** is the entire group of individuals or cases about which we want information. A **sample** is a subset of the population from which we actually collect data. A **census** is a study that attempts to collect data from every member of the population.

Each research question refers to a target population. In the first question, the target population is all swordfish in the Atlantic Ocean, and each fish represents a case. Oftentimes it is not feasible to collect data for every case in a population. A census is difficult because it may be too expensive, or it might be difficult or impossible to even identify every member of the population. Instead, we take a sample. Ideally, a sample is a manageable fraction of the population. For instance, 60 swordfish might be selected, and this sample data may be used to estimate the population average mercury content.

For the second and third research questions above, identify the target population and what represents an individual case.

Show answer

For the question about time to complete a degree, the population includes all UW–La Crosse undergraduates who graduated in the last five years (we can only compute an average for students who actually finished). Each graduate is an individual case. For the question about the heart disease drug, the population includes all people with severe heart disease, and each person represents a case.

3.1.1 Parameters and statistics

In most statistical analyses, the research question boils down to understanding a numerical summary — perhaps a quantity you already know (like the average) or one you will learn about in this course.

A numerical summary can be calculated on either the sample or the entire population. However, measuring every unit in the population is usually impossible. So we calculate the summary from a sample and use it to estimate the corresponding population value.

A **population parameter** is a numerical summary calculated from (or defined for) the entire population. A **sample statistic** is a numerical summary calculated from the sample data.

We use specific terms to distinguish between numbers calculated from a sample (sample statistics) and numbers that describe the entire population (population parameters). These terms will be used extensively in later chapters when we discuss making inferences about populations.

A poll of 1,000 likely voters finds that 54% plan to vote for Candidate A. Identify the population parameter and the sample statistic.

Show answer

The sample statistic is 54% — this is the proportion observed in the sample of 1,000 voters. The population parameter is the true proportion of *all* likely voters who plan to vote for Candidate A, which is unknown. The sample statistic (54%) is our best estimate of that unknown parameter.

3.1.2 Anecdotal evidence

Consider the following possible responses to the three research questions from above:

1. A man on the news got mercury poisoning from eating swordfish, so the average mercury concentration in swordfish must be dangerously high.
2. I met two students who took more than 7 years to graduate from UW–La Crosse, so it must take longer to graduate here than at many other schools.
3. My friend's dad had a heart attack and died after they gave him a new heart disease drug, so the drug must not work.

Each conclusion is based on data. However, there are two problems. First, the data only represent one or two cases. Second, and more importantly, it is unclear whether these cases are actually representative of the population. Data collected in this haphazard fashion are called **anecdotal evidence**.

Be careful of anecdotal evidence. Such evidence may be true and verifiable, but it may only represent extraordinary cases and therefore not be a good representation of the population.

Anecdotal evidence typically consists of unusual cases that we recall based on their striking characteristics. For instance, we are more likely to remember the two people who took 7 years to graduate than the six others who graduated in four years. Instead of focusing on the most unusual cases, we should examine a sample of many cases that better represent the population.

3.2 Sampling from a population

We might try to estimate the time to graduation for UW–La Crosse undergraduates by collecting a sample of graduates. All graduates in the last five years represent the *population*, and graduates who are selected for review are collectively called the *sample*. In general, we always seek to *randomly* select a sample from a population. The most basic type of random selection is equivalent to how raffles are conducted. For example, we could write each graduate's name on a raffle ticket and draw 10 tickets. The selected names would represent a random sample of 10 graduates.

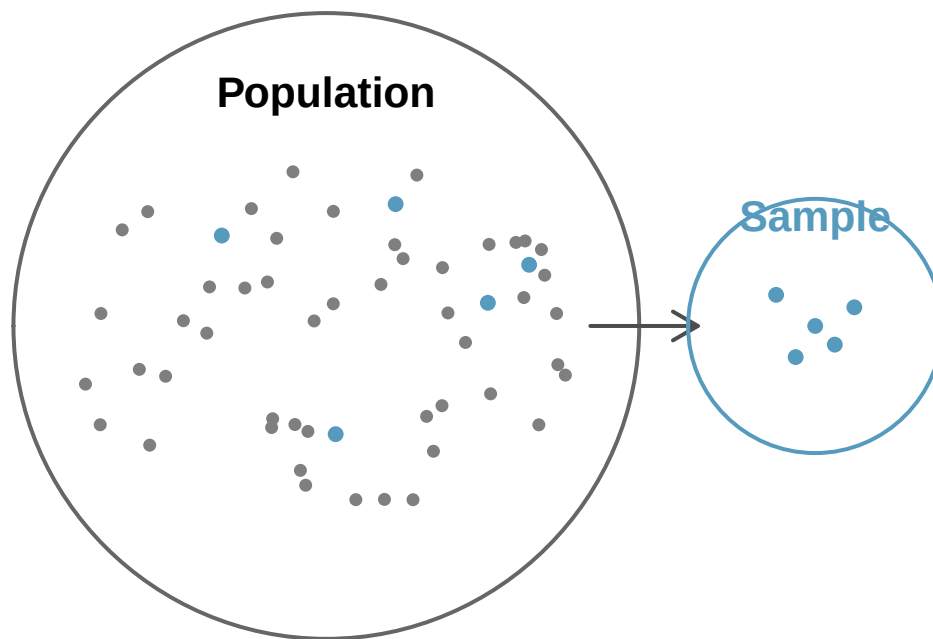


Figure 3.1: Sampling from a population. Individuals are randomly selected from the population (large circle) to be included in the sample (small circle).

Suppose we ask a student who happens to be majoring in nutrition to select several graduates for the study. Which students do you think they might pick? Do you think their sample would be representative of all graduates?

They might pick a disproportionate number of graduates from health-related fields. When selecting samples by hand, we run the risk of picking a **biased** sample, even if our bias is unintentional.



Figure 3.2: A biased sample. A nutrition major asked to select graduates might inadvertently pick a disproportionate number from health-related fields.

If someone is permitted to pick and choose exactly which individuals are included in the sample, it is entirely possible that the sample will overrepresent that person's interests — even if the bias is unintentional. Sampling randomly helps address this problem.

A **simple random sample (SRS)** is a sample drawn so that each case in the population has an equal chance of being included, and the selection of any one case is independent of the selection of any other case. This is equivalent to drawing names out of a hat.

The act of taking a simple random sample helps minimize bias. However, bias can crop up in other ways. Even when people are selected at random — for example, for surveys — caution must be exercised if the **non-response rate** is high. If only 30% of the people randomly sampled for a survey actually respond, then it is unclear whether the results are **representative** of the entire population. This **non-response bias** can skew results.

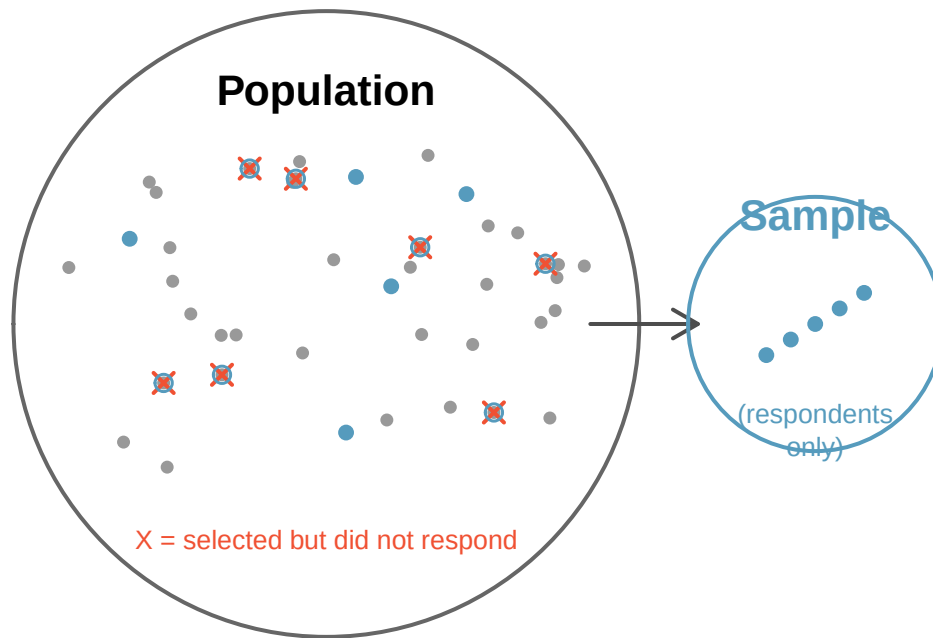


Figure 3.3: Non-response bias. Survey studies may only reach a certain group within the population. Non-response can make even a random sample unrepresentative.

Another common problem is a **convenience sample**, where individuals who are easily accessible are more likely to be included. For instance, if a political survey is conducted by stopping people on a single street corner, the results will not represent the broader city. It is often difficult to discern what sub-population a convenience sample actually represents.

We can easily access ratings for products, sellers, and companies on websites like Amazon. These ratings come only from people who go out of their way to provide a rating. If 50% of online reviews for a product are negative, do you think this means that 50% of buyers are dissatisfied with the product? Why or why not?

Show answer

Probably not. People tend to be more motivated to leave a review when they have a negative experience than when a product meets expectations. This creates a negative bias in product ratings — dissatisfied customers are overrepresented among reviewers. This is a form of non-response bias (satisfied customers are less likely to respond).

3.3 Sampling methods

Almost all statistical methods are based on the notion of randomness. If data are not collected using a random process from a population, then the estimates and the errors associated with those estimates are not reliable. Here we consider four random sampling techniques: simple random sampling, stratified sampling, cluster sampling, and multistage sampling.

3.3.1 Simple random sampling

Simple random sampling is the most intuitive form of random sampling. Consider the salaries of Major League Baseball (MLB) players, where each player belongs to one of the league's 30 teams. To

take a simple random sample of 120 baseball players and their salaries, we could write the names of all players onto slips of paper, drop the slips into a bucket, mix them thoroughly, then draw out 120 slips. In general, a sample is called “simple random” if each case in the population has an equal chance of being included in the final sample *and* knowing that a particular case is included tells us nothing about which other cases are included.

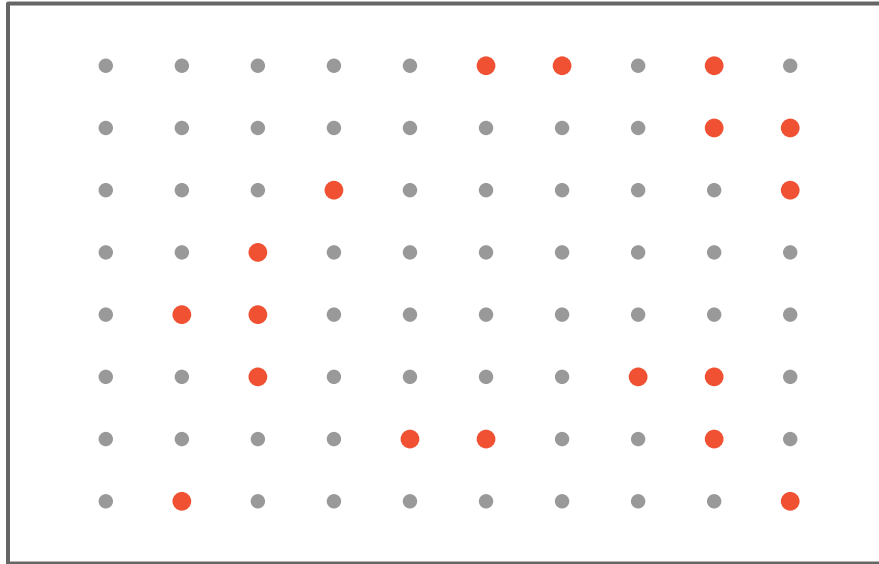


Figure 3.4: Simple random sampling. 18 cases (shown in red) are randomly selected from the population.

3.3.2 Stratified sampling

Stratified sampling is a divide-and-conquer strategy. The population is divided into groups called **strata**. The strata are chosen so that similar cases are grouped together, then a second sampling method (usually simple random sampling) is used within each stratum. In the baseball salary example, each of the 30 teams could represent a stratum, since some teams have much larger payrolls than others. We might randomly sample 4 players from each team to get our sample of 120.

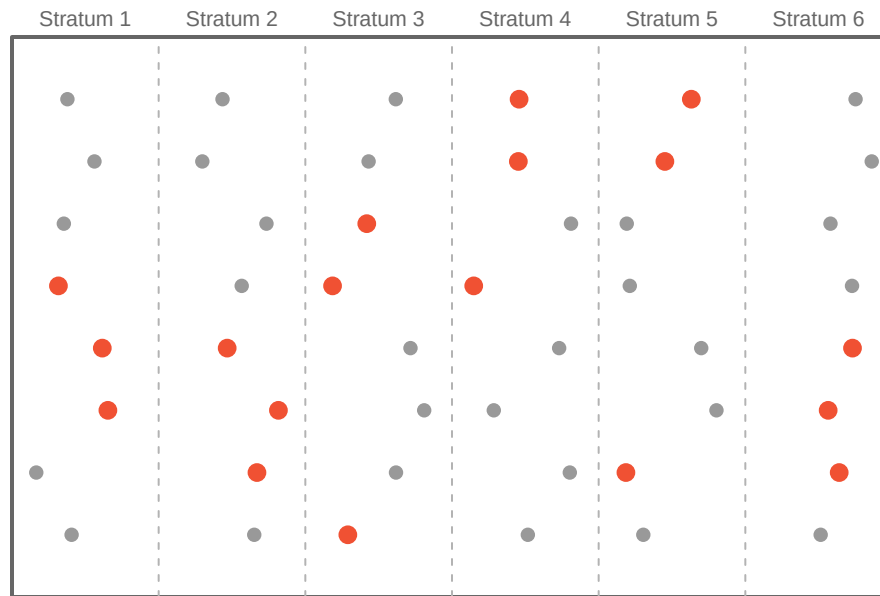


Figure 3.5: Stratified sampling. Cases are first grouped into strata, then simple random sampling is used within each stratum to select 3 cases per stratum.

Stratified sampling is especially useful when cases within each stratum are very similar with respect to the outcome of interest.

Why is it beneficial for cases within each stratum to be very similar?

If cases within a stratum are similar, we can get a very stable (precise) estimate for each subgroup. When we combine these stratum-level estimates into a single estimate for the full population, the overall estimate will tend to be more precise because each individual component is itself more precise.

3.3.3 Cluster sampling

In a **cluster sample**, we break up the population into many groups called **clusters**, then randomly select a fixed number of clusters and include *all* observations from each selected cluster in the sample.

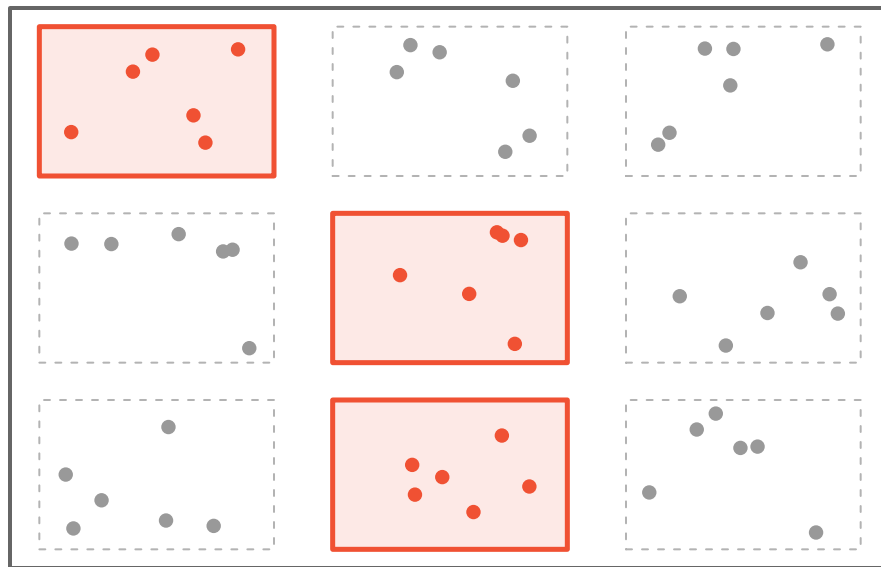


Figure 3.6: Cluster sampling. The population is divided into nine clusters. Three clusters are randomly selected, and all observations within those clusters are included in the sample.

3.3.4 Multistage sampling

A **multistage sample** is like a cluster sample, but instead of keeping all observations in each selected cluster, we take a random sample *within* each selected cluster.



Figure 3.7: Multistage sampling. Three clusters are randomly selected, then a random sample of observations is drawn from within each selected cluster.

Cluster and multistage sampling can be more economical than other techniques. Unlike stratified sampling, these approaches work best when there is a lot of case-to-case variability *within* a cluster but the clusters themselves are fairly similar to one another. For example, if neighborhoods represent clusters, then cluster sampling works best when the residents within each neighborhood are diverse.

Suppose we want to estimate the malaria rate in a densely tropical portion of rural Indonesia. There are 30 villages in the region, each more or less like the next, but the distances between villages are substantial. We want to test 150 individuals for malaria. What sampling method should we use?

A simple random sample would likely draw individuals from all 30 villages, making data collection very expensive given the travel distances. Stratified sampling would be difficult because it is unclear how to group individuals into similar strata. However, cluster or multistage sampling are excellent choices. With multistage sampling, we could randomly select, say, 15 of the 30 villages, then randomly select 10 people from each village. This would substantially reduce data collection costs while still yielding reliable information.

3.3.5 Comparing the four methods

The table below summarizes the key features of each sampling method:

Method	How it works	Best used when...
Simple random	Every case has an equal chance of selection	Population is relatively homogeneous and accessible
Stratified	Divide into similar groups (strata), then SRS within each	Known subgroups differ on the outcome of interest
Cluster	Randomly select entire groups (clusters)	Clusters are internally diverse but similar to each other; travel/cost is a concern
Multistage	Randomly select clusters, then SRS within each	Same as cluster, but clusters are large

3.4 Experiments

An **experiment** is a study where the researchers actively assign treatments to cases. When this assignment uses randomization (e.g., a coin flip to decide which treatment a patient receives), it is called a **randomized experiment**. Randomized experiments are fundamentally important when trying to establish a causal connection between two variables.

3.4.1 Principles of experimental design

Well-designed experiments incorporate four key principles:

1. **Controlling.** Researchers assign treatments to cases and do their best to **control** any other differences between the groups. For example, when patients take a drug in pill form, some patients might take the pill with a sip of water while others drink a full glass. To control for the effect of water consumption, a doctor might instruct every patient to drink a 12-ounce glass of water with the pill.
2. **Randomization.** Researchers randomly assign subjects to treatment groups to account for variables that cannot be controlled. For example, some patients may be more susceptible to a disease than others due to their dietary habits. Randomly assigning patients to treatment or control groups helps even out such differences across groups.

3. **Replication.** The more cases researchers observe, the more accurately they can estimate the effect of the explanatory variable on the response. In a single study, we **replicate** by collecting a sufficiently large sample. At a minimum, we want multiple subjects per treatment group. Replication also refers to repeating an entire study to verify earlier findings. (The **replication crisis** refers to the ongoing problem in several scientific disciplines where past findings have failed to be replicated in new studies.)
4. **Blocking.** Researchers sometimes know or suspect that variables other than the treatment influence the response. They may first group individuals based on this variable into **blocks** and then randomize cases within each block to the treatment groups. For instance, in studying the effect of a drug on heart attacks, researchers might first split patients into low-risk and high-risk blocks, then randomly assign half the patients from each block to the treatment and half to the control group.

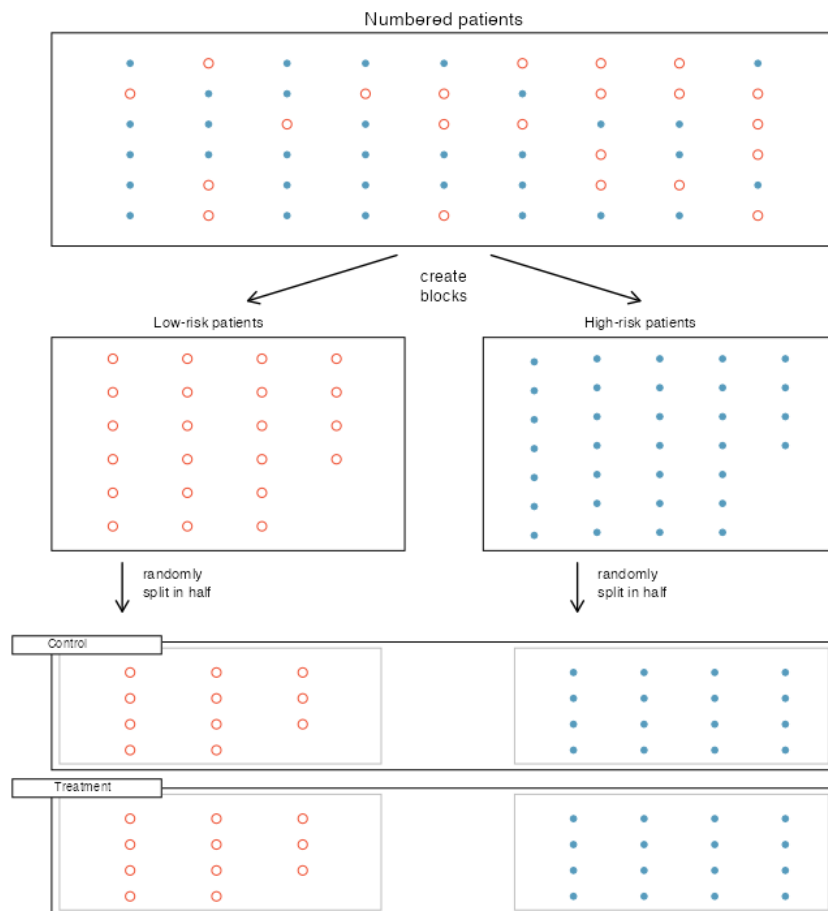


Figure 3.8: Blocking for patient risk. Patients are first divided into low-risk and high-risk blocks, then patients in each block are evenly randomized into treatment groups.

It is important to incorporate at least the first three principles into any study. Blocking is a slightly more advanced technique that adds precision when relevant grouping variables are known.

3.4.2 Reducing bias in human experiments

Randomized experiments are considered the gold standard for establishing cause-and-effect, but they do not automatically eliminate all bias. Human studies are a perfect example of where bias can unintentionally arise.

Consider a study where a new drug is tested to see if it reduces deaths in heart attack patients. Researchers designed a randomized experiment: study volunteers were randomly placed into two groups. The **treatment group** received the drug. The **control group** did not receive any drug treatment.

Put yourself in the place of a person in the study. If you are in the treatment group, you receive a promising new drug and may feel hopeful. A person in the control group receives nothing and might feel anxious. These emotional effects could influence health outcomes independently of the drug itself. This means there are actually two effects at play: the drug's real effectiveness, and the emotional impact of knowing (or not knowing) that you are receiving treatment.

A study is **blind** (or **single-blind**) if the patients do not know which treatment group they are in. A **placebo** is a fake treatment (such as a sugar pill) given to the control group so that patients cannot tell whether they are receiving the real treatment. The **placebo effect** is the phenomenon where patients show improvement simply because they believe they are receiving treatment. A study is **double-blind** if neither the patients nor the doctors and researchers interacting with them know who is receiving which treatment.

To prevent emotional effects from biasing the study, researchers want patients to be unaware of their group assignment — that is, the study should be *blind*. But if a patient in the control group receives nothing at all, they will know they are not getting treated. The solution is a placebo. An effective placebo is the key to making a study truly blind.

The patients are not the only ones who should be blinded. Doctors and researchers can also unintentionally bias a study. A doctor who knows a patient is receiving the real treatment might give that patient more attention or care. To guard against this, most modern studies use a **double-blind** setup.

Think back to the stent study described in Section 2.1. Is it an experiment? Was the study blinded? Was it double-blinded?

Show answer

The researchers assigned patients to treatment groups, so it is an experiment. However, patients could distinguish what treatment they received because a stent is a surgical procedure. There is no simple equivalent of a “placebo stent” (though sham surgeries do exist), so the study was likely not fully blinded. If the study was not blind, it could not be double-blind either.

For the stent study, could the researchers have employed a placebo? If so, what would it have looked like?

Show answer

In principle, researchers could use a **sham surgery** — the patient undergoes a surgical procedure but does not actually receive the stent. Sham surgeries do exist in medical research, but they raise serious ethical questions because they expose control-group patients to the risks of surgery without the potential benefit of the treatment.

There are always multiple viewpoints on experiments and placebos, and rarely is it obvious which approach is ethically “correct.” Is it ethical to use a sham surgery that creates risk for the patient?

On the other hand, if we do not use one, we might promote a costly treatment that has no real effect, diverting resources from treatments that actually work.

3.5 Observational studies

An **observational study** is a study where no treatment has been explicitly applied or withheld by the researchers. The researchers simply observe and record what happens naturally.

Making causal conclusions based on experiments is often reasonable because researchers can randomly assign treatments. However, making causal conclusions from observational data is much more dangerous and is generally not recommended. Observational studies are typically only sufficient to show **associations** or to form hypotheses that can later be tested with experiments.

Suppose an observational study tracked sunscreen use and skin cancer, and found that people who used more sunscreen were *more* likely to have skin cancer. Does this mean sunscreen causes skin cancer?

No! Previous research tells us that sunscreen actually *reduces* skin cancer risk. The missing piece is **sun exposure**. People who spend a lot of time in the sun are more likely to use sunscreen *and* more likely to get skin cancer. Sun exposure is a **confounding variable** — it is associated with both sunscreen use and skin cancer, creating a misleading association between the two.

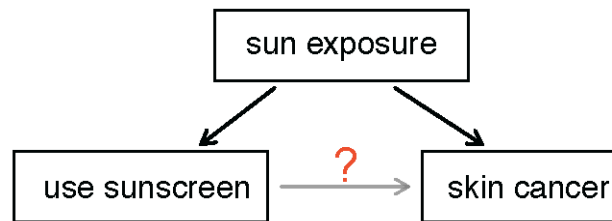


Figure 3.9: Sun exposure as a confounding variable. Sun exposure causes both increased sunscreen use and increased skin cancer risk, creating a spurious association between sunscreen use and skin cancer.

3.6 Confounding variables

A **confounding variable** is a variable that is associated with *both* the explanatory variable and the response variable. Because of this dual association, a confounding variable makes it impossible to determine whether the explanatory variable truly caused the observed response.

Confounding variables may or may not be measured as part of the study. Regardless, drawing cause-and-effect conclusions from observational studies is difficult because of the ever-present possibility of confounding.

Consider a classic example: ice cream sales and drowning rates both increase during summer months. If we looked only at the association between ice cream sales and drownings, we might (absurdly) conclude that ice cream causes drowning. The confounding variable is **temperature** (or season) — warm weather drives both increased ice cream consumption and increased swimming, which leads to more drowning incidents.

Suppose data show a negative association between homeownership rates and the percentage of housing units that are in multi-unit structures across U.S. counties. Is it reasonable to conclude that building more apartments *causes* homeownership rates to drop? Suggest a confounding variable.

Show answer

No, a causal conclusion is not warranted from observational data. A likely confounding variable is **population density**. Dense urban counties tend to have more multi-unit housing (apartments, condos) because of limited space, and they also tend to have lower homeownership rates because property values are higher. Population density is associated with both variables.

3.6.1 Prospective and retrospective studies

Observational studies come in two forms:

- A **prospective study** identifies individuals and collects information as events unfold going forward. For example, medical researchers might identify and follow a group of patients over many years to assess the influence of behavior on cancer risk. The famous Nurses' Health Study, which started in 1976 and has collected data on over 275,000 nurses, is a prospective study.
- A **retrospective study** collects data after events have already taken place — for example, by reviewing past medical records.

Some studies contain both prospective and retrospective elements. For instance, a medical study might gather information about participants' histories (retrospective) and then follow them forward in time (prospective).

3.7 Random assignment and causation

The distinction between random *sampling* and random *assignment* is crucial. They serve different purposes and affect what conclusions we can draw.

- **Random sampling** (selecting subjects randomly from a population) allows us to **generalize** our findings to the broader population.
- **Random assignment** (randomly assigning subjects to treatment groups in an experiment) allows us to draw **causal** conclusions.

Random sampling and random assignment are independent concepts. A study can have one, both, or neither. The type of randomization determines the scope of conclusions we can draw.

A company wants to know whether a new keyboard layout increases typing speed. They recruit 40 volunteers from their office, randomly assign 20 to train on the new layout and 20 to continue with the standard layout, then measure typing speed after one month.

Can the company conclude that the new layout *causes* faster typing? Can they generalize the results to all office workers?

Because the study used **random assignment** to treatment groups, the company can make a causal conclusion: any observed difference in typing speed can be attributed to the keyboard layout rather than to pre-existing differences between the groups. However, the 40 volunteers were not randomly selected from all office workers — they are a convenience sample of volunteers from one company. Therefore, the results may not generalize to all office workers.

3.8 Scope of inference

The scope of inference — what conclusions we can draw from a study — depends on how data were collected. The following table summarizes the four possible scenarios:

	Random assignment (to treatments)	No random assignment
Random sampling (from population)	Can generalize to the population and can draw causal conclusions	Can generalize to the population, but cannot draw causal conclusions
No random sampling	Cannot generalize to the population, but can draw causal conclusions (for subjects like those in the study)	Cannot generalize and cannot draw causal conclusions

		Random Assignment	
		Yes	No
Random Sampling	Yes	Generalize & Causation	Generalize only
	No	Causation only	Association only

Figure 3.10: Scope of inference. The type of sampling determines whether results generalize to the population; the type of assignment determines whether causal conclusions are possible.

Very few studies achieve both random sampling *and* random assignment. In practice, experiments typically use volunteers (not random samples from the population), so they fall in the bottom-left cell: they can establish causation, but the results may not generalize beyond the type of people who volunteered. Observational studies based on random samples (top-right cell) can generalize to the population but cannot establish cause-and-effect.

A researcher randomly selects 500 adults from a city's voter rolls and surveys them about exercise habits and depression symptoms. She finds that people who exercise more report fewer depression symptoms. What conclusions can she draw?

Show answer

Because the researcher used **random sampling** (from the voter rolls), she can generalize her findings to the city's adult voter population. However, because she did not randomly **assign** people to exercise or not exercise (this is an observational study), she cannot conclude that exercise *causes* fewer depression symptoms. There may be confounding variables — for example, people with fewer depression symptoms may be more motivated to exercise in the first place.

A pharmaceutical company recruits 200 volunteers with chronic pain and randomly assigns half to receive a new pain medication and half to receive a placebo. The group receiving the medication reports significantly less pain. What is the scope of inference?

Show answer

Because the study used **random assignment**, the company can conclude that the medication *caused* the reduction in pain (a causal conclusion). However, because the participants were volunteers rather than a random sample of all chronic pain patients, the results may not generalize to the entire population of chronic pain sufferers — only to people similar to those who volunteered.

3.9 Chapter review

3.9.1 Summary

In this chapter, we explored how data are collected and why the method of collection matters for the conclusions we can draw.

Key concepts:

- The **population** is the entire group of interest; a **sample** is a subset from which we collect data. A **census** collects data on every member of the population but is usually impractical.
- **Anecdotal evidence** — conclusions drawn from one or a few non-representative cases — is unreliable.
- **Sampling methods** include simple random sampling, stratified sampling, cluster sampling, and multistage sampling. Each has advantages depending on the population's structure and practical constraints.
- **Bias** can enter through non-random sampling (selection bias), non-response, or convenience sampling.
- In an **experiment**, researchers actively assign treatments to subjects. In an **observational study**, researchers simply observe without intervening.
- **Random assignment** to treatment groups is the key to establishing **causation**.
- **Confounding variables** are associated with both the explanatory and response variables and can create misleading associations.
- **Blinding** (single and double) and **placebos** help reduce bias in human experiments.
- The **scope of inference** depends on whether the study used random sampling (which enables generalization) and random assignment (which enables causal conclusions).

3.9.2 Key terms

Term	Definition
Population	The entire group of individuals about which we want information
Sample	A subset of the population from which we collect data
Census	An attempt to collect data from every member of the population
Population parameter	A numerical summary of the entire population
Sample statistic	A numerical summary calculated from sample data
Anecdotal evidence	Data collected in a haphazard fashion, often representing unusual cases
Simple random sample (SRS)	A sample where every case has an equal chance of selection, independently

Term	Definition
Stratified sampling	Dividing the population into strata, then taking an SRS within each
Cluster sampling	Randomly selecting entire clusters and including all cases within them
Multistage sampling	Randomly selecting clusters, then taking an SRS within each selected cluster
Bias	Systematic favoritism in data collection that makes a sample unrepresentative
Non-response bias	Bias introduced when sampled individuals do not respond
Convenience sample	A sample where easily accessible individuals are overrepresented
Experiment	A study where the researchers assign treatments to subjects
Randomized experiment	An experiment where treatments are assigned using randomization
Observational study	A study where researchers observe without assigning treatments
Confounding variable	A variable associated with both the explanatory and response variables
Treatment group	The group receiving the experimental treatment
Control group	The group not receiving the treatment (or receiving a placebo)
Placebo	A fake treatment given to the control group
Placebo effect	Improvement in patients who believe they are receiving treatment
Blind (single-blind)	Subjects do not know their treatment assignment
Double-blind	Neither subjects nor researchers know the treatment assignments
Blocking	Grouping subjects by a known variable before random assignment
Prospective study	An observational study that follows subjects forward in time
Retrospective study	An observational study that looks back at past data
Scope of inference	The extent to which study results can be generalized and/or support causal claims

3.9.3 Exercises

Exercises for this chapter are under development.

Chapter 4

Data Visualization

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

This chapter focuses on **visualizing data** using graphs and charts. Good visualizations reveal patterns, highlight unusual observations, and communicate findings clearly. We begin with displays for categorical data (bar plots and pie charts), then move to displays for numerical data (dot plots, histograms, and density plots), explore distribution shape, and introduce box plots for comparing groups. We close with principles for creating effective, honest data visualizations.

StatBench tools for this chapter: Use the Descriptive Statistics Explorer to build histograms, dot plots, and box plots interactively. Upload any CSV dataset and experiment with different bin widths and display options.

4.1 Visualizing categorical data

4.1.1 Bar plots

A **bar plot** is the most common way to display the distribution of a single categorical variable. Each category gets its own bar, and the height (or length) of the bar represents either the count or the proportion of observations in that category.

Consider a dataset of 10,000 loans from a lending platform. Each borrower's homeownership status is recorded as rent, mortgage, or own. The frequency table below summarizes the data:

Homeownership	Count
Rent	3,858
Mortgage	4,789
Own	1,353
Total	10,000

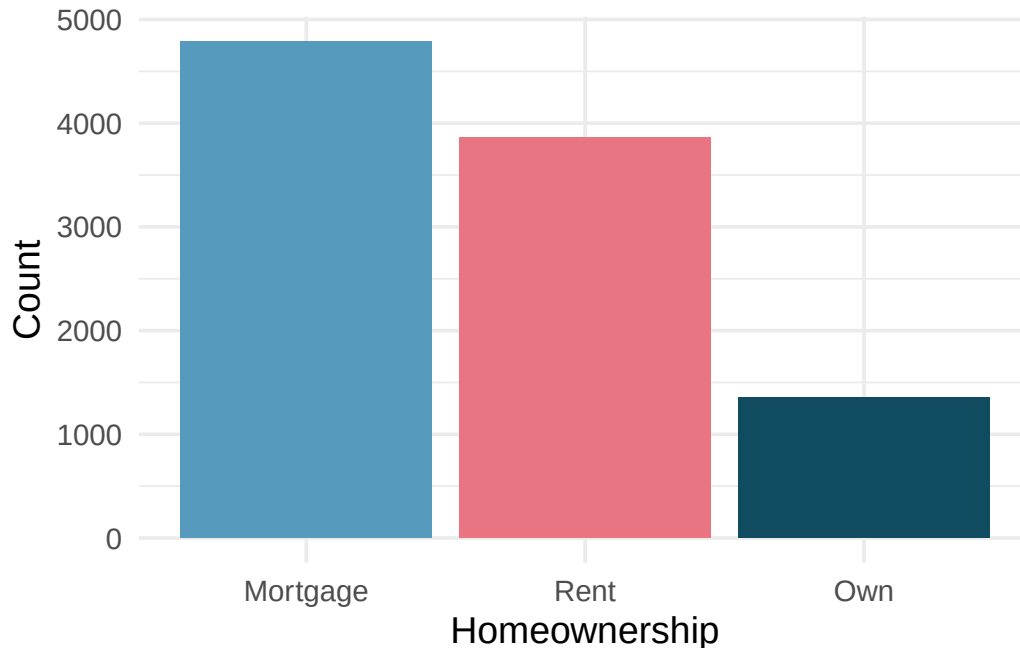


Figure 4.1: Bar plot showing counts of homeownership status. Mortgage is the tallest bar, followed by rent, then own.

A bar plot of proportions conveys the same information but rescales the vertical axis so the bars represent the fraction of observations in each category. In the homeownership example, about 47.9% of borrowers have a mortgage, 38.6% rent, and 13.5% own their home outright.

In the homeownership bar plot, which category is most common? Which is least common? About what fraction of borrowers rent?

Show answer

Mortgage is the most common category and own is the least common. About 38.6% of borrowers rent their home.

4.1.2 Pie charts

A **pie chart** divides a circle into slices, where each slice represents a category and the size of the slice corresponds to the proportion of observations in that category.

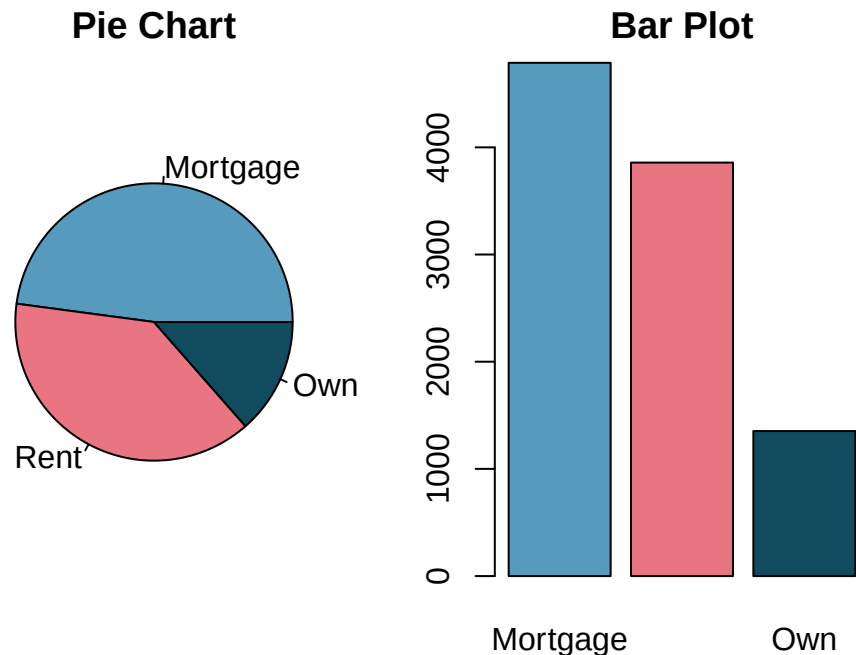


Figure 4.2: A pie chart and bar plot displaying the same homeownership data side by side.

Pie charts can give a quick high-level overview, especially when a variable has only a few levels or when each level represents a simple fraction (one-half, one-quarter, etc.). However, bar plots are almost always easier to read because our eyes are better at comparing lengths than angles or areas.

Bar plots vs. pie charts. When a categorical variable has many levels, bar plots are far superior to pie charts for making comparisons. Reserve pie charts for situations where the variable has very few categories and the goal is to show how a whole breaks into parts.

4.1.3 Waffle charts

A **waffle chart** is another way to visualize proportions. It uses a grid (often 10 by 10 = 100 squares) where each square represents 1% of the data. Squares are shaded according to the categories. Like pie charts, waffle charts work best when the number of categories is small, but they can make it easier to compare proportions that do not correspond to simple fractions.

Why might a waffle chart be easier to read than a pie chart when comparing two categories with similar proportions (say, 38% and 42%)?

Show answer

In a waffle chart, you can count the number of shaded squares for each category, making it straightforward to see the difference. In a pie chart, the angle difference between 38% and 42% is very subtle and hard to judge visually.

4.2 Visualizing numerical data

4.2.1 Dot plots

A **dot plot** is the simplest display for a single numerical variable. Each observation is represented by a dot placed above its value on a number line. When observations share the same value (or are

rounded to the same value), the dots stack vertically.

Consider the interest rates on 50 loans. In a dot plot, each of the 50 loans is a single dot.

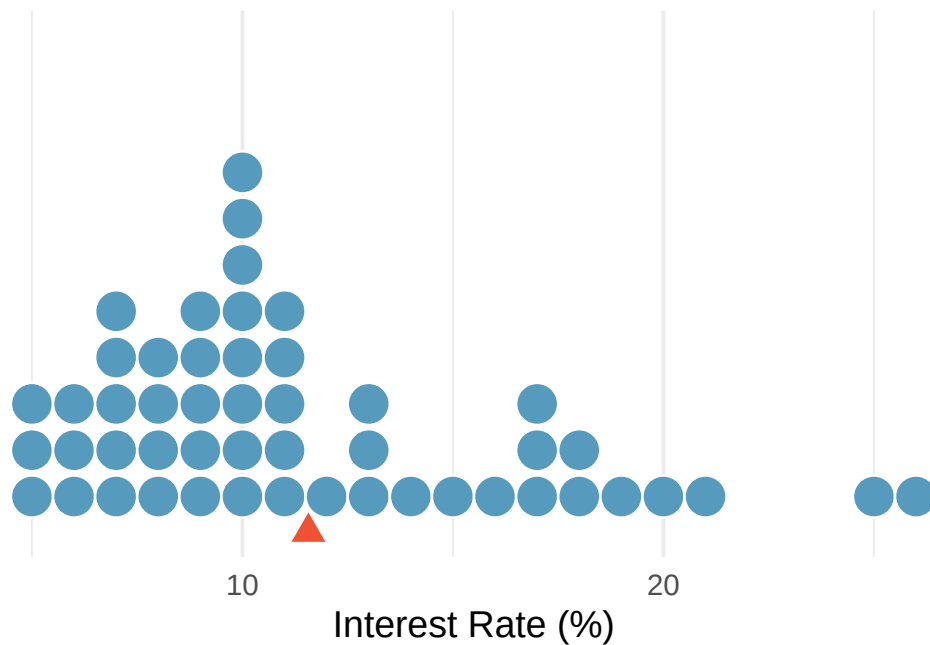


Figure 4.3: A dot plot of interest rates for 50 loans. Rates range from about 5% to 26%, with most values clustered between 5% and 15%.

Dot plots are most useful for small datasets (up to about 50-100 observations) where you want to see every individual value. For larger datasets, histograms provide a better summary.

4.2.2 Histograms

A **histogram** groups numerical data into bins (intervals) and displays the count or proportion of observations in each bin as a bar. Unlike bar plots for categorical data, histogram bars touch each other because the bins represent a continuous range of values.

To build a histogram, we:

1. Choose a bin width (e.g., 2.5 percentage points for interest rates)
2. Count how many observations fall in each bin
3. Draw bars whose heights equal the counts

Interest rate bin	Count
(5.0% - 7.5%]	11
(7.5% - 10.0%]	15
(10.0% - 12.5%]	8
(12.5% - 15.0%]	4
(15.0% - 17.5%]	5
(17.5% - 20.0%]	4
(20.0% - 22.5%]	1
(22.5% - 25.0%]	1

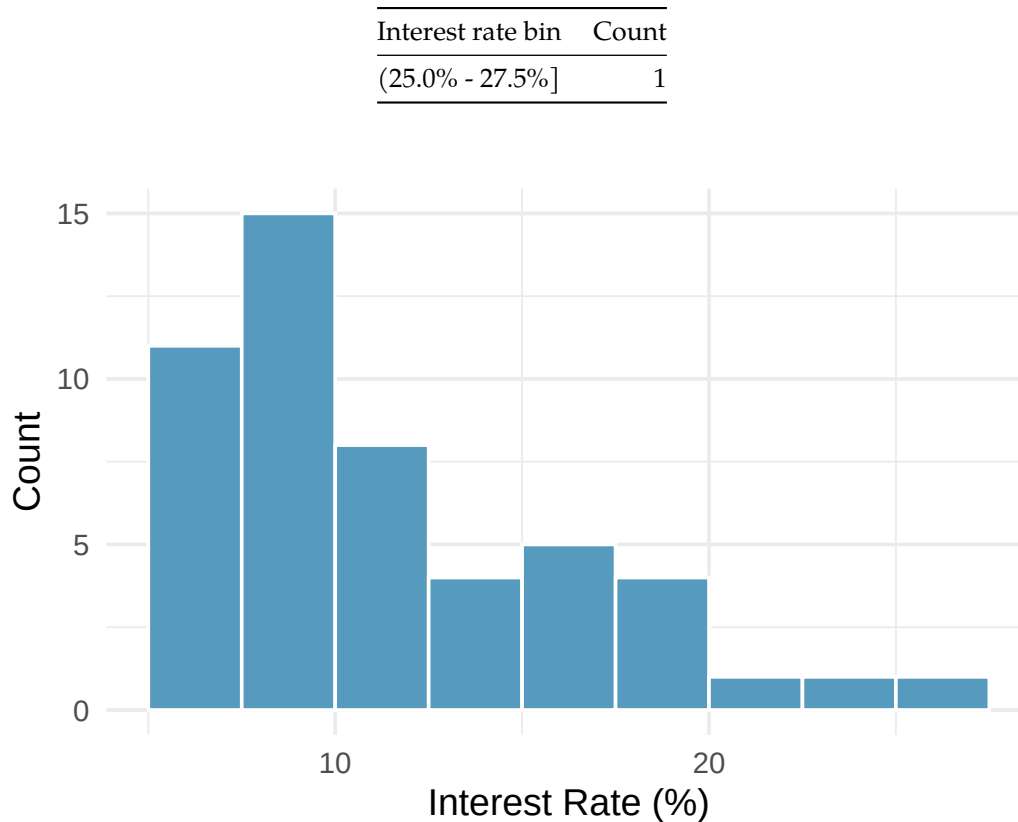


Figure 4.4: A histogram of interest rates with bin width 2.5 percentage points. The distribution is right skewed.

Histograms provide a view of the **data density**: higher bars indicate where data values are more common, and lower bars indicate where they are less common.

Choosing bin width. The bin width affects the appearance of a histogram. Too few bins (wide bins) can obscure important features; too many bins (narrow bins) can create a jagged display that is hard to interpret. There is no single “correct” bin width – experiment with different choices and look for the display that best reveals the shape of the data.

4.2.3 Density plots

A **density plot** is a smoothed version of a histogram. Instead of bars, it draws a smooth curve that estimates the distribution of the data. The area under the entire curve equals 1.

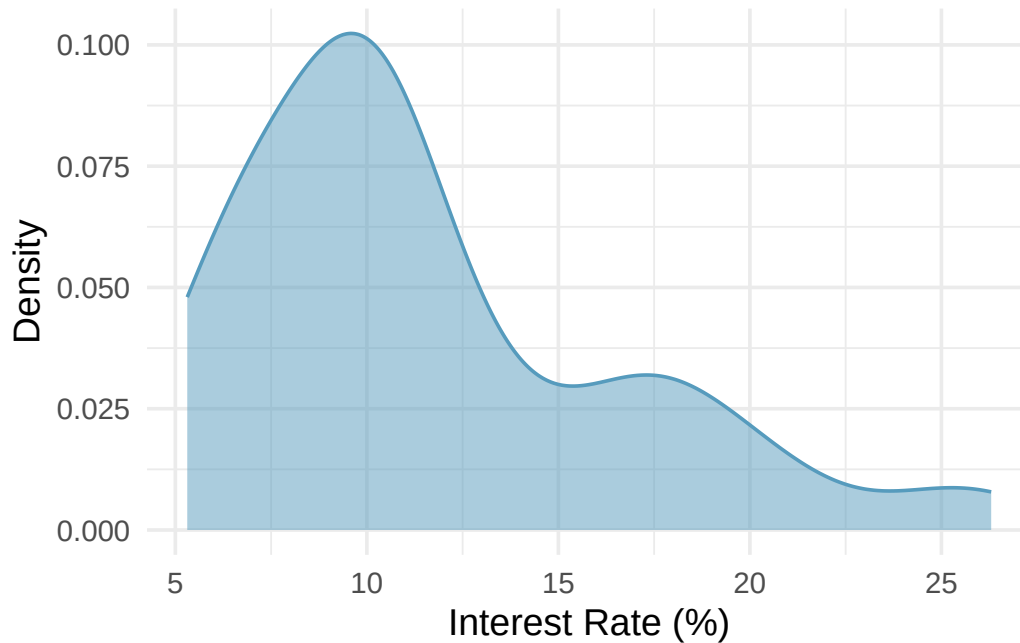


Figure 4.5: A density plot of interest rates, showing the same right-skewed shape as the histogram.

Density plots are particularly useful for comparing the shapes of distributions because overlapping curves are easier to read than overlapping histogram bars.

4.3 Histograms and distribution shape

Histograms are especially useful for identifying the **shape** of a distribution. We describe shape using two characteristics: symmetry and modality.

4.3.1 Skewness

A distribution is **symmetric** if the left and right sides are approximately mirror images of each other.

A distribution is **right skewed** (or skewed to the right) if the right **tail** — the thinner end of the distribution — is longer. Most observations are on the left, with a few large values pulling the tail to the right.

A distribution is **left skewed** (or skewed to the left) if the left tail is longer. Most observations are on the right, with a few small values pulling the tail to the left.

Identifying skew. The direction of skew refers to the direction of the *long tail*, not the direction where most data are concentrated. A distribution with a long tail stretching to the right is right skewed, even though the bulk of the data is on the left side.

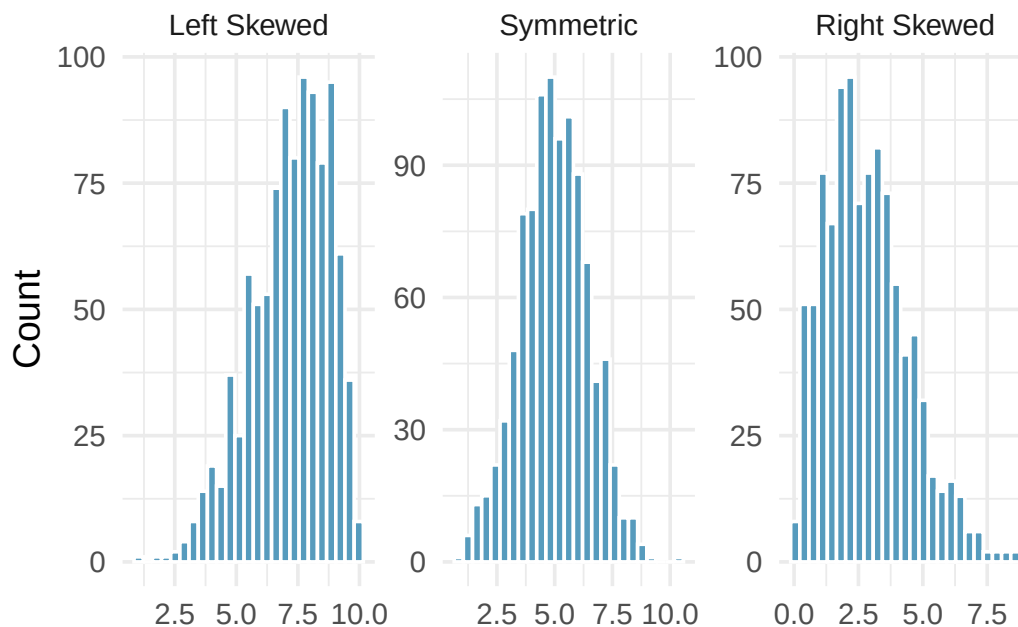


Figure 4.6: Three distributions illustrating left skewed, symmetric, and right skewed shapes.

The histogram of interest rates from the loan dataset shows that most loans have rates between 5% and 15%, with only a few loans above 20%. How would you describe the shape?

The distribution is right skewed because there is a long tail extending toward the higher interest rates. The bulk of the data is concentrated on the left side of the histogram.

Consider the distribution of home prices in a city. Would you expect this distribution to be symmetric, left skewed, or right skewed? Explain your reasoning.

Show answer

Home prices tend to be right skewed. Most homes fall in a moderate price range, but a small number of very expensive homes create a long right tail. This is common with income, wealth, and price data.

4.3.2 Modality

A **mode** is a prominent peak in a distribution. A distribution with a single prominent peak is called **unimodal**. A distribution with two prominent peaks is **bimodal**, and a distribution with more than two prominent peaks is **multimodal**.

A distribution with no prominent peaks — where all values are roughly equally likely — is called **uniform**.

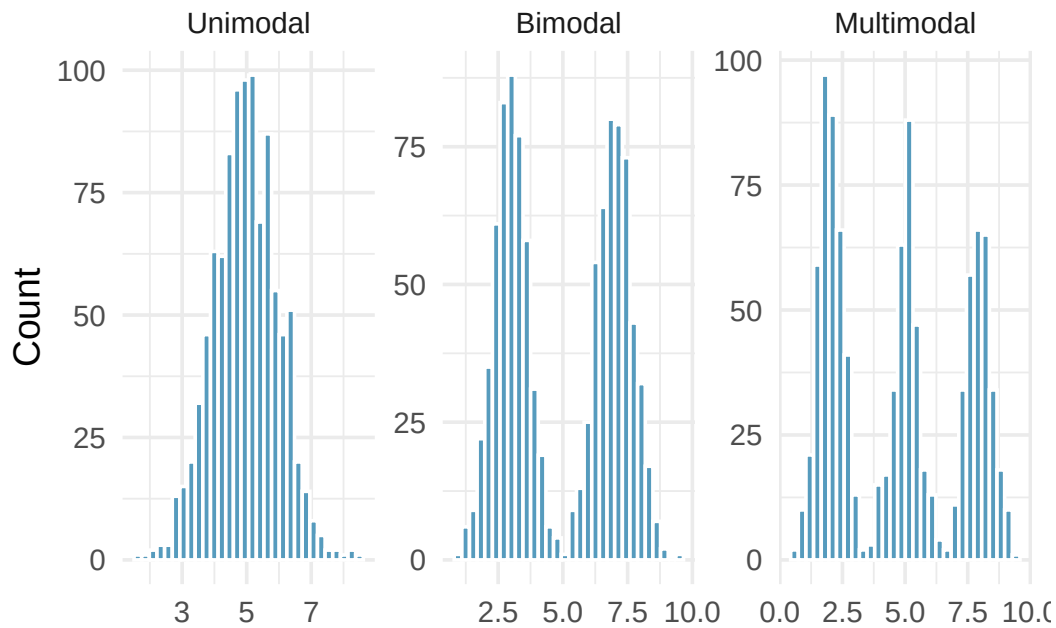


Figure 4.7: Three distributions: unimodal, bimodal, and multimodal.

Height measurements were collected from a group of young students and adult teachers at an elementary school. How many modes would you expect in a histogram of these heights?

You would likely expect two modes (bimodal): one centered on the typical height of children and another centered on the typical height of adults.

A histogram of interest rates shows one prominent peak near 7%. There is a very small bump near 20%, but it differs from its neighboring bins by only one or two observations. Would you call this distribution unimodal or bimodal?

Show answer

Unimodal. The small bump near 20% is not a prominent peak — it differs only slightly from its neighbors. We look for prominent peaks when identifying modes.

4.3.3 Describing a distribution

When describing the distribution of a numerical variable, a complete description should mention:

1. **Shape:** Is the distribution symmetric or skewed? How many modes does it have?
2. **Center:** Where is the “typical” value? (We will define this precisely in Chapter 4.)
3. **Spread:** How variable are the data? Are values tightly clustered or widely dispersed?
4. **Unusual features:** Are there outliers, gaps, or clusters?

Always describe these features **in context** — using the units and variable name rather than abstract terms.

Describe the distribution of interest rates from the loan dataset.

The distribution of interest rates is unimodal and right skewed. Most rates fall between 5% and 15%, with a center around 10-11%. The rates range from about 5% to 26%, and there are a few unusually high rates above 20%.

4.4 Box plots

A **box plot** (also called a box-and-whisker plot) provides a compact summary of a distribution using five numbers: the minimum, first quartile (Q_1), median, third quartile (Q_3), and maximum — with special handling for outliers.

4.4.1 Constructing a box plot

The components of a box plot are:

1. **The box:** Spans from Q_1 (the 25th percentile) to Q_3 (the 75th percentile). This box contains the middle 50% of the data. The length of the box is the **interquartile range** ($IQR = Q_3 - Q_1$).
2. **The median line:** A line inside the box marks the median (the 50th percentile), which divides the data in half.
3. **The whiskers:** Lines extending from the box toward the smallest and largest observations that are *not* outliers. Specifically, the whiskers extend to the most extreme data points within $1.5 \times IQR$ of the box.
4. **Outlier points:** Any observations beyond $1.5 \times IQR$ from the edges of the box are plotted individually as dots. These are potential outliers.

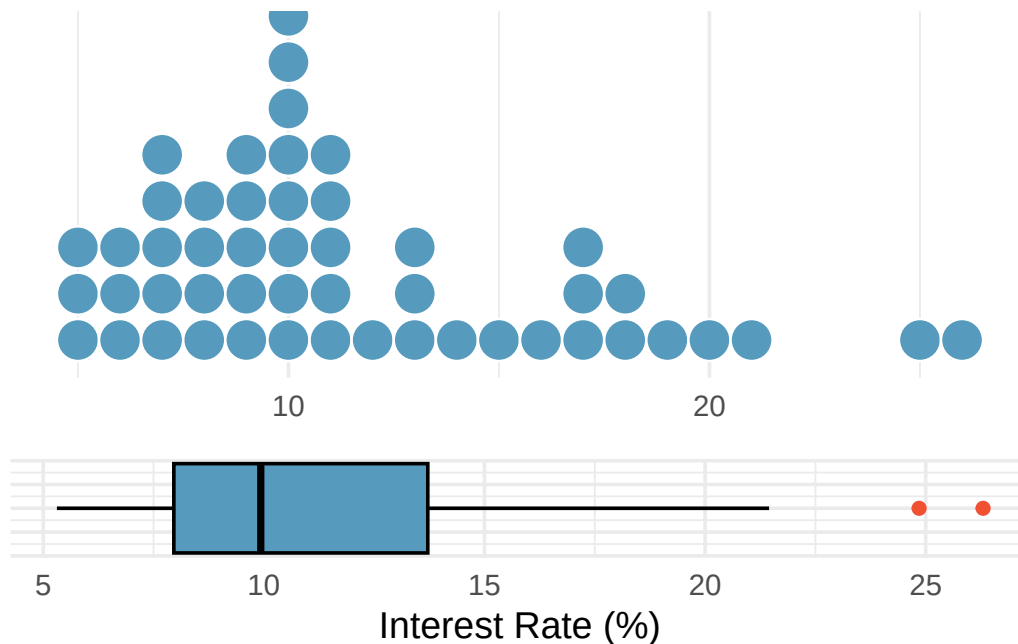


Figure 4.8: A dot plot and box plot of the same interest rate data, shown one above the other for comparison.

The **median** is the value in the middle when data are arranged from smallest to largest. If there are an even number of observations, the median is the average of the two middle values.

The **first quartile** (Q_1) is the 25th percentile — 25% of the data fall below this value. The **third quartile** (Q_3) is the 75th percentile — 75% of the data fall below this value.

The **interquartile range** (IQR) is the range of the middle 50% of the data: $\text{IQR} = Q_3 - Q_1$.

4.4.2 Interpreting a box plot

A box plot of interest rates shows: the left edge of the box at about 8%, the median line at about 10%, and the right edge of the box at about 14%. Two dots appear beyond the right whisker at about 25% and 26%. What can we learn from this display?

The middle 50% of interest rates fall between about 8% and 14% (the IQR is about 6 percentage points). The median rate is about 10%. The right whisker extends to about 20%, and the two dots beyond the whisker at 25% and 26% indicate unusually high interest rates. The box plot also reveals right skew, since the median is closer to the left edge of the box and the right whisker is longer than the left whisker.

What percentage of observations fall between Q_1 and the median? What percentage fall between the median and Q_3 ?

Show answer

25% of the data fall between Q_1 and the median, and another 25% fall between the median and Q_3 . Together, the middle 50% of the data lie within the box.

4.4.3 Comparing groups with side-by-side box plots

Side-by-side box plots display box plots for different groups on the same scale, making it easy to compare centers, spreads, and skewness across groups.

Consider comparing median household income for US counties, split by whether the county gained population or lost population from 2010 to 2017:

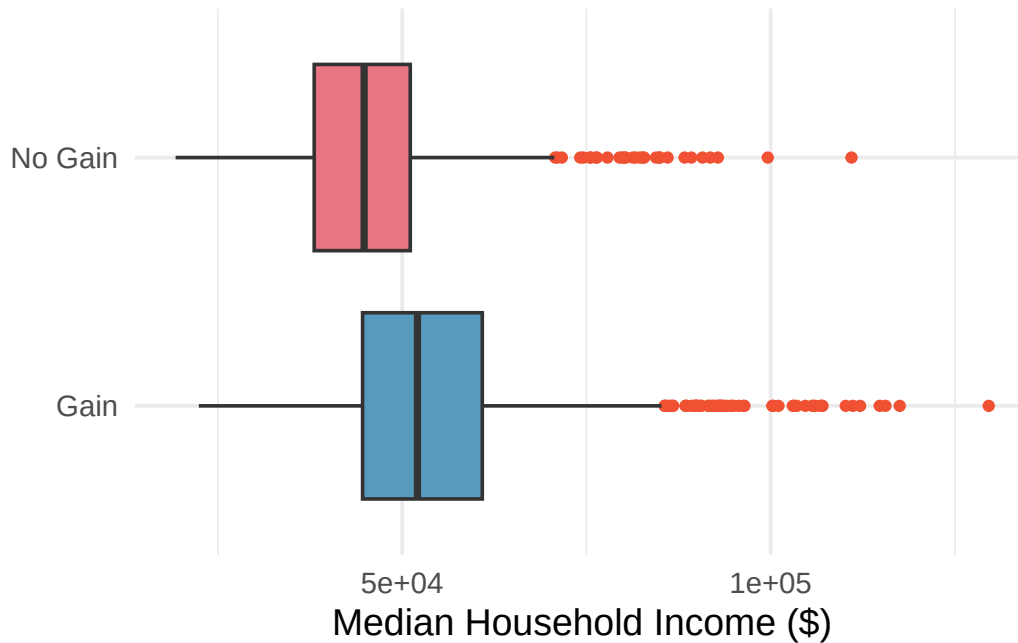


Figure 4.9: Side-by-side box plots of median household income for counties with population gain versus no gain.

Using the side-by-side box plots above, compare the income distributions for counties with and without population gain. Discuss center, spread, and shape.

Show answer

Counties with population gains tend to have higher median household incomes (about \$45,000 vs. \$40,000). The IQR is also somewhat larger for the gain group, indicating more variability. Both distributions appear right skewed, with outliers on the high end.

When are side-by-side box plots particularly useful? When might histograms be better?

Side-by-side box plots are excellent for comparing centers (medians) and spreads (IQRs) across groups in a compact display. However, they cannot reveal details about shape such as bimodality. Histograms or density plots are better when you want to see the full shape of each distribution, including the number of modes and the exact nature of the skew.

4.5 Effective data visualization

Graphs can powerfully communicate ideas directly and quickly. However, there are times when a visualization conveys a message that is inaccurate or misleading. This section provides guiding principles for creating clear, honest, and effective visualizations.

4.5.1 Keep it simple

Colors should be used purposefully — to group items or differentiate levels in meaningful ways. Colors used only for decoration can be distracting.

Consider a company that has summarized its costs into five categories. A three-dimensional pie chart makes it hard to compare the sizes of the slices, and the unnecessary 3D effect distorts the proportions.

A simple bar plot conveys the same information much more clearly.

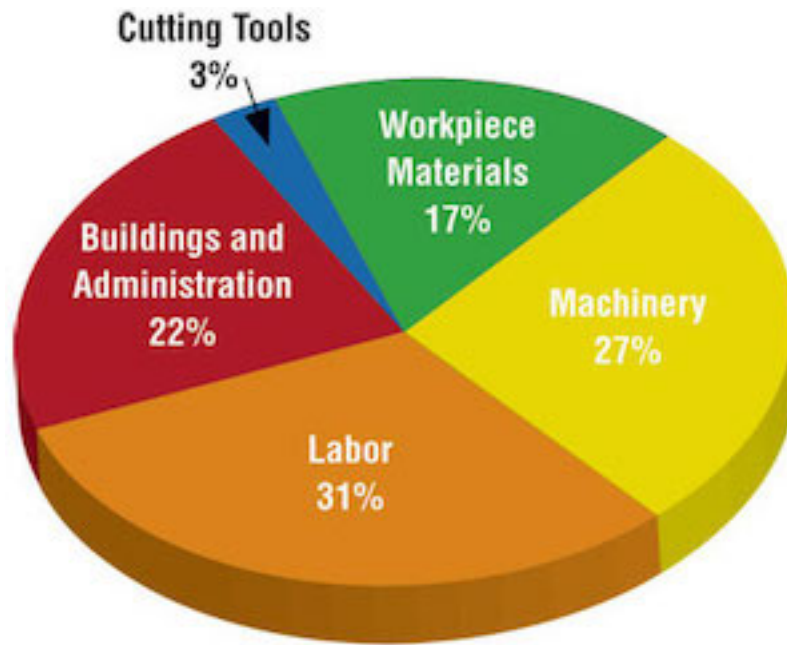


Figure 4.10: A 3D pie chart next to a bar plot, both showing the same cost breakdown. The bar plot is much easier to read.

4.5.2 Use color to draw attention

An important principle is to use color **to draw attention** to the most important part of your graphic. Default rainbow coloring often adds no meaning and can confuse readers.

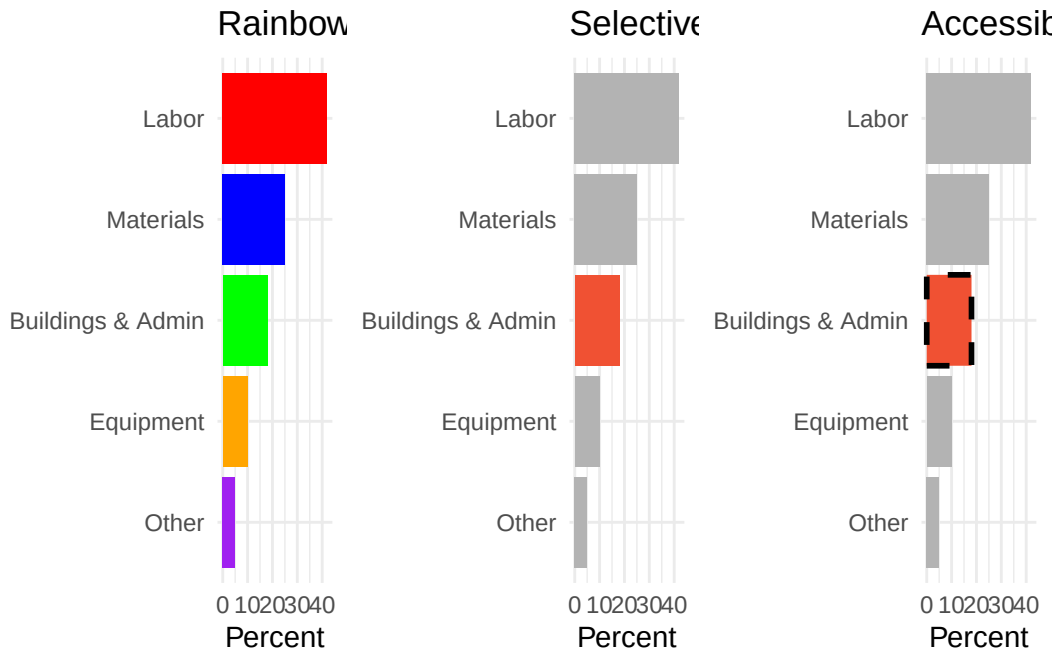


Figure 4.11: Three versions of the same bar chart. The first uses distracting rainbow colors. The second highlights one important bar in red against gray bars. The third adds both color and a dashed border for accessibility.

Accessibility matters. Not everyone perceives color the same way. When using color to draw attention, also consider adding a secondary visual cue (such as a pattern, line style, or border) so that the highlighted feature is distinguishable even for people with color vision deficiency.

4.5.3 Tell a story

Effective graphs often include **annotations** — text, lines, or labels that provide context beyond the raw data. For example, adding a note that “July is the start of the fiscal year” to a time series of monthly hiring can explain why there is a spike every July.

4.5.4 Order matters

The arrangement of categories in a bar plot can help or hinder understanding. Alphabetical order is rarely the most informative choice. Consider ordering bars by frequency (tallest to shortest) or by a meaningful sequence (such as the order the options appeared in a survey question).

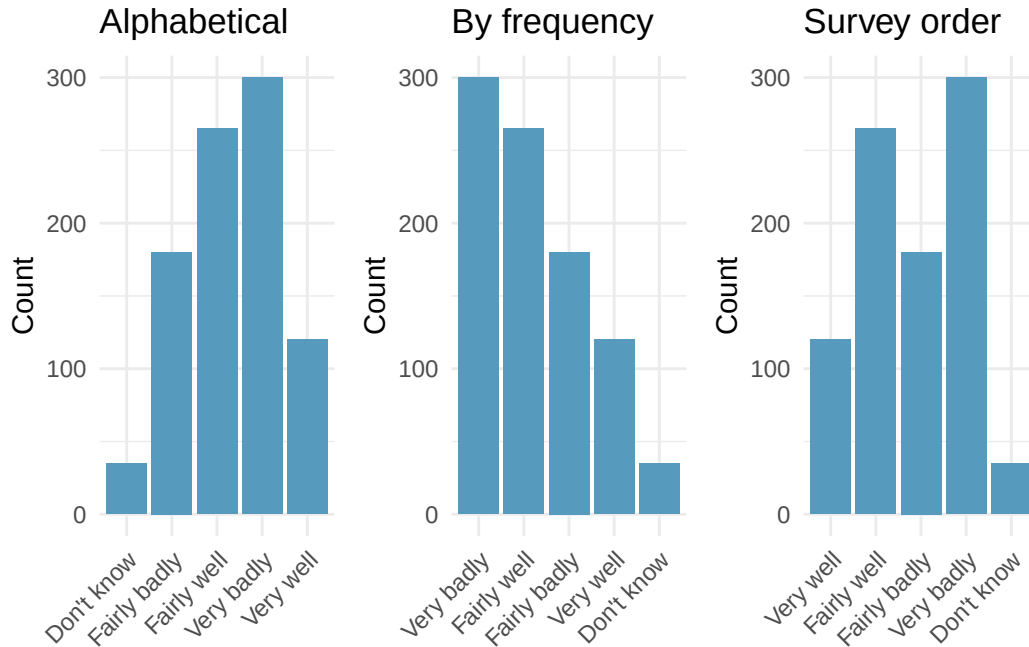


Figure 4.12: Three versions of a bar chart showing survey responses ordered alphabetically, by frequency, and by the original survey order.

4.5.5 Make labels readable

When category labels are long, consider using horizontal bars (where labels appear along the vertical axis with plenty of space) rather than vertical bars (where labels must be angled or truncated).

4.5.6 Pick a purpose

Every graphical decision should be made with a purpose. Before choosing a plot type, ask:

- What is the key comparison I want the reader to make?
- Which plot type makes that comparison easiest?
- Am I showing proportions, counts, distributions, or relationships?

A survey asks residents of five regions about their opinion on a policy, with responses ranging from “Very well” to “Very badly.” You want to compare how opinions differ across regions. What visualization would you choose?

A standardized (filled) bar plot would be useful if the main comparison is about proportions within each region. A dodged bar plot would work if comparing raw counts across regions matters. A faceted bar plot — one panel per region — would allow the most detailed comparison. The best choice depends on which comparison is most important to the story you are telling.

4.5.7 Select meaningful colors

Default or rainbow color schemes are not always the best choice. Perceptually uniform color scales (such as the viridis or cividis scales) are designed to be distinguishable by people with a variety of color vision abilities and to be readable when printed in grayscale. For ordinal data, a sequential color scale (light to dark) communicates the ordering effectively.

Summary of data visualization principles:

1. **Keep it simple** — avoid unnecessary 3D effects, decorative colors, and chart junk
2. **Use color with purpose** — to group, differentiate, or draw attention
3. **Include context** — titles, axis labels, annotations, and data sources
4. **Choose appropriate ordering** — by frequency, natural order, or meaningful sequence
5. **Make labels readable** — flip axes if labels are long; avoid angled text
6. **Consider your audience** — use accessible color schemes and secondary visual cues
7. **Match the plot to the question** — each plot type has strengths; choose deliberately

4.6 Chapter review

4.6.1 Summary

This chapter introduced the fundamental tools for visualizing data. For categorical data, bar plots are the workhorse visualization, with pie charts reserved for simple cases. For numerical data, dot plots reveal individual values, histograms show the shape and density of a distribution, and density plots provide smoothed alternatives. Box plots are powerful tools for summarizing distributions and comparing groups. Distribution shape can be described using symmetry (left skewed, symmetric, right skewed) and modality (unimodal, bimodal, multimodal). Effective visualizations follow principles of simplicity, purposeful use of color, meaningful ordering, and accessibility.

4.6.2 Key terms

- Bar plot, pie chart, waffle chart
- Dot plot, histogram, density plot, box plot
- Data density, bin width
- Symmetric, left skewed, right skewed, tail
- Unimodal, bimodal, multimodal, uniform
- Median, quartiles (Q_1 , Q_3), interquartile range (IQR)
- Whiskers, outlier
- Side-by-side box plot
- Stacked bar plot, standardized (filled) bar plot, dodged bar plot

4.6.3 Exercises

Exercises for this chapter are under development.

Chapter 5

Describing Distributions

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In Chapter 3 we learned to *visualize* distributions. Now we develop the numerical tools to *describe* them. A histogram or box plot gives a picture, but we also need precise numbers to summarize center, spread, and unusual features. This chapter covers the mean and median (measures of center), the standard deviation and IQR (measures of spread), and the concept of robust statistics. We close with the empirical rule, which connects the standard deviation to the shape of a distribution.

StatBench tools for this chapter: Use the Descriptive Statistics Explorer to compute means, medians, standard deviations, IQR, and quartiles for any dataset. Experiment with adding or removing outliers and watch how the statistics change.

5.1 Measures of center

When we look at a distribution, one of the first questions is: *where is the center?* Two statistics give us an answer.

5.1.1 The mean

The **mean** (also called the **average**) is computed by adding up all the observed values and dividing by the number of observations. If we denote our observations as $x_1, x_2, \dots, x_n$, the sample mean is:

Mean.

The sample mean can be calculated as the sum of the observed values divided by the number of observations:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

The notation $\bar{x}$ (read “x-bar”) represents the sample mean. The letter x is a placeholder for the variable of interest, and the bar indicates we are computing an average.

Consider the interest rates (in percent) for 50 loans. After adding up all 50 values and dividing by 50, the sample mean is $\bar{x} = 11.57\%$. What does this number tell us?

The mean interest rate of 11.57% represents the “balancing point” of the distribution. If we think of the dot plot as a seesaw, placing a fulcrum at 11.57% would balance it. The mean gives us a single number to represent the center of the interest rate distribution.

It’s useful to think of the mean as the balancing point of a distribution. On a dot plot, the mean is the position where the data would balance if placed on a seesaw.

In the formula for the mean, what does x_1 correspond to? And x_2 ? What does x_i represent in general?

Show answer

x_1 is the value of the variable for the first observation in the dataset, x_2 is the value for the second observation, and x_i is the value for the i th observation. For example, if the variable is interest rate, x_4 is the interest rate of the fourth loan.

The population mean is denoted by the Greek letter μ (pronounced “mew”). We often cannot measure μ directly because we rarely have access to every member of the population. Instead, we estimate μ using the sample mean $\bar{x}$.

Suppose we want to compute the average income per person in the US. We might think to take the mean of per capita incomes across all 3,142 counties. What is wrong with this approach?

Each county represents a different number of people. If we simply average the county-level incomes, we treat a county with 5,000 residents the same as one with 5,000,000 residents. Instead, we should compute the total income for each county, add up all counties’ totals, and divide by the total population. This is called a **weighted mean**. Failing to weight by population would underestimate the true average because larger, higher-income urban counties would be underrepresented.

5.1.2 The median

The **median** is the value in the middle of the ordered data. It divides the dataset in half: 50% of the observations fall below the median and 50% fall above it.

Median: the number in the middle.

If the data are ordered from smallest to largest, the **median** is the observation right in the middle. If there are an even number of observations, the median is the average of the two middle values.

Consider the interest rates for 50 loans arranged in order from smallest to largest. Since there are 50 observations (an even number), the median is the average of the 25th and 26th values. Both of these values happen to be 9.93%, so the median is 9.93%.

How does this compare to the mean of 11.57%?

The median (9.93%) is lower than the mean (11.57%). This makes sense because the distribution is right skewed — a few loans with very high interest rates pull the mean upward. The median is less affected by these extreme values.

For a dataset with 7 observations arranged in order, which observation is the median?

Show answer

The 4th observation. With an odd number of observations, the median is the single middle value. There are 3 observations below it and 3 above it.

5.1.3 The mode

The **mode** is the most frequently occurring value in a dataset. While useful for categorical data (e.g., the most common eye color), the mode is less useful for continuous numerical data because many values may occur only once.

In practice, when we refer to the “mode” of a numerical distribution, we usually mean the location of a prominent *peak* in a histogram or density plot (as discussed in Section 4.3), rather than a single repeated value.

5.1.4 When to use the mean vs. the median

The mean and median measure different things. The choice between them depends on the context:

A company wants to understand its profit per customer. Should it use the mean or the median?

It depends on the question:

- If the concern is overall profitability (total profit divided by number of customers), the **mean** is the right statistic. A company could have a positive median profit per customer and still be unprofitable if a few customers generate large losses.
- If the concern is understanding what the *typical* customer experience looks like, the **median** tells you more about the profit from a representative customer.

Mean vs. median: which to choose?

- The **mean** is best when you care about the *total* or *aggregate* (e.g., total revenue, total time spent).
- The **median** is best when you care about the *typical* observation, especially when the distribution is skewed or contains outliers.
- For symmetric distributions without extreme outliers, the mean and median are approximately equal and either is appropriate.

The distribution of home prices in a city is strongly right skewed. If you wanted to describe the price of a typical home, should you report the mean or the median? Why?

Show answer

The median. In a right-skewed distribution, a few very expensive homes pull the mean upward, making it unrepresentative of a typical home price. The median is not affected by these extreme values and better represents what a typical buyer would encounter.

5.2 Measures of spread

Knowing the center of a distribution is important, but it tells only part of the story. Two distributions can have the same center but look very different if one is tightly clustered and the other is widely dispersed. We need measures of **variability** (also called spread or dispersion).

5.2.1 Range

The simplest measure of spread is the **range**: the difference between the maximum and minimum values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

While easy to compute, the range has a major weakness: it depends entirely on the two most extreme observations. A single outlier can dramatically increase the range, giving a misleading impression of the overall variability. For this reason, we usually prefer the IQR or standard deviation.

5.2.2 The interquartile range (IQR)

The **interquartile range** measures the spread of the middle 50% of the data:

Interquartile range (IQR).

The IQR is the length of the box in a box plot. It is computed as:

$$\text{IQR} = Q_3 - Q_1$$

where Q_1 is the first quartile (25th percentile) and Q_3 is the third quartile (75th percentile).

The IQR tells us how spread out the middle half of the data is. A larger IQR indicates more variability among the “typical” observations.

For the 50-loan interest rate data, $Q_1 \approx 8\%$ and $Q_3 \approx 14\%$. What is the IQR, and what does it tell us?

$\text{IQR} = 14\% - 8\% = 6$ percentage points. This means the middle 50% of loans have interest rates spanning a range of 6 percentage points. Half of all loans have rates between about 8% and 14%.

5.2.3 Variance and standard deviation

The **standard deviation** is the most commonly used measure of spread. It roughly describes how far the typical observation is from the mean.

We start with the concept of a **deviation**: the distance of an observation from the mean.

$$\text{deviation of } x_i = x_i - \bar{x}$$

Some deviations are positive (observations above the mean) and some are negative (observations below the mean). If we simply averaged the deviations, the positives and negatives would cancel out, giving us zero. To avoid this, we *square* each deviation before averaging.

The **variance** (denoted s^2) is the average of the squared deviations, using $n - 1$ in the denominator:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}$$

The **standard deviation** (denoted s) is the square root of the variance:

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

We divide by $n - 1$ rather than n when computing a sample’s variance. The mathematical reasons are beyond this course, but the practical effect is that this makes the statistic slightly more reliable as an estimate of the population variance.

Why do we square the deviations instead of just taking the absolute value? What are the two consequences of squaring?

Show answer

Squaring the deviations does two things: (1) it eliminates negative signs (since any number squared is non-negative), and (2) it makes large deviations relatively *much* larger than small ones. An observation that is 10 units from the mean contributes 100 to the sum of squared deviations, while one that is 2 units away contributes only 4. This means the variance and standard deviation are especially sensitive to outliers.

For the 50-loan interest rate data, the variance is $s^2 = 25.52$ and the standard deviation is $s = \sqrt{25.52} = 5.05$ percentage points.

What does a standard deviation of 5.05 percentage points mean in context?

The standard deviation tells us that interest rates typically deviate about 5 percentage points from the mean (11.57%). Most loans have rates that are within one standard deviation of the mean — roughly between 6.5% and 16.6%.

Like the mean, the population values for variance and standard deviation have special symbols: σ^2 for the variance and σ for the standard deviation. The Greek letter σ is pronounced “sigma.”

Variance and standard deviation.

The variance is the average squared distance from the mean. The standard deviation is the square root of the variance. The standard deviation is useful when considering how far the data are distributed from the mean.

The standard deviation represents the *typical* deviation of observations from the mean. Often about 68% of the data will be within one standard deviation of the mean and about 95% will be within two standard deviations. However, these percentages are not strict rules.

5.3 Describing shape

A complete description of a distribution should address three things: **shape**, **center**, and **spread**. We introduced shape terminology in Section 4.3. Here we connect shape to our measures of center and spread.

5.3.1 How skewness affects the mean and median

In a **symmetric** distribution, the mean and median are approximately equal.

In a **right-skewed** distribution, the mean is pulled toward the right tail and is larger than the median.

In a **left-skewed** distribution, the mean is pulled toward the left tail and is smaller than the median.

Relationship between mean, median, and skewness.

- Right skewed: mean $>$ median
- Symmetric: mean $\approx$ median
- Left skewed: mean $<$ median

The direction of skew pulls the mean toward the tail. The median is more resistant to this pull.

A complete description of the interest rate distribution from the loan dataset:

The distribution of interest rates is **unimodal** and **right skewed**. The **center** is around 10-12% (median = 9.93%, mean = 11.57%). The **spread** is moderate: the standard deviation is 5.05 percentage points and the IQR is about 6 percentage points. There are a few **unusually high** interest rates above 20%, which contribute to the right skew and the gap between the mean and median.

5.3.2 Writing a good description

A good description of a distribution should always:

1. Name the **shape**: symmetric or skewed (and direction), and the number of modes
2. Give the **center**: report the mean and/or median with units
3. Give the **spread**: report the standard deviation and/or IQR with units
4. Note any **unusual features**: outliers, gaps, clusters
5. Use **context**: refer to the actual variable and its units, not just abstract numbers

Example: “The distribution of commute times is unimodal and right skewed, with a median of 22 minutes and an IQR of 15 minutes. A few commuters have commute times exceeding 90 minutes.”

A description says: “The mean is 15 and the standard deviation is 3.” What is missing from this description?

Show answer

The description lacks context (what variable? what units?), shape information (symmetric? skewed? unimodal?), and any mention of unusual features. A better version might be: “The distribution of wait times (in minutes) is approximately symmetric and unimodal, with a mean of 15 minutes and a standard deviation of 3 minutes. There are no obvious outliers.”

5.4 Outliers and robust statistics

5.4.1 Identifying outliers

An **outlier** is an observation that appears extreme relative to the rest of the data.

Examining data for outliers serves several purposes:

- Identifying strong skew in the distribution
- Identifying possible data collection or data entry errors
- Providing insight into interesting or unusual properties of the data

Not all outliers are errors, and not all errors are outliers. Some datasets naturally have long tails and outlying points are legitimate observations. **Never remove an outlier without a substantive reason** (e.g., confirmed data entry error). Blindly removing outliers can lead to misleading conclusions.

A common rule for identifying outliers in the context of box plots is the **1.5 IQR rule**: any observation more than $1.5 \times \text{IQR}$ below Q_1 or above Q_3 is flagged as a potential outlier. This is the rule used to determine which points appear as dots beyond the whiskers of a box plot.

5.4.2 Robust statistics

A statistic is called **robust** (or **resistant**) if extreme observations have little effect on its value.

Consider what happens when we change a single extreme observation in the interest rate data. The table below shows how the median, IQR, mean, and standard deviation respond when the most extreme value (26.3%) is moved:

Scenario	Median	IQR	Mean	SD
Original data	9.93	5.75	11.57	5.05
Move 26.3% to 15%	9.93	5.75	11.34	4.61
Move 26.3% to 35%	9.93	5.75	11.74	5.68

In the table above, the median and IQR are unchanged across all three scenarios, but the mean and standard deviation shift. Why?

The median and IQR are determined by the values near the middle of the ordered data (Q_1 , the median, and Q_3). Moving a single extreme value does not change the positions of these middle values. The mean and standard deviation, however, use every data value in their calculations. The mean adds up all values, so changing one extreme value changes the sum. The standard deviation squares deviations from the mean, so extreme values have an outsized effect.

Robust vs. non-robust statistics.

Robust (resistant to outliers)	Not robust (sensitive to outliers)
Median	Mean
IQR	Standard deviation
	Range

When a distribution is skewed or contains outliers, robust statistics (median and IQR) often give a more representative summary of the “typical” observation. When a distribution is roughly symmetric with no extreme outliers, the mean and standard deviation work well and carry additional mathematical advantages.

The distribution of salaries at a company is right skewed because a few executives earn much more than most employees. Which pair of summary statistics would better represent the “typical” salary and its variability: (mean, standard deviation) or (median, IQR)?

Show answer

The median and IQR. In a right-skewed distribution, the mean is pulled upward by the high executive salaries, making it unrepresentative of the typical employee. The median is not affected by these extreme values, and the IQR captures the spread of the middle 50% of salaries.

The distribution of loan amounts is right skewed. If you wanted to understand the typical loan size, should you be more interested in the mean or the median?

Show answer

If the goal is to understand what a typical individual loan looks like, the median is probably more useful. However, if the goal is to understand something that scales — such as the total amount of money needed on hand to fund 1,000 loans — then the mean would be more useful.

5.5 The empirical rule and Chebyshev’s theorem

5.5.1 The empirical rule (68-95-99.7 rule)

For distributions that are approximately **bell-shaped** (symmetric and unimodal), the standard deviation provides a useful ruler:

The empirical rule.

For a roughly bell-shaped distribution:

- About **68%** of the data fall within 1 standard deviation of the mean: $(\bar{x} - s, \bar{x} + s)$
- About **95%** of the data fall within 2 standard deviations of the mean: $(\bar{x} - 2s, \bar{x} + 2s)$
- About **99.7%** of the data fall within 3 standard deviations of the mean: $(\bar{x} - 3s, \bar{x} + 3s)$

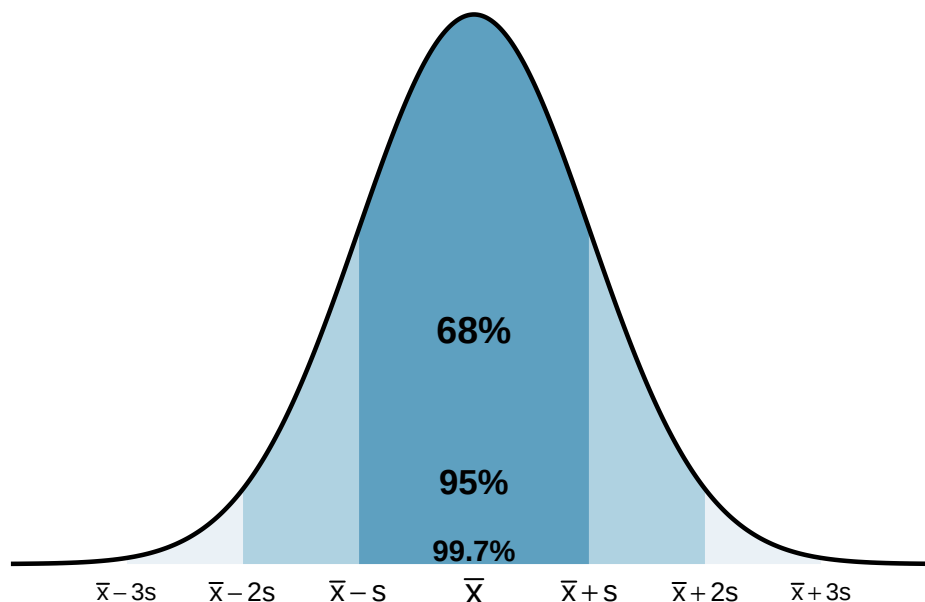


Figure 5.1: A bell-shaped distribution with shaded regions showing 68%, 95%, and 99.7% of the data within 1, 2, and 3 standard deviations of the mean, respectively.

The heights of adult women in the US are approximately bell-shaped with a mean of 64.5 inches and a standard deviation of 2.5 inches. Use the empirical rule to estimate the range of heights for the middle 95% of women.

By the empirical rule, about 95% of the data fall within 2 standard deviations of the mean:

$$64.5 - 2(2.5) = 59.5 \text{ inches} \quad \text{to} \quad 64.5 + 2(2.5) = 69.5 \text{ inches}$$

So about 95% of adult women in the US are between 59.5 inches (4'11.5") and 69.5 inches (5'9.5") tall.

Using the same height distribution (mean = 64.5 inches, SD = 2.5 inches), what percentage of women are shorter than 59.5 inches?

Show answer

By the empirical rule, about 95% of women are between 59.5 and 69.5 inches. That means about 5% are outside this range. Since the distribution is symmetric, about half of that 5% (i.e., about 2.5%) are below 59.5 inches.

The empirical rule is only an approximation, and it works best for distributions that are roughly bell-shaped. For skewed or multimodal distributions, the percentages can be quite different. For example,

in a right-skewed distribution, more than 68% of the data may fall within one standard deviation of the mean because the bulk of the data is concentrated on one side.

Three distributions all have the same mean (0) and the same standard deviation (1), but one is bimodal, one is bell-shaped, and one is right skewed. Would the empirical rule work well for all three? Why or why not?

Show answer

No. The empirical rule works well only for the bell-shaped distribution. The bimodal distribution would have more data near the edges (near the two peaks) and less in the middle. The right-skewed distribution would have more data close to the mean on one side and a long tail on the other. In both cases, the 68-95-99.7 percentages would be poor approximations.

5.5.2 Chebyshev's theorem

While the empirical rule applies only to bell-shaped distributions, **Chebyshev's theorem** provides a guarantee that works for *any* distribution, regardless of shape:

Chebyshev's theorem.

For **any** distribution (regardless of shape), at least $(1 - \frac{1}{k^2}) \times 100\%$ of the data fall within k standard deviations of the mean, for any $k > 1$.

Standard deviations (k)	At least this % of data
2	75%
3	89%
4	93.75%

Chebyshev's theorem is weaker than the empirical rule (it guarantees "at least 75%" within 2 standard deviations, while the empirical rule says "about 95%" for bell-shaped data), but it has the advantage of applying to every distribution. It is most useful when you cannot assume the distribution is bell-shaped.

5.6 Chapter review

5.6.1 Summary

This chapter introduced the numerical tools for describing distributions. The **mean** and **median** measure center, with the mean being the balancing point and the median being the middle value. The **standard deviation** and **IQR** measure spread, with the standard deviation roughly representing the typical deviation from the mean and the IQR representing the range of the middle 50%. The **median** and **IQR** are robust to outliers, while the **mean** and **standard deviation** are sensitive to extreme values. The **empirical rule** connects the standard deviation to the distribution shape for bell-shaped data: about 68%, 95%, and 99.7% of observations fall within 1, 2, and 3 standard deviations of the mean. For any distribution, Chebyshev's theorem guarantees at least 75% of data within 2 standard deviations.

5.6.2 Key terms

- Mean ($\bar{x}$), median, mode
- Population mean (μ), weighted mean
- Deviation, variance (s^2), standard deviation (s)

- Population variance (σ^2), population standard deviation (σ)
- Range, interquartile range (IQR)
- First quartile (Q_1), third quartile (Q_3), percentile
- Outlier, robust statistic (resistant statistic)
- Empirical rule (68-95-99.7 rule), Chebyshev's theorem
- Variability, data density

5.6.3 Exercises

Exercises for this chapter are under development.

Chapter 6

Relationships Between Variables

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

The preceding chapters focused on exploring one variable at a time. But the most interesting statistical questions involve **relationships between variables**. Does higher education lead to higher income? Is a drug more effective for one group than another? Are two categorical variables associated? This chapter introduces tools for exploring relationships: contingency tables and mosaic plots for two categorical variables, side-by-side box plots and faceted histograms for a categorical and a numerical variable, and scatterplots for two numerical variables. We close with the critical distinction between association and causation.

StatBench tools for this chapter: Use the Two-Variable Explorer to create scatterplots, side-by-side box plots, and contingency tables. Upload a CSV dataset and select two variables to explore their relationship interactively.

6.1 Two categorical variables

6.1.1 Contingency tables

When we want to explore the relationship between two categorical variables, we organize the data in a **contingency table** (also called a two-way table or cross-tabulation). Each cell in the table shows the count of observations that belong to a particular combination of categories.

A **contingency table** summarizes data for two categorical variables. The rows represent the levels of one variable, the columns represent the levels of the other, and each cell contains the count of observations in that combination. **Row totals** and **column totals** give the marginal distributions.

Consider a dataset of 10,000 loans. Each loan has a homeownership status (rent, mortgage, or own) and an application type (joint or individual). The contingency table below summarizes these two variables:

	Rent	Mortgage	Own	Total
Joint	362	950	183	1,495
Individual	3,496	3,839	1,170	8,505
Total	3,858	4,789	1,353	10,000

Rent	Mortgage	Own	Total
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Each cell represents a count. For example, 3,496 loans were made by individuals who rent their home. The row totals tell us that 1,495 loans were joint applications and 8,505 were individual applications. The column totals tell us the overall distribution of homeownership.

What does the value 950 in the table represent? What does the column total 4,789 represent?

Show answer

The value 950 represents the number of loans that were joint applications from borrowers with a mortgage. The column total 4,789 represents the total number of borrowers (both joint and individual) who have a mortgage.

6.1.2 Row and column proportions

Raw counts can be hard to compare when group sizes differ. Instead, we can compute **row proportions** or **column proportions** to see conditional relationships.

Row proportions are computed by dividing each count by its row total. They show how a variable is distributed *within each level* of the row variable.

Column proportions are computed by dividing each count by its column total. They show how a variable is distributed *within each level* of the column variable.

Both row and column proportions are examples of **conditional proportions** — the proportion of observations in one category, conditional on (given) the level of the other variable.

Row proportions (each count divided by its row total):

	Rent	Mortgage	Own	Total
Joint	0.242	0.635	0.122	1.000
Individual	0.411	0.451	0.138	1.000

For example, $3,496/8,505 = 0.411$, meaning 41.1% of individual applicants rent their home.

Column proportions (each count divided by its column total):

	Rent	Mortgage	Own
Joint	0.094	0.198	0.135
Individual	0.906	0.802	0.865
Total	1.000	1.000	1.000

The value 0.906 tells us that 90.6% of renters applied individually. This rate is higher than for borrowers with mortgages (80.2%) or who own their home (86.5%). Because these rates vary across the levels of homeownership, there is evidence that application type and homeownership may be **associated**.

What does 0.451 represent in the row proportions table? What does 0.802 represent in the column proportions table?

Show answer

0.451 represents the proportion of individual applicants who have a mortgage. 0.802 represents the fraction of applicants with mortgages who applied as individuals.

Data scientists use statistics to build email spam filters. One characteristic examined is email format (HTML or plain text). A contingency table of email format and spam status is shown below:

	HTML	Text	Total
Not spam	2,568	986	3,554
Spam	158	209	367
Total	2,726	1,195	3,921

Which would be more useful for classifying email: row or column proportions?

A data scientist would want to know how the proportion of spam changes depending on email format. This corresponds to **column proportions**: $209/1,195 = 17.5\%$ of plain text emails are spam, compared to $158/2,726 = 5.8\%$ of HTML emails. Plain text emails are more likely to be spam. Row or column proportions are not equivalent — before computing, consider which variable you want to “condition on” (usually the explanatory variable).

In the spam example, which variable would you consider the explanatory variable and which the response?

Show answer

Email format (HTML vs. text) is the explanatory variable — it is a characteristic of the email that might help explain or predict whether the email is spam. Spam status (spam vs. not spam) is the response variable.

6.1.3 Bar plots for two categorical variables

Bar plots can be extended to display the relationship between two categorical variables. There are three common variants:

A **stacked bar plot** stacks the bars for the second variable on top of each other within each level of the first variable. It shows both the total count and the breakdown by the second variable.

A **standardized bar plot** (also called a **filled bar plot**) is like a stacked bar plot, but the bars are scaled to the same height (100%). It shows the *proportion* of each category of the second variable within each level of the first variable.

A **dodged bar plot** (also called a **grouped bar plot**) places bars for the second variable side by side within each level of the first variable. It makes it easy to compare counts across groups.

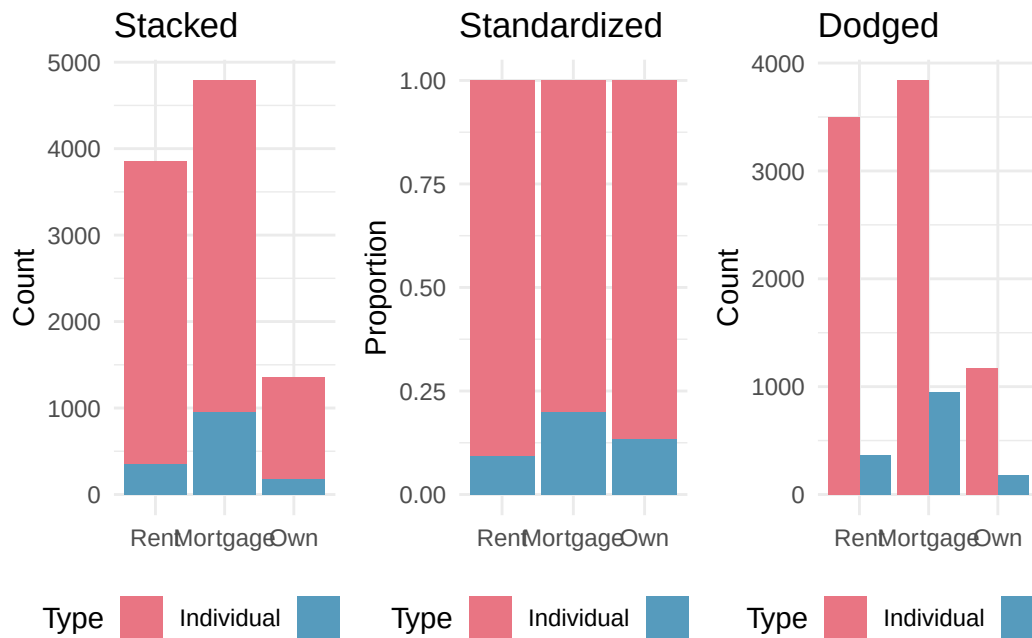


Figure 6.1: Three bar plots showing homeownership and application type: stacked, standardized (filled), and dodged.

When is each type of bar plot most useful?

- The **stacked bar plot** is most useful when you want to see both the total count for each group and the breakdown by the second variable. It works well when one variable is clearly the explanatory variable.
- The **standardized bar plot** is helpful when comparing proportions across groups, especially when the groups have very different sizes. However, it sacrifices information about the total count.
- The **dodged bar plot** makes it easy to compare individual counts directly. It works well for showing both variables without implying one is the explanatory variable. However, it requires more horizontal space and can feel cramped when there are many categories.

6.1.4 Mosaic plots

A **mosaic plot** is a visualization for contingency tables that improves on the standardized bar plot by also encoding the *size* of each group. In a mosaic plot:

- The *width* of each column is proportional to the number of observations in that category of the first variable
- The *height* of each section within a column is proportional to the conditional proportion of the second variable

This means the *area* of each rectangle is proportional to the count in that cell of the contingency table.

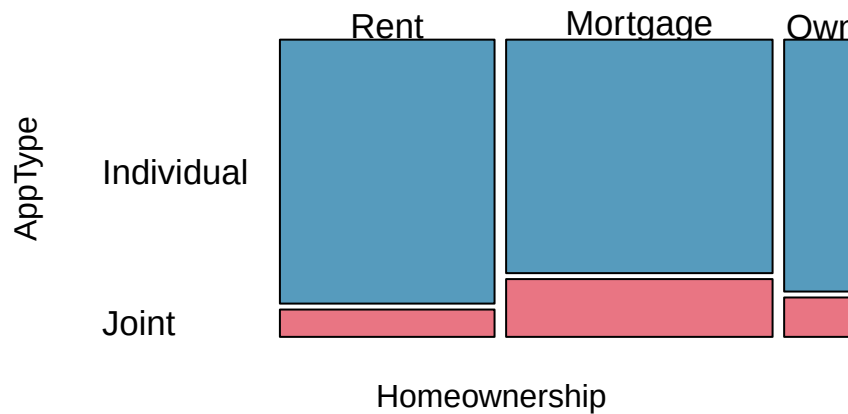


Figure 6.2: A mosaic plot of homeownership and application type. The width of each column reflects the number of borrowers in each homeownership category, and the height of each section reflects the proportion of joint vs. individual applications.

In the mosaic plot, how can you tell whether homeownership and application type are associated?

Show answer

If the variables were not associated, the division between joint and individual would occur at the same height in every column. Since the columns are divided at different heights — mortgage holders have a higher proportion of joint applications than renters or owners — the variables appear to be associated.

Detecting association in a mosaic plot or standardized bar plot. If the proportional breakdown of the response variable is the *same* across all levels of the explanatory variable, then the variables are **not** associated. If the breakdown *differs* across levels, the variables are associated. In a mosaic plot, this means the horizontal dividing lines would be at different heights.

6.2 One categorical and one numerical variable

Some of the most interesting investigations involve comparing a numerical variable across groups defined by a categorical variable.

6.2.1 Side-by-side box plots

Side-by-side box plots place a box plot for each group on the same scale, making it easy to compare centers, spreads, and shapes.

Consider comparing median household income for US counties, grouped by whether the county experienced population growth from 2010 to 2017:

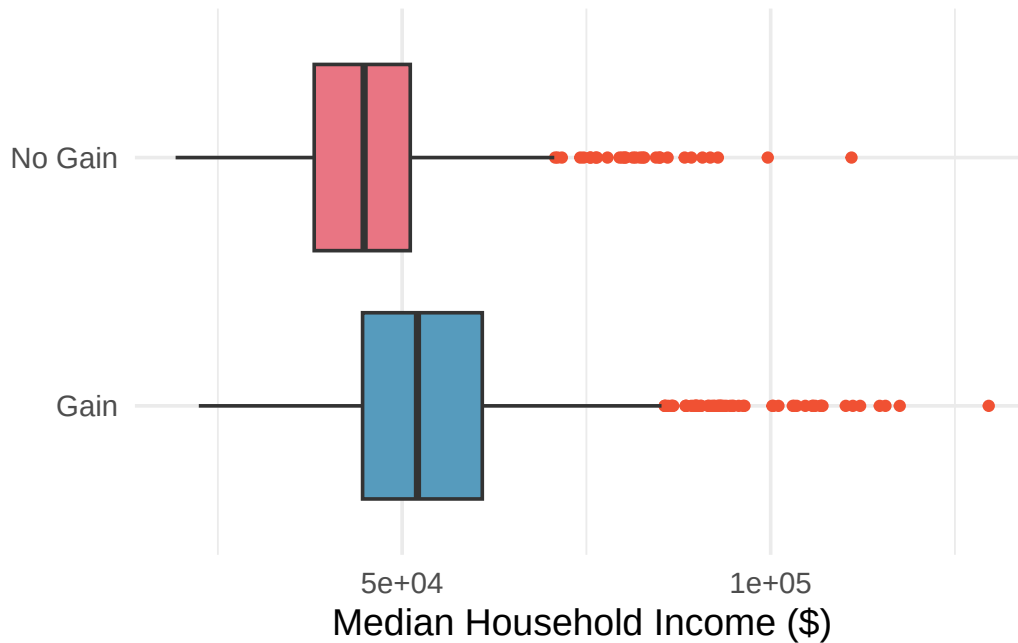


Figure 6.3: Side-by-side box plots of median household income by population change status (gain vs. no gain). Counties with population gain have a higher median income.

Using the box plots, compare the income distributions for the two groups. What do you notice about the centers, spreads, and shapes?

Show answer

Counties with population gains tend to have higher income (median about \$45,000) compared to counties without gains (median about \$40,000). The variability is also slightly larger for the gain group, as seen in the wider IQR. Both distributions show slight to moderate right skew, with outliers on the high end.

6.2.2 Faceted histograms

Faceting splits a graphical display into panels based on the levels of a categorical variable. Each panel shows the distribution for one group, using the same scale for easy comparison.

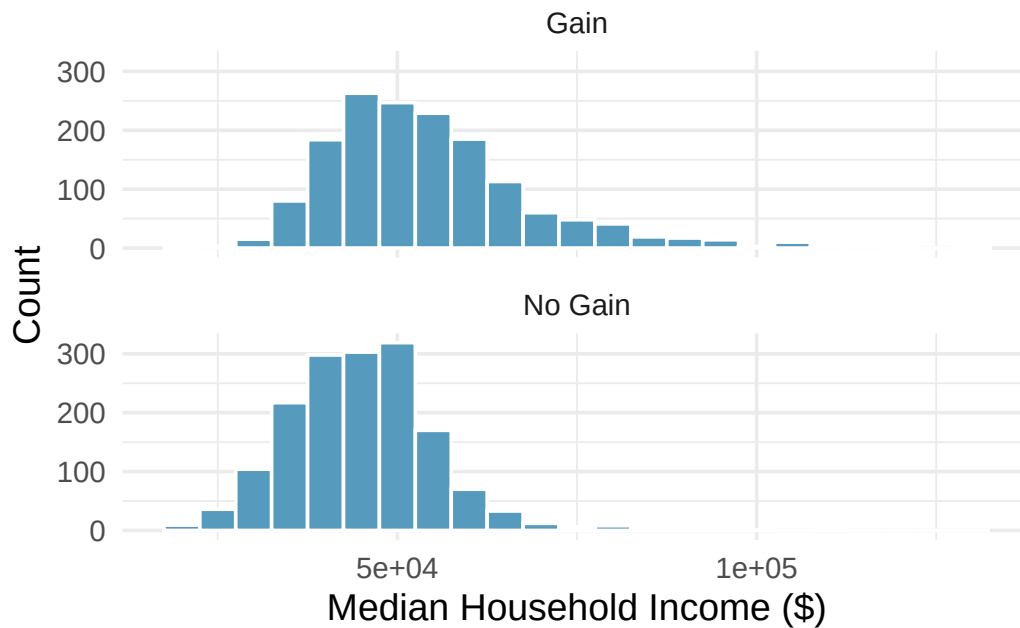


Figure 6.4: Faceted histograms of median household income, separated into panels for counties with population gain and those with no gain.

Faceting extends naturally to multiple categorical variables. For example, we could split the income data by both population change *and* whether the county is in a metropolitan area, creating a 2-by-2 grid of histograms.

What are the advantages of faceted histograms compared to side-by-side box plots?

Faceted histograms reveal the full shape of each distribution, including modality (unimodal vs. bimodal), the exact nature of any skewness, and gaps or clusters. Box plots, on the other hand, are more compact and make it easier to compare medians and IQRs at a glance, but they cannot show features like bimodality. The two displays complement each other well.

6.2.3 Ridge plots

A **ridge plot** (also called a joy plot) is a set of overlapping density plots, one for each group, drawn on the same horizontal scale but offset vertically. Ridge plots are useful for comparing the shapes of many distributions at once while maintaining a compact display.

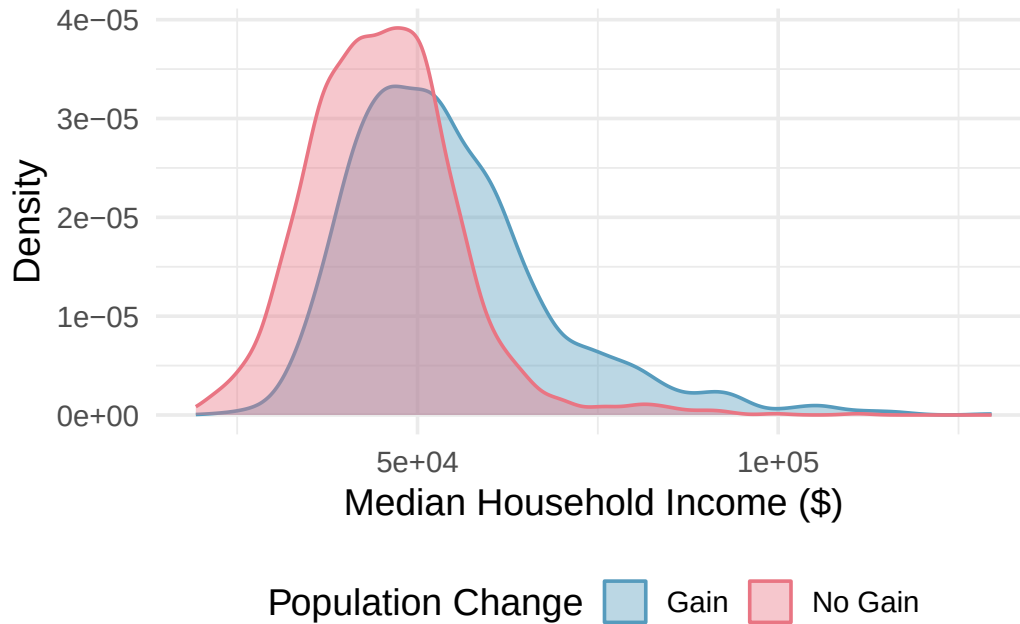


Figure 6.5: A ridge plot comparing median household income for counties with and without population gain. Each group gets its own smoothed density curve.

What does a ridge plot reveal that side-by-side box plots do not?

Show answer

Ridge plots show the full shape of each distribution, including modality and skewness, similar to histograms. Box plots compress all of this shape information into a five-number summary. Ridge plots are especially useful when you want to compare shapes across many groups in a compact space.

6.3 Two numerical variables

6.3.1 Scatterplots

A **scatterplot** displays the relationship between two numerical variables. Each observation is represented as a point, with one variable on the horizontal axis and the other on the vertical axis.

A **scatterplot** provides a case-by-case view of data for two numerical variables. Each point represents one observation. The horizontal axis shows the values of one variable and the vertical axis shows the values of the other.

Consider a dataset of 50 loans, where we plot total income on the horizontal axis and loan amount on the vertical axis:

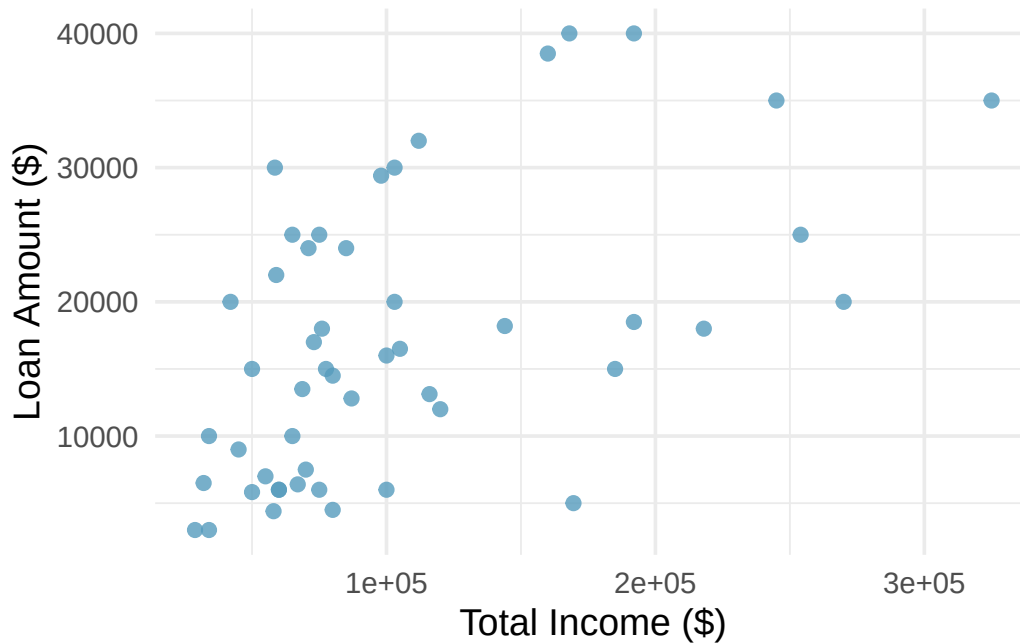


Figure 6.6: A scatterplot of loan amount versus total income for 50 loans. There is a moderate positive association.

What do scatterplots reveal about the data, and how are they useful?

Show answer

Scatterplots are helpful for quickly spotting associations between two numerical variables — whether they tend to increase together, decrease together, or show no pattern. They can also reveal whether the relationship is linear or curved, how strong the association is, and whether there are any unusual points.

6.3.2 Describing associations

When examining a scatterplot, we describe the association using four characteristics:

Describing the association between two numerical variables:

1. **Direction:** Is the association **positive** (both variables tend to increase together) or **negative** (one increases as the other decreases)?
2. **Form:** Is the relationship approximately **linear** (following a straight-line pattern) or **nonlinear** (curved)?
3. **Strength:** Is the association **strong** (points closely follow a pattern), **moderate**, or **weak** (points are loosely scattered)?
4. **Unusual observations:** Are there any **outliers** — points that deviate markedly from the overall pattern?

A scatterplot shows the poverty rate vs. median household income for over 3,000 US counties. How would you describe the association?

The association is **negative** (as poverty rate increases, median household income tends to decrease), **nonlinear** (the relationship curves, with income decreasing more steeply at lower poverty rates and

leveling off at higher poverty rates), and **moderately strong** (the points follow the curving pattern fairly closely, though there is noticeable scatter). There do not appear to be any extreme outliers, though counties with very high poverty rates or very high incomes stand out somewhat.

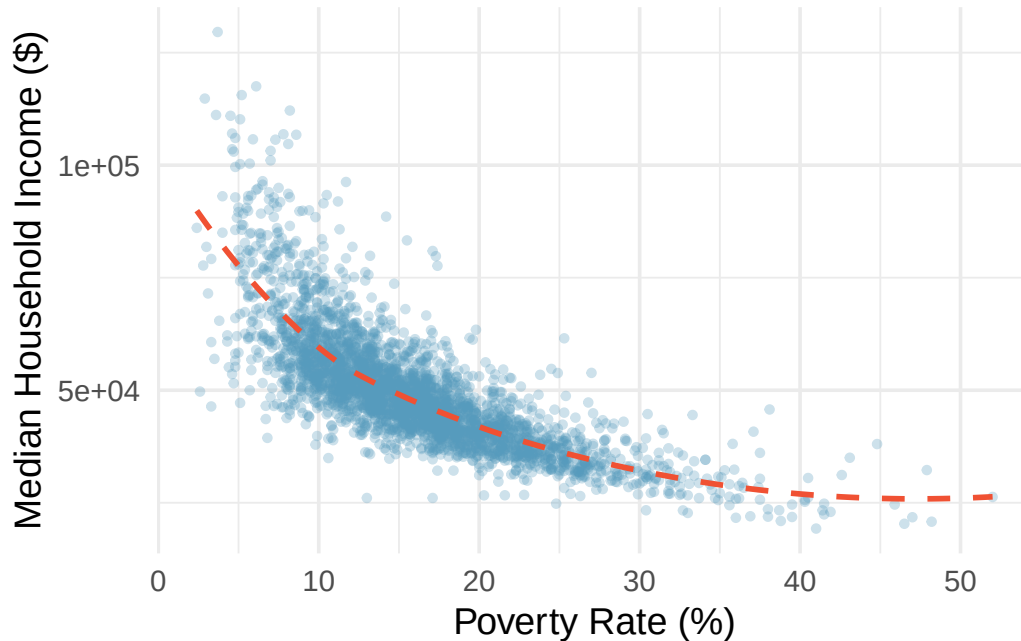


Figure 6.7: A scatterplot of median household income versus poverty rate for US counties, with a curved trend line. The relationship is negative and nonlinear.

Describe two variables that would have a horseshoe-shaped ($\cap$) association in a scatterplot.

Show answer

Consider a situation where the vertical axis represents something “good” and the horizontal axis represents something that is good only in moderation. For example, health and water consumption: we need some water to survive, but consuming too much becomes dangerous. Athletic performance and practice time is another example — some practice improves performance, but overtraining leads to injury and decline.

For the 50-loan dataset, the scatterplot of total income versus loan amount shows that borrowers with higher incomes tend to take out larger loans. Describe this association.

The association is **positive** (higher income is associated with higher loan amounts), approximately **linear** (the general trend follows a straight line, though the pattern is loose), and **weak to moderate** (there is considerable scatter — many borrowers with similar incomes take out very different loan amounts). There are a few borrowers with income above \$250,000 who stand out on the right side of the plot.

6.3.3 Scatterplots with a third variable

We can add a third variable to a scatterplot by using color, shape, or size to represent a categorical or numerical variable. For example, in a scatterplot of income vs. poverty rate, we could color the points

by whether the county is in a metropolitan area. This technique helps reveal whether the relationship between two numerical variables differs across groups.

6.4 Association vs. causation

Throughout this chapter, we have explored associations between variables. But association is not the same as causation.

Association does not imply causation.

When two variables are associated (correlated), it means they tend to vary together — but that does not necessarily mean one *causes* the other. There may be a **confounding variable** (a third variable) that is related to both, creating the illusion of a direct relationship.

A study finds a strong positive association between ice cream sales and drowning deaths. Does eating ice cream cause drowning?

Of course not. The association exists because both ice cream sales and drowning are related to a confounding variable: **temperature** (or season). In hot weather, people buy more ice cream *and* more people go swimming, which increases the risk of drowning. Temperature is the lurking variable that drives both.

A **confounding variable** (also called a **lurking variable**) is a variable that is related to both the explanatory and response variables. Confounding variables can create misleading associations between two variables that have no direct causal relationship.

A study finds that students who eat breakfast tend to have higher GPAs than students who skip breakfast. Can we conclude that eating breakfast causes higher GPAs?

Show answer

Not from this information alone. This is likely an observational study, and there could be confounding variables. For example, students who eat breakfast may come from more supportive home environments, may have better time management skills, or may generally take better care of their health — all of which could independently contribute to higher GPAs. To establish a causal connection, we would need a randomized experiment.

6.4.1 When can we infer causation?

Establishing a causal relationship between two variables generally requires:

1. **A randomized experiment:** Randomly assigning subjects to treatment and control groups helps eliminate confounding variables. If a difference is observed between groups, we can more confidently attribute it to the treatment.
2. **Temporal precedence:** The cause must precede the effect in time.
3. **Ruling out confounders:** We must be able to argue that no lurking variables explain the observed association.

In observational studies (where we observe but do not control the variables), we can identify associations but generally cannot draw causal conclusions. This is one of the most important distinctions in statistics.

Be especially skeptical of causal claims from observational data. Media reports often say “X causes Y” when the underlying study only found an association. Always ask: Was this a randomized experiment? Could there be confounding variables?

6.5 Chapter review

6.5.1 Summary

This chapter introduced tools for exploring relationships between variables. For **two categorical variables**, contingency tables organize counts, row and column proportions reveal conditional relationships, and bar plots (stacked, standardized, dodged) and mosaic plots provide visual comparisons. For **one categorical and one numerical variable**, side-by-side box plots, faceted histograms, and ridge plots compare distributions across groups. For **two numerical variables**, scatterplots reveal the direction, form, strength, and unusual features of an association. We closed with the fundamental principle that **association does not imply causation** — confounding variables can create misleading associations, and establishing causation generally requires a randomized experiment.

6.5.2 Key terms

- Contingency table, row totals, column totals
- Row proportions, column proportions, conditional proportions
- Stacked bar plot, standardized (filled) bar plot, dodged bar plot
- Mosaic plot
- Side-by-side box plot, faceted plot, ridge plot
- Scatterplot
- Positive association, negative association
- Linear, nonlinear
- Direction, form, strength (of an association)
- Confounding variable (lurking variable)
- Association vs. causation

6.5.3 Exercises

Exercises for this chapter are under development.

PART II

SIMULATE

How can we draw conclusions about a population from a single sample? We answer this question through *simulation*. By resampling data thousands of times, we build an intuitive understanding of sampling variability, hypothesis testing, and confidence intervals — without formulas or distribution tables.

This is the heart of the simulation-first approach: understand the *logic* of inference before learning the mathematical shortcuts.

Chapter 7

Sampling Variability

So far in this course, you have learned how to collect data, visualize it, and compute summary statistics. But statistics is about more than describing the data you have — it is about using data to learn about a larger population. The central challenge is this: **every sample is different**. If you took another sample from the same population, you would get different results. This chapter introduces the foundational ideas that make statistical inference possible: the distinction between parameters and statistics, the concept of a sampling distribution, and the standard error as a measure of how much sample statistics vary. These ideas are the bridge from describing data to drawing conclusions about the world.

7.1 Parameters vs. statistics

When we collect data, we typically have a larger group in mind. A political poll surveys 1,000 likely voters, but the real interest is in all voters. A quality engineer measures 50 widgets from an assembly line, but the real interest is in all widgets the line produces. The larger group is the **population**; the subset we actually observe is the **sample**.

A **population parameter** is a numerical summary of the entire population. It is a fixed number, but it is almost always unknown because we cannot observe the whole population.

A **sample statistic** (also called a **point estimate**) is a numerical summary computed from sample data. It is our best guess at the population parameter. Because the statistic is computed from a sample, it is known — but it changes from sample to sample.

We use specific notation to distinguish parameters from statistics:

Quantity	Parameter (population)	Statistic (sample)
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$
Difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$

The “hat” notation ($\hat{p}$) is a common convention: it signals that we are estimating the corresponding parameter.

A medical consultant has facilitated 62 liver donor surgeries, and 3 of her patients experienced complications. She uses this track record to attract new patients.

Let p represent the true complication rate for liver donors working with this consultant. What is the point estimate $\hat{p}$?

The sample proportion is $\hat{p} = 3/62 = 0.048$. That is, about 4.8% of her patients experienced complications. This is our point estimate of the true complication rate p .

A Stanford University graduate student recruited 120 tappers and 120 listeners for a study on communication. Only 3 out of 120 listeners were able to guess the tune being tapped. What is the point estimate $\hat{p}$ for the true proportion of listeners who can guess correctly?

Show answer

The point estimate is $\hat{p} = 3/120 = 0.025$, or 2.5%. This is our best single estimate of the true proportion p of listeners who can guess the tune.

In a study of sex discrimination, 48 male bank supervisors each reviewed a personnel file and decided whether the candidate should be promoted. The files were identical except that half listed the candidate as male and half as female. The results are summarized below:

	Promoted	Not promoted	Total
Male	21	3	24
Female	14	10	24
Total	35	13	48

What is the point estimate for the difference in promotion rates between male and female candidates?

The promotion rate for male candidates is $\hat{p}_M = 21/24 = 0.875$ and for female candidates is $\hat{p}_F = 14/24 = 0.583$. The point estimate of the difference is:

$$\hat{p}_M - \hat{p}_F = 0.875 - 0.583 = 0.292$$

The observed difference is 29.2 percentage points in favor of male candidates.

A point estimate is almost never exactly equal to the population parameter. This is not because we did anything wrong — it is an inherent consequence of working with a subset of the population. The critical question is: *how far off might the estimate be?*

7.2 Sampling distributions

To understand how far a statistic might be from its parameter, we need to understand what happens when we take **many samples** from the same population.

Imagine a large bowl containing thousands of beads, 60% red and 40% white. The proportion of red beads in the bowl is the population parameter: $p = 0.60$. Now suppose you reach in and scoop out a sample of $n = 50$ beads. You count 28 red beads, giving $\hat{p} = 28/50 = 0.56$. Your friend scoops a different sample of 50 and counts 32 red, giving $\hat{p} = 32/50 = 0.64$. Another friend gets 30 red for $\hat{p} = 0.60$.

Each person got a different sample proportion. No one did anything wrong — this is **sampling variability** in action.

Now imagine hundreds of people each scooping their own sample of 50 beads. If we recorded every $\hat{p}$ and made a histogram of all those values, we would see the **sampling distribution** of $\hat{p}$.

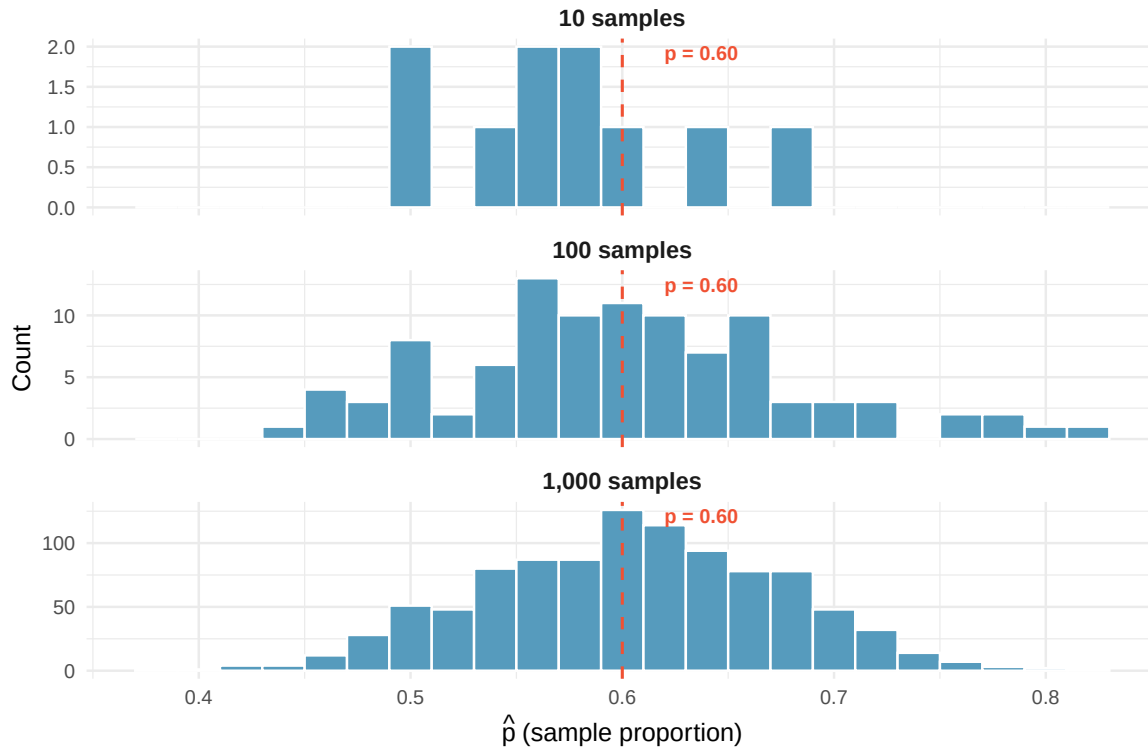


Figure 7.1: Building a sampling distribution. Each dot represents the proportion of red beads from one sample of $n = 50$. With more samples, the bell-shaped pattern centered at $p = 0.60$ becomes clear.

The **sampling distribution** of a statistic is the distribution of all possible values of that statistic computed from all possible samples of a given size from a given population.

A sampling distribution is *not* the distribution of individual data values. It is the distribution of a *summary statistic* (like $\hat{p}$ or $\bar{x}$) across many hypothetical samples.

Key properties of sampling distributions. For many common statistics, the sampling distribution has three important features:

1. **Center:** The sampling distribution is centered at the population parameter. On average, the sample statistic equals the parameter.
2. **Spread:** The sampling distribution has a measurable spread that decreases as sample size increases. Larger samples give more precise estimates.
3. **Shape:** For many statistics, the sampling distribution is approximately bell-shaped (normal) when the sample size is large enough.

Here is an analogy that may help. Imagine you are trying to determine the average depth of a lake. You drop a weighted line at a random spot and measure the depth — that is one sample. If you did this at many random spots and recorded all those measurements, you would get a distribution of individual depth measurements (the *data distribution*). But if instead, for each spot you visited, you dropped the line 30 times in a small cluster and computed the average of those 30 measurements, then

moved to a new random spot and did it again, the collection of those averages would form a *sampling distribution* of the mean.

The two distributions are fundamentally different:

- The **data distribution** shows how individual observations vary.
- The **sampling distribution** shows how a *statistic* (like the mean) varies from sample to sample.

7.3 Variability of sample statistics

Different samples from the same population produce different statistics. This variability is not a flaw — it is a fundamental feature of sampling that we must understand and quantify.

Suppose the true proportion of all US adults who support a particular policy is $p = 0.52$. Three polling organizations each take a random sample of $n = 1,000$ adults and compute $\hat{p}$. Their results might look like:

Poll	Sample proportion $\hat{p}$
Poll A	0.49
Poll B	0.54
Poll C	0.51

All three polls were conducted correctly, yet they produced different results. Poll A even shows less than 50% support! This is sampling variability: the unavoidable variation that occurs when we estimate a population parameter from a sample.

If the three polls above each used a sample of $n = 10,000$ instead of $n = 1,000$, would you expect the three $\hat{p}$ values to be closer together or farther apart? Why?

Show answer

We would expect them to be closer together. Larger samples contain more information about the population, so the statistic is more precise — meaning less variability from sample to sample.

7.3.1 What determines how much a statistic varies?

Two factors control how much a sample statistic bounces around from sample to sample:

1. **Sample size (n):** Larger samples produce statistics that are closer to the parameter. This is intuitive — a sample of 1,000 tells you more about the population than a sample of 10.
2. **Population variability:** If the population is very homogeneous (everyone is similar), then any sample will look like any other sample, and the statistic won't vary much. If the population is very heterogeneous (lots of individual differences), then different samples can look quite different, and the statistic will vary more.

Of these two, sample size is the one we can control, and it turns out to be very powerful. Quadrupling the sample size cuts the variability of most statistics roughly in half.

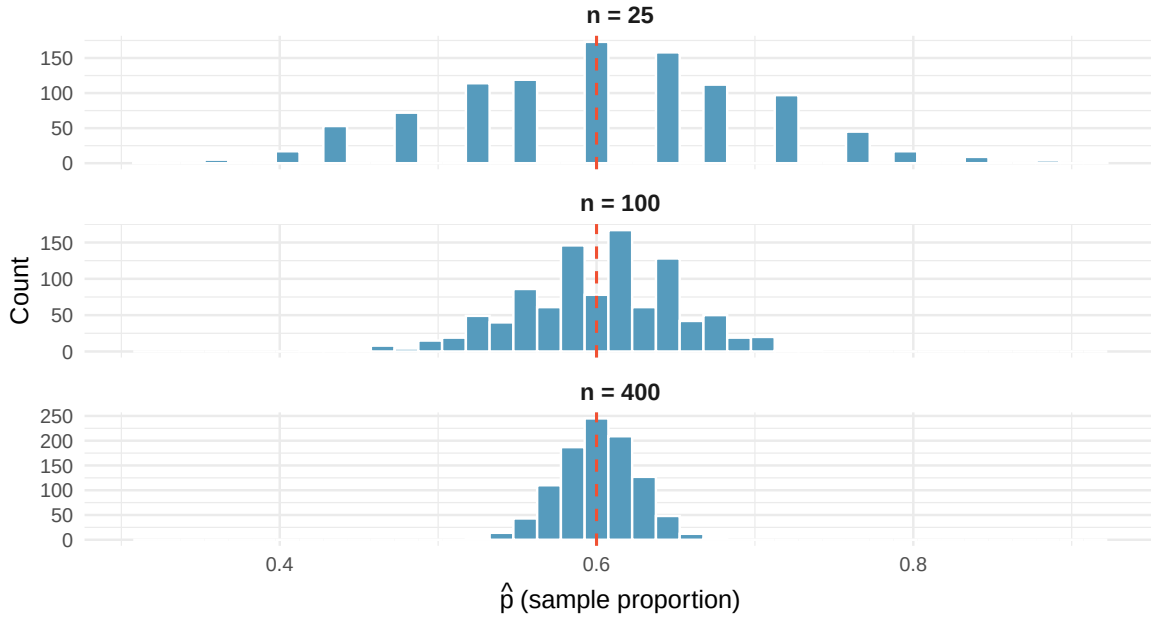


Figure 7.2: Sampling distributions of the sample proportion for three different sample sizes. All three are centered at $p = 0.60$, but the spread decreases dramatically as the sample size increases.

Figure 7.2 shows this dramatically: with $n = 25$, individual sample proportions range from about 0.35 to 0.85. With $n = 400$, they are tightly clustered between 0.55 and 0.65.

Explore sampling variability! Use StatBench’s Sampling Distribution Simulator to see this in action:

[Launch Sampling Distribution Simulator](#)

1. Set the population proportion to $p = 0.60$ and the sample size to $n = 25$.
2. Draw 500 samples and observe the resulting sampling distribution.
3. Now change the sample size to $n = 100$ and draw 500 more samples.
4. Compare the two sampling distributions. How do their centers compare? Their spreads? Their shapes?

7.4 Standard error

We need a single number that quantifies how much a statistic varies from sample to sample. That number is the **standard error**.

The **standard error** (SE) of a statistic is the standard deviation of its sampling distribution. It measures the typical distance between a sample statistic and the population parameter.

A small standard error means the statistic tends to fall close to the parameter (precise estimate). A large standard error means the statistic could plausibly fall far from the parameter (imprecise estimate).

Do not confuse the standard error with the standard deviation of the data:

- The **standard deviation** (s or σ) measures how much *individual observations* vary around the mean.
- The **standard error** (SE) measures how much a *sample statistic* varies around the population parameter from sample to sample.

The standard error is always smaller than the standard deviation (for samples larger than $n = 1$) because averaging reduces variability.

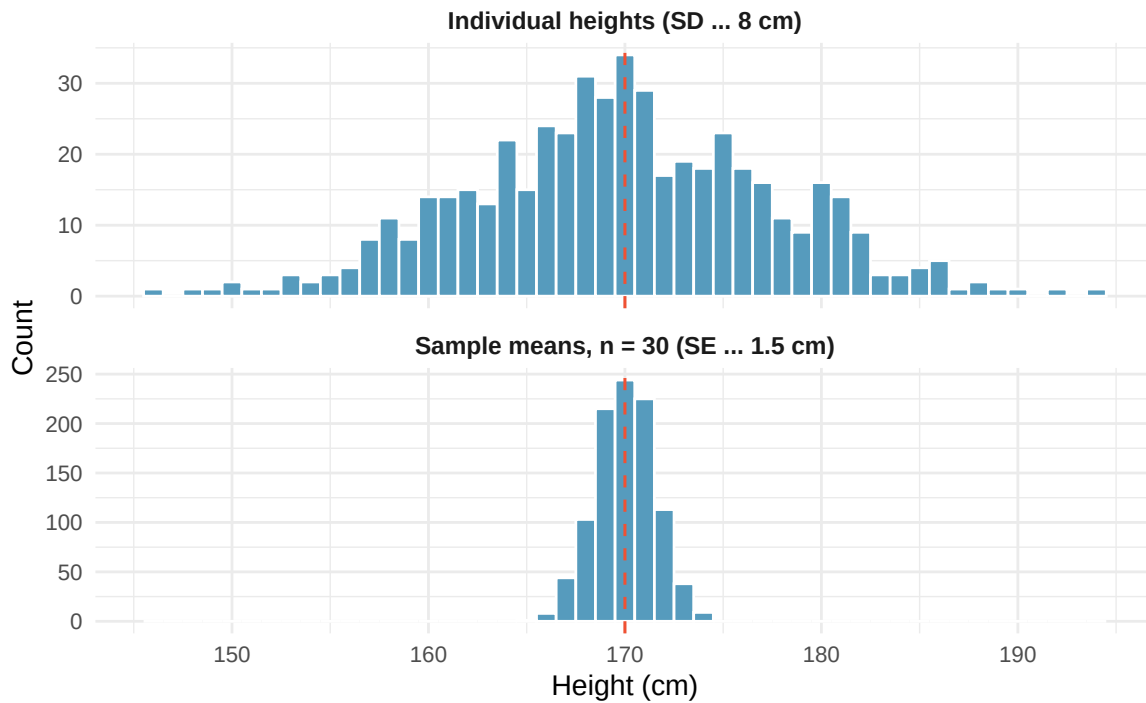


Figure 7.3: Comparing the data distribution (top) with the sampling distribution of the mean (bottom). Both are centered near the same value, but the sampling distribution is much narrower — its spread is the standard error, not the standard deviation.

In Figure 7.3, the top panel shows how individual heights vary ($SD \approx 8$ cm), while the bottom panel shows how sample means vary ($SE \approx 1.5$ cm). Both are centered near $\mu = 170$, but the sampling distribution of the mean is much narrower.

In the medical consultant example, $\hat{p} = 3/62 = 0.048$. If we could take many samples of 62 patients from this consultant's true patient population and compute $\hat{p}$ each time, the standard deviation of all those $\hat{p}$ values would be the standard error of $\hat{p}$.

While we usually cannot literally take repeated samples, simulation methods (which we will explore in Chapters 7 and 8) allow us to approximate the standard error. In this case, bootstrap simulations suggest that the standard error of $\hat{p}$ is approximately 0.027. This means that if we could repeat the study many times, the sample complication rate would typically differ from the true complication rate by about 2.7 percentage points.

7.4.1 Why the standard error matters

The standard error is the engine of statistical inference. Here is why:

- **Confidence intervals** are built by going a certain number of standard errors above and below the point estimate. A rough 95% confidence interval is:

$$\text{statistic} \pm 2 \times SE$$

- **Hypothesis tests** compare an observed statistic to what we would expect under the null hypothesis. The comparison is made in units of standard errors: “Is the observed result more than 2 standard errors from the null value?”
- **Sample size planning** uses the standard error to determine how many observations are needed to achieve a desired level of precision.

In the coming chapters, we will use simulation to *estimate* the standard error. In later chapters (Part III), we will use mathematical formulas to *calculate* it directly.

Two researchers both estimate the proportion of college students who prefer online textbooks. Researcher A surveys 100 students and finds $\hat{p}_A = 0.62$ with $SE_A = 0.049$. Researcher B surveys 400 students and finds $\hat{p}_B = 0.58$ with $SE_B = 0.025$.

- Whose estimate is more precise? How do you know?
- Why is Researcher B’s standard error smaller?

Show answer

- Researcher B’s estimate is more precise because the standard error is smaller ($0.025 < 0.049$), meaning the sample statistic is expected to be closer to the true parameter. (b) Researcher B surveyed four times as many students, and increasing the sample size reduces the standard error.

7.5 Looking ahead

This chapter has established the core vocabulary and ideas that drive all of statistical inference:

- We want to know about **parameters** but can only observe **statistics**.
- Statistics vary from sample to sample — this is **sampling variability**.
- The pattern of this variation is described by the **sampling distribution**.
- The **standard error** quantifies the typical amount of sampling variability.

The fundamental question of inference is: *Given the sampling variability we expect, is the pattern in our data real, or could it have happened by chance?*

In the next three chapters, we will answer this question using three different approaches — all rooted in simulation:

- **Chapter 7 (Randomization Tests)**: Shuffle the data to see what differences look like when there is no real effect. If the observed difference is extreme compared to the shuffled results, we have evidence of a real effect.
- **Chapter 8 (Bootstrap Confidence Intervals)**: Resample from the data to approximate the sampling distribution, and use it to construct a range of plausible values for the parameter.
- **Chapter 9 (Sampling Distributions via Simulation)**: Connect the simulation results from Chapters 7 and 8 to mathematical theory, setting the stage for the formula-based methods in Part III.

The goal of Part II is to build your *intuition* for inference. The formulas will come later; the ideas come first.

7.6 Chapter review

7.6.1 Summary

- A **population parameter** is a fixed but unknown numerical summary of the population. A **sample statistic** (point estimate) is computed from data and serves as our best guess of the parameter.
- Different samples from the same population yield different statistics. This is **sampling variability** — it is not a mistake, but a fundamental feature of working with samples.
- The **sampling distribution** of a statistic describes how the statistic varies across all possible samples of a given size. It is typically centered at the parameter, and its spread decreases as sample size increases.
- The **standard error** is the standard deviation of the sampling distribution. It measures the precision of the statistic as an estimate of the parameter.
- The standard error is not the same as the standard deviation of the data. The standard deviation measures variability of individual observations; the standard error measures variability of a statistic.
- Larger samples produce smaller standard errors, meaning more precise estimates.

7.6.2 Key terms

Term	Definition
Population parameter	A fixed numerical summary of the entire population (e.g., p, μ)
Sample statistic	A numerical summary computed from sample data (e.g., $\hat{p}, \bar{x}$)
Point estimate	Another name for sample statistic; our best guess at the parameter
Sampling variability	The fact that different samples produce different statistics
Sampling distribution	The distribution of a statistic over all possible samples of a given size
Standard error (SE)	The standard deviation of the sampling distribution

7.6.3 Exercises

Exercises for this chapter are under development.

Chapter 8

Randomization Tests

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Statistical inference is primarily concerned with understanding and quantifying the uncertainty of parameter estimates. While the equations and details change depending on the setting, the foundations for inference are the same throughout all of statistics. In this chapter, we introduce the **hypothesis testing framework** — a formal method for evaluating competing claims about a population using data. We learn how to build **randomization distributions** by shuffling data under the assumption that there is no effect, and we use these distributions to compute **p-values** that quantify the strength of the evidence.

8.1 The hypothesis testing framework

Throughout this course so far, you have worked with data in a variety of contexts. You have learned how to summarize and visualize data as well as how to describe relationships between variables. But more often than not, data have been collected to answer a research question about a larger group — and the data are a (hopefully) representative subset.

There is almost always variability in data — one dataset will not be identical to a second dataset even if they are both collected from the same population using the same methods. However, quantifying the variability in the data is neither obvious nor easy. Answering the question “*how* different is one dataset from another?” is not trivial.

Suppose your professor splits the students in your class into two groups: students who sit on the left side of the classroom and students who sit on the right side. If $\hat{p}_L$ represents the proportion of left-side students who prefer reading on a screen and $\hat{p}_R$ represents the proportion of right-side students who prefer reading on a screen, would you be surprised if $\hat{p}_L$ did not *exactly* equal $\hat{p}_R$?

While the proportions $\hat{p}_L$ and $\hat{p}_R$ would probably be close to each other, it would be unusual for them to be exactly the same. We would probably observe a small difference due to *chance*.

If we do not think the side of the room a person sits on in class is related to whether they prefer to read books on a screen, what assumption are we making about the relationship between these two variables?

Show answer

We would be assuming that these two variables are **independent** — knowing which side of the room someone sits on tells us nothing about their reading preference.

The question at the heart of hypothesis testing is: *Could the pattern we observe in the data have arisen from chance alone, or is there something real going on?*

Hypothesis testing.

A **hypothesis test** is a statistical technique used to evaluate competing claims using data. The process involves:

1. **State the hypotheses:** Define a null hypothesis (H_0) and an alternative hypothesis (H_A).
2. **Collect data:** Gather evidence that bears on the hypotheses.
3. **Assess the evidence:** Determine how likely the observed data would be if the null hypothesis were true.
4. **Draw a conclusion:** Decide whether the evidence is strong enough to reject the null hypothesis.

If the null hypothesis and the data notably disagree, then we reject the null hypothesis in favor of the alternative hypothesis. There are many nuances to hypothesis testing, and we will discuss these ideas throughout this chapter and those that follow.

8.2 Null and alternative hypotheses

The **null hypothesis** (H_0) often represents a skeptical perspective or a claim of “no difference” or “no effect.” It assumes that any observed pattern in the data is due to chance alone.

The **alternative hypothesis** (H_A) represents the claim under investigation — typically that there *is* a real difference, a real effect, or a real relationship. The alternative hypothesis is usually the reason the research was conducted in the first place.

The US court system considers two possible claims about a defendant: they are either innocent or guilty. If we set these claims up in a hypothesis framework, which would be the null hypothesis and which the alternative?

The jury considers whether the evidence is so convincing that there is no reasonable doubt regarding the person’s guilt. The skeptical perspective (null hypothesis) is that the person is innocent until evidence is presented that convinces the jury the person is guilty (alternative hypothesis).

H_0 : The defendant is innocent.

H_A : The defendant is guilty.

Notice that if a jury finds a defendant *not guilty*, this does not necessarily mean the jury is confident in the person’s innocence. They are simply not convinced of the alternative. This is also the case with hypothesis testing: **even if we fail to reject the null hypothesis, we do not accept the null hypothesis as truth.** Failing to find evidence in favor of the alternative is not the same as finding evidence that the null hypothesis is true.

8.3 Case study: sex discrimination

We consider a study investigating sex discrimination in the 1970s, set in the context of personnel decisions within a bank. The research question: “Are individuals who identify as female discriminated

against in promotion decisions made by their managers who identify as male?”

8.3.1 Observed data

The participants were 48 bank supervisors who identified as male, attending a management institute at the University of North Carolina in 1972. They were asked to assume the role of personnel director of a bank and were given a personnel file to judge whether the person should be promoted to a branch manager position. The files were identical, except that half indicated the candidate identified as male and the other half indicated the candidate identified as female. These files were randomly assigned to the bank managers.

Is this an observational study or an experiment? How does the type of study impact what can be inferred from the results?

Show answer

The study is an experiment, as subjects were randomly assigned a “male” file or a “female” file (remember, all the files were actually identical in content). Since this is an experiment, the results can be used to evaluate a causal relationship between the sex of a candidate and the promotion decision.

The results are summarized in the table below:

	Promoted	Not promoted	Total
Male	21	3	24
Female	14	10	24
Total	35	13	48

When looking at the rates of promotion in this study, why might we be tempted to immediately conclude that individuals identifying as female are being discriminated against?

The large difference in promotion rates (58.3% for female personnel versus 87.5% for male personnel) suggests there might be discrimination against women in promotion decisions. However, we cannot yet be sure if the observed difference represents discrimination or is just due to random chance when there is no discrimination occurring. Since we wouldn’t expect the sample proportions to be *exactly* equal, even if the truth was that the promotion decisions were independent of sex, we can’t rule out random chance as a possible explanation.

The observed difference in promotion rates is:

$$\hat{p}_M - \hat{p}_F = \frac{21}{24} - \frac{14}{24} = 0.875 - 0.583 = 0.292$$

This is 29.2 percentage points. The point estimate is large, but the sample size is small, making it unclear if this represents real discrimination or simply chance variation. We label two competing claims:

- H_0 : **Null hypothesis.** The variables sex and decision are independent. The difference in promotion rates of 29.2% was due to natural variability inherent in the random assignment.
- H_A : **Alternative hypothesis.** The variables sex and decision are *not* independent. The difference in promotion rates of 29.2% was not due to natural variability, and equally qualified female personnel are less likely to be promoted than male personnel.

8.4 Building a randomization distribution

What would it mean if the null hypothesis were true? It would mean each banker would decide whether to promote the candidate without regard to the sex indicated on the file. The 35 promotions and 13 non-promotions were determined by factors *other than sex*, and the difference we observed was just bad luck in how the files happened to be assigned.

We can test this by performing a **randomization** — simulating what would have happened if the bankers' decisions had been independent of sex but the files had been distributed differently.

8.4.1 The card-shuffling metaphor

Think of the data as 48 cards:

- 35 red cards (promoted)
- 13 white cards (not promoted)

In the actual study, 24 cards were dealt to the “male” pile and 24 to the “female” pile. Under the null hypothesis, the color of each card (the promotion decision) has nothing to do with which pile it lands in.

To simulate what the null hypothesis looks like, we:

1. **Thoroughly shuffle** all 48 cards together.
2. **Deal** 24 into a “male” pile and 24 into a “female” pile.
3. **Count** the number of red (promoted) cards in each pile and compute the difference in proportions.

This gives us one simulated difference *under the null hypothesis* — a difference that arose purely from the random shuffle, not from any real discrimination.

8.4.2 One simulation

Suppose after shuffling and dealing, we get the following result:

	Promoted	Not promoted	Total
Male	18	6	24
Female	17	7	24
Total	35	13	48

The simulated difference is $18/24 - 17/24 = 0.042$, or about 4.2% — much smaller than the observed 29.2%.

What is the difference in promotion rates in the simulated data above? How does this compare to the observed difference of 29.2% from the actual study?

Show answer

The simulated difference is $18/24 - 17/24 = 0.042$ or about 4.2% in favor of male personnel. This difference due to chance is much smaller than the 29.2% difference observed in the actual groups.

8.4.3 Many simulations

One simulation is not enough. We need to repeat the shuffle-and-deal process many times to build up a picture of what *chance alone* looks like. Using a computer, we can do this hundreds or thousands of times.

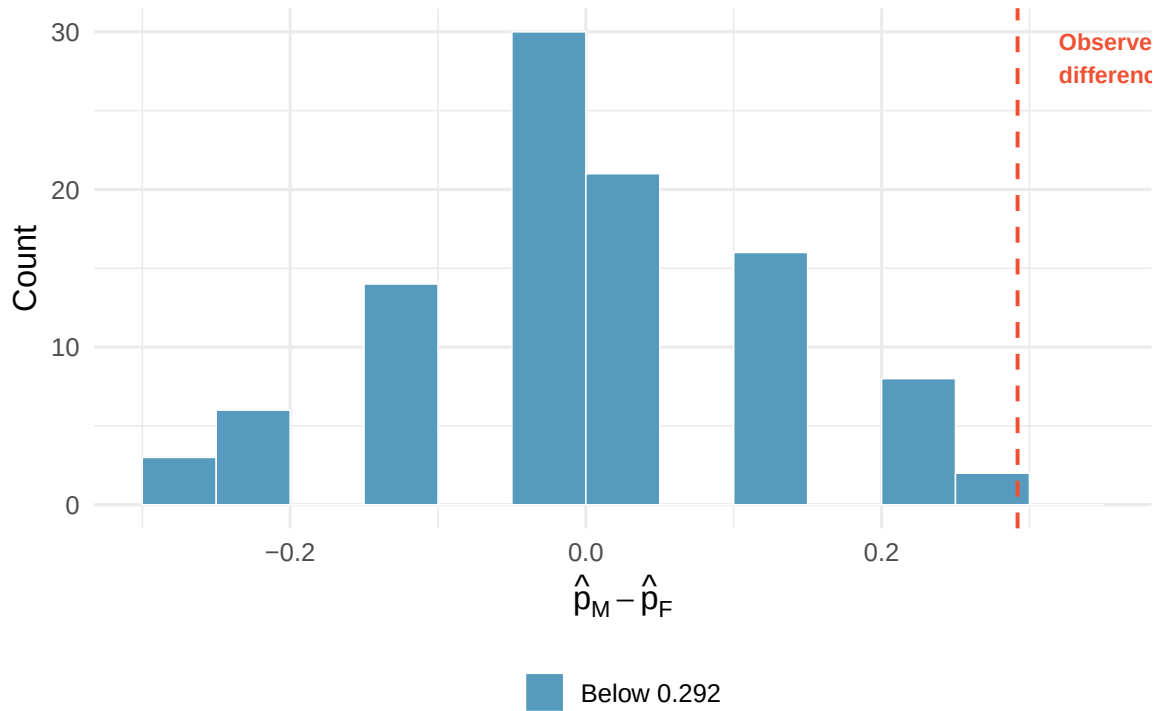


Figure 8.1: A histogram of differences from 100 simulations produced under the null hypothesis. The simulated sex and decision are independent. Differences at least as large as the observed 0.292 are highlighted in red.

The distribution of these simulated differences is centered around 0. This makes sense: under the null hypothesis, which makes no distinction between male and female files, we would expect the difference to be near zero with some random fluctuation.

How often would you observe a difference of at least 29.2% according to the randomization distribution? Often, sometimes, rarely, or never?

It appears that a difference of at least 29.2% under the null hypothesis would only happen about 2% of the time. Such a low probability indicates that observing such a large difference from chance alone is rare.

8.4.4 The conclusion

The difference of 29.2% is a rare event if there really is no effect of sex on promotion decisions. This provides us with two possible interpretations:

- If H_0 is true: Sex has no effect on promotion decisions, and we observed a difference that is so large it would only happen rarely by chance.
- If H_A is true: Sex has an effect on promotion decisions, and what we observed was actually due to equally qualified female candidates being discriminated against.

When we conduct formal studies, we reject a null position if the data strongly conflict with it. Since there was only about a 2% probability of obtaining a difference at least as large as 29.2% under the null hypothesis, we conclude that the data provide strong evidence of sex discrimination. We reject the null hypothesis in favor of the alternative.

Try it yourself! Use StatBench’s Permutation Test Simulator to replicate the sex discrimination analysis:

[Launch Randomization Test: Difference in Proportions](#)

1. Enter the sex discrimination data (or use the built-in dataset).
2. Generate 1,000 shuffled differences.
3. Find how many simulated differences are at least as large as 0.292.
4. What proportion of simulations produced a difference this extreme or more?

8.5 Case study: opportunity cost

How rational and consistent is the behavior of the typical American college student? Here, we explore whether reminding students that money not spent now can be spent later causes them to be thriftier.

One hundred and fifty students were recruited for the study. All were given a scenario about purchasing a video on sale for \$14.99. Half (the control group) were given these options:

- (A) Buy this entertaining video.
- (B) Not buy this entertaining video.

The other half (the treatment group) saw a slightly modified option (B):

- (A) Buy this entertaining video.
- (B) Not buy this entertaining video. Keep the \$14.99 for other purchases.

The hypotheses are:

- H_0 : Reminding students that they can save money for later purchases will not have any impact on their spending decisions.
- H_A : Reminding students that they can save money for later purchases will reduce the chance they will continue with a purchase.

8.5.1 Observed data

The results are summarized below:

	Buy video	Not buy video	Total
Control	56	19	75
Treatment	41	34	75
Total	97	53	150

We define a “success” as a student who chooses *not* to buy the video. The difference in the proportion choosing not to buy is:

$$\hat{p}_T - \hat{p}_C = \frac{34}{75} - \frac{19}{75} = 0.453 - 0.253 = 0.200$$

The proportion of students who chose not to buy was 20 percentage points higher in the treatment group. Is this 20% difference so prominent that it is unlikely to have occurred from chance alone?

8.5.2 Simulating under the null

Using the same randomization technique, we start with 150 index cards: 53 labeled “not buy video” and 97 labeled “buy video.” We shuffle all cards together and deal them into two stacks of 75.

If we randomly assign the cards into simulated treatment and control groups, how many “not buy video” cards would we expect in each group? What would be the expected difference in proportions?

Since the groups are equal in size, we would expect about $53/2 \approx 26$ or 27 “not buy video” cards in each group, yielding a difference of roughly 0%. Due to random chance, we might sometimes observe a little more or fewer than 26–27.

After running 1,000 simulations, the results show that the distribution of simulated differences is centered at 0, as expected under the null hypothesis.

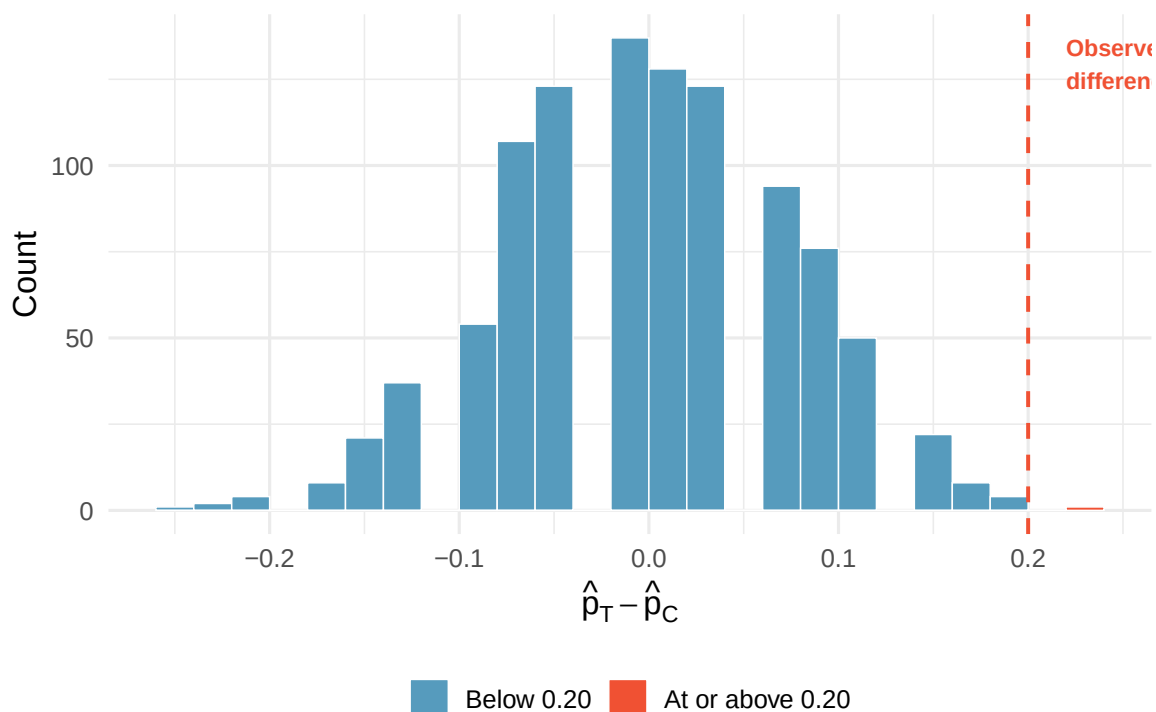


Figure 8.2: A histogram of 1,000 simulated differences produced under the null hypothesis. Differences at least as large as the observed 0.20 are highlighted in red.

Under the null hypothesis, we would observe a difference of at least +20% about 0.6% of the time. That is very rare! We conclude the data provide strong evidence there is a treatment effect: reminding students that they could spend money later lowers their chance of making a purchase. Because this is an experiment (with random assignment), we can make a causal claim.

8.6 P-values

The probability we computed in the previous sections — the chance of observing a result as extreme as or more extreme than the one actually observed, *if the null hypothesis were true* — has a name.

The **p-value** is the probability of observing data at least as favorable to the alternative hypothesis as our current dataset, if the null hypothesis were true. We typically use a summary statistic of the data, such as a difference in proportions, to compute the p-value. This summary value is called the **test statistic**.

In the sex discrimination study, the test statistic was the difference in promotion rates. What was the test statistic in the opportunity cost study?

The test statistic in the opportunity cost study was the difference in the proportion of students who decided against the video purchase in the treatment and control groups: $\hat{p}_T - \hat{p}_C = 0.200$.

Interpreting the p-value.

- A **small p-value** (e.g., below 0.05) means that the observed result would be rare if the null hypothesis were true. This is evidence *against* the null hypothesis.
- A **large p-value** (e.g., above 0.10) means that the observed result is not particularly unusual under the null hypothesis. The data do not provide strong evidence against the null hypothesis.
- The p-value is **not** the probability that the null hypothesis is true. It is the probability of the observed data (or more extreme data) *given that* the null hypothesis is true.

A common mistake is to interpret the p-value as the probability that the null hypothesis is true (e.g., “there is a 2% chance that sex has no effect on promotions”). This is incorrect. The p-value tells us how *surprising* the data would be *if* the null hypothesis were true, not how likely the null hypothesis is to be true.

8.7 Statistical significance

We say that the data provide **statistically significant** evidence against the null hypothesis if the p-value is less than a predetermined threshold called the **significance level**, denoted α (the Greek letter alpha).

The most commonly used significance level is $\alpha = 0.05$, meaning we reject the null hypothesis when the p-value is less than 5%. However, there is nothing magical about 0.05 — it is a convention, and in some fields or applications a stricter (e.g., 0.01) or more lenient (e.g., 0.10) threshold may be appropriate.

In the opportunity cost study, we found that a difference of at least 20% would occur only about 6-in-1,000 times if the reminder had no influence on student decisions. The p-value is approximately 0.006. Would you classify this as statistically significant?

Yes. Since $p\text{-value} = 0.006 < 0.05 = \alpha$, the data provide statistically significant evidence that US college students were actually influenced by the reminder.

What “statistically significant” does and does not mean.

“Statistically significant” means the p-value fell below the chosen significance level. It does **not** necessarily mean the effect is large or practically important. A very large sample can detect a tiny difference as “statistically significant” even if the difference is too small to matter in practice.

Conversely, “not statistically significant” does not mean there is no effect. It may simply mean the sample was too small to detect the effect.

8.7.1 Decision framework

The logic of hypothesis testing follows a consistent pattern:

If the p-value is...	Then we...	Because...
Less than α	Reject H_0	The data are unlikely under H_0 ; evidence supports H_A
Greater than or equal to α	Fail to reject H_0	The data are not sufficiently unusual under H_0

A researcher tests whether a new teaching method improves exam scores compared to the standard method. The p-value from the randomization test is 0.12. Using $\alpha = 0.05$, what conclusion should the researcher draw?

Show answer

Since $p\text{-value} = 0.12 > 0.05 = \alpha$, the researcher fails to reject the null hypothesis. The data do not provide sufficient evidence that the new teaching method improves exam scores. This does not mean the new method is ineffective — only that this study did not produce enough evidence to conclude it is effective.

8.8 Two-sided tests

In the sex discrimination study, we looked for differences in *one direction only* — whether female candidates were promoted at a lower rate. This is called a **one-sided** (or one-tailed) test. The alternative hypothesis specified a direction:

$$H_A : p_M - p_F > 0 \quad (\text{males promoted at a higher rate})$$

Sometimes, however, we are interested in detecting a difference in *either direction*. For example, if we are studying whether a new drug changes blood pressure (it could increase it *or* decrease it), we would use a **two-sided** (or two-tailed) test:

$$H_A : p_1 \neq p_2 \quad (\text{the proportions are different, in either direction})$$

In a **one-sided test**, the alternative hypothesis specifies a direction (greater than or less than). The p-value is computed as the proportion of the randomization distribution in one tail.

In a **two-sided test**, the alternative hypothesis does not specify a direction (just “different”). The p-value is computed as the proportion of the randomization distribution in *both* tails — values at least as extreme as the observed statistic in either direction.

Suppose we are testing whether a coin is fair. We flip it 100 times and get 60 heads.

- **One-sided** ($H_A : p > 0.5$): The p-value counts only simulations with 60 or more heads.
- **Two-sided** ($H_A : p \neq 0.5$): The p-value counts simulations with 60 or more heads *and* simulations with 40 or fewer heads (equally extreme in the other direction).

For a symmetric randomization distribution, the two-sided p-value is approximately twice the one-sided p-value. If the one-sided p-value is 0.03, the two-sided p-value is approximately 0.06.

A researcher wants to know if a new fertilizer changes plant growth compared to no fertilizer. Should this be a one-sided or two-sided test? Write the hypotheses using appropriate notation, where μ_T is the mean growth with fertilizer and μ_C is the mean growth without.

Show answer

This should be a two-sided test because the fertilizer could either increase or decrease growth (perhaps it has a toxic effect in some soils). The hypotheses are: $H_0 : \mu_T - \mu_C = 0$ (no difference in mean growth) and $H_A : \mu_T - \mu_C \neq 0$ (the fertilizer changes mean growth, in either direction).

Choosing one-sided vs. two-sided.

- Use a **one-sided test** when you have a specific directional hypothesis *before looking at the data*. For example, “we believe this drug *lowers* blood pressure.”
- Use a **two-sided test** when you want to detect a difference in either direction, or when you are not sure which direction the difference might go.
- When in doubt, use a two-sided test. It is the more conservative choice and is less susceptible to bias from choosing the direction after seeing the data.

8.9 The randomization test procedure: a summary

We can summarize the randomization test procedure as follows:

1. **Frame the research question in terms of hypotheses.** The null hypothesis (H_0) usually represents no difference or no effect. The alternative hypothesis (H_A) represents the claim of interest.
2. **Collect data.** If the research question focuses on associations but not causation, use an observational study. If you want to establish a causal connection, use an experiment with random assignment.
3. **Model what would happen under the null hypothesis.** Shuffle (randomize) the data to break any association between the explanatory and response variables. Compute the test statistic for each shuffle. Repeat many times to build the **randomization distribution** (also called the null distribution).
4. **Compute the p-value.** Determine what proportion of the randomization distribution is at least as extreme as the observed test statistic.
5. **Draw a conclusion.** If the p-value is small (less than α), reject H_0 and conclude there is evidence for H_A . If the p-value is not small, fail to reject H_0 . Write the conclusion in context, in plain language.

Practice the full procedure! Use StatBench’s Permutation Test Simulator for different scenarios:

[Launch Randomization Test: Difference in Proportions](#)

Try entering your own data and running the full randomization test. Observe how the p-value changes when you adjust the sample size or the magnitude of the observed difference.

8.10 Chapter review

8.10.1 Summary

- A **hypothesis test** evaluates two competing claims: the null hypothesis (H_0), which represents no effect or no difference, and the alternative hypothesis (H_A), which represents the claim under investigation.

- A **randomization test** (or permutation test) assesses the null hypothesis by shuffling the data to simulate what would happen if there were truly no effect. The collection of simulated test statistics forms the **randomization distribution**.
- The **test statistic** is the summary value computed from the data (e.g., a difference in proportions) that we use to evaluate the hypotheses.
- The **p-value** is the probability of observing a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. It quantifies the strength of the evidence against H_0 .
- We reject H_0 when the p-value is less than the **significance level** α (commonly 0.05). This is called a **statistically significant** result.
- A **one-sided test** looks for effects in a specified direction; a **two-sided test** looks for effects in either direction.
- Failing to reject H_0 is not the same as proving H_0 is true.
- Statistical significance does not imply practical importance.

8.10.2 Key terms

Term	Definition
Hypothesis test	A formal procedure for evaluating competing claims using data
Null hypothesis (H_0)	The claim of no effect or no difference; the skeptical position
Alternative hypothesis (H_A)	The claim that there is an effect or difference
Randomization test	A hypothesis test that builds a null distribution by shuffling data
Randomization distribution	The distribution of the test statistic under repeated randomizations
Test statistic	The summary value (e.g., difference in proportions) used to evaluate hypotheses
P-value	Probability of observing a result as extreme as or more extreme than the data, if H_0 is true
Significance level (α)	The threshold below which the p-value leads us to reject H_0
Statistically significant	A result whose p-value falls below α
One-sided test	Alternative hypothesis specifies a direction
Two-sided test	Alternative hypothesis does not specify a direction

8.10.3 Exercises

Exercises for this chapter are under development.

Chapter 9

Bootstrap Confidence Intervals

In Chapter 6, we saw that sample statistics vary from sample to sample. This variability is not a nuisance — it's the key to quantifying uncertainty. In this chapter, we learn **bootstrapping**: a simulation-based method for constructing confidence intervals that requires no formulas and no assumptions about the shape of the population.

9.1 The big idea: resampling

Imagine you want to estimate the average height of all UW-La Crosse students. You take a random sample of 30 students and compute the sample mean: $\bar{x} = 170.2$ cm.

How confident should you be that $\bar{x} = 170.2$ is close to the true population mean μ ? If you took a different sample, you'd get a different $\bar{x}$. But you can't keep taking new samples from the population — you only have the one sample you collected.

The bootstrap idea: Treat your sample as a stand-in for the population and resample *from it*. Each **bootstrap resample** is drawn with replacement from your original sample, at the same sample size n .

A **bootstrap resample** is a sample of size n drawn **with replacement** from the original sample. The statistic computed from a bootstrap resample is called a **bootstrap statistic**. The collection of many bootstrap statistics forms the **bootstrap distribution**.

Why does this work? If the original sample is representative of the population, then resampling from it mimics what would happen if we could repeatedly sample from the population itself. The variability we see in the bootstrap distribution approximates the true sampling variability of our statistic.

9.2 Bootstrapping a sample mean

Let's build a bootstrap confidence interval step by step using a concrete example.

9.2.1 Step 1: Collect data

Suppose we have a random sample of $n = 30$ heights (in cm) from UW-La Crosse students:

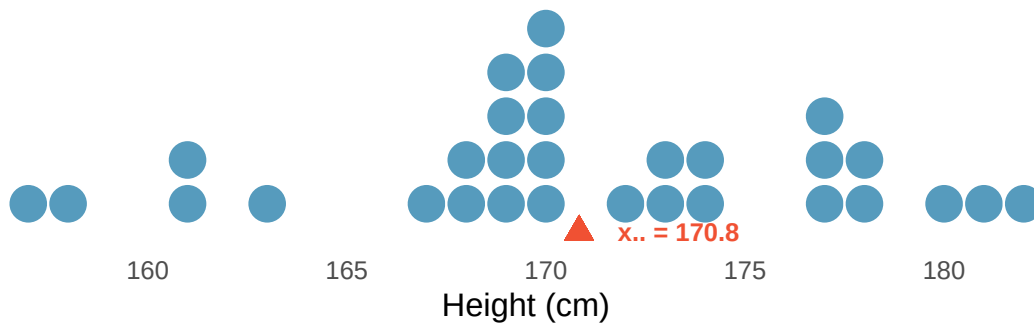


Figure 9.1: Dot plot of our original sample of 30 student heights. The sample mean is shown as a red triangle.

The sample mean is $\bar{x} = 170.8$ cm. This is our **point estimate** of the population mean μ . But how precise is this estimate?

9.2.2 Step 2: Resample with replacement

We draw a new sample of 30 values from the original 30, *with replacement*. Some values will appear more than once; others will be left out. Each such draw is one **bootstrap resample**, and the mean of that resample is a **bootstrap statistic** $\bar{x}^*$.

9.2.3 Step 3: Repeat many times

We repeat Step 2 a large number of times — typically $B = 1,000$ or more — and record the bootstrap mean $\bar{x}^*$ each time. This gives us a **bootstrap distribution**.

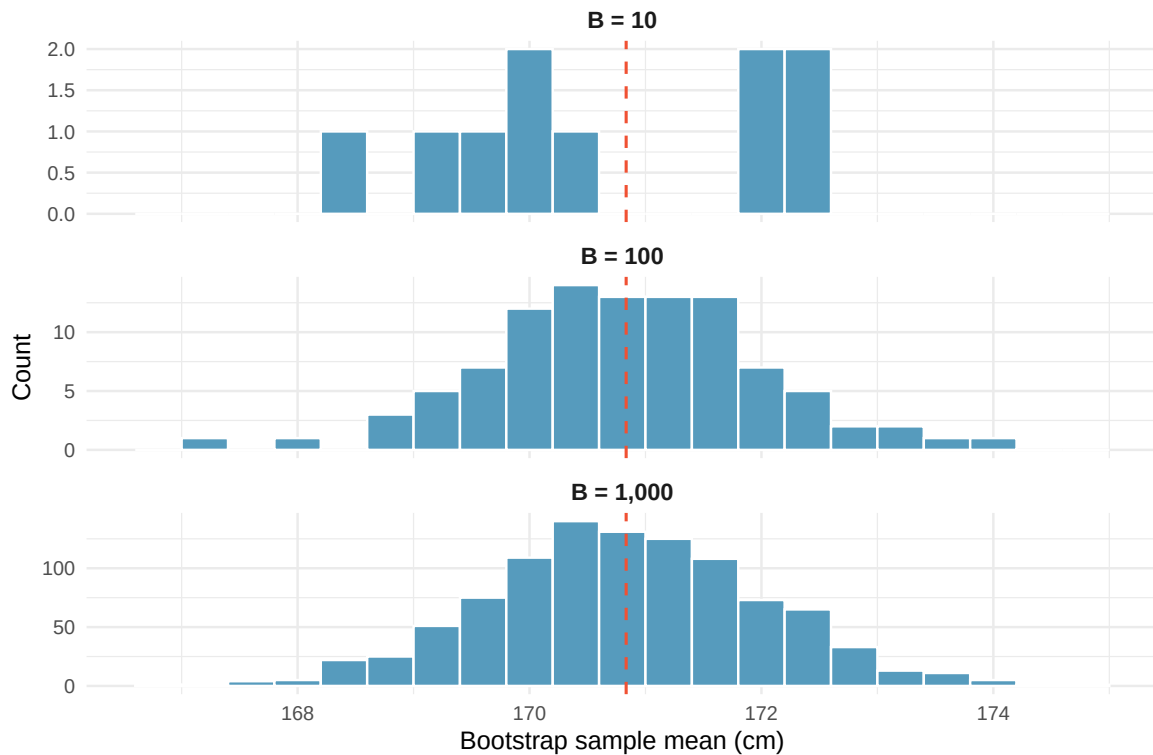


Figure 9.2: Building up the bootstrap distribution. With 10 resamples we see only a rough sketch; by 1,000 resamples the shape of the distribution is clear.

Notice that the bootstrap distribution is centered near the original sample mean (dashed red line). This is expected: the bootstrap resamples come from the original data, so their means cluster around $\bar{x}$.

9.2.4 Step 4: Find the confidence interval

For a 95% confidence interval, we find the middle 95% of the bootstrap distribution by taking the 2.5th and 97.5th percentiles.

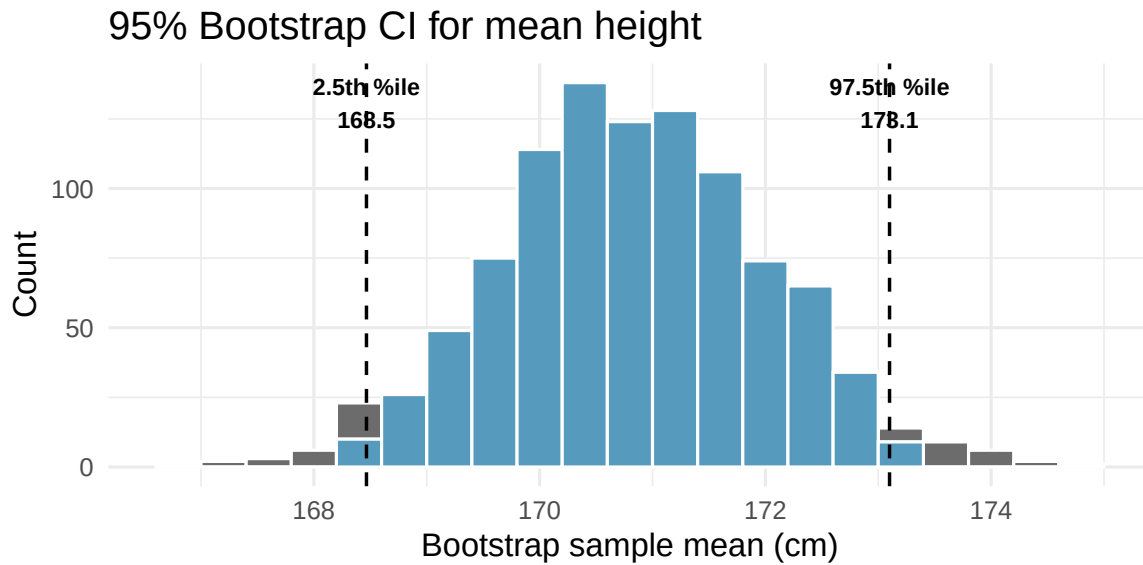


Figure 9.3: Bootstrap distribution of 1,000 sample means with the middle 95% shaded in blue. The 2.5th and 97.5th percentiles form the 95% confidence interval.

The 95% bootstrap confidence interval for the mean height is approximately (168.5, 173.1) cm.

Try it yourself! Use StatBench to generate a bootstrap distribution and confidence interval:

[Launch Bootstrap CI for One Sample](#)

1. Paste the heights data (or use StatBench's built-in dataset).
2. Click **+1000** to generate 1,000 bootstrap resamples.
3. Observe the bootstrap distribution forming.
4. Read the 95% confidence interval from the output.

How does the bootstrap distribution relate to the original sample mean?

9.3 Interpreting a confidence interval

A **confidence interval** is a range of plausible values for the population parameter. A **95% confidence interval** is constructed using a procedure that, when applied repeatedly to many different samples, captures the true parameter in approximately 95% of all intervals produced.

What "95% confident" means: If we repeated the entire study many times — collecting a new sample each time and computing a new 95% confidence interval — approximately 95% of those intervals would contain the true population parameter. Any single interval either contains the parameter or it doesn't; the 95% refers to the long-run success rate of the *procedure*.

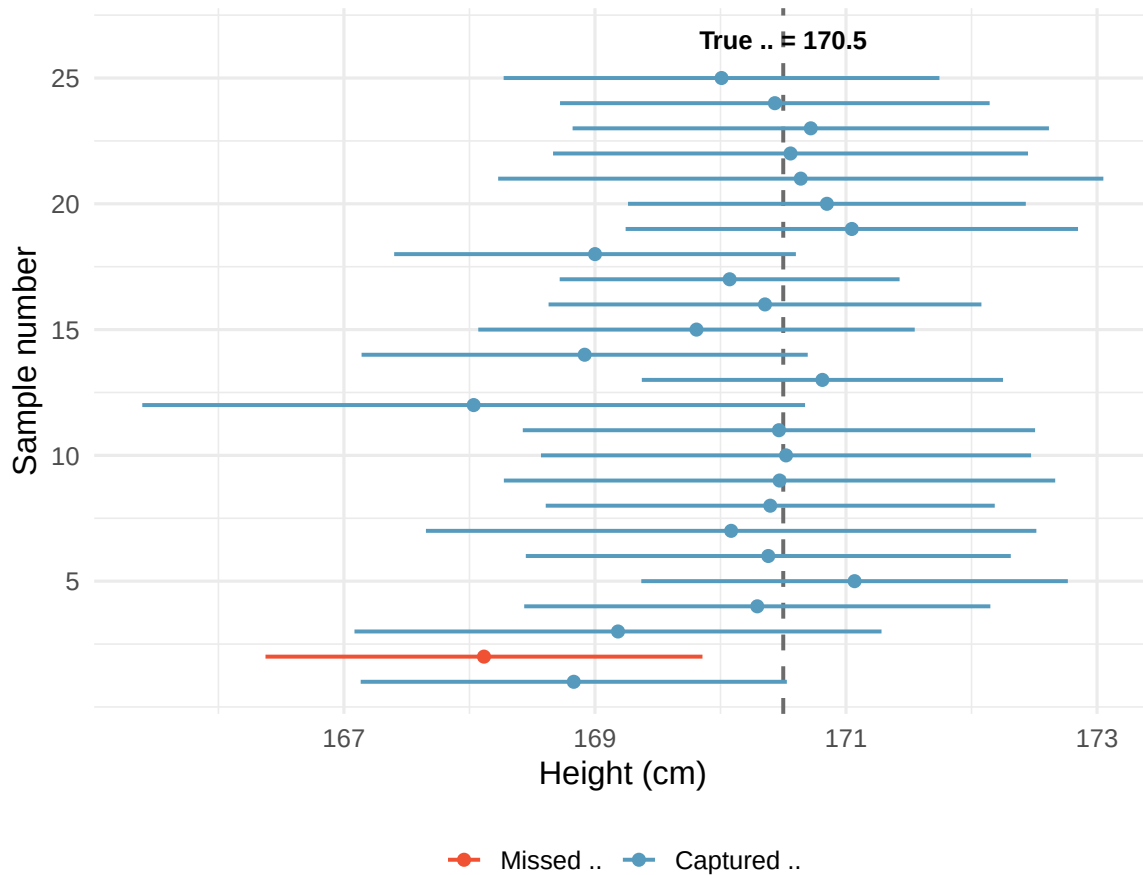


Figure 9.4: Twenty-five 95% confidence intervals from 25 different random samples. The true population mean is shown as a vertical dashed line. Approximately 95% of intervals capture the true mean; a few (shown in red) miss it.

Each horizontal line in Figure 9.4 represents a 95% CI from a different sample. Most intervals (blue) capture the true mean, but a few (red) miss it entirely. Over many samples, roughly 95% of the intervals succeed. This is the meaning of “95% confidence.”

Common misinterpretation: “There is a 95% probability that the true mean is in this interval.” This statement treats the population mean as random and the interval as fixed. In fact, the population mean is fixed (we just don’t know it) and the interval is random (it depends on which sample we happened to collect).

A 95% bootstrap CI for the mean height of UW-La Crosse students is (167.8, 174.6). Which of the following interpretations is correct?

- 95% of all UW-La Crosse students have heights between 167.8 and 174.6 cm.
- There is a 95% probability that the true mean height is between 167.8 and 174.6.
- We are 95% confident that the true mean height of all UW-La Crosse students is between 167.8 and 174.6 cm.

Show answer

Only (c) is correct. Statement (a) describes individual heights, not the mean. Statement (b) assigns a probability to the fixed parameter, which is incorrect. Statement (c) correctly describes confidence

in the procedure.

9.4 The percentile method

The approach we've used is called the **percentile method** for bootstrap confidence intervals.

For a $(1 - \alpha) \times 100\%$ confidence interval using the percentile method:

- **Lower bound** = the $(\alpha/2) \times 100$ th percentile of the bootstrap distribution
- **Upper bound** = the $(1 - \alpha/2) \times 100$ th percentile of the bootstrap distribution

For a 95% CI ($\alpha = 0.05$): use the 2.5th and 97.5th percentiles. For a 90% CI ($\alpha = 0.10$): use the 5th and 95th percentiles. For a 99% CI ($\alpha = 0.01$): use the 0.5th and 99.5th percentiles.

If you increase the confidence level from 95% to 99%, what happens to the width of the confidence interval? Why?

Show answer

The interval gets wider. A 99% CI must capture the true parameter more often than a 95% CI, so it must cast a wider net. The 0.5th percentile is farther from the center than the 2.5th percentile.

9.5 Medical consultant case study

People providing an organ for donation sometimes seek the help of a special medical consultant. These consultants assist the patient in all aspects of the surgery, with the goal of reducing the possibility of complications during the medical procedure and recovery.

One consultant tried to attract patients by noting the average complication rate for liver donor surgeries in the US is about 10%, but her clients have had only 3 complications in the 62 liver donor surgeries she has facilitated. She claims this is strong evidence that her work reduces complications.

We estimate the true complication rate p using the sample proportion: $\hat{p} = 3/62 = 0.048$ (4.8%).

Is it possible to assess the consultant's claim — that the *reduction* in complications is *due to her work* — using these data?

No. The claim is causal, but the data are observational. Patients who can afford a medical consultant may also afford better medical care overall, which could lead to a lower complication rate. We cannot establish causation here. But we *can* estimate the consultant's true complication rate using a confidence interval.

To build a bootstrap CI for a proportion, we treat the 62 patients as marbles in a bag: 3 red (complication) and 59 white (no complication). We draw 62 marbles with replacement, compute $\hat{p}_{boot}^*$, and repeat 10,000 times.

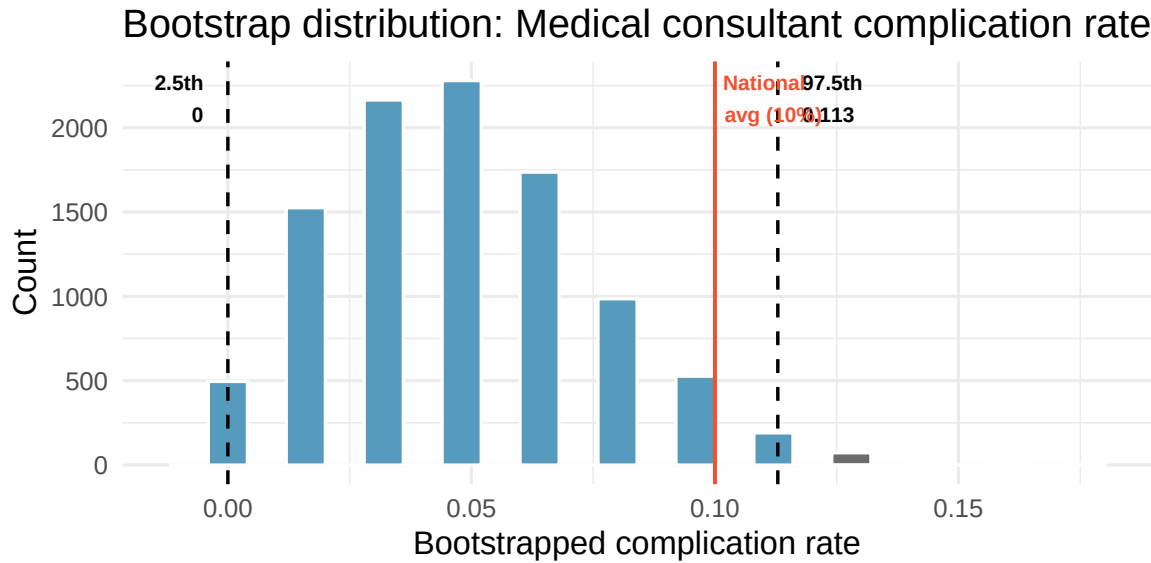


Figure 9.5: Bootstrap distribution of 10,000 resampled complication rates. The 95% CI extends from the 2.5th percentile to the 97.5th percentile. Because the interval includes 10%, we cannot conclude the consultant’s rate is lower than the national average.

The 95% bootstrap CI is approximately (0, 0.113). Because the interval includes 0.10 (the national average), we cannot conclude that the consultant’s complication rate is lower than the national rate.

Suppose the consultant had facilitated 620 surgeries (instead of 62) with the same complication rate of 4.8% (about 30 complications). How would the bootstrap CI change? Would the conclusion change?

Show answer

With 10 times as much data, the bootstrap distribution would be much narrower. The 95% CI would likely fall entirely below 10%, providing strong evidence that the consultant’s rate is lower. Larger samples give more precise estimates.

9.6 Tappers and listeners case study

A Stanford University graduate student named Elizabeth Newton conducted an experiment using a simple game: one person (the “tapper”) taps out a well-known song on a desk, and another person (the “listener”) tries to guess the song. About 50% of tappers expected the listener would guess correctly.

In Newton’s study, only 3 out of 120 listeners ($\hat{p} = 3/120 = 0.025$) guessed the tune correctly. What is the true proportion of listeners who can guess the song?

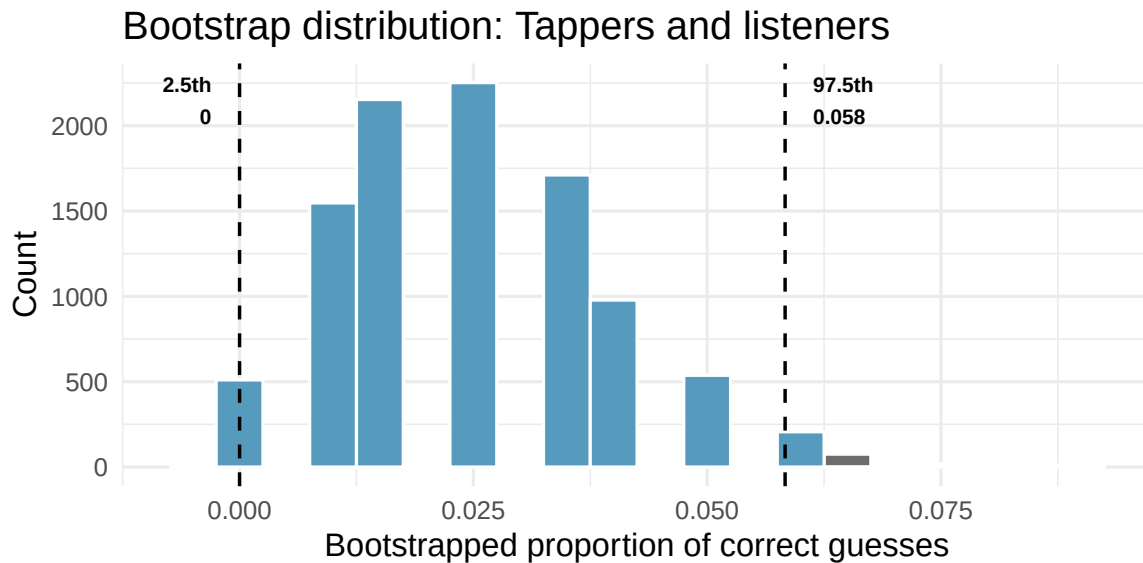


Figure 9.6: Bootstrap distribution for the proportion of listeners who guessed correctly. The 95% CI is far below the tappers' expected success rate of 50%, providing very convincing evidence against the tappers' expectation.

The 95% bootstrap CI is approximately (0, 0.058). The tappers' expectation of 50% is nowhere near this interval — it provides **very convincing evidence** that listeners are far worse at guessing than tappers expect. This is an example of the “curse of knowledge”: once you know something, it's hard to imagine not knowing it.

Do the data provide convincing evidence against the claim that 50% of listeners can guess the tapper's tune?

Show answer

Yes, very convincing evidence. The value 50% (0.50) is far outside the 95% CI, and even far from the largest bootstrapped proportion. The data strongly suggest that the true success rate is much lower than 50%.

9.7 What affects the width of a CI?

Three factors control how wide a confidence interval is:

1. **Sample size (n):** Larger samples → narrower intervals (more information about the population)
2. **Variability in the data:** More spread → wider intervals (less precision)
3. **Confidence level:** Higher confidence → wider intervals (more certainty requires a wider net)

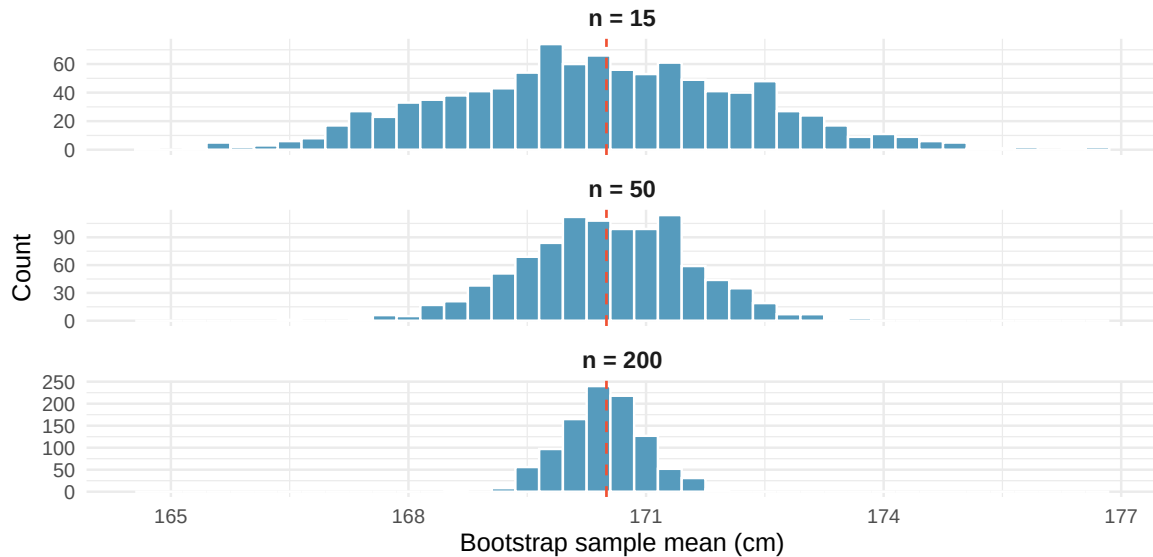


Figure 9.7: The effect of sample size on the bootstrap distribution. Larger samples produce narrower bootstrap distributions, leading to narrower confidence intervals.

As sample size increases, the bootstrap distribution becomes tighter around the sample mean. This is why large studies produce more precise estimates than small ones.

Explore it! Return to the [Bootstrap CI tool](#) and experiment:

- Try different sample sizes. How does the CI width change?
- Try 90% vs 95% vs 99% confidence levels. What happens?

9.8 When does bootstrapping work?

Bootstrapping is remarkably flexible, but it does require some conditions to produce reliable intervals:

1. **Independence:** The observations in the sample should be independent of one another. This is usually satisfied when data come from a random sample or a randomized experiment.
2. **Representative sample:** The sample should be reasonably representative of the population. If the sample is biased (e.g., only volunteers), the bootstrap CI will reflect that bias.
3. **Sufficient sample size:** Very small samples (e.g., $n < 10$) may not contain enough information about the population's shape. With small n , the bootstrap distribution can be unreliable.

Bootstrapping does **not** require the population to be normally distributed. This is one of its greatest strengths compared to traditional formula-based methods. The bootstrap lets the data “speak for themselves.”

A researcher surveys 8 of her friends about their study hours and computes a bootstrap CI. What condition is most clearly violated?

Show answer

Two conditions are violated: (1) the sample is a convenience sample of friends, not a random sample, so it may not be representative of the population, and (2) $n = 8$ is quite small, making the bootstrap distribution potentially unreliable.

9.9 Bootstrap CI for other statistics

The beauty of bootstrapping is its **generality**. The same resampling logic works for *any* statistic, not just the mean:

- **Median:** Resample → compute median → repeat → percentile CI
- **Proportion:** Resample → compute proportion → repeat → percentile CI
- **Standard deviation:** Resample → compute SD → repeat → percentile CI
- **Difference in means:** Resample from each group → compute difference → repeat → percentile CI

This is a powerful advantage. Many statistics (like the median or the difference in medians) don't have convenient formulas for their standard error. Bootstrapping handles them all the same way.

StatBench provides bootstrap CI tools for several statistics:

- [Bootstrap CI: One Proportion](#)
- [Bootstrap CI: Paired Differences](#)
- [Bootstrap CI: Two Samples](#)

9.10 Chapter review

9.10.1 Summary

- The **bootstrap** constructs a confidence interval by resampling with replacement from the original sample.
- The **bootstrap distribution** approximates the sampling distribution of the statistic.
- The **percentile method** uses the middle $(1 - \alpha) \times 100\%$ of the bootstrap distribution as the CI.
- A **95% confidence interval** means that if we repeated the study many times, about 95% of the resulting intervals would capture the true parameter.
- CI width depends on sample size, data variability, and confidence level.
- Bootstrap CIs work for any statistic — means, medians, proportions, differences, and more.
- Bootstrapping requires independent observations from a representative sample with sufficient sample size, but does **not** require normality.

9.10.2 Key terms

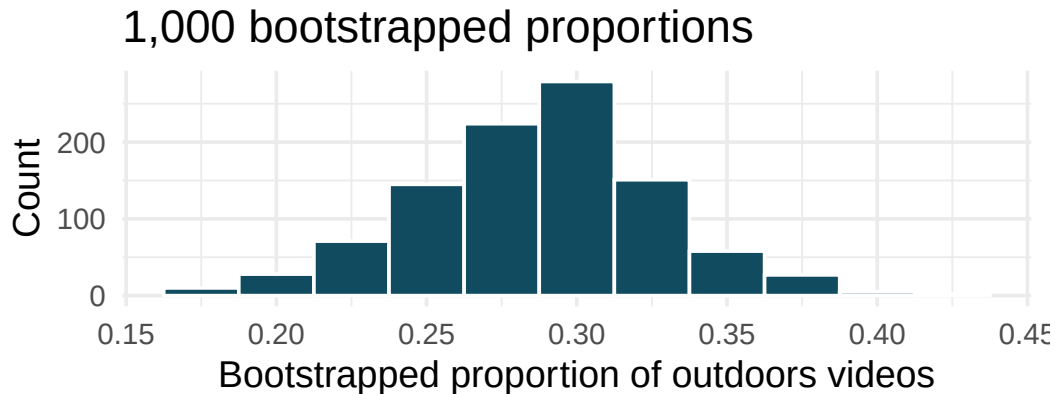
- Bootstrap resample
- Bootstrap statistic
- Bootstrap distribution
- Confidence interval
- Confidence level
- Percentile method
- Point estimate
- Parameter
- Sampling with replacement

9.10.3 Exercises

Answers to odd-numbered exercises are provided inline.

1. **Outside YouTube videos.** Let's say that you want to estimate the proportion of YouTube videos which take place outside (define "outside" to be if any part of the video takes place outdoors). You take a random sample of 128 YouTube videos and determine that 37 of them take place

outside. You'd like to estimate the proportion of all YouTube videos which take place outside, so you decide to create a bootstrap interval from the original sample of 128 videos.

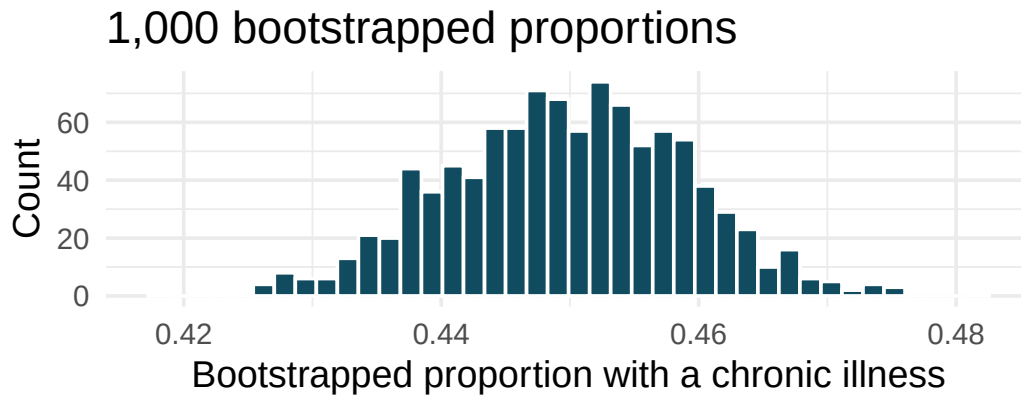


- Describe in words the relevant statistic and parameter for this problem. If you know the numerical value for either one, provide it. If you don't know the numerical value, explain why the value is unknown.
- What notation is used to describe, respectively, the statistic and the parameter?
- If using software to bootstrap the original dataset, what is the statistic calculated on each bootstrap sample?
- When creating a bootstrap sampling distribution (histogram) of the bootstrapped sample proportions, where should the center of the histogram lie?
- The histogram provides a bootstrap sampling distribution for the sample proportion (with 1,000 bootstrap repetitions). Using the histogram, estimate a 90% confidence interval for the proportion of YouTube videos which take place outdoors.
- Interpret the confidence interval in context of the data.

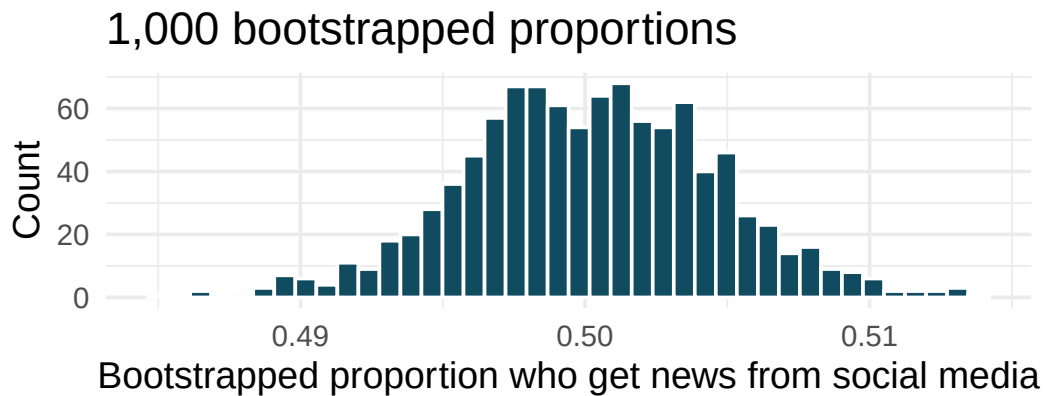
Show answer

- The statistic is the sample proportion $\hat{p} = 37/128 = 0.289$; the parameter is the true population proportion p , which is unknown. (b) $\hat{p}$ and p . (c) The bootstrap sample proportion. (d) At the observed sample proportion, 0.289. (e) Roughly (0.22, 0.35). (f) We are 90% confident that between 22% and 35% of all YouTube videos take place outdoors.

2. **Chronic illness.** In 2012 the Pew Research Foundation reported that “45% of US adults report that they live with one or more chronic conditions.” However, this value was based on a sample, so it may not be a perfect estimate for the population parameter of interest on its own. The study was based on a sample of 3,014 adults. Below is a distribution of 1,000 bootstrapped sample proportions from the Pew dataset. Using the distribution of 1,000 bootstrapped proportions, approximate a 92% confidence interval for the true proportion of US adults who live with one or more chronic conditions and interpret it.



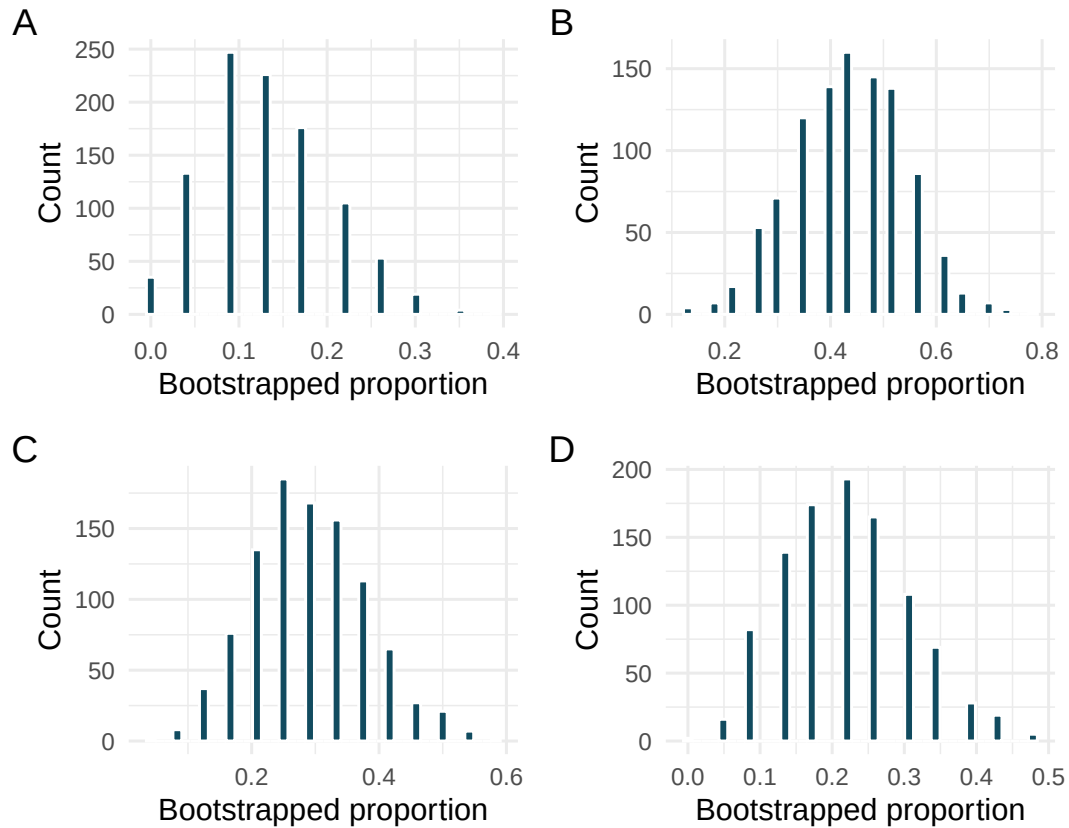
3. **Social media users and news, bootstrapping.** A poll conducted in 2022 found that 50% of U.S. adults get news from social media sometimes or often. However, the value was based on a sample, so it may not be a perfect estimate for the population parameter of interest on its own. The study was based on a sample of 12,147 adults. Below is a distribution of 1,000 bootstrapped sample proportions from the Pew dataset. Using the distribution of 1,000 bootstrapped proportions, approximate a 98% confidence interval for the true proportion of US adult social media users (in 2022) who get at least some of their news from social media. Interpret the interval in the context of the problem.



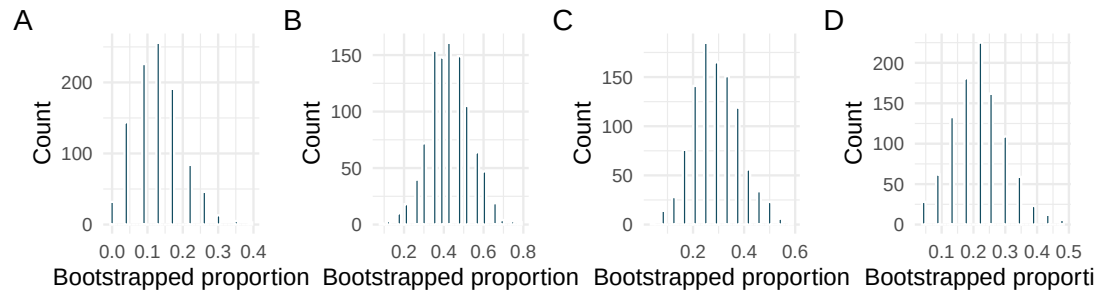
Show answer

With 98% confidence, the true proportion of all US adults (in 2022) who get news from social media sometimes or often is between approximately 0.487 and 0.513.

4. **Bootstrap distributions of $\hat{p}$, I.** Each of the following four distributions was created using a different dataset. Each dataset was based on $n = 23$ observations. The original datasets had the following proportions of successes: $\hat{p} = 0.13$, $\hat{p} = 0.22$, $\hat{p} = 0.30$, and $\hat{p} = 0.43$. Match each histogram with the original data proportion of success.



5. **Bootstrap distributions of $\hat{p}$, II.** Each of the following four distributions was created using a different dataset. Each dataset was based on $n = 23$ observations.



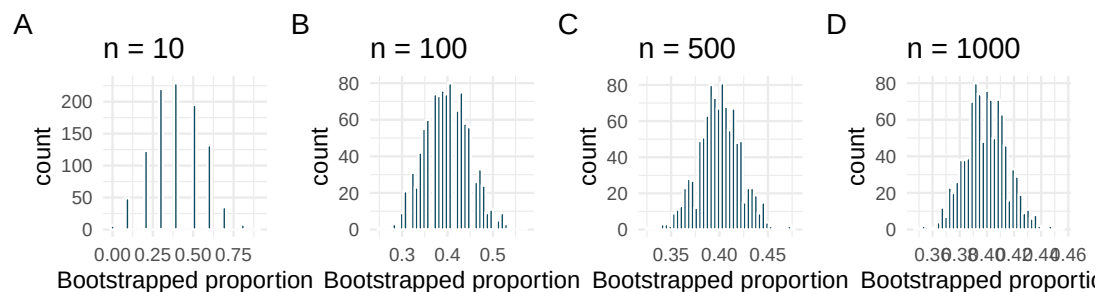
Consider each of the following values for the true population proportion p . Datasets A, B, C, D were bootstrapped 1,000 times, with bootstrap proportions as given in the histograms provided. For each parameter value, list the datasets which could plausibly have come from that population. (Hint: there may be more than one dataset for each parameter value.)

- $p = 0.05$
- $p = 0.25$
- $p = 0.45$
- $p = 0.55$
- $p = 0.75$

Show answer

(a) A or perhaps D. (b) A, B, C, or D. (c) B or C. (d) B. (e) None.

6. **Bootstrap distributions of $\hat{p}$, III.** Each of the following four distributions was created using a different dataset. Each dataset had the same proportion of successes ($\hat{p} = 0.4$) but a different sample size. The four datasets had $n = 10$, $n = 100$, $n = 500$, and $n = 1000$.



Consider each of the following values for the true population proportion p . Datasets A, B, C, D were bootstrapped 1,000 times, with bootstrap proportions as given in the histograms provided. For each parameter value, list the datasets which could plausibly have come from that population. (Hint: there may be more than one dataset for each parameter value.)

- $p = 0.05$
- $p = 0.25$
- $p = 0.45$
- $p = 0.55$
- $p = 0.75$

7. **Cyberbullying rates.** Teens were surveyed about cyberbullying, and 54% to 64% reported experiencing cyberbullying (95% confidence interval). Answer the following questions based on this interval.
- A newspaper claims that a majority of teens have experienced cyberbullying. Is this claim supported by the confidence interval? Explain your reasoning.
 - A researcher conjectured that 70% of teens have experienced cyberbullying. Is this claim supported by the confidence interval? Explain your reasoning.
 - Without actually calculating the interval, determine if the claim of the researcher from part (b) would be supported based on a 90% confidence interval?

Show answer

- (a) This claim is reasonable, since the entire interval lies above 50%. (b) The value of 70% lies outside of the interval, so we have convincing evidence that the researcher's conjecture is wrong. (c) A 90% confidence interval will be narrower than a 95% confidence interval. Even without calculating the interval, we can tell that 70% would not fall in the interval, and we would reject the researcher's conjecture based on a 90% confidence level as well.
8. **Waiting at an ER.** A 95% confidence interval for the mean waiting time at an emergency room (ER) is (128 minutes, 147 minutes). Answer the following questions based on this interval.
- A local newspaper claims that the average waiting time at this ER exceeds 3 hours. Is this claim supported by the confidence interval? Explain your reasoning.
 - The Dean of Medicine at this hospital claims the average wait time is 2.2 hours. Is this claim supported by the confidence interval? Explain your reasoning.
 - Without actually calculating the interval, determine if the claim of the Dean from part (b) would be supported based on a 99% confidence interval?

Chapter 10

Sampling Distributions via Simulation

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In Chapters 7 and 8, we used simulation to answer two fundamental questions of inference: *Is there an effect?* (randomization tests) and *How big is the parameter?* (bootstrap confidence intervals). Both methods relied on the computer to generate distributions — randomization distributions and bootstrap distributions — that helped us quantify uncertainty. In this chapter, we step back to see the bigger picture. We examine the shape, center, and spread of these simulated distributions and discover a remarkable pattern: under certain conditions, they all look approximately **normal** (bell-shaped). This observation, formalized as the **Central Limit Theorem**, is the bridge from the simulation-based methods of Part II to the formula-based methods of Part III.

10.1 From simulation to theory

In Chapters 7 and 8, we encountered several case studies:

- **Sex discrimination** (Chapter 7): We shuffled promotion decisions and computed the difference in promotion rates 10,000 times.
- **Opportunity cost** (Chapter 7): We shuffled buying decisions and computed the difference in proportions 1,000 times.
- **Medical consultant** (Chapter 8): We bootstrapped the complication rate from 62 surgeries 10,000 times.
- **Tappers and listeners** (Chapter 8): We bootstrapped the proportion of correct guesses from 120 listeners 10,000 times.

Each of these simulations produced a distribution of a sample statistic. While they differed in their settings, outcomes, and the simulation technique used, they all shared something in common: the general shape of the distribution.

Think back to the randomization and bootstrap distributions you have seen. What shape did most of them have? Were they skewed, uniform, or something else?

Show answer

In general, the distributions were approximately symmetric and bell-shaped. The medical consultant example was the only one with noticeable skew (skewed right), which makes sense because the proportion was very close to zero.

This is no coincidence. The bell shape keeps appearing because of a deep mathematical result that underlies most of statistics. Whether we shuffle labels, resample with replacement, or imagine taking repeated samples from a population, the distribution of the resulting statistic tends to look the same — symmetric, unimodal, and bell-shaped.

In the simulation chapters, we let the computer show us this pattern directly. Now, we formalize it.

10.2 The Central Limit Theorem

The **Central Limit Theorem (CLT)** states that when we take sufficiently large random samples from a population, the sampling distribution of many common statistics (such as $\hat{p}$ or $\bar{x}$) is approximately **normal** (bell-shaped), regardless of the shape of the underlying population distribution.

This is a stunning result. Even if the population distribution is skewed, bimodal, or has some other non-normal shape, the distribution of the *sample statistic* will be approximately normal — provided the sample is large enough.

Central Limit Theorem: the key ideas.

1. The sampling distribution of a sample statistic is approximately normal when the sample size is large enough.
2. The center of the sampling distribution is at the population parameter.
3. The spread of the sampling distribution (the standard error) decreases as sample size increases.

These three facts together mean: with a large enough sample, the sample statistic will be close to the population parameter, and we can use the normal distribution to describe *how* close.

10.2.1 Conditions for the CLT

The normal approximation does not apply in all situations. Two conditions must hold:

1. **Independent observations.** The observations in the sample must be independent of each other. This is typically guaranteed by random sampling from a population, or by random assignment in an experiment.
2. **Large enough sample.** The sample size cannot be too small. What counts as “large enough” depends on the context:
 - For proportions: we generally need at least 10 expected successes and 10 expected failures.
 - For means: the rule of thumb depends on how skewed the population is. For roughly symmetric populations, $n \geq 30$ is often sufficient. For highly skewed populations, larger samples may be needed.

When these conditions are not met, the normal approximation may be poor and simulation-based methods (randomization tests, bootstrap CIs) are the better choice. This is one of the great advantages of simulation: it works even when mathematical conditions are not satisfied.

10.2.2 Visualizing the CLT

Imagine a population that is distinctly non-normal — say, a right-skewed distribution like household income. If we take a single random sample of $n = 5$ people and compute the mean income, that mean might be quite far from the population mean μ . But if we do this thousands of times (each time taking a new sample of 5 and computing the mean), the distribution of those thousands of sample means will already start to look somewhat bell-shaped, even though the underlying population is skewed.

Now increase the sample size to $n = 30$. The distribution of sample means looks even more bell-shaped. At $n = 100$, it is nearly indistinguishable from a perfect normal curve.

This progression — from skewed population, to somewhat normal at small n , to very normal at larger n — is the Central Limit Theorem in action.

Sample size	Shape of sampling distribution
$n = 5$	Retains some skew from the population
$n = 30$	Approximately normal for most populations
$n = 100$	Very close to normal for virtually all populations

The speed at which the sampling distribution approaches normality depends on how non-normal the population is. Symmetric populations converge quickly; highly skewed populations take longer.

See the CLT in action! Use StatBench's Sampling Distribution Simulator:

[Launch Sampling Distribution Simulator](#)

1. Select a skewed population (e.g., exponential or right-skewed).
2. Set the sample size to $n = 5$ and draw 1,000 samples. Observe the shape.
3. Increase to $n = 30$ and draw 1,000 more. How does the shape change?
4. Try $n = 100$. How close is the distribution to a normal curve?

10.3 Comparing randomization, bootstrap, and mathematical approaches

We now have three tools for statistical inference, each grounded in the same core ideas but using different mechanisms:

	Randomization	Bootstrap	Mathematical model
Question answered	Is there an effect? (hypothesis test)	How big is the parameter? (confidence interval)	Both
How variability is generated	Shuffle labels to simulate H_0	Resample with replacement from sample	Use formulas based on the normal distribution
What it models	Variability due to random assignment	Variability due to random sampling	Variability predicted by mathematical theory
Center of distribution	Null hypothesis value	Sample statistic	Depends on context (H_0 for tests, $\hat{p}$ for CIs)
Conditions required	Minimal	Minimal	Independence + large sample
Computational cost	Moderate (need many simulations)	Moderate (need many resamples)	Low (just plug into formula)

Key insight. All three methods answer the same fundamental question: *How much does the statistic vary?* They just measure that variability in different ways.

- **Randomization** asks: “How much would the statistic vary if the null hypothesis were true?”
- **Bootstrap** asks: “How much would the statistic vary across different samples from this population?”
- **Mathematical models** ask: “How much does theory predict the statistic would vary?”

When conditions are met, all three give similar answers. The mathematical approach is faster and requires no simulation, which is why it is widely used. But the simulation approaches are more flexible and work even when conditions for the mathematical model are not met.

10.3.1 Which method should I use?

Here is a practical guide:

1. **When conditions for the mathematical model are clearly met** (independent observations, large sample, not too skewed), the formula-based approach is convenient and widely accepted. This will be our primary tool in Part III.
2. **When conditions are questionable** (small sample, very skewed data, unusual statistic), simulation-based methods are safer. The bootstrap and randomization test make fewer assumptions.
3. **When you want to build intuition**, simulation is invaluable. Seeing 10,000 shuffles or resamples gives you a concrete feel for what “random variability” looks like — something a formula alone cannot provide.
4. **In practice**, many statisticians use simulation to check their formula-based results. If the two approaches agree, confidence in the conclusion is strengthened.

10.4 When is the normal approximation good enough?

The mathematical approach to inference (which we will develop in Part III) replaces simulation with the normal distribution. Instead of generating thousands of shuffles or resamples, we use a formula to compute the standard error and then rely on the bell curve to find p-values or construct confidence intervals.

But when can we trust this shortcut?

10.4.1 The success-failure condition for proportions

For a sample proportion $\hat{p}$, the sampling distribution is approximately normal when:

$$np \geq 10 \quad \text{and} \quad n(1 - p) \geq 10$$

where n is the sample size and p is the proportion of interest. In practice, since p is unknown, we use $\hat{p}$ as a stand-in:

$$n\hat{p} \geq 10 \quad \text{and} \quad n(1 - \hat{p}) \geq 10$$

This is called the **success-failure condition**.

In the medical consultant example, $n = 62$ and $\hat{p} = 3/62 = 0.048$. Check the success-failure condition.

-
- Expected successes: $n\hat{p} = 62 \times 0.048 = 3$ (less than 10)
 - Expected failures: $n(1 - \hat{p}) = 62 \times 0.952 = 59$ (at least 10)

The success-failure condition is **not met** because there are fewer than 10 expected successes. The normal approximation would be unreliable here. This is exactly the kind of situation where the bootstrap (as used in Chapter 8) is the better approach.

In the tappers and listeners study, $n = 120$ and $\hat{p} = 3/120 = 0.025$. Check the success-failure condition. Should we use the normal approximation?

Show answer

Expected successes: $120 \times 0.025 = 3$ (less than 10). Expected failures: $120 \times 0.975 = 117$ (at least 10). The condition is not met because there are too few successes. The normal approximation would not be appropriate. Bootstrap methods are preferred.

A poll of $n = 1,000$ voters finds that $\hat{p} = 0.52$ support a policy. Check the success-failure condition.

-
- Expected successes: $n\hat{p} = 1,000 \times 0.52 = 520$ (well above 10)
 - Expected failures: $n(1 - \hat{p}) = 1,000 \times 0.48 = 480$ (well above 10)

The success-failure condition is comfortably met. The normal approximation is reliable here.

10.4.2 The sample size condition for means

For a sample mean $\bar{x}$, the CLT tells us the sampling distribution is approximately normal when:

- The sample size is “large enough,” and
- The observations are independent.

A common guideline is $n \geq 30$, but this is only a rough rule of thumb. If the population distribution is nearly symmetric, the normal approximation can work well even for smaller samples. If the population is highly skewed or has extreme outliers, a larger sample may be needed.

Population shape	Minimum n for normal approximation
Symmetric, no outliers	$n \geq 15$ or even smaller
Slightly skewed	$n \geq 30$
Highly skewed or heavy-tailed	$n \geq 60$ or more

When in doubt, use simulation (bootstrap) to check whether the normal approximation is reasonable.

10.4.3 Simulation as a check on theory

One of the most powerful uses of simulation is as a **reality check** on mathematical results. Here is the workflow:

1. Compute a confidence interval or p-value using the formula-based (normal) approach.
2. Compute the same quantity using simulation (bootstrap CI or randomization test).
3. Compare. If the two results are similar, you can be confident in the mathematical approach. If they differ substantially, the conditions for the normal approximation may not be met, and you should rely on the simulation result.

This “trust but verify” approach is standard practice among professional statisticians and data scientists.

10.5 The normal distribution: a preview

Since the normal distribution will be our primary tool in Part III, let us briefly introduce its key features here.

10.5.1 The 68-95-99.7 rule

The normal distribution has a very useful property: the percentage of observations falling within 1, 2, and 3 standard deviations of the mean is always approximately the same.

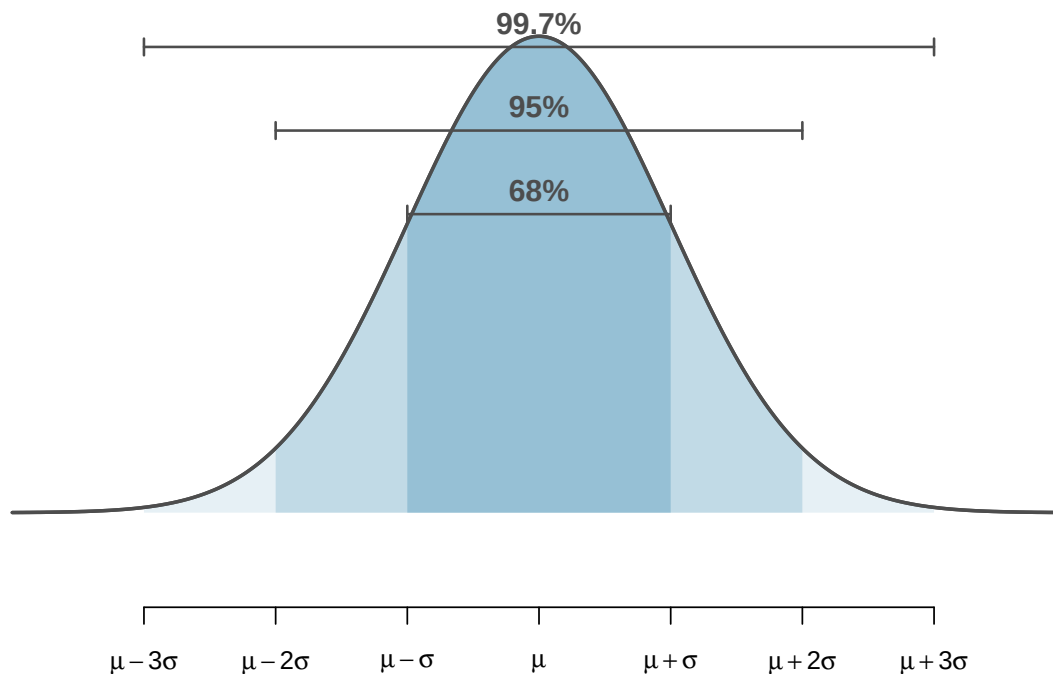


Figure 10.1: The 68-95-99.7 rule for normal distributions. Approximately 68% of values fall within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3.

The 68-95-99.7 rule.

For any normal distribution:

- About **68%** of values fall within 1 standard deviation of the mean: $\mu \pm \sigma$
- About **95%** of values fall within 2 standard deviations of the mean: $\mu \pm 2\sigma$
- About **99.7%** of values fall within 3 standard deviations of the mean: $\mu \pm 3\sigma$

This rule has an immediate connection to confidence intervals. In Chapter 8, we found that a 95% bootstrap confidence interval captures the middle 95% of the bootstrap distribution. If the bootstrap distribution is approximately normal, then the middle 95% corresponds to approximately ± 2 standard errors. This gives us the quick approximation:

$$95\% \text{ CI} \approx \text{statistic} \pm 2 \times SE$$

A poll of $n = 1,000$ voters finds $\hat{p} = 0.52$ with a standard error of $SE = 0.016$. Use the quick approximation to construct a 95% confidence interval.

$$\hat{p} \pm 2 \times SE = 0.52 \pm 2(0.016) = 0.52 \pm 0.032 = (0.488, 0.552)$$

We are 95% confident that the true proportion of voters who support the policy is between 48.8% and 55.2%.

Why is it “approximately 2” standard errors rather than exactly 2? What is the more precise multiplier for a 95% confidence interval based on the normal distribution?

Show answer

The more precise multiplier is 1.96, not 2. The value 1.96 is the Z-score that puts exactly 2.5% in each tail of the standard normal distribution, leaving 95% in the middle. The approximation of 2 is close enough for quick mental calculations.

10.5.2 Z scores

A **Z score** measures how many standard deviations an observation is above or below the mean:

$$Z = \frac{x - \mu}{\sigma}$$

In the context of sampling distributions, we can compute a Z score for an observed statistic:

$$Z = \frac{\text{statistic} - \text{parameter}}{SE}$$

This Z score tells us how unusual the observed statistic is, measured in standard-error units. In Part III, we will use Z scores (and their cousins, t-scores) extensively to compute p-values and confidence intervals without simulation.

In the sex discrimination study, the observed difference in promotion rates was 0.292. Suppose the standard error of the difference (computed under the null hypothesis) is 0.12. Compute the Z score.

Under H_0 , the parameter (the true difference) is 0. So:

$$Z = \frac{0.292 - 0}{0.12} = 2.43$$

The observed difference is 2.43 standard errors above the null value of 0. Since values beyond 2 standard deviations are unusual for a normal distribution (the 95% rule), this is strong evidence against the null hypothesis — consistent with our earlier simulation result.

10.6 Recap: connecting the three approaches

Types of distributions (a reference guide).

You have now encountered several types of distributions in this course. Understanding the differences among them is essential:

- A **data distribution** (or sample distribution) describes the shape, center, and variability of the observed data.
- A **population distribution** describes the entire population — almost always unknown.
- A **sampling distribution** describes how a *sample statistic* varies from sample to sample. The CLT tells us this is approximately normal under certain conditions.
- A **randomization distribution** describes how a test statistic varies under random shuffling of the treatment variable. It is centered at the null hypothesis value.
- A **bootstrap distribution** describes how a statistic varies under resampling from the observed data. It is centered at the observed statistic.

10.6.1 The big picture of Part II

Part II has built your intuition for statistical inference using simulation:

Chapter	Method	Question	How variability is generated
6	Conceptual foundation	How much do statistics vary?	Thought experiments
7	Randomization test	Is there an effect?	Shuffle labels
8	Bootstrap CI	How big is the parameter?	Resample with replacement
9	CLT and normal model	Can we skip the simulation?	Mathematical formula

The key takeaway: **the simulation methods and the mathematical methods are answering the same questions.** The CLT tells us *when* the mathematical shortcuts are valid. When they are, we can replace thousands of computer simulations with a single formula.

In Part III, we will develop these formula-based methods for specific settings: one proportion, one mean, two proportions, two means, and more. But you now have the conceptual foundation to understand *why* those formulas work: they are mathematical descriptions of the same sampling variability that you observed through simulation.

10.7 Chapter review

10.7.1 Summary

- Randomization distributions, bootstrap distributions, and theoretical sampling distributions all tend to be approximately **bell-shaped** (normal) under common conditions. This is the **Central Limit Theorem** (CLT).
- The CLT states that for sufficiently large samples with independent observations, the sampling distribution of many common statistics is approximately normal, regardless of the shape of the population.
- The **success-failure condition** ($n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$) is the practical check for whether the normal approximation is appropriate for proportions.
- The **68-95-99.7 rule** describes the area under a normal curve within 1, 2, and 3 standard deviations of the mean.
- A **Z score** measures how many standard deviations (or standard errors) an observation or statistic falls from the mean: $Z = (x - \mu)/\sigma$.

- The three inference approaches — randomization, bootstrap, and mathematical models — answer the same fundamental questions. When conditions are met, they produce similar results. Mathematical models are faster; simulation methods are more flexible.
- This chapter bridges Part II (simulation-based inference) and Part III (formula-based inference). The CLT is the theoretical justification for the formulas we will use in Part III.

10.7.2 Key terms

Term	Definition
Central Limit Theorem (CLT)	The sampling distribution of a statistic is approximately normal for large samples
Normal distribution	A symmetric, bell-shaped distribution described by its mean and standard deviation
Success-failure condition	For proportions: need $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$
68-95-99.7 rule	About 68%, 95%, 99.7% of normal values fall within 1, 2, 3 SDs of the mean
Z score	Number of standard deviations an observation falls from the mean: $Z = (x - \mu)/\sigma$
Null distribution	The sampling distribution of the test statistic under H_0
Sampling distribution	Distribution of a statistic over all possible samples of a given size

10.7.3 Exercises

Exercises for this chapter are under development.

PART III

CALCULATE

You've built sampling distributions and computed confidence intervals through simulation. Now we introduce the mathematical tools — the normal distribution, the t -distribution — that approximate what simulation does. These theoretical methods are faster to compute and appear in published research, but they rest on the same logic you already understand.

Chapter 11

Normal Approximation

In Chapters 7–9, we used simulation-based methods – randomization tests and bootstrapping – to make inferences about population parameters. Those methods are powerful and flexible, but they share a limitation: every new problem requires running thousands of simulations. In this chapter, we introduce the **normal distribution**, a mathematical model that describes the bell-shaped pattern we keep seeing in our sampling distributions. When certain conditions are met, the normal distribution lets us calculate probabilities and construct confidence intervals using formulas instead of simulations. This is not a replacement for simulation-based thinking – it is a complement. The normal model provides the theoretical justification for *why* those simulations work the way they do.

StatBench tools for this chapter: **Normal Distribution Explorer** – Use it to visualize different normal curves, shade areas, compute probabilities, and see how changing μ and σ affects the distribution.

[Launch Normal Distribution Explorer](#)

11.1 The normal distribution

Among all the distributions encountered in statistics, one is overwhelmingly the most common. The symmetric, unimodal, bell-shaped curve is so ubiquitous that it goes by many names: the **normal curve**, the **normal model**, or the **normal distribution**.

The **normal distribution** is a symmetric, unimodal, bell-shaped distribution described by two **parameters**: the mean μ (which determines the center) and the standard deviation σ (which determines the spread). We write the distribution in shorthand as $N(\mu, \sigma)$.

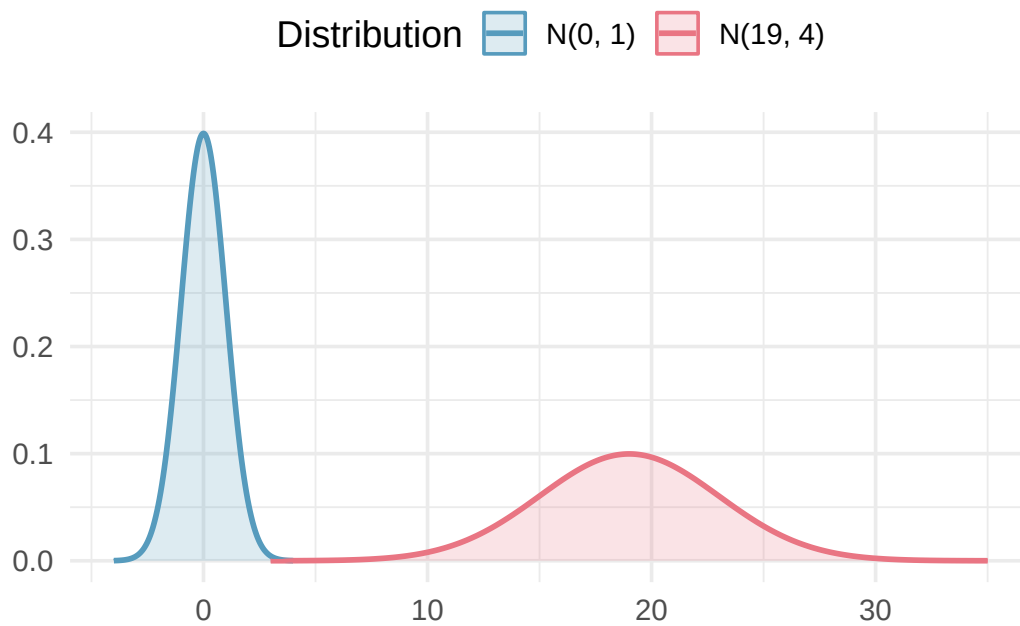


Figure 11.1: Two normal distributions with different parameters. The left curve has $\mu = 0$ and $\sigma = 1$; the right curve has $\mu = 19$ and $\sigma = 4$. Changing μ shifts the curve left or right; changing σ stretches or compresses it.

Key properties of the normal distribution:

- The curve is perfectly symmetric about the mean μ .
- The total area under the curve is exactly 1 (representing 100% of observations).
- The tails extend infinitely in both directions but get very close to zero far from the center.
- Changing μ shifts the entire curve left or right without changing its shape.
- Increasing σ makes the curve wider and shorter; decreasing σ makes it narrower and taller.

Normal distribution facts. Many variables are *nearly* normal, but none are *exactly* normal. The normal distribution, while not perfect for any single problem, is extremely useful for a wide variety of problems. Variables like SAT scores, adult heights, and blood pressure measurements closely follow the normal distribution.

11.1.1 The standard normal distribution

The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is called the **standard normal distribution**, written $N(0, 1)$. It serves as a reference distribution that we will use throughout the course.

Write down the shorthand for a normal distribution with each set of parameters.

- a. Mean 5 and standard deviation 3
- b. Mean -100 and standard deviation 10
- c. Mean 2 and standard deviation 9

-
- a. $N(\mu = 5, \sigma = 3)$
 - b. $N(\mu = -100, \sigma = 10)$
 - c. $N(\mu = 2, \sigma = 9)$

11.2 The 68-95-99.7 rule

One of the most useful facts about the normal distribution is how observations spread out around the mean. This relationship is captured by the **68-95-99.7 rule** (sometimes called the **empirical rule**).

The 68-95-99.7 rule. For any normal distribution $N(\mu, \sigma)$:

- About **68%** of observations fall within **1 standard deviation** of the mean: between $\mu - \sigma$ and $\mu + \sigma$.
- About **95%** of observations fall within **2 standard deviations** of the mean: between $\mu - 2\sigma$ and $\mu + 2\sigma$.
- About **99.7%** of observations fall within **3 standard deviations** of the mean: between $\mu - 3\sigma$ and $\mu + 3\sigma$.

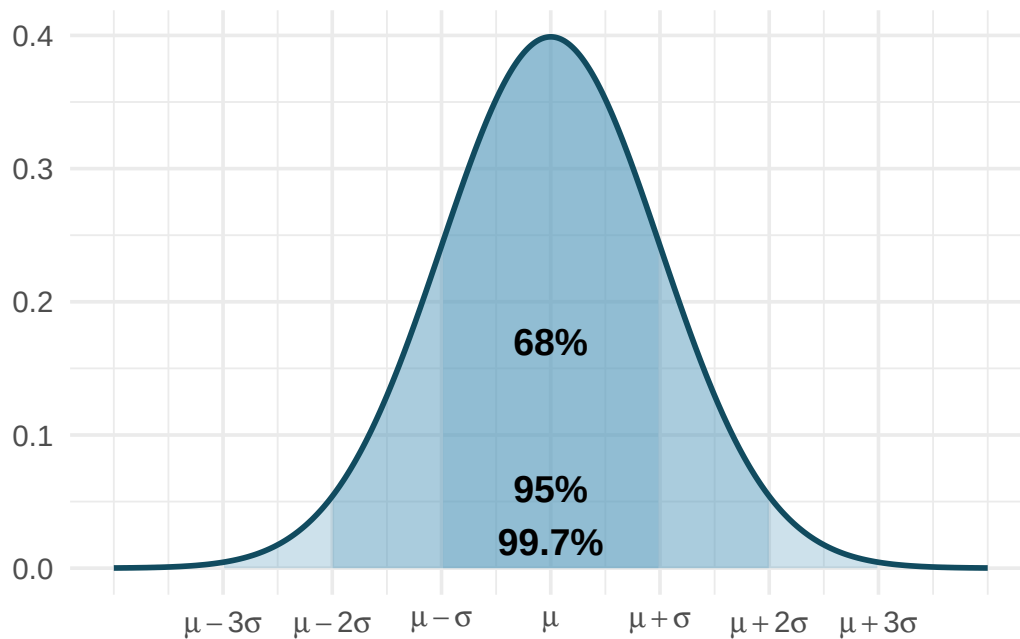


Figure 11.2: The 68-95-99.7 rule illustrated on a normal curve. The innermost shaded region (68%) extends from $\mu - \sigma$ to $\mu + \sigma$. The middle region (95%) extends from $\mu - 2\sigma$ to $\mu + 2\sigma$. The outermost region (99.7%) extends from $\mu - 3\sigma$ to $\mu + 3\sigma$.

This rule gives quick, approximate answers without needing a calculator or software.

SAT scores closely follow the normal distribution with mean $\mu = 1500$ and standard deviation $\sigma = 300$.

- About what percent of test takers score between 1200 and 1800?
- About what percent score between 1500 and 2100?

-
- 1200 is one standard deviation below the mean ($1500 - 300 = 1200$), and 1800 is one standard deviation above the mean ($1500 + 300 = 1800$). By the 68-95-99.7 rule, about **68%** of test takers score between 1200 and 1800.
 - The range from 900 to 2100 covers $\mu \pm 2\sigma$, which contains about 95% of scores. Since the normal distribution is symmetric, the range from 1500 to 2100 (the upper half) contains about $\frac{95\%}{2} =$

47.5% of scores.

Adult male heights in the US follow approximately a normal distribution with mean 70 inches and standard deviation 3.3 inches. Using the 68-95-99.7 rule, approximately what percentage of adult males are between 63.4 and 76.6 inches tall?

Show answer

63.4 inches is $70 - 2(3.3) = 63.4$ and 76.6 inches is $70 + 2(3.3) = 76.6$. These are each 2 standard deviations from the mean, so by the 68-95-99.7 rule, approximately 95% of adult males have heights in this range.

Continuing with SAT scores ($\mu = 1500$, $\sigma = 300$), about what percent of test takers score above 2100?

Show answer

2100 is 2 standard deviations above the mean. By the 68-95-99.7 rule, about 95% of scores are within 2 standard deviations, leaving about 5% outside. Since the distribution is symmetric, about 2.5% are above 2100.

11.3 Z-scores and standardizing

When comparing observations from different distributions, raw values are not directly comparable. A score of 1800 on the SAT and a score of 24 on the ACT look very different, but we need a common scale to determine which performance is better relative to other test takers. The solution is **standardizing** the observations.

The **Z-score** of an observation x from a distribution with mean μ and standard deviation σ is the number of standard deviations it falls above or below the mean:

$$Z = \frac{x - \mu}{\sigma}$$

- Positive Z-scores indicate observations **above** the mean.
- Negative Z-scores indicate observations **below** the mean.
- A Z-score of 0 means the observation equals the mean.

If the observation x comes from a normal distribution $N(\mu, \sigma)$, then its Z-score follows the standard normal distribution $N(0, 1)$. Standardizing shifts the center to 0 and rescales the spread to 1, but preserves the normal shape.

SAT scores follow $N(\mu = 1500, \sigma = 300)$ and ACT scores follow $N(\mu = 21, \sigma = 5)$. Nel scored 1800 on the SAT and Sian scored 24 on the ACT. Who performed better relative to other test takers?

We compute Z-scores for each student.

Nel's Z-score: $Z_{\text{Nel}} = \frac{1800-1500}{300} = \frac{300}{300} = 1.00$

Sian's Z-score: $Z_{\text{Sian}} = \frac{24-21}{5} = \frac{3}{5} = 0.60$

Nel is 1 standard deviation above the SAT mean, while Sian is only 0.6 standard deviations above the ACT mean. Nel performed better relative to other test takers.

Head lengths of brushtail possums follow a nearly normal distribution with mean 92.6 mm and standard deviation 3.6 mm. Compute the Z-scores for possums with head lengths of 95.4 mm and 85.8 mm. Which possum has the more unusual head length?

Show answer

For $x_1 = 95.4$ mm: $Z_1 = \frac{95.4-92.6}{3.6} = \frac{2.8}{3.6} = 0.78$. For $x_2 = 85.8$ mm: $Z_2 = \frac{85.8-92.6}{3.6} = \frac{-6.8}{3.6} = -1.89$. The second possum has a more unusual head length because $|-1.89| > |0.78|$.

Let X represent a random variable from $N(\mu = 3, \sigma = 2)$, and suppose we observe $x = 5.19$.

- Find the Z-score.
- How many standard deviations above or below the mean is this observation?

a. $Z = \frac{x-\mu}{\sigma} = \frac{5.19-3}{2} = \frac{2.19}{2} = 1.095$

- b. The observation is 1.095 standard deviations *above* the mean. We know it is above because the Z-score is positive.

11.4 Finding probabilities from the normal distribution

One of the most important skills in statistics is finding the probability that a normally distributed variable falls in a specified range. This is equivalent to finding the **area under the normal curve** over that range.

Strategy for normal probability problems. Always follow these steps:

- Draw a picture.** Sketch the normal curve, mark the mean, and shade the area you want to find.
- Find the Z-score.** Standardize the value(s) of interest using $Z = \frac{x-\mu}{\sigma}$.
- Look up the area.** Use a normal probability table, statistical software, or the StatBench Normal Distribution Explorer to find the area to the **left** of the Z-score.
- Adjust if needed.** If you need a right-tail area, subtract the left-tail area from 1. For an area between two values, subtract the smaller left-tail area from the larger.

11.4.1 Left-tail probabilities (percentiles)

The area to the left of a value on the normal curve gives the **percentile** – the proportion of observations that fall below that value.

The **percentile** of an observation is the percentage of values in the distribution that fall at or below that observation.

Nel scored 1800 on the SAT ($Z = 1.00$). What percentile is this?

We need the area to the left of $Z = 1.00$ on the standard normal curve.

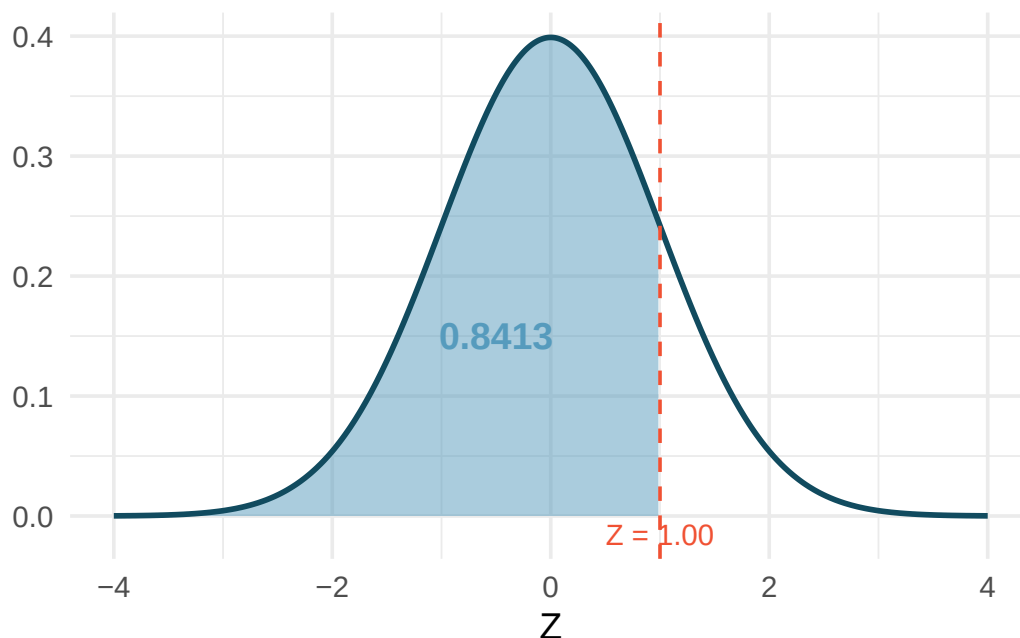


Figure 11.3: The standard normal curve with the area to the left of $Z = 1.00$ shaded. This area represents Nel's percentile.

Using a normal probability table or software, the area to the left of $Z = 1.00$ is **0.8413**. Nel is at the **84th percentile** – 84.13% of SAT takers scored lower.

11.4.2 Right-tail probabilities

Shannon is a randomly selected SAT taker. What is the probability Shannon scores at least 1630?

Step 1: Draw a picture. We shade the area above 1630 on the $N(1500, 300)$ curve.

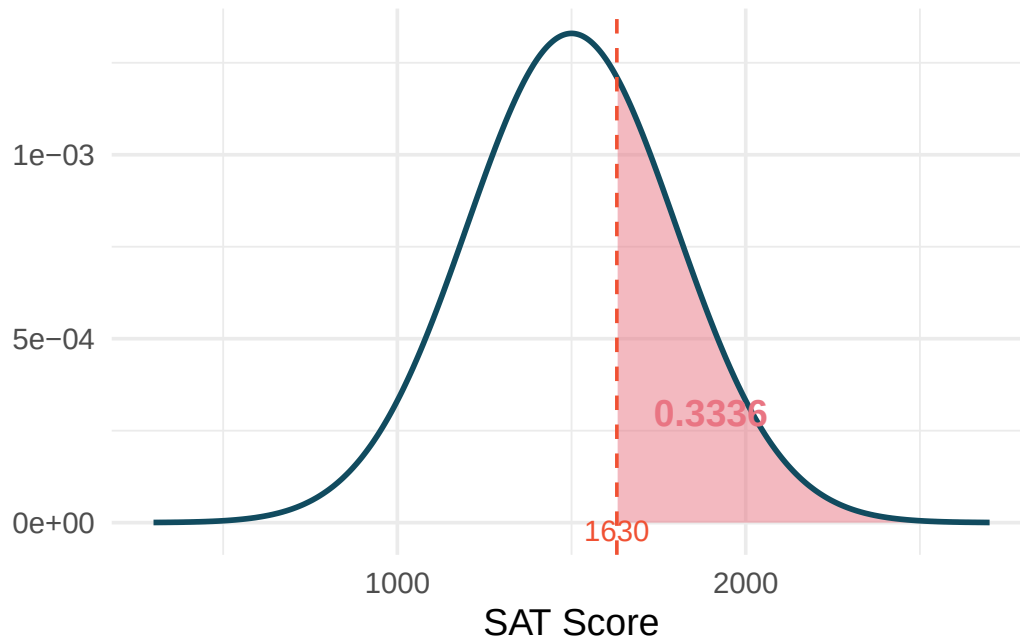


Figure 11.4: A normal curve with mean 1500 and standard deviation 300. The area to the right of 1630 is shaded.

Step 2: Find the Z-score.

$$Z = \frac{1630 - 1500}{300} = \frac{130}{300} = 0.43$$

Step 3: Look up the left-tail area. The area to the left of $Z = 0.43$ is 0.6664.

Step 4: Adjust. The area to the *right* is $1 - 0.6664 = 0.3336$.

The probability Shannon scores at least 1630 is **0.3336**, or about 33.4%.

If the probability of Shannon scoring at least 1630 is 0.3336, what is the probability Shannon scores *less than* 1630?

Show answer

The probability of scoring less than 1630 is the area to the left: 0.6664, or about 66.6%. The left-tail and right-tail areas always sum to 1.

11.4.3 Probabilities between two values

Adult male heights follow $N(\mu = 70, \sigma = 3.3)$ inches. What is the probability that a randomly selected adult male is between 5'9" (69 inches) and 6'2" (74 inches)?

Step 1: Draw a picture. Shade the area between 69 and 74 on the normal curve.

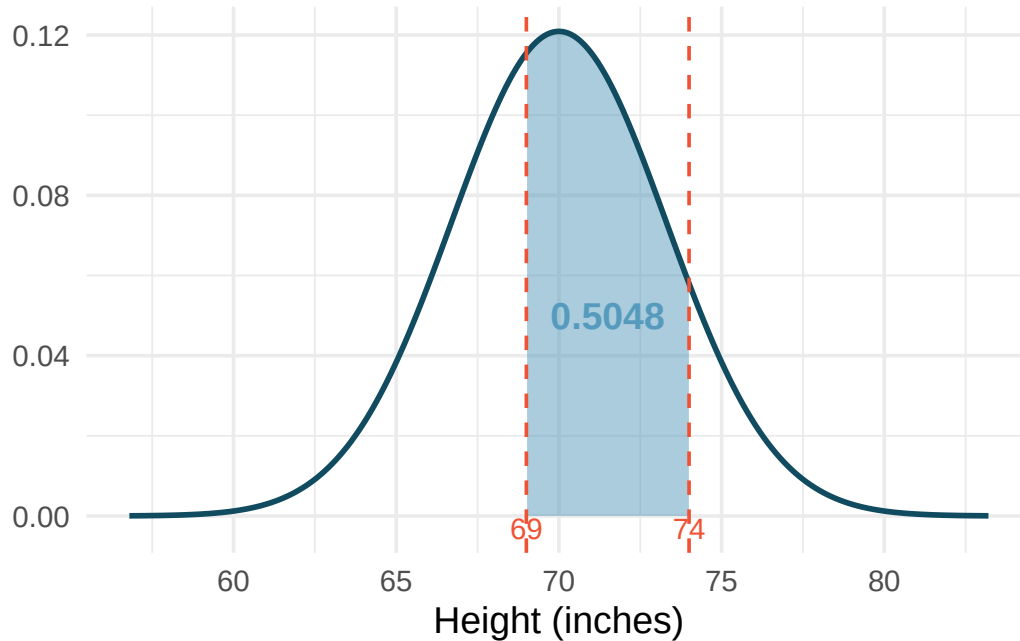


Figure 11.5: A normal curve centered at 70 with standard deviation 3.3. The area between 69 and 74 is shaded.

Step 2: Find both Z-scores.

$$Z_1 = \frac{69 - 70}{3.3} = -0.30 \quad Z_2 = \frac{74 - 70}{3.3} = 1.21$$

Step 3: Look up both left-tail areas.

- Area to the left of $Z_1 = -0.30$: 0.3821
- Area to the left of $Z_2 = 1.21$: 0.8869

Step 4: Subtract.

$$P(69 < X < 74) = 0.8869 - 0.3821 = 0.5048$$

About **50.5%** of adult males are between 5'9" and 6'2".

What percent of SAT takers score between 1500 and 2000? (SAT scores follow $N(1500, 300)$.)

Show answer

First find the Z-scores: $Z_1 = \frac{1500-1500}{300} = 0.00$ and $Z_2 = \frac{2000-1500}{300} = 1.67$. The left-tail areas are 0.5000 and 0.9525. The area between them is $0.9525 - 0.5000 = 0.4525$, so about 45.25% score between 1500 and 2000.

11.4.4 Finding values from percentiles

Sometimes we want to work backwards: given a percentile, find the observation.

Adult male heights follow $N(70, 3.3)$. Yousef's height is at the 40th percentile. How tall is Yousef?

Step 1: Draw a picture. The 40th percentile means 40% of the area is to the left.

Step 2: Find the Z-score for the 40th percentile. Using a normal probability table (look for 0.40 in the body of the table) or software, $Z = -0.25$.

Step 3: Solve for x . Using the Z-score formula:

$$-0.25 = \frac{x - 70}{3.3}$$

$$x = 70 + (-0.25)(3.3) = 70 - 0.825 = 69.2 \text{ inches}$$

Yousef is approximately 69.2 inches (about 5'9") tall.

What is the 95th percentile for SAT scores?

Show answer

The Z-score for the 95th percentile is $Z = 1.645$. Then $x = 1500 + 1.645 \times 300 = 1500 + 493.5 = 1993.5$. The 95th percentile for SAT scores is approximately 1994.

11.4.5 The normal probability table

A **normal probability table** (also called a Z-table) lists areas to the left of Z-scores for the standard normal distribution. The table is organized with the ones and tenths digits of the Z-score in the rows, and the hundredths digit in the columns.

Common Z-scores and areas worth remembering:

Table 11.1: Commonly used Z-scores and their corresponding left-tail areas.

Z-score	Left-tail area	Interpretation
-1.96	0.0250	2.5th percentile
-1.645	0.0500	5th percentile
-1.00	0.1587	15.87th percentile
0.00	0.5000	50th percentile (the mean)
1.00	0.8413	84.13th percentile
1.645	0.9500	95th percentile
1.96	0.9750	97.5th percentile
2.576	0.9950	99.5th percentile

Practice with the Normal Distribution Explorer. Use StatBench to verify the table entries above. Enter a Z-score and check that the shaded area matches. Then try working backwards: enter a percentile and find the corresponding Z-score.

[Launch Normal Distribution Explorer](#)

11.5 The normal approximation to sampling distributions

In Chapter 9, we used simulation to explore sampling distributions – the distributions of sample statistics computed from many repeated samples. A striking pattern emerged: regardless of whether we were looking at sample means, sample proportions, or differences in means or proportions, the sampling distributions tended to be **bell-shaped and symmetric**. This is not a coincidence.

11.5.1 The Central Limit Theorem

The Central Limit Theorem (CLT). When observations are independent and the sample size is sufficiently large, the sampling distribution of many common statistics (sample means, sample proportions, and their differences) is approximately normal, regardless of the shape of the population distribution.

Specifically:

- The sampling distribution of $\bar{x}$ is approximately $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$.
- The sampling distribution of $\hat{p}$ is approximately $N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$.

The CLT is one of the most remarkable results in all of mathematics. It explains why simulation-based null distributions and bootstrap distributions looked bell-shaped: they were approximating the normal distribution all along.

Two conditions for the CLT to apply:

1. **Independence.** The observations must be independent of one another. This is satisfied when data come from a random sample or a randomized experiment.
2. **Sample size.** The sample must be large enough. What counts as “large enough” depends on the context:
 - For **proportions**: at least 10 expected successes and 10 expected failures ($np \geq 10$ and $n(1-p) \geq 10$). This is called the **success-failure condition**.
 - For **means**: $n \geq 30$ is a common guideline, though smaller samples can work if the population is not strongly skewed.

11.5.2 Standard error: quantifying the variability of a statistic

The CLT tells us the sampling distribution is approximately normal. But how wide is that normal distribution? The answer is given by the **standard error**.

The **standard error (SE)** of a statistic is the standard deviation of its sampling distribution. It measures how much the statistic varies from sample to sample.

The standard error depends on the parameter of interest:

Table 11.2: Standard error formulas for common parameters.

Parameter	Point estimate	Standard error
Population mean μ	$\bar{x}$	$SE = \frac{s}{\sqrt{n}}$
Population proportion p	$\hat{p}$	$SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Difference in means $\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
Difference in proportions $p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$	$SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

Margin of error. The distance $z^* \times SE$ is called the **margin of error**. For a 95% confidence interval, $z^* = 1.96$ (often rounded to 2), giving a margin of error of approximately $2 \times SE$. This is directly connected to the 68-95-99.7 rule: 95% of sample statistics fall within about 2 standard errors of the true parameter value.

11.5.3 Connecting simulation and normal approximation

Let's see the connection between simulation-based methods and the normal approximation in a concrete example.

In Chapter 7, we tested whether opportunity cost reminders reduce spending. We simulated 10,000 randomization samples and found a null distribution that was bell-shaped with a center of 0 and a spread described by $SE \approx 0.078$. The observed difference was 0.20.

Simulation approach: Count how many of the 10,000 simulated differences were at least as extreme as 0.20. The proportion gives the p-value: approximately 0.006.

Normal approximation approach: Model the null distribution as $N(0, 0.078)$. Compute the Z-score:

$$Z = \frac{0.20 - 0}{0.078} = 2.56$$

The area to the right of $Z = 2.56$ is 0.0052.

Both approaches give nearly the same p-value (0.006 vs. 0.0052) and the same conclusion: reject H_0 . The normal approximation works because the conditions are met – independent observations and a sufficiently large sample.

11.5.4 Confidence intervals using the normal model

The normal approximation also lets us build confidence intervals without bootstrapping.

General confidence interval formula. When the sampling distribution of a statistic is approximately normal with standard error SE , a confidence interval takes the form:

$$\text{point estimate} \pm z^* \times SE$$

where z^* is chosen based on the desired confidence level:

Table 11.3: Critical values z^* for common confidence levels.

Confidence level	z^*
90%	1.645
95%	1.960
99%	2.576

A study examined whether stents reduce stroke risk. The observed difference in 30-day stroke rates (treatment minus control) was $\hat{p}_T - \hat{p}_C = 0.090$, with $SE = 0.028$. Construct a 95% confidence interval.

The conditions for the normal model have been verified. Using $z^* = 1.96$:

$$0.090 \pm 1.96 \times 0.028 = 0.090 \pm 0.055 = (0.035, 0.145)$$

We are 95% confident that implanting a stent increased the 30-day stroke rate by between 3.5 and 14.5 percentage points. Since the interval does not contain 0, we have evidence of a real difference.

11.6 When does the normal model apply – and when doesn't it?

The normal distribution is powerful but not universal. Here we summarize when it is and is not appropriate.

When the normal model is a good approximation:

- Sample proportions $\hat{p}$, when the success-failure condition is met ($np \geq 10$ and $n(1 - p) \geq 10$).
- Sample means $\bar{x}$, when the sample size is at least 30 or the population is approximately normal (we will use the t -distribution for means – see Chapter 11).
- Differences in proportions or means, when both groups satisfy the above conditions.

When the normal model is NOT a good approximation:

- Small samples with a skewed population distribution.
- Proportions near 0 or 1 with small sample sizes (the success-failure condition fails).
- Data with strong dependence between observations.

A medical consultant claims that only 3 of their 62 liver transplant clients had complications, compared to the national rate of 10%. Is it appropriate to use the normal approximation to test whether this consultant's rate is lower?

Under the null hypothesis ($p_0 = 0.10$), we check the success-failure condition:

- Expected successes (complications): $np_0 = 62 \times 0.10 = 6.2$
- Expected failures (no complications): $n(1 - p_0) = 62 \times 0.90 = 55.8$

Since the expected number of successes (6.2) is less than 10, the **success-failure condition is not met**. The normal approximation would underestimate the variability and give an inaccurate p-value. A simulation-based approach (as in Chapter 7) would be more appropriate here.

When conditions fail, use simulation. If the success-failure condition is not met or the sample size is too small for the normal approximation, we should fall back on the simulation-based methods from Chapters 7 and 8. The bootstrap and randomization approaches work well even when the normal model does not. This is one of the great advantages of simulation-based inference: it is more broadly applicable.

11.7 Chapter review

11.7.1 Summary

This chapter introduced the normal distribution, the most commonly encountered distribution in statistics. Key concepts include:

- The normal distribution is described by two parameters: the mean μ (center) and standard deviation σ (spread), written as $N(\mu, \sigma)$.
- The **68-95-99.7 rule** provides quick estimates: about 68%, 95%, and 99.7% of observations fall within 1, 2, and 3 standard deviations of the mean.
- A **Z-score** standardizes an observation by measuring how many standard deviations it lies from the mean: $Z = \frac{x - \mu}{\sigma}$.
- Normal probabilities are found by computing Z-scores and looking up areas under the standard normal curve using tables, software, or StatBench.
- The **Central Limit Theorem** guarantees that sampling distributions of means and proportions are approximately normal when the sample is large enough and observations are independent.

- The **standard error** measures how much a statistic varies from sample to sample and is the key ingredient in confidence intervals and hypothesis tests.
- The normal approximation and simulation-based methods give similar results when conditions are met, but simulation is more flexible when conditions fail.

11.7.2 Key terms

Normal distribution, standard normal distribution, parameters (μ and σ), 68-95-99.7 rule, Z-score, standardizing, percentile, Central Limit Theorem (CLT), standard error (SE), margin of error, success-failure condition.

11.7.3 Exercises

Exercises for this chapter are under development.

Chapter 12

Confidence Intervals for Means

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In Chapter 8, we used bootstrapping to construct confidence intervals – a flexible, simulation-based approach that works well in many settings. In Chapter 10, we saw that sampling distributions are often well approximated by the normal distribution. For means, however, we encounter a complication: the population standard deviation σ is almost never known, and estimating it with the sample standard deviation s introduces extra uncertainty. To account for this, we use a slightly different bell-shaped distribution called the ***t*-distribution**. In this chapter, we develop *t*-based confidence intervals for one mean, the difference between two independent means, and the mean of paired differences.

StatBench tools for this chapter: Use the ***t*-Distribution Explorer** to compare the *t*-distribution to the standard normal at various degrees of freedom.

[Launch *t*-Distribution Explorer](#)

12.1 The *t*-distribution

12.1.1 Why not the normal distribution?

In Chapter 10, we used the normal distribution to model sampling distributions. For proportions, this works well because the standard error formula $SE = \sqrt{\hat{p}(1 - \hat{p})/n}$ depends only on the sample proportion and the sample size – both of which are known. For means, the standard error is $SE = \sigma/\sqrt{n}$, which depends on σ , the population standard deviation. Since we rarely know σ , we estimate it using the sample standard deviation s :

$$SE \approx \frac{s}{\sqrt{n}}$$

This substitution introduces additional uncertainty, especially in small samples. When n is small, s can be a poor estimate of σ , and the normal model underestimates how variable the statistic really is. The *t*-distribution corrects for this.

12.1.2 Properties of the t -distribution

The t -distribution is a bell-shaped, symmetric distribution that is similar to the normal distribution but has **heavier tails** (more probability in the extremes). The exact shape of the t -distribution depends on a parameter called the **degrees of freedom** (df).

- When df is small, the t -distribution has noticeably heavier tails than the normal – extreme values are more likely.
- As df increases, the t -distribution approaches the standard normal distribution.
- For $df \geq 30$, the t -distribution is nearly indistinguishable from the normal.

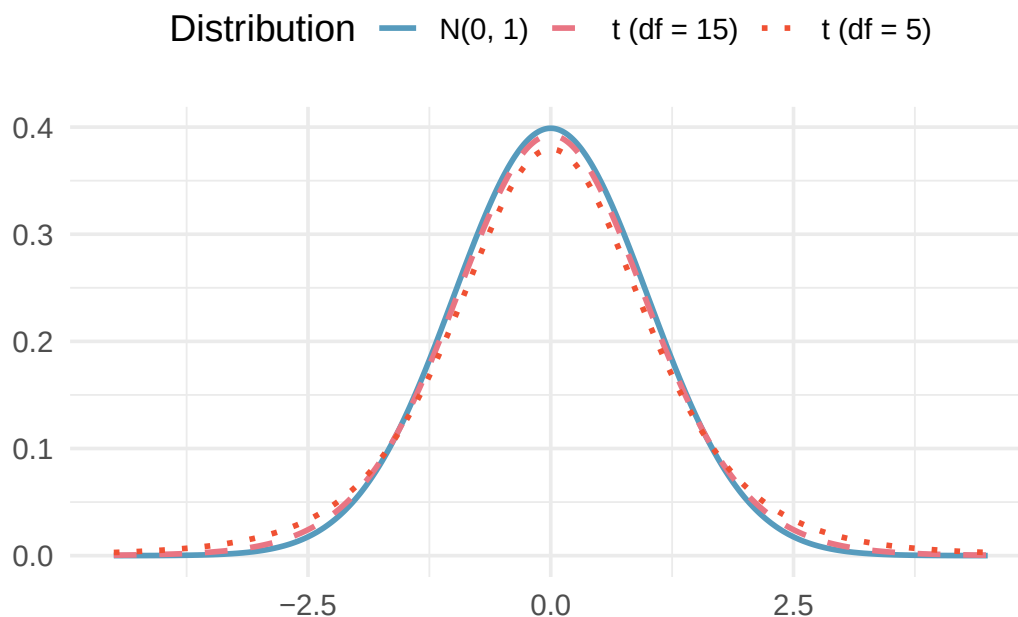


Figure 12.1: Three distributions plotted on the same axes: the standard normal distribution $N(0, 1)$, a t -distribution with $df = 5$, and a t -distribution with $df = 15$. All are centered at 0 and bell-shaped, but the t -distributions have heavier tails, especially when df is small.

Why heavier tails matter. Heavier tails mean that extreme values (large t -scores) are more probable under the t -distribution than under the normal distribution. In practice, this leads to **wider confidence intervals** when sample sizes are small – exactly the extra caution we need when σ is estimated by s .

How does the t -distribution compare to the normal distribution when $df = 2$? When $df = 100$?

Show answer

When $df = 2$, the t -distribution has very heavy tails – values far from 0 are much more likely than under the normal curve. When $df = 100$, the t -distribution is nearly identical to the standard normal. In general, the t -distribution becomes more like the normal as degrees of freedom increase.

12.1.3 Critical values from the t -distribution

Just as we use z^* for the normal distribution, we use t_{df}^* for the t -distribution. Because the t -distribution has heavier tails, t^* values are larger than the corresponding z^* values (especially for small df).

Selected critical values for 95% confidence intervals:**Table 12.1:** Critical values t_{df}^* for a 95% confidence interval. As df increases, t^* approaches $z^* = 1.96$.

df	t_{df}^* (95% CI)	z^* (for comparison)
5	2.571	1.960
10	2.228	1.960
20	2.086	1.960
30	2.042	1.960
50	2.009	1.960
100	1.984	1.960
∞	1.960	1.960

Why is the critical value $t_5^* = 2.571$ so much larger than $z^* = 1.96$?

Show answer

With only $df = 5$, there is substantial uncertainty in estimating σ with s . The t -distribution accounts for this by having heavier tails, which pushes the critical values further from zero. This results in a wider confidence interval, reflecting the greater uncertainty.

12.2 One-sample t confidence interval

12.2.1 The setup

We want to estimate the population mean μ based on a single sample. The point estimate is the sample mean $\bar{x}$, and the standard error is $SE = s/\sqrt{n}$.

One-sample t confidence interval for μ .

$$\bar{x} \pm t_{df}^* \times \frac{s}{\sqrt{n}}$$

where $df = n - 1$.

Conditions:

1. **Independence.** The observations are independent (e.g., from a random sample).
2. **Normality / sample size.** Either the population distribution is approximately normal, or the sample size is large enough ($n \geq 30$). For smaller samples, check for strong skewness or extreme outliers.

12.2.2 Checking conditions

The normality condition is about the shape of the **population**, not the sample. In practice:

- If $n < 30$: Check a histogram or dotplot of the sample data. If there is no strong skewness and no extreme outliers, the t -methods are reasonable.
- If $n \geq 30$: The Central Limit Theorem ensures that $\bar{x}$ is approximately normal, even if the population is moderately skewed. However, be cautious with extreme outliers, which can distort $\bar{x}$ and s .

Outliers and the t -test. The sample mean $\bar{x}$ and sample standard deviation s are sensitive to outliers. A single extreme observation can substantially shift $\bar{x}$ and inflate s , leading to misleading confidence

intervals. When outliers are present, consider whether they represent data errors. If the outliers are genuine, a bootstrap confidence interval (Chapter 8) may be more reliable.

12.2.3 Worked example: Mercury levels in Risso's dolphins

Researchers measured mercury concentrations (in $\mu\text{g/g}$) in the muscle tissue of 19 Risso's dolphins from the Adriatic Sea. The sample mean was $\bar{x} = 4.4 \mu\text{g/g}$ and the sample standard deviation was $s = 2.3 \mu\text{g/g}$. Construct a 95% confidence interval for the mean mercury level in the population of Risso's dolphins.

Check conditions.

1. *Independence.* The dolphins were sampled from different locations, so it is reasonable to treat the measurements as independent.
2. *Normality.* With $n = 19$, we need to check for strong skewness or outliers. Suppose a histogram of the data shows a roughly symmetric distribution with no extreme outliers. The condition is satisfied.

Compute the interval. With $df = 19 - 1 = 18$, the critical value for a 95% CI is $t_{18}^* = 2.101$.

$$SE = \frac{s}{\sqrt{n}} = \frac{2.3}{\sqrt{19}} = \frac{2.3}{4.359} = 0.528$$

$$\bar{x} \pm t_{18}^* \times SE = 4.4 \pm 2.101 \times 0.528 = 4.4 \pm 1.109 = (3.29, 5.51)$$

We are 95% confident that the mean mercury concentration in Risso's dolphins is between 3.29 and 5.51 $\mu\text{g/g}$.

A sample of 35 UW-La Crosse students reported their weekly study hours. The sample mean is $\bar{x} = 15.2$ hours and $s = 6.8$ hours. Construct a 90% confidence interval for the mean weekly study hours of all UW-La Crosse students. (Use $t_{34}^* = 1.691$ for a 90% CI.)

Show answer

First check conditions: the sample is a random sample (independence), and $n = 35 \geq 30$ (normality via CLT). Then $SE = 6.8/\sqrt{35} = 1.149$. The 90% CI is $15.2 \pm 1.691 \times 1.149 = 15.2 \pm 1.943 = (13.26, 17.14)$. We are 90% confident the mean weekly study hours for all UW-La Crosse students is between 13.26 and 17.14 hours.

12.3 Two-sample t confidence interval

12.3.1 The setup

When comparing the means of two independent groups, we estimate the difference $\mu_1 - \mu_2$ using $\bar{x}_1 - \bar{x}_2$.

Two-sample t confidence interval for $\mu_1 - \mu_2$.

$$(\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Degrees of freedom: The exact formula for df is complex and typically computed by software. A conservative (safe) approximation is:

$$df = \min(n_1 - 1, n_2 - 1)$$

Conditions:

1. **Independence (extended).** Observations are independent within each group *and* between groups (e.g., from two independent random samples or a randomized experiment).
2. **Normality / sample size.** Each group's data should be approximately normal or have $n \geq 30$. Check each group separately for strong skewness or extreme outliers.

12.3.2 Worked example: Embryonic stem cells and heart function

Researchers studied whether embryonic stem cell (ESC) treatment improves heart function in sheep after a heart attack. Nine sheep received the ESC treatment and nine were in the control group. The change in pumping capacity was measured:

Table 12.2: Summary statistics for the stem cell study.

Group	n	$\bar{x}$	s
ESC	9	3.50	5.17
Control	9	-4.33	2.76

Construct a 95% confidence interval for the difference in mean change in pumping capacity, $\mu_{ESC} - \mu_{Control}$.

Check conditions.

1. *Independence.* The sheep were randomly assigned to groups, so independence within and between groups is satisfied.
2. *Normality.* With $n = 9$ per group (small samples), we check histograms. Suppose they show no extreme outliers in either group. The condition is reasonably met.

Compute the interval.

$$SE = \sqrt{\frac{5.17^2}{9} + \frac{2.76^2}{9}} = \sqrt{\frac{26.73}{9} + \frac{7.62}{9}} = \sqrt{2.970 + 0.847} = \sqrt{3.817} = 1.954$$

Using $df = \min(9 - 1, 9 - 1) = 8$, the critical value for a 95% CI is $t_8^* = 2.306$.

$$\text{Point estimate} = 3.50 - (-4.33) = 7.83$$

$$7.83 \pm 2.306 \times 1.954 = 7.83 \pm 4.506 = (3.32, 12.34)$$

We are 95% confident that the ESC treatment increases heart pumping capacity by between 3.32% and 12.34% compared to the control. Because the interval is entirely above zero, we have evidence that the ESC treatment improves heart function. Since this was a randomized experiment, we can attribute the improvement to the treatment.

Researchers collected birth weight data from a random sample of 1,000 US births. Nonsmoker mothers ($n = 867$) had babies with $\bar{x}_n = 7.27$ lbs ($s_n = 1.23$), and smoker mothers ($n = 114$) had babies with $\bar{x}_s = 6.68$ lbs ($s_s = 1.60$). Is there evidence that babies born to nonsmoker mothers weigh more on average? Compute a 95% confidence interval for the difference $\mu_n - \mu_s$.

Show answer

The point estimate is $7.27 - 6.68 = 0.59$ lbs. The SE is $\sqrt{1.23^2/867 + 1.60^2/114} = \sqrt{0.001745 + 0.02246} = \sqrt{0.02420} = 0.156$. Using $df = \min(866, 113) = 113$ and $t_{113}^* \approx 1.981$, the CI is $0.59 \pm 1.981 \times 0.156 = 0.59 \pm 0.309 = (0.28, 0.90)$. We are 95% confident that babies born to nonsmoker mothers weigh between 0.28 and 0.90 lbs more on average than babies born to smoker mothers.

12.4 Paired data confidence interval

12.4.1 What makes data “paired”?

Sometimes two measurements are made on the same individual (e.g., before and after a treatment) or on naturally matched pairs (e.g., the same textbook priced at two stores). In these cases, the two groups are **not independent** – they are **paired**.

Paired data. Two sets of observations are **paired** if each observation in one set has a special correspondence or connection with exactly one observation in the other set. Common examples include:

- Before/after measurements on the same individual
- Two treatments applied to the same subject (e.g., tires on the same car)
- Matched pairs in an experiment (e.g., twins assigned to different treatments)
- The same item measured under two conditions (e.g., textbook prices at two stores)

Table 12.3: Examples of paired designs.

Observational unit	Comparison groups	Measurement	Value of interest
Car	Smooth Turn vs. Quick Spin	Tire tread after 1,000 miles	Difference in tread
Textbook	UCLA Bookstore vs. Amazon	Price of new textbook	Difference in price
Individual	Pre-course vs. Post-course	Exam score	Difference in score

12.4.2 The key idea: analyze the differences

The key to working with paired data is simple: **compute the difference for each pair, then analyze those differences as a single sample**. This converts a two-sample problem into a one-sample problem.

Paired t confidence interval for μ_{diff} .

For paired data, compute the difference $d_i = x_{1,i} - x_{2,i}$ for each pair. Then the confidence interval for the mean difference μ_{diff} is:

$$\bar{x}_{diff} \pm t_{df}^* \times \frac{s_{diff}}{\sqrt{n_{diff}}}$$

where $\bar{x}_{diff}$ is the sample mean of the differences, s_{diff} is the sample standard deviation of the differences, n_{diff} is the number of pairs, and $df = n_{diff} - 1$.

Conditions:

1. **Independence of pairs.** The pairs themselves are independent (e.g., different cars, different textbooks).
2. **Normality of differences.** The distribution of the *differences* should be approximately normal, or the number of pairs should be large ($n_{diff} \geq 30$). Check for extreme outliers.

12.4.3 Worked example: Textbook prices

Researchers sampled 68 UCLA courses and compared textbook prices at the UCLA Bookstore to prices on Amazon. Summary statistics for the price differences (UCLA — Amazon) are:

Table 12.4: Summary statistics for the 68 price differences.

n_{diff}	$\bar{x}_{diff}$	s_{diff}
68	\$3.58	\$13.42

Construct a 95% confidence interval for the mean price difference.

Check conditions.

1. *Independence of pairs.* The textbooks were randomly sampled, so the pairs are independent.
2. *Normality of differences.* With $n_{diff} = 68$, the CLT ensures the sampling distribution of $\bar{x}_{diff}$ is approximately normal. Though the distribution of individual differences is right-skewed, the large sample size and absence of extreme outliers make the t -methods reasonable.

Compute the interval.

$$SE = \frac{s_{diff}}{\sqrt{n_{diff}}} = \frac{13.42}{\sqrt{68}} = \frac{13.42}{8.246} = 1.628$$

With $df = 68 - 1 = 67$, the critical value for a 95% CI is $t_{67}^* \approx 2.00$.

$$3.58 \pm 2.00 \times 1.628 = 3.58 \pm 3.256 = (0.32, 6.84)$$

We are 95% confident that UCLA Bookstore prices are, on average, between \$0.32 and \$6.84 higher than Amazon prices for UCLA course books.

A fitness program enrolled 25 participants and measured their resting heart rates before and after 8 weeks of training. The mean difference (before — after) was $\bar{x}_{diff} = 4.2$ beats per minute with $s_{diff} = 6.1$ bpm. Construct a 90% confidence interval for the mean decrease in resting heart rate. (Use $t_{24}^* = 1.711$.)

Show answer

First check conditions: pairs (individuals) were randomly selected (independence), and with $n = 25$ and no extreme outliers reported, the normality condition is reasonable. Then $SE = 6.1/\sqrt{25} = 6.1/5 = 1.22$. The 90% CI is $4.2 \pm 1.711 \times 1.22 = 4.2 \pm 2.09 = (2.11, 6.29)$. We are 90% confident the training program reduces resting heart rate by between 2.11 and 6.29 bpm on average.

12.4.4 Why not use a two-sample t interval for paired data?

Never use a two-sample procedure for paired data. A two-sample t interval treats the two groups as independent, which they are not. Ignoring the pairing throws away valuable information about the within-pair relationship and typically produces a wider (less precise) confidence interval. Always compute the differences first and then use the one-sample (paired) t procedure.

12.5 Connecting t -based and bootstrap confidence intervals

The t -based confidence intervals in this chapter and the bootstrap confidence intervals from Chapter 8 are answering the same question: “What is a plausible range for the population parameter?” When conditions are met, they give similar results.

When do they agree?

- Large samples with no extreme outliers: the t -interval and bootstrap interval will be nearly identical.
- Moderate samples from approximately normal populations: close agreement.

When might they differ?

- Small samples from skewed populations: the bootstrap can capture asymmetry in the sampling distribution, while the t -interval is always symmetric around $\bar{x}$.
- Presence of outliers: the bootstrap is somewhat more robust because it does not rely on s being a good estimate of σ .

Which should you use?

- If conditions are met, the t -interval is simpler to compute and has well-understood properties.
- If conditions are questionable (small sample, skewness, outliers), prefer the bootstrap.
- In practice, reporting both and noting any discrepancies is informative.

12.6 Chapter review

12.6.1 Summary

This chapter developed formula-based confidence intervals for means using the t -distribution.

- The **t -distribution** accounts for the extra uncertainty from estimating σ with s . It has heavier tails than the normal distribution, especially for small degrees of freedom.
- **One-sample t CI:** $\bar{x} \pm t_{n-1}^* \times \frac{s}{\sqrt{n}}$. Requires independent observations and approximate normality (or large n).
- **Two-sample t CI:** $(\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$. Requires independence within and between groups, and approximate normality or large samples in each group.
- **Paired t CI:** Compute differences for each pair, then apply the one-sample t CI to the differences: $\bar{x}_{diff} \pm t_{n_{diff}-1}^* \times \frac{s_{diff}}{\sqrt{n_{diff}}}$.
- Always check conditions before using t -methods: independence, normality/sample size, and absence of extreme outliers.
- When conditions are met, t -based and bootstrap confidence intervals give similar results. When conditions are questionable, the bootstrap may be more reliable.

12.6.2 Key terms

t -distribution, degrees of freedom (df), critical value (t^*), one-sample t confidence interval, two-sample t confidence interval, paired data, paired t confidence interval, standard error of the mean, independence condition, normality condition.

12.6.3 Exercises

Exercises for this chapter are under development.

Chapter 13

Hypothesis Tests for Means

In Chapter 7, we developed the logic of hypothesis testing using simulation: shuffle the data under the null hypothesis, build a null distribution, and measure how extreme the observed statistic is. In Chapter 11, we introduced the t -distribution as a mathematical model for the sampling distribution of means. Now we bring these ideas together. In this chapter, we use the t -distribution to conduct hypothesis tests for a single mean, the difference between two independent means, and the mean of paired differences. Along the way, we will see that the t -test and simulation-based tests generally agree when conditions are met, and we will discuss what to do when they disagree.

13.1 The t -test framework

Every hypothesis test follows the same four-step structure, whether we use simulation or a mathematical model:

1. **State hypotheses.** Write H_0 (null hypothesis) and H_A (alternative hypothesis) in both words and symbols.
2. **Check conditions.** Verify that the mathematical model is appropriate.
3. **Compute the test statistic and p-value.** Calculate how far the observed data are from what H_0 predicts, and determine how unusual that distance is.
4. **Draw a conclusion.** Compare the p-value to the significance level α and state the conclusion in context.

For means, the test statistic is a T score – the standardized distance from the observed sample mean to the hypothesized value, measured in units of standard errors.

13.2 One-sample t -test

13.2.1 Setup and hypotheses

The one-sample t -test is used when we have a single sample of numerical data and want to test a claim about the population mean μ .

The test statistic for a single mean is a T .

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x}$ is the sample mean, μ_0 is the hypothesized value from H_0 , s is the sample standard deviation, and n is the sample size.

When the null hypothesis is true and the conditions are met, T follows a t -distribution with $df = n - 1$.

Conditions:

- **Independent observations** (e.g., random sample).
- **Normality / sample size:** The population distribution should be approximately normal, or n should be large enough ($n \geq 30$). Check for strong skewness or extreme outliers.

13.2.2 Worked example: Cherry Blossom race times

The average finishing time for all runners in the Cherry Blossom 10-mile race in 2006 was $\mu = 93.29$ minutes. A random sample of 100 runners from a more recent year had a mean finishing time of $\bar{x} = 97.32$ minutes with $s = 16.98$ minutes. Is there evidence that the average finishing time has changed?

Step 1: State hypotheses.

- $H_0: \mu = 93.29$ (the average finishing time has not changed)
- $H_A: \mu \neq 93.29$ (the average finishing time has changed)

This is a two-sided test because we are looking for a change in either direction.

Step 2: Check conditions.

- *Independence.* The sample is a random sample of runners, so observations are independent.
- *Sample size.* With $n = 100 \geq 30$, the CLT guarantees the sampling distribution of $\bar{x}$ is approximately normal, even if the population distribution of finishing times is skewed.

Step 3: Compute the test statistic and p-value.

$$SE = \frac{s}{\sqrt{n}} = \frac{16.98}{\sqrt{100}} = 1.698$$

$$T = \frac{\bar{x} - \mu_0}{SE} = \frac{97.32 - 93.29}{1.698} = \frac{4.03}{1.698} = 2.37$$

With $df = 100 - 1 = 99$, we find the area to the right of $T = 2.37$ on the t -distribution: approximately 0.0099. Since this is a two-sided test, the p-value is $2 \times 0.0099 = 0.0198$.

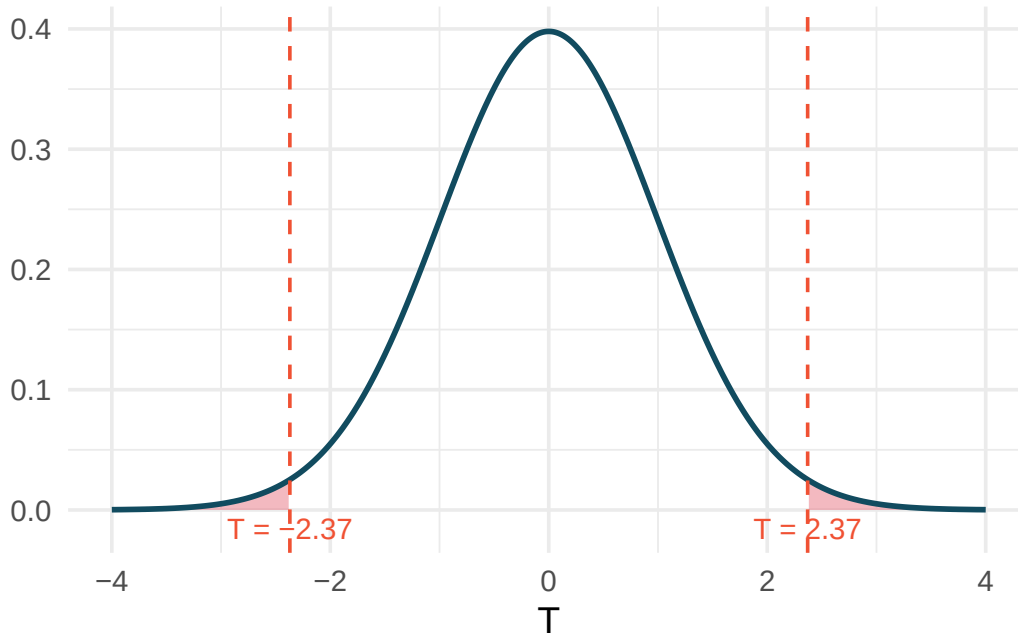


Figure 13.1: A t -distribution with $df = 99$ showing the two-sided p -value. The shaded areas in both tails represent the probability of observing a test statistic at least as extreme as 2.37 in either direction.

Step 4: Conclusion. The p -value of 0.0198 is less than $\alpha = 0.05$, so we reject H_0 . The data provide statistically significant evidence that the average finishing time has changed from the 2006 average of 93.29 minutes. Based on the sample, runners appear to be finishing more slowly on average.

A nutrition study claims that the average American adult consumes 2,000 mg of sodium per day. A random sample of 45 adults shows $\bar{x} = 3,120$ mg and $s = 1,100$ mg. Test whether the data provide evidence that the true average sodium intake differs from 2,000 mg. Use $\alpha = 0.05$.

Show answer

Hypotheses: $H_0 : \mu = 2000$, $H_A : \mu \neq 2000$. Conditions: random sample (independence), $n = 45 \geq 30$ (normality via CLT). Test statistic: $SE = 1100/\sqrt{45} = 163.9$, $T = (3120 - 2000)/163.9 = 6.83$. With $df = 44$, the two-tailed p -value is essentially 0 (far less than 0.05). We reject H_0 . There is very strong evidence that the average sodium intake is different from (and appears much higher than) 2,000 mg per day.

13.3 Two-sample t -test

13.3.1 Setup and hypotheses

The two-sample t -test compares the means of two independent groups. The parameter of interest is $\mu_1 - \mu_2$, and the null hypothesis typically claims no difference.

The test statistic for comparing two independent means is a T .

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

When the null hypothesis is true and conditions are met, T follows a t -distribution with $df = \min(n_1 - 1, n_2 - 1)$ (conservative approximation; software computes the exact df).

Conditions:

- **Independence (extended).** Observations are independent within and between the two groups.
- **Normality / sample size.** Each group should be approximately normal or have a large enough sample size. Check each group separately for strong skewness or extreme outliers.

13.3.2 Worked example: Birth weights and smoking

Is there evidence that newborns of mothers who smoke during pregnancy have a different average birth weight than newborns of non-smoking mothers? Data from a random sample of 1,000 US births in 2014:

Table 13.1: Summary statistics for birth weights by smoking status.

Group	n	Mean (lbs)	SD (lbs)
Nonsmoker	867	7.27	1.23
Smoker	114	6.68	1.60

Step 1: State hypotheses.

- $H_0: \mu_n - \mu_s = 0$ (no difference in average birth weight)
- $H_A: \mu_n - \mu_s \neq 0$ (there is a difference in average birth weight)

Step 2: Check conditions.

- *Independence.* The data come from a simple random sample, so observations are independent within and between groups.
- *Normality.* Both groups have large sample sizes ($n_n = 867$ and $n_s = 114$), both well above 30. Histograms of the data show no extreme outliers. Conditions are met.

Step 3: Compute the test statistic and p-value.

$$\bar{x}_n - \bar{x}_s = 7.27 - 6.68 = 0.59$$

$$SE = \sqrt{\frac{1.23^2}{867} + \frac{1.60^2}{114}} = \sqrt{\frac{1.5129}{867} + \frac{2.56}{114}} = \sqrt{0.001745 + 0.02246} = \sqrt{0.02420} = 0.156$$

$$T = \frac{0.59 - 0}{0.156} = 3.78$$

Using the conservative $df = \min(867 - 1, 114 - 1) = 113$, the one-tail area for $T = 3.78$ is approximately 0.00012. The two-tailed p-value is $2 \times 0.00012 = 0.00024$.

Step 4: Conclusion. With a p-value of 0.00024, which is far less than $\alpha = 0.05$, we reject H_0 . The data provide strong evidence that there is a difference in average birth weight between babies born to smoking and non-smoking mothers. Specifically, babies born to non-smoking mothers weigh about 0.59 lbs more on average.

13.3.3 Worked example: Exam versions

An instructor created two versions of an exam and randomly distributed them. Summary statistics:

Table 13.2: Summary statistics for exam versions.

Version	n	Mean	SD	Min	Max
A	58	75.1	13.9	44	100
B	55	72.0	13.8	38	100

Test whether there is convincing evidence that one version is more difficult. Use $\alpha = 0.01$.

Step 1: State hypotheses.

- $H_0: \mu_A - \mu_B = 0$ (the exams are equally difficult)
- $H_A: \mu_A - \mu_B \neq 0$ (one exam is more difficult)

Step 2: Check conditions.

- *Independence.* Exams were randomly assigned, so independence is satisfied.
- *Normality.* Both groups have $n > 30$ and the summary statistics suggest roughly symmetric distributions. Conditions are met.

Step 3: Compute the test statistic and p-value.

$$\bar{x}_A - \bar{x}_B = 75.1 - 72.0 = 3.1$$

$$SE = \sqrt{\frac{13.9^2}{58} + \frac{13.8^2}{55}} = \sqrt{\frac{193.21}{58} + \frac{190.44}{55}} = \sqrt{3.331 + 3.463} = \sqrt{6.794} = 2.607$$

$$T = \frac{3.1}{2.607} = 1.19$$

Using $df = \min(57, 54) = 54$, the one-tail area for $T = 1.19$ is about 0.12. The two-tailed p-value is approximately 0.24.

Step 4: Conclusion. The p-value of 0.24 is much larger than $\alpha = 0.01$, so we fail to reject H_0 . The data do not provide convincing evidence that one exam version is more difficult than the other.

Does the conclusion above mean that the two exams are equally difficult?

Show answer

No. Failing to reject H_0 does not prove that H_0 is true. The data are consistent with the exams being equally difficult, but they are also consistent with exam A being about 3.1 points easier. The test simply did not find strong enough evidence to distinguish these possibilities.

13.4 Paired t -test

13.4.1 Setup and hypotheses

When data are paired (see Chapter 11 for what makes data “paired”), we compute the difference within each pair and test whether the mean difference is zero.

The test statistic for a paired mean difference is a T .

$$T = \frac{\bar{x}_{diff} - 0}{s_{diff} / \sqrt{n_{diff}}}$$

When the null hypothesis is true and conditions are met, T follows a t -distribution with $df = n_{diff} - 1$.

Conditions:

- **Independently sampled pairs.**
- **Normality of differences.** The differences should be approximately normal, or n_{diff} should be large. Check for extreme outliers in the differences.

13.4.2 Worked example: Textbook prices

Is there evidence that UCLA Bookstore prices differ from Amazon prices, on average? Summary statistics for the 68 price differences (UCLA – Amazon):

Table 13.3: Summary statistics for the 68 price differences.

n_{diff}	$\bar{x}_{diff}$	s_{diff}
68	\$3.58	\$13.42

Step 1: State hypotheses.

- $H_0: \mu_{diff} = 0$ (no difference in average textbook price)
- $H_A: \mu_{diff} \neq 0$ (there is a difference in average price)

Step 2: Check conditions.

- *Independence.* The textbooks were a simple random sample of courses; pairs are independent.
- *Normality.* With $n_{diff} = 68$, the CLT applies. While some outliers exist (a few textbooks with large price differences), none are extreme enough to invalidate the approach given the large sample size.

Step 3: Compute the test statistic and p-value.

$$SE = \frac{s_{diff}}{\sqrt{n_{diff}}} = \frac{13.42}{\sqrt{68}} = 1.628$$

$$T = \frac{3.58 - 0}{1.628} = 2.20$$

With $df = 68 - 1 = 67$, the one-tail area for $T = 2.20$ is approximately 0.0156. The two-tailed p-value is $2 \times 0.0156 = 0.0312$.

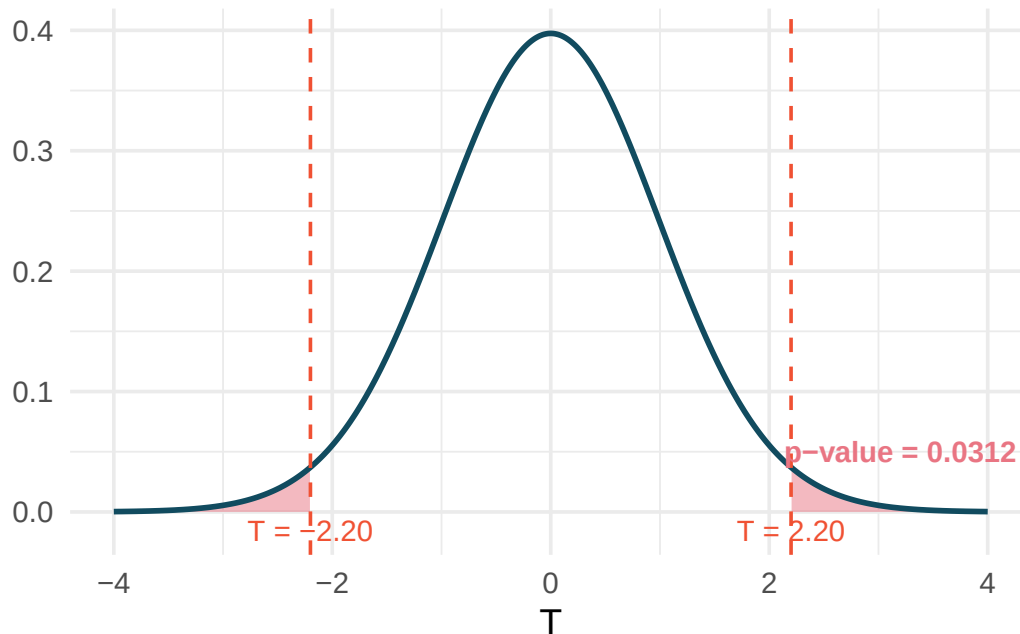


Figure 13.2: A t -distribution with $df = 67$ showing the two-sided p -value. The shaded areas in both tails represent the p -value of 0.0312.

Step 4: Conclusion. The p -value of 0.0312 is less than $\alpha = 0.05$, so we reject H_0 . The data provide evidence that UCLA Bookstore prices are different from Amazon prices on average. Based on the positive mean difference, the UCLA Bookstore tends to charge more.

Researchers measured tire tread (in cm) for 25 cars, each equipped with one Smooth Turn and one Quick Spin tire. The mean tread difference (Smooth Turn – Quick Spin) was $\bar{x}_{diff} = 0.002$ cm with $s_{diff} = 0.004$ cm. Test whether there is a difference in average tread wear between the two brands. Use $\alpha = 0.05$.

Show answer

Hypotheses: $H_0 : \mu_{diff} = 0$, $H_A : \mu_{diff} \neq 0$. Conditions: cars are independent, $n = 25$ and no extreme outliers reported. $SE = 0.004/\sqrt{25} = 0.0008$. $T = 0.002/0.0008 = 2.50$. With $df = 24$, the two-tailed p -value is approximately 0.020. Since $0.020 < 0.05$, we reject H_0 and conclude there is evidence of a difference in tread wear between the two tire brands.

13.5 Connecting simulation-based and t -based tests

An important theme throughout this course is that simulation-based methods and mathematical models are complementary approaches to the same questions. When conditions are met, they should give similar answers. Here we compare the two approaches explicitly.

13.5.1 When they agree

For the birth weights example:

- **Randomization test:** Shuffle smoking/nonsmoking labels 10,000 times, compute the difference in means each time, and find the proportion of shuffled differences as extreme as 0.59 lbs. This gives a p -value around 0.001.

- ***t*-test:** The *p*-value was 0.00024.

Both approaches lead to the same conclusion (reject H_0) by a wide margin. This agreement is typical when sample sizes are moderate to large and conditions are met.

13.5.2 When they might disagree

Disagreement arises primarily in two situations:

1. **Small samples from skewed populations.** The *t*-test assumes the sampling distribution is symmetric, which may not hold for small samples from skewed distributions. A simulation-based approach can capture the asymmetry.
2. **Extreme outliers.** The *t*-test relies on $\bar{x}$ and *s*, both of which are sensitive to outliers. A randomization test, which works with ranks and permutations, can be more robust.

13.5.3 Practical guidance

- **When conditions are clearly met** (large samples, no extreme outliers), use the *t*-test – it is simpler and faster.
- **When conditions are borderline** (moderate samples, mild skewness), consider running both approaches. If they agree, report the *t*-test. If they disagree, investigate why and consider the simulation-based result more trustworthy.
- **When conditions are clearly violated** (small samples, strong skewness, extreme outliers), prefer the simulation-based approach.

13.6 Practical vs. statistical significance

Statistical significance $\neq$ practical significance.

A result is **statistically significant** if the *p*-value is below the chosen significance level α . But statistical significance only means the observed effect is unlikely to be due to chance alone. It says nothing about whether the effect is *large enough to matter in practice*.

With very large samples, even trivially small differences can be statistically significant. With very small samples, even large differences may not be statistically significant (due to low power).

A large study of 50,000 students found that students who eat breakfast score an average of 0.5 points higher on a 100-point exam than students who skip breakfast ($p < 0.001$). Is this result practically significant?

The result is statistically significant (the *p*-value is very small), but the effect size of 0.5 points on a 100-point exam is tiny. A teacher or student would likely not consider half a point to be a meaningful difference. This is an example where statistical significance does not imply practical significance.

13.6.1 Effect sizes

To bridge the gap between statistical and practical significance, researchers often report **effect sizes** – standardized measures of how large the observed effect is.

Cohen's *d* is a common measure of effect size for comparing two means. It expresses the difference in means as a multiple of the pooled standard deviation:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}$$

where s_p is the pooled standard deviation. General guidelines for interpreting d :

Table 13.4: Guidelines for interpreting Cohen's d .

$ d $	Interpretation
0.2	Small effect
0.5	Medium effect
0.8	Large effect

These guidelines are rough benchmarks. What constitutes a “meaningful” effect depends on the context. In medicine, even a small effect size might be important if it concerns a life-threatening condition. In education, a medium effect might be required for a costly intervention to be worthwhile.

In the birth weights study, the difference was 0.59 lbs with a pooled standard deviation of approximately 1.30 lbs. Compute Cohen's d and interpret it.

Show answer

$d = 0.59/1.30 = 0.45$. This is between a small and medium effect. While the difference is statistically significant, the practical impact is moderate – about half a standard deviation in birth weight.

13.7 Chapter review

13.7.1 Summary

This chapter presented hypothesis tests for means using the t -distribution:

- **One-sample t -test:** Tests whether a population mean μ equals a hypothesized value μ_0 . Test statistic: $T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ with $df = n - 1$.
- **Two-sample t -test:** Tests whether two population means are equal. Test statistic: $T = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$ with $df = \min(n_1 - 1, n_2 - 1)$.
- **Paired t -test:** Tests whether the mean of paired differences equals zero. Compute differences first, then apply the one-sample t -test to the differences: $T = \frac{\bar{x}_{diff}}{s_{diff}/\sqrt{n_{diff}}}$ with $df = n_{diff} - 1$.
- All three tests require **independence** and **approximate normality or large sample size**.
- **Simulation-based and t -based tests** generally agree when conditions are met. When conditions are questionable, simulation is more reliable.
- **Statistical significance** (small p -value) does not imply **practical significance** (large enough to matter). Effect sizes like **Cohen's d** help assess practical importance.

13.7.2 Key terms

T test statistic, one-sample t -test, two-sample t -test, paired t -test, degrees of freedom, p -value, significance level (α), statistical significance, practical significance, effect size, Cohen's d .

13.7.3 Exercises

Exercises for this chapter are under development.

Chapter 14

Inference for Proportions

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In earlier chapters, we used simulation-based methods to draw conclusions about population proportions. In Chapters 7 and 8, we built randomization distributions and bootstrap confidence intervals; in Chapter 10, we learned that the normal distribution provides a mathematical model for these sampling distributions when certain conditions are met. In this chapter, we bring everything together for proportions. We develop normal-based confidence intervals and hypothesis tests for a single proportion and for the difference between two proportions. Throughout, we compare these formula-based results to the simulation-based results from earlier chapters, reinforcing the idea that these are two routes to the same destination.

StatBench tools for this chapter: Use the **Bootstrap CI for One Proportion** and **Permutation Test Simulator** alongside the formula-based methods in this chapter to see both approaches side by side.

[Launch Bootstrap CI Tool](#)

14.1 Normal approximation for proportions

14.1.1 The sampling distribution of $\hat{p}$

In Chapter 10, we saw that the Central Limit Theorem guarantees that the sampling distribution of $\hat{p}$ is approximately normal when certain conditions are met. Let's state these conditions precisely.

Sampling distribution of $\hat{p}$.

The sampling distribution of the sample proportion $\hat{p}$, based on a sample of size n from a population with true proportion p , is approximately normal when:

1. **Independence.** The observations are independent (e.g., from a simple random sample).
2. **Success-failure condition.** We expect to see at least 10 successes and 10 failures: $np \geq 10$ and $n(1 - p) \geq 10$.

When these conditions are met, the sampling distribution is approximately:

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

The **success-failure condition** requires that the expected number of successes (np) and the expected number of failures ($n(1 - p)$) are both at least 10. This ensures the sampling distribution of $\hat{p}$ is not too skewed for the normal approximation to work well.

A medical consultant facilitated 62 liver transplant surgeries, with 3 complications. Under the null hypothesis that the true complication rate is $p_0 = 0.10$, is it appropriate to use the normal approximation?

Check the success-failure condition using $p_0 = 0.10$:

- Expected successes (complications): $np_0 = 62 \times 0.10 = 6.2$
- Expected failures: $n(1 - p_0) = 62 \times 0.90 = 55.8$

The expected number of successes (6.2) is less than 10. The **success-failure condition is not met**, so the normal approximation should not be used here. A simulation-based approach (such as the parametric bootstrap from Chapter 8) would be more appropriate.

A poll surveys 826 randomly selected payday loan borrowers and finds that 51% support a new regulation. Check whether the normal approximation can be used to model $\hat{p}$ for a hypothesis test with $H_0 : p = 0.50$.

Show answer

Check the success-failure condition using $p_0 = 0.50$: $np_0 = 826 \times 0.50 = 413$ and $n(1 - p_0) = 826 \times 0.50 = 413$. Both are well above 10, so the success-failure condition is met. Combined with the random sample (independence), the normal approximation is appropriate.

14.1.2 The standard error of $\hat{p}$

Standard error of $\hat{p}$.

When conditions are met, the variability of $\hat{p}$ is described by:

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

Since we rarely know the true p :

- **For confidence intervals**, use $\hat{p}$ as the best guess: $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
- **For hypothesis tests**, use p_0 (the value from H_0): $SE = \sqrt{\frac{p_0(1-p_0)}{n}}$

Consider polls of size 300 where approximately 2/3 of voters support legalized marijuana. Calculate the standard error of $\hat{p}$.

Show answer

Using $p \approx 2/3$: $SE = \sqrt{\frac{(2/3)(1/3)}{300}} = \sqrt{\frac{2/9}{300}} = \sqrt{0.000741} = 0.027$.

14.2 Confidence interval for a single proportion

14.2.1 The formula

Confidence interval for a single proportion p .

$$\hat{p} \pm z^* \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where z^* is the critical value for the desired confidence level (e.g., $z^* = 1.96$ for 95%).

Steps:

1. Check conditions: independence and success-failure using $\hat{p}$.
2. Compute SE using $\hat{p}$.
3. Apply the confidence interval formula.

14.2.2 Worked example: Payday loan regulation

A simple random sample of 826 payday loan borrowers was surveyed, and 70% supported new regulations on payday lenders. Construct a 95% confidence interval for the true proportion who support the regulations.

Step 1: Check conditions.

- *Independence.* The data are a random sample.
- *Success-failure condition.* Using $\hat{p} = 0.70$: $n\hat{p} = 826 \times 0.70 = 578$ and $n(1 - \hat{p}) = 826 \times 0.30 = 248$. Both are well above 10.

Step 2: Compute SE.

$$SE = \sqrt{\frac{0.70 \times 0.30}{826}} = \sqrt{\frac{0.21}{826}} = \sqrt{0.000254} = 0.016$$

Step 3: Construct the interval.

$$0.70 \pm 1.96 \times 0.016 = 0.70 \pm 0.031 = (0.669, 0.731)$$

We are 95% confident that between 66.9% and 73.1% of all payday loan borrowers support the new regulation.

14.2.3 Changing the confidence level

The confidence level is controlled by the choice of z^* :

Table 14.1: Critical values for common confidence levels.

Confidence level	z^*	Interval width
90%	1.645	Narrower
95%	1.960	Moderate
99%	2.576	Wider

Higher confidence requires a wider interval – we must cast a wider net to be more confident we have captured the true parameter.

Using the payday loan data ($\hat{p} = 0.70$, $n = 826$, $SE = 0.016$), construct a 99% confidence interval.

Show answer

Using $z^* = 2.576$: $0.70 \pm 2.576 \times 0.016 = 0.70 \pm 0.041 = (0.659, 0.741)$. The 99% interval is wider than the 95% interval, as expected.

A study on stent surgery found a point estimate of 0.090 (the difference in stroke rates) with $SE = 0.028$. Construct a 90% confidence interval.

Show answer

Using $z^* = 1.645$: $0.090 \pm 1.645 \times 0.028 = 0.090 \pm 0.046 = (0.044, 0.136)$. We are 90% confident that stent implantation increases the 30-day stroke rate by between 4.4% and 13.6%.

General confidence interval formula. If a point estimate follows a normal model with standard error SE , then a confidence interval for the population parameter is:

$$\text{point estimate} \pm z^* \times SE$$

where z^* corresponds to the chosen confidence level.

14.3 Hypothesis test for a single proportion

14.3.1 The Z test for a proportion

The test statistic for a single proportion is a Z.

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$$

where $\hat{p}$ is the sample proportion and p_0 is the hypothesized proportion from H_0 .

When H_0 is true and conditions are met, Z follows a standard normal distribution $N(0, 1)$.

Conditions:

- Independent observations
- $np_0 \geq 10$ and $n(1 - p_0) \geq 10$

Note that for hypothesis tests, the standard error is computed using p_0 (not $\hat{p}$), because we are evaluating the data under the assumption that H_0 is true.

14.3.2 Worked example: Support for credit checks

From a random sample of 826 payday loan borrowers, 51% said they would support a regulation requiring lenders to pull credit reports. Is there convincing evidence that a majority (more than 50%) of borrowers support this regulation?

Step 1: State hypotheses.

- $H_0: p = 0.50$ (no majority support)
- $H_A: p > 0.50$ (majority support)

This is a one-sided test.

Step 2: Check conditions.

- *Independence.* Random sample.
- *Success-failure.* Using $p_0 = 0.50$: $np_0 = 826 \times 0.50 = 413$ and $n(1 - p_0) = 413$. Both ≥ 10 .

Step 3: Compute the test statistic and p-value.

$$SE = \sqrt{\frac{0.50 \times 0.50}{826}} = \sqrt{\frac{0.25}{826}} = 0.017$$

$$Z = \frac{0.51 - 0.50}{0.017} = \frac{0.01}{0.017} = 0.59$$

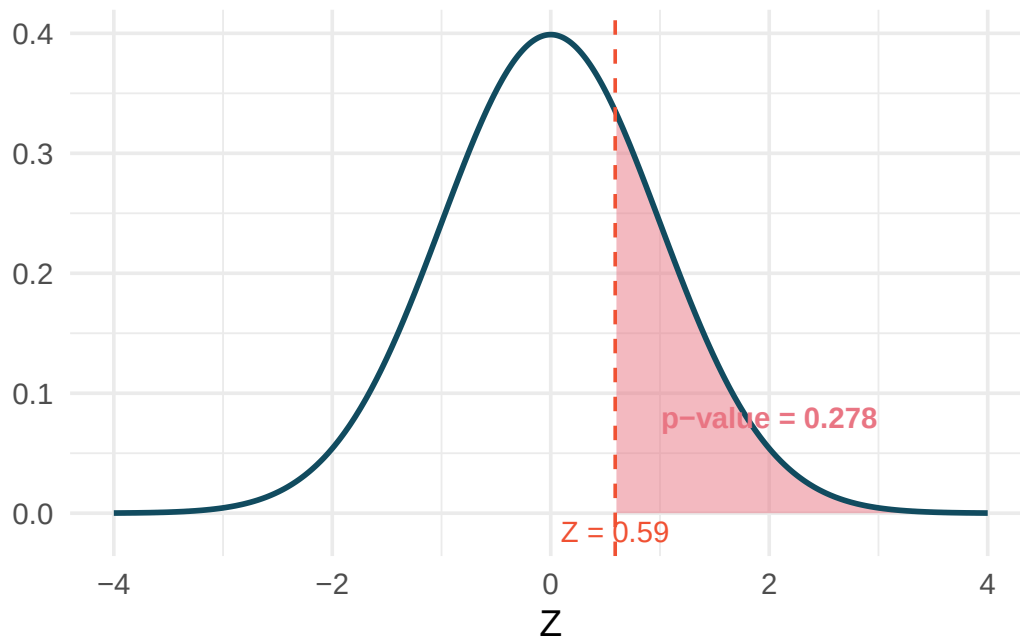


Figure 14.1: A standard normal curve with the area to the right of $Z = 0.59$ shaded, representing the one-sided p-value.

The area to the right of $Z = 0.59$ is 0.278.

Step 4: Conclusion. The p-value of 0.278 is much larger than $\alpha = 0.05$, so we fail to reject H_0 . The poll does not provide convincing evidence that a majority of payday loan borrowers support the regulation requiring credit checks.

A candidate claims that more than 60% of voters in a district support her. A random sample of 500 voters finds $\hat{p} = 0.64$. Test the claim at $\alpha = 0.05$.

Show answer

Hypotheses: $H_0 : p = 0.60$, $H_A : p > 0.60$. Success-failure: $500(0.60) = 300$ and $500(0.40) = 200$, both ≥ 10 . $SE = \sqrt{0.60 \times 0.40/500} = \sqrt{0.000480} = 0.0219$. $Z = (0.64 - 0.60)/0.0219 = 1.83$. The right-tail area is 0.034. Since $0.034 < 0.05$, we reject H_0 . There is sufficient evidence that more than 60% of voters support the candidate.

14.3.3 When conditions fail

When the success-failure condition is not met, the normal model can underestimate the variability of $\hat{p}$, leading to inaccurate p-values and confidence intervals. In these cases, use simulation-based methods:

- **Bootstrap hypothesis test:** Simulate samples under H_0 by drawing from a population with proportion p_0 (the parametric bootstrap from Chapter 8).
- **Bootstrap confidence interval:** Resample from the original data to estimate the variability of $\hat{p}$.

14.4 Difference of two proportions

14.4.1 The parameter and point estimate

When comparing two groups, the parameter of interest is the difference in population proportions: $p_1 - p_2$. The point estimate is $\hat{p}_1 - \hat{p}_2$.

14.4.2 Confidence interval for a difference in proportions

Confidence interval for $p_1 - p_2$.

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \times \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

Conditions:

1. **Independence (extended).** Observations are independent within and between groups.
2. **Success-failure condition (each group separately).** $n_1\hat{p}_1 \geq 10$, $n_1(1 - \hat{p}_1) \geq 10$, $n_2\hat{p}_2 \geq 10$, and $n_2(1 - \hat{p}_2) \geq 10$.

In a CPR study, patients who had a heart attack were randomly assigned to receive a blood thinner (treatment) or not (control). The survival rates after 24 hours:

Table 14.2: CPR study results.

Group	n	Survived	$\hat{p}$
Treatment	50	14	0.280
Control	40	11	0.275

Construct a 90% confidence interval for the difference in survival rates, $p_T - p_C$.

Check conditions.

- *Independence.* Patients were randomly assigned, so groups are independent.
- *Success-failure.* Treatment: $50(0.28) = 14 \geq 10$ and $50(0.72) = 36 \geq 10$. Control: $40(0.275) = 11 \geq 10$ and $40(0.725) = 29 \geq 10$. All conditions met.

Compute the interval.

$$\hat{p}_T - \hat{p}_C = 0.280 - 0.275 = 0.005$$

$$SE = \sqrt{\frac{0.280(0.720)}{50} + \frac{0.275(0.725)}{40}} = \sqrt{\frac{0.2016}{50} + \frac{0.1994}{40}} = \sqrt{0.004032 + 0.004985} = \sqrt{0.009017} = 0.095$$

Using $z^* = 1.645$ for 90% confidence:

$$0.005 \pm 1.645 \times 0.095 = 0.005 \pm 0.156 = (-0.151, 0.161)$$

We are 90% confident that the difference in 24-hour survival rates is between -15.1% and $+16.1\%$. Because the interval contains 0, we cannot conclude that the blood thinner affects survival rates.

14.4.3 Hypothesis test for a difference in proportions

The test statistic for comparing two proportions is a Z .

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where the **pooled proportion** is:

$$\hat{p}_{pool} = \frac{\text{total successes in both groups}}{\text{total sample size}} = \frac{x_1 + x_2}{n_1 + n_2}$$

When H_0 is true ($p_1 = p_2$), both groups share the same population proportion, so we pool the data to get a single best estimate.

Conditions:

- Independence within and between groups.
- $n_1\hat{p}_{pool} \geq 10$, $n_1(1 - \hat{p}_{pool}) \geq 10$, $n_2\hat{p}_{pool} \geq 10$, $n_2(1 - \hat{p}_{pool}) \geq 10$.

Does the blood thinner improve 24-hour survival after CPR? Using the data from the previous example (treatment: 14/50 survived; control: 11/40 survived), test at $\alpha = 0.05$.

Step 1: State hypotheses.

- $H_0: p_T - p_C = 0$ (the blood thinner has no effect)
- $H_A: p_T - p_C \neq 0$ (the blood thinner has some effect)

Step 2: Check conditions.

The pooled proportion is $\hat{p}_{pool} = \frac{14+11}{50+40} = \frac{25}{90} = 0.278$. Check:

- $50(0.278) = 13.9 \geq 10$, $50(0.722) = 36.1 \geq 10$
- $40(0.278) = 11.1 \geq 10$, $40(0.722) = 28.9 \geq 10$

Conditions are met.

Step 3: Compute the test statistic and p-value.

$$SE = \sqrt{0.278(0.722) \left(\frac{1}{50} + \frac{1}{40} \right)} = \sqrt{0.2007 \times 0.045} = \sqrt{0.009032} = 0.095$$

$$Z = \frac{0.005 - 0}{0.095} = 0.05$$

The two-sided p-value for $Z = 0.05$ is $2 \times P(Z > 0.05) = 2 \times 0.480 = 0.960$.

Step 4: Conclusion. The p-value of 0.960 is far larger than 0.05. We fail to reject H_0 . There is no evidence that the blood thinner affects 24-hour survival rates.

14.4.4 Why pool for hypothesis tests but not for confidence intervals?

You may have noticed a subtle difference:

- **Hypothesis tests** use the **pooled proportion** $\hat{p}_{pool}$ in the standard error. This is because under H_0 , we assume $p_1 = p_2$, so the best estimate of this common proportion comes from combining both samples.
- **Confidence intervals** use the **individual sample proportions** $\hat{p}_1$ and $\hat{p}_2$ separately. We are not assuming the proportions are equal – we are estimating how different they are.

This distinction applies only to proportions. For means, we use separate sample standard deviations in both tests and confidence intervals.

14.5 Comparing simulation-based and normal-based methods

Throughout this course, we have developed two parallel approaches to inference:

Table 14.3: Comparison of simulation-based and normal approximation approaches.

Feature	Simulation-based	Normal approximation
Requires conditions?	Minimal (mainly independence)	Independence + success-failure
Small samples?	Works well	May be inaccurate
Computational cost	Higher (needs many simulations)	Lower (one formula)
Conceptual clarity	High (directly simulates the process)	Moderate (requires distribution theory)
Precision	Approximate (varies by number of simulations)	Approximate (varies by how well conditions hold)

14.5.1 An example comparison

Consider the opportunity cost study from Chapter 7. The observed difference in proportions was 0.20 ($\hat{p}_T - \hat{p}_C$), and we wanted to test $H_0 : p_T - p_C = 0$.

Randomization test (Chapter 7): We shuffled the group labels 10,000 times. The p-value was approximately 0.006.

Normal approximation (this chapter): Using $SE = 0.078$ and $Z = 0.20/0.078 = 2.56$, the p-value was 0.005.

Both methods give essentially the same answer and the same conclusion. This is reassuring and typical when conditions are met.

14.5.2 When do they disagree?

The methods can give different answers when:

1. **The success-failure condition fails.** The normal approximation may undercount the probability in the tails, leading to p-values that are too small. The simulation approach remains valid.
2. **The sampling distribution is noticeably skewed.** The normal model is symmetric by definition, while the true sampling distribution of $\hat{p}$ near 0 or 1 is skewed.

The medical consultant example illustrates this: with only 6.2 expected complications, the normal model gave a p-value of 0.085, while the simulation gave 0.122 – nearly 45% higher. The simulation-based p-value is more trustworthy in this case.

14.5.3 Practical recommendation

- **When conditions are met:** Both methods work. Use the normal approximation for its simplicity, or use simulation for its conceptual directness.
- **When conditions are borderline or not met:** Use simulation. The bootstrap and randomization approaches make fewer assumptions and are more reliable in these settings.
- **Always check conditions first.** The success-failure condition is a simple, quick check that helps you decide which approach to use.

14.6 Chapter review

14.6.1 Summary

This chapter developed normal-based inference methods for proportions:

- **Success-failure condition.** The normal approximation for $\hat{p}$ requires at least 10 expected successes and 10 expected failures. For confidence intervals, check with $\hat{p}$; for hypothesis tests, check with p_0 .
- **CI for one proportion:** $\hat{p} \pm z^* \times \sqrt{\hat{p}(1-\hat{p})/n}$. Use $\hat{p}$ in the SE.
- **Hypothesis test for one proportion:** $Z = \frac{\hat{p}-p_0}{\sqrt{p_0(1-p_0)/n}}$. Use p_0 in the SE.
- **CI for difference of proportions:** $(\hat{p}_1 - \hat{p}_2) \pm z^* \times SE$, where SE uses separate sample proportions.
- **Hypothesis test for difference of proportions:** $Z = \frac{\hat{p}_1 - \hat{p}_2}{SE}$, where SE uses the **pooled proportion** $\hat{p}_{pool}$.
- **Confidence level** is controlled by z^* : higher confidence means wider intervals.
- **When conditions fail** (especially the success-failure condition), use simulation-based methods from Chapters 7 and 8 instead.
- **Normal-based and simulation-based methods agree** when conditions are met and can give different answers when conditions are violated.

14.6.2 Key terms

Success-failure condition, standard error of $\hat{p}$, Z test for a proportion, pooled proportion, confidence interval for a proportion, confidence interval for a difference of proportions, margin of error, critical value (z^*), confidence level.

14.6.3 Exercises

Exercises for this chapter are under development.

PART IV

RELATIONSHIPS

You now have the full inference toolkit — simulation-based and formula-based. In this part, we apply that toolkit to new data types and richer questions: testing categorical distributions with chi-square, comparing multiple groups with ANOVA, and modeling relationships with correlation and linear regression.

Chapter 15

Chi-Square Goodness of Fit

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In previous chapters, we have developed inferential methods for comparing proportions across two groups. But what if we want to assess whether a single categorical variable with multiple levels follows a particular distribution? For instance, we might want to know if a jury pool is racially representative of the community, or whether the days of the week on which accidents occur are equally likely. The **chi-square goodness of fit test** is designed for exactly this purpose. It compares a set of observed counts to a set of expected counts derived from a hypothesized distribution, using a single test statistic that summarizes all of the differences at once.

15.1 Observed versus expected counts

Consider data from a random sample of 275 jurors in a small county. Jurors identified their racial group, and we would like to determine if these jurors are racially representative of the county's population of registered voters. The observed counts and the population proportions are shown below.

	White	Black	Hispanic	Other	Total
Observed jurors	205	26	25	19	275
Population proportion	0.72	0.07	0.12	0.09	1.00

While the proportions in the jury sample do not precisely match the population proportions, it is unclear whether these data provide convincing evidence that the sample is not representative. If the jurors really were randomly sampled from the registered voters, we might expect small differences due to chance. However, unusually large differences may provide convincing evidence that the juries were not representative.

15.1.1 Computing expected counts

If the jurors were randomly sampled from the population (i.e., if the null hypothesis is true), we would expect the sample proportions to roughly match the population proportions. The **expected count** for each category is found by multiplying the sample size by the hypothesized proportion for that category.

Of the 275 people who served on a jury, about how many would we expect to be White if jurors are randomly selected from the population? How many would we expect to be Black?

About 72% of the population is White, so we would expect about 72% of the jurors to be White: $0.72 \times 275 = 198$. Similarly, we would expect about 7% of the jurors to be Black, which would correspond to about $0.07 \times 275 = 19.25$ Black jurors.

Twelve percent of the population is Hispanic and 9% represent other races. How many of the 275 jurors would we expect to be Hispanic? How many from other races?

Show answer

We would expect $0.12 \times 275 = 33$ Hispanic jurors and $0.09 \times 275 = 24.75$ jurors from other races.

The complete table of observed and expected counts is:

	White	Black	Hispanic	Other	Total
Observed counts	205	26	25	19	275
Expected counts	198	19.25	33	24.75	275

Computing expected counts for a goodness of fit test.

If the null hypothesis specifies proportions $p_1, p_2, \dots, p_k$ for k categories, and the total sample size is n , then the expected count for category i is:

$$E_i = n \times p_i$$

15.1.2 Setting up hypotheses

The sample proportions from each racial group were not a precise match for the population proportions. We need to test whether the differences are strong enough to provide convincing evidence that the jurors are not a random sample. These ideas can be organized into hypotheses:

- H_0 : The jurors are a random sample from the population. The observed counts reflect natural sampling fluctuation.
- H_A : The jurors are not randomly sampled from the population. There is bias in juror selection.

To evaluate these hypotheses, we need to quantify how different the observed counts are from the expected counts. Strong evidence for the alternative hypothesis would come in the form of unusually large deviations from what would be expected based on sampling variation alone.

15.2 The chi-square statistic

To build a test statistic, we first compute the standardized difference between each observed and expected count. For each category i :

$$Z_i = \frac{O_i - E_i}{\sqrt{E_i}}$$

where O_i is the observed count and E_i is the expected count. The denominator $\sqrt{E_i}$ serves as the standard error for the count.

For the jury data:

$$\begin{aligned} Z_{\text{White}} &= \frac{205 - 198}{\sqrt{198}} = 0.50 \\ Z_{\text{Black}} &= \frac{26 - 19.25}{\sqrt{19.25}} = 1.54 \\ Z_{\text{Hispanic}} &= \frac{25 - 33}{\sqrt{33}} = -1.39 \\ Z_{\text{Other}} &= \frac{19 - 24.75}{\sqrt{24.75}} = -1.16 \end{aligned}$$

We would like to use a single test statistic to determine if these four standardized differences are unusually far from zero as a group. We do this by squaring each standardized difference and adding them up:

$$X^2 = Z_1^2 + Z_2^2 + Z_3^2 + Z_4^2 = 0.50^2 + 1.54^2 + (-1.39)^2 + (-1.16)^2 = 5.89$$

Squaring each standardized difference before adding them together does two things:

1. Any standardized difference that is squared will be positive, so negative and positive deviations both contribute to the test statistic.
2. Differences that are already unusual (e.g., a standardized difference of 2.5) become much larger after squaring, which gives them more weight.

Chi-square test statistic.

The **chi-square test statistic** is the sum of the squared standardized differences between observed and expected counts:

$$X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

where the sum is over all k categories, O_i is the observed count in category i , and E_i is the expected count in category i under the null hypothesis.

15.3 Simulation-based goodness of fit test

Before turning to the mathematical chi-square distribution, it is instructive to see how a simulation-based approach works for goodness of fit tests.

The idea is the same as in other randomization tests: if the null hypothesis is true (the data come from the hypothesized distribution), we can simulate many datasets that would arise under the null hypothesis and compute the chi-square statistic for each. This gives us a null distribution of the chi-square statistic, against which we can compare our observed statistic.

How the simulation works:

1. Under H_0 , each observation has probability p_i of falling into category i .
2. For each simulated sample of size n , randomly assign each observation to a category according to the null probabilities.
3. Compute the chi-square statistic X^2 for the simulated data.
4. Repeat many times (e.g., 1,000 or 10,000 times) to build a null distribution.

5. Compare the observed X^2 to the null distribution. The p-value is the proportion of simulated statistics that are as large or larger than the observed statistic.

For the jury example, each simulated dataset would randomly assign 275 jurors to the four racial categories with probabilities 0.72, 0.07, 0.12, and 0.09. The resulting chi-square statistics from the simulated datasets form the null distribution.

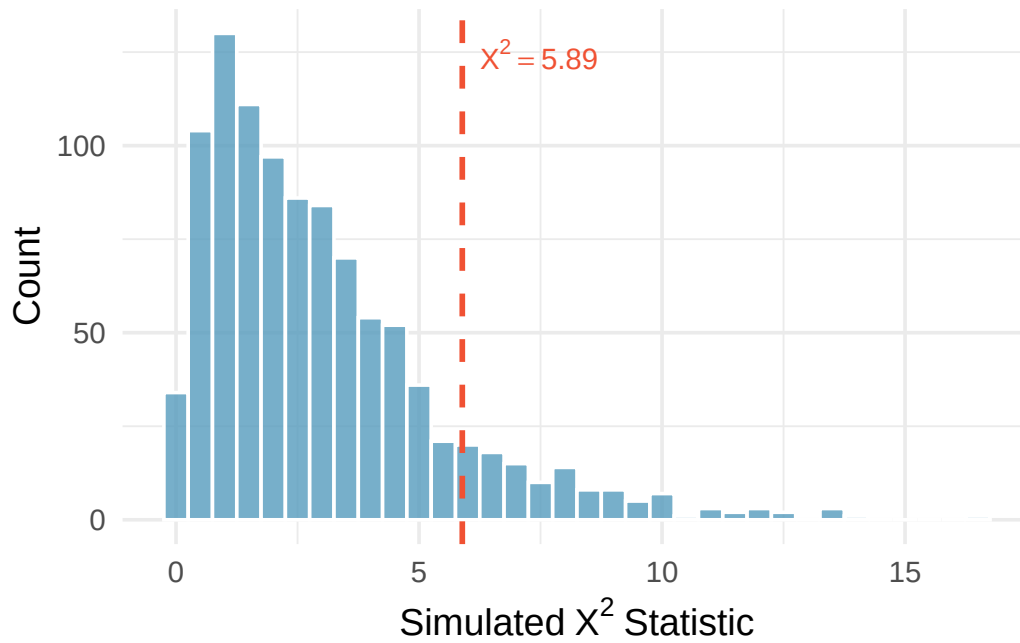


Figure 15.1: Histogram of chi-square statistics from 1,000 simulations under the null hypothesis for the jury example. The observed statistic of 5.89 is marked. Most simulated values fall below 5.89, but a meaningful fraction exceed it.

The simulation-based approach confirms what we will find with the mathematical model: the observed chi-square statistic of 5.89 is not unusual enough to reject the null hypothesis at a 0.05 significance level.

StatBench: Categorical Data Explorer. Use the Categorical Data Explorer to enter observed and expected counts and run a simulation-based chi-square goodness of fit test. The tool will generate a null distribution and compute a p-value.

15.4 The chi-square distribution

The chi-square test statistic X^2 follows a **chi-square distribution** when the null hypothesis is true and the sample size conditions are met. The chi-square distribution is characterized by a single parameter called the **degrees of freedom** (df), which influences the shape, center, and spread of the distribution.

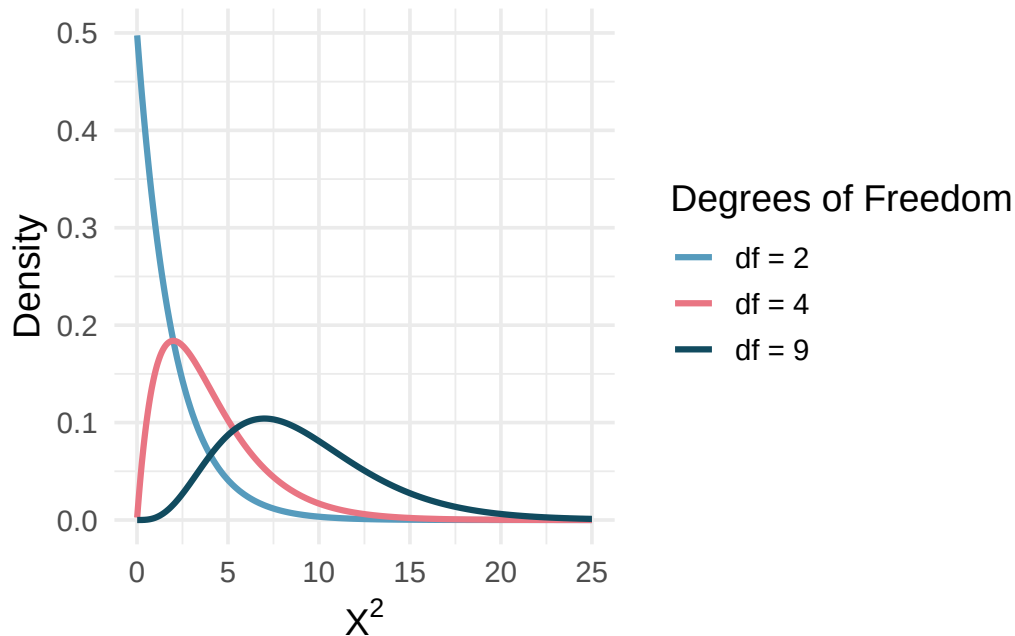


Figure 15.2: Three chi-square distributions with degrees of freedom 2, 4, and 9. As the degrees of freedom increase, the distribution becomes more symmetric, the center moves to the right, and the variability increases.

Key properties of the chi-square distribution:

- It is always non-negative (the chi-square statistic is a sum of squared values).
- It is right-skewed, especially for small degrees of freedom.
- As the degrees of freedom increase, the distribution becomes more symmetric and shifts to the right.
- The mean of the distribution equals the degrees of freedom.

15.4.1 Degrees of freedom for goodness of fit

For a goodness of fit test with k categories, the degrees of freedom are:

$$df = k - 1$$

How many categories were there in the juror example? How many degrees of freedom should be associated with the chi-square distribution?

There were $k = 4$ categories: White, Black, Hispanic, and Other. The degrees of freedom are $df = k - 1 = 3$.

15.5 Finding a p-value

If the null hypothesis is true, X^2 follows a chi-square distribution with $k - 1$ degrees of freedom. Because larger values of X^2 correspond to greater deviations from the null hypothesis, the p-value is always found from the **upper tail** of the distribution.

For the jury data, the test statistic is $X^2 = 5.89$ and the degrees of freedom are $df = 3$. Find the p-value.

Using a chi-square distribution with 3 degrees of freedom, the area to the right of 5.89 is approximately 0.117.

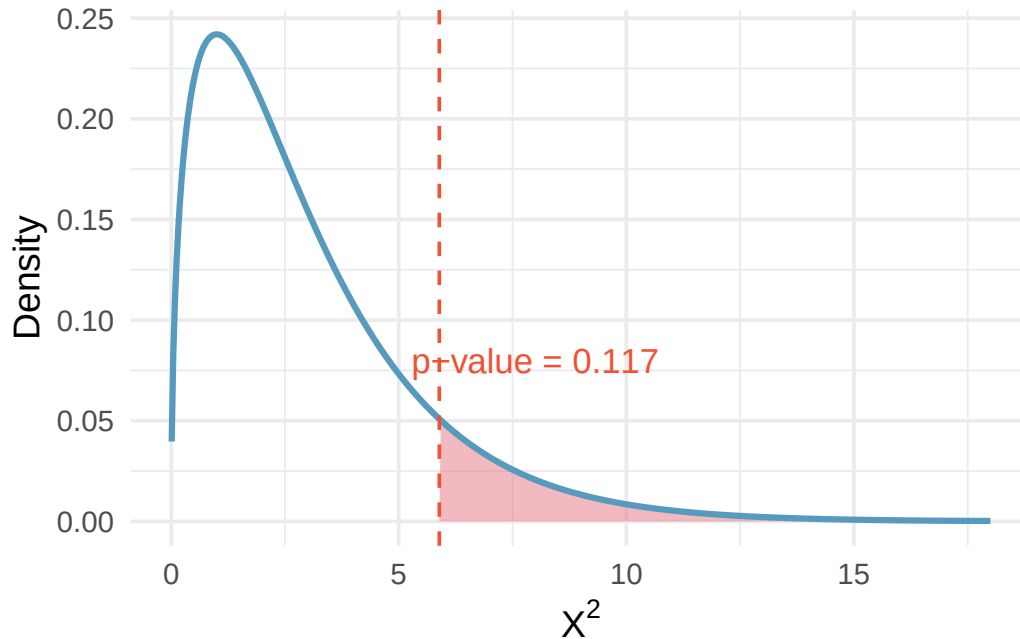


Figure 15.3: Chi-square distribution with 3 degrees of freedom, with the area above 5.89 shaded to represent the p-value of approximately 0.117.

Since the p-value of 0.117 is larger than $\alpha = 0.05$, we do not reject the null hypothesis. The data do not provide convincing evidence of racial bias in the juror selection process.

Consider a chi-square distribution with 5 degrees of freedom and a test statistic of 5.1. Find the p-value.

Using statistical software or a chi-square table, the area to the right of 5.1 on a chi-square distribution with 5 degrees of freedom is approximately 0.404. This is a very large p-value, so we would not reject the null hypothesis.

A chi-square distribution with 4 degrees of freedom has a test statistic of 10. Find the approximate p-value.

Show answer

Using statistical software, the area to the right of 10 on a chi-square distribution with 4 degrees of freedom is approximately 0.040. Since this is less than 0.05, we would reject the null hypothesis.

15.6 Chi-square goodness of fit test

We can now summarize the complete procedure for a chi-square goodness of fit test.

Chi-square test for goodness of fit.

Suppose we want to evaluate whether the observed counts $O_1, O_2, \dots, O_k$ in k categories are consistent with a hypothesized distribution that specifies expected counts $E_1, E_2, \dots, E_k$.

Hypotheses:

- H_0 : The data follow the hypothesized distribution (the observed counts are consistent with the expected counts, up to sampling variability).
- H_A : The data do not follow the hypothesized distribution.

Test statistic:

$$X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

When the null hypothesis is true and the conditions are met, X^2 follows a chi-square distribution with $df = k - 1$.

The p-value is found from the upper tail of the chi-square distribution.

Conditions:

- **Independence.** Each case that contributes a count must be independent of all other cases.
- **Sample size.** Each expected count must be at least 5.

15.7 Evaluating goodness of fit for a distribution

The goodness of fit test can also be used to evaluate whether data follow a particular theoretical distribution.

Suppose we observe the daily change in a stock index over 1,362 trading days. We classify each day according to how many consecutive days of increase preceded it, and compare the observed frequencies to what we would expect if each day's movement were independent of the previous day's (a geometric distribution model). The observed and expected counts are shown below.

Days since last increase	1	2	3	4	5	6	7+
Observed	717	369	155	69	28	14	10
Expected	743	338	154	70	32	14	12

Are these data consistent with the geometric distribution model?

We set up the hypotheses:

- H_0 : Stock returns are independent from one day to the next and follow a geometric model.
- H_A : Stock returns show a pattern inconsistent with the geometric model.

The chi-square statistic is:

$$X^2 = \frac{(717 - 743)^2}{743} + \frac{(369 - 338)^2}{338} + \frac{(155 - 154)^2}{154} + \frac{(69 - 70)^2}{70} + \frac{(28 - 32)^2}{32} + \frac{(14 - 14)^2}{14} + \frac{(10 - 12)^2}{12} = 4.61$$

With $k = 7$ categories, $df = 6$. The p-value for $X^2 = 4.61$ with 6 degrees of freedom is approximately 0.595.

Since the p-value is very large, we do not reject the null hypothesis. The data are consistent with the geometric distribution model, suggesting that stock movements may be independent from day to day over this period.

15.8 Chapter review

15.8.1 Summary

In this chapter, we developed the chi-square goodness of fit test, which assesses whether a sample of categorical data is consistent with a hypothesized distribution. The test uses the chi-square statistic, which measures how far the observed counts deviate from the expected counts. When the null hypothesis is true and the conditions are met, this statistic follows a chi-square distribution with $k - 1$ degrees of freedom, where k is the number of categories. The p-value is always computed from the upper tail of the chi-square distribution, since larger values indicate greater discrepancy between the observed and expected counts.

15.8.2 Key terms

- Expected count
- Chi-square statistic (X^2)
- Chi-square distribution
- Degrees of freedom (goodness of fit)
- Goodness of fit test

Chapter 16

Chi-Square Test of Independence

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In the previous chapter, we used the chi-square statistic to test whether a single categorical variable follows a particular distribution. In this chapter, we extend the chi-square framework to two-way tables, where we have two categorical variables and want to assess whether they are associated. The key question becomes: is the distribution of one variable the same across all levels of the other variable, or does the distribution shift depending on the group? As with our other inference methods, we begin with a simulation-based approach (randomization) and then introduce the mathematical chi-square approximation.

Note that with two-way tables, there is not an obvious single parameter of interest. Instead, research questions focus on how the proportions of the response variable change (or not) across the different levels of the explanatory variable. Because there is no single population parameter to estimate, we focus on hypothesis testing (and not confidence intervals).

16.1 Randomization test of independence

16.1.1 Case study: The iPod experiment

We all buy used products – cars, computers, textbooks – and we sometimes assume the sellers will be forthright about any underlying problems. Researchers recruited 219 participants in a study where they would sell a used iPod that was known to have frozen twice in the past. The participants were incentivized to get as much money as they could. The researchers wanted to understand what types of questions would elicit the seller to disclose the freezing issue.

The buyers (who were collaborating with the researchers) asked one of three scripted questions:

- **General:** “What can you tell me about it?”
- **Positive Assumption:** “It does not have any problems, does it?”
- **Negative Assumption:** “What problems does it have?”

The results are shown below:

Question	Disclose problem	Hide problem	Total
General	2	71	73

Question	Disclose problem	Hide problem	Total
Positive assumption	23	50	73
Negative assumption	36	37	73
Total	61	158	219

The data suggest that asking “What problems does it have?” was the most effective at getting the seller to disclose the past freezing issues. But could we see these results due to chance alone, or is this evidence that some questions are more effective at getting to the truth?

16.1.2 Expected counts in two-way tables

The hypothesis test is about assessing whether there is convincing evidence that the question asked was related to whether the seller disclosed the problem. In other words, we want to check whether the buyer’s question was **independent** of whether the seller disclosed a problem.

From the experiment, the overall proportion of sellers who disclosed the freezing problem was $61/219 = 0.2785$. If there really is no difference among the questions and 27.85% of sellers were going to disclose regardless, how many of the 73 people in the General group would we have expected to disclose?

We would predict that $0.2785 \times 73 = 20.33$ sellers would disclose the problem. We observed far fewer than this, though it is not yet clear if that is due to chance variation or because the questions truly differ in effectiveness.

If the questions were actually equally effective, meaning about 27.85% of respondents would disclose the freezing issue regardless of what question they were asked, about how many sellers in the Positive Assumption group would we expect to *hide* the freezing problem?

Show answer

We would expect $(1 - 0.2785) \times 73 = 52.67$.

We can compute the expected number for all cells using the row totals, column totals, and table total. For instance, the proportion who disclosed overall was $61/219$. Multiplying by each row total:

$$\frac{61}{219} \times 73 = 20.33 \quad (\text{for each question group})$$

More generally, this can be rewritten as:

$$\text{Expected Count}_{\text{row } i, \text{col } j} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{\text{table total}}$$

Computing expected counts in a two-way table.

To calculate the expected count for the i^{th} row and j^{th} column, compute:

$$\text{Expected Count}_{\text{row } i, \text{col } j} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{\text{table total}}$$

Using this formula, the observed counts with expected counts in parentheses are:

Question	Disclose problem	Hide problem	Total
General	2 (20.33)	71 (52.67)	73
Positive assumption	23 (20.33)	50 (52.67)	73
Negative assumption	36 (20.33)	37 (52.67)	73
Total	61	158	219

16.1.3 The observed chi-square statistic

The chi-square test statistic for a two-way table is found by computing the squared standardized difference for every cell in the table and summing them:

$$\begin{array}{ll}
 \text{General formula} & \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}} \\
 \text{Row 1, Col 1} & \frac{(2 - 20.33)^2}{20.33} = 16.53 \\
 \text{Row 2, Col 1} & \frac{(23 - 20.33)^2}{20.33} = 0.35 \\
 \vdots & \vdots \\
 \text{Row 3, Col 2} & \frac{(37 - 52.67)^2}{52.67} = 4.66
 \end{array}$$

Adding the computed value for each cell gives the chi-square test statistic:

$$X^2 = 16.53 + 0.35 + \cdots + 4.66 = 40.13$$

16.1.4 Variability of the statistic

Is 40.13 a big number? Does it indicate that the observed and expected values are really different, or is it a value we might expect to see just due to natural variability?

Assuming that the individuals would disclose or hide the problems **regardless** of the question they were given (i.e., that the null hypothesis is true), we can randomize the data by reassigning the 61 disclosed problems and 158 hidden problems to the three question groups at random. This is exactly the same type of randomization we used in previous chapters.

One such randomization might produce:

Question	Disclose problem	Hide problem	Total
General	29	44	73
Positive assumption	15	58	73
Negative assumption	17	56	73
Total	61	158	219

Computing the chi-square statistic for this randomized table:

$$X^2 = \frac{(29 - 20.33)^2}{20.33} + \frac{(15 - 20.33)^2}{20.33} + \cdots + \frac{(56 - 52.67)^2}{52.67} = 8$$

This is much smaller than 40.13.

16.1.5 Observed statistic vs. null chi-square statistics

One randomization is not sufficient. We generate many chi-square statistics under the null hypothesis to build a null distribution.

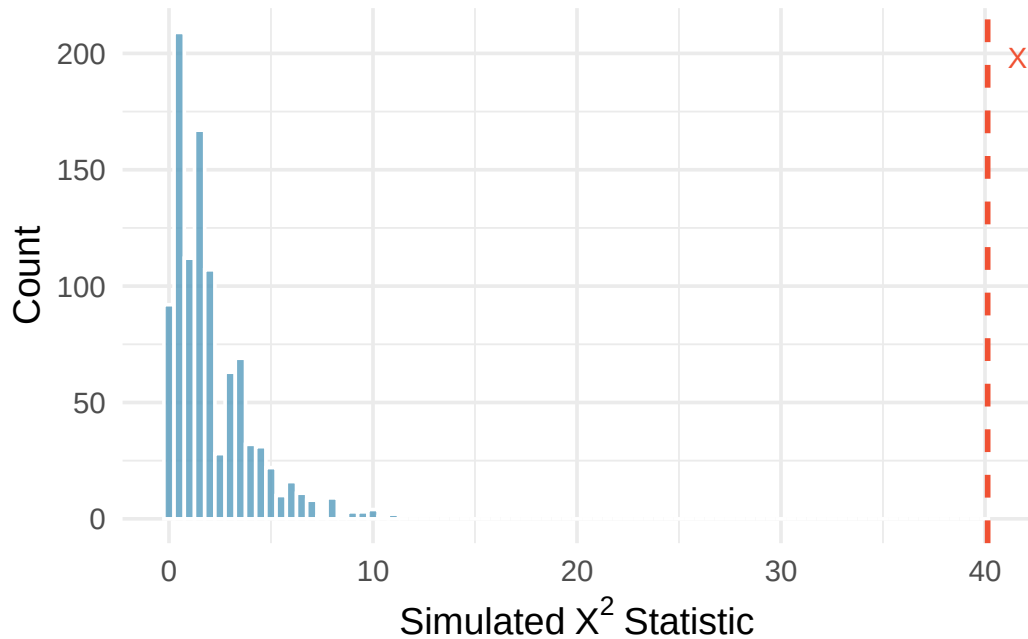


Figure 16.1: A histogram of chi-square statistics from 1,000 simulations produced under the null hypothesis, where the question is independent of the response. The observed statistic of 40.13 is marked by a red line. None of the 1,000 simulations had a chi-square value anywhere close to 40.13.

The observed value is so far from the null statistics that the simulated p-value is essentially zero. We conclude that the decision of whether to disclose the iPod's problem **is changed** by the question asked. We use the causal language of "changed" because the study was a randomized experiment.

Note that with a chi-square test, we only know that the two variables (question type and response) are related (i.e., not independent). We are not able to claim which specific type of question causes which specific type of response.

16.2 Mathematical model for test of independence

16.2.1 The chi-square test of independence

The chi-square test statistic follows a **chi-square distribution** when the null hypothesis is true and the conditions are met. For two-way tables, the degrees of freedom are:

$$df = (R - 1) \times (C - 1)$$

where R is the number of rows and C is the number of columns.

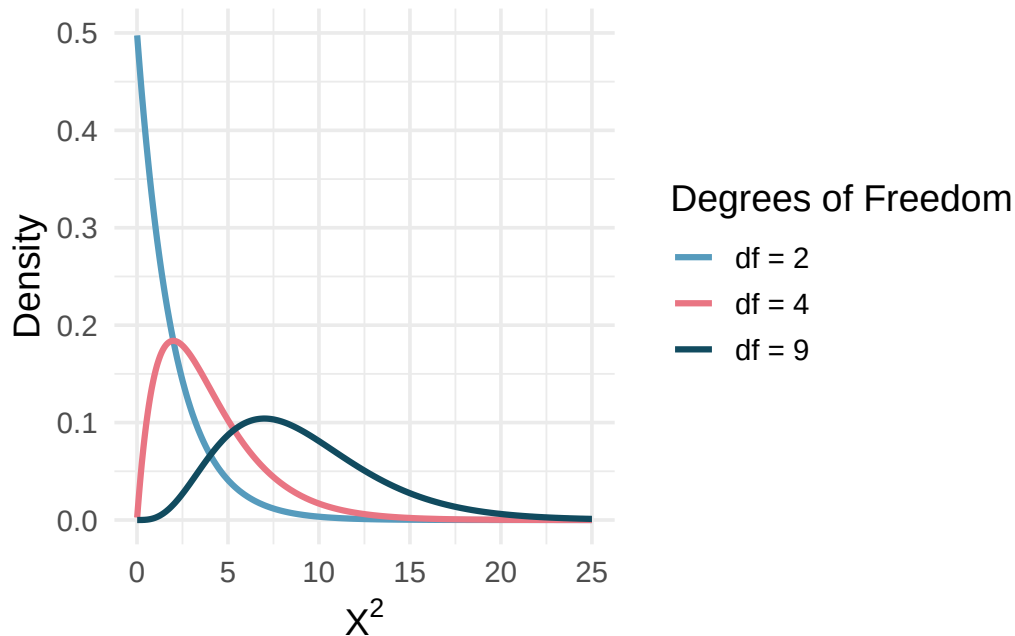


Figure 16.2: The chi-square distribution for different degrees of freedom. The larger the degrees of freedom, the longer the right tail extends. The smaller the degrees of freedom, the more peaked the mode on the left becomes.

16.2.2 Variability of the chi-square statistic

As it turns out, the chi-square test statistic follows a chi-square distribution when the null hypothesis is true. For two-way tables, the degrees of freedom equal $(R - 1) \times (C - 1)$. In the iPod example, this is $(3 - 1) \times (2 - 1) = 2$.

The test statistic for assessing independence between two categorical variables is X^2 .

The X^2 statistic is a ratio of how the observed counts differ from the expected counts, standardized by the expected counts:

$$X^2 = \sum_{i,j} \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}}$$

When the null hypothesis is true and the conditions are met, X^2 has a chi-square distribution with $df = (R - 1) \times (C - 1)$.

Conditions:

- Independent observations
- Large samples: at least 5 expected counts in each cell

Computing degrees of freedom for a two-way table.

When applying the chi-square test to a two-way table, use $df = (R - 1) \times (C - 1)$ where R is the number of rows and C is the number of columns.

16.2.3 Finding the p-value

We can safely assume that the observations in the iPod experiment are independent (the question groups were randomly assigned). Each expected count exceeds 5, so the conditions for using the chi-square distribution are met. The test statistic $X^2 = 40.13$ follows a chi-square distribution with 2 degrees of freedom.

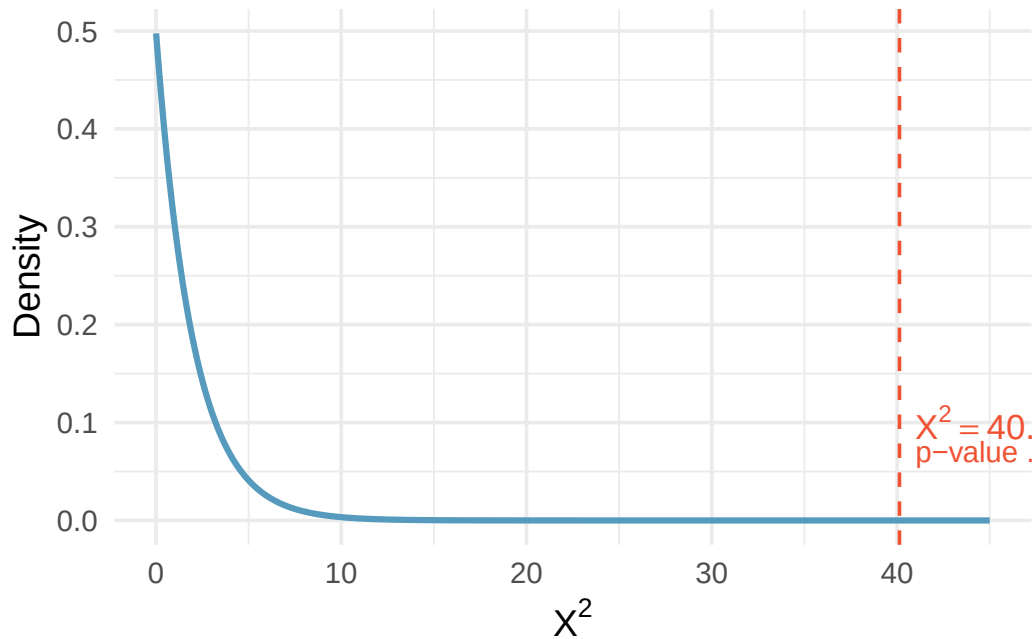


Figure 16.3: The chi-square distribution with $df = 2$, showing the area to the right of $X^2 = 40.13$. The p-value is so small it is not visible on the plot.

Find the p-value and draw a conclusion about whether the question affects the seller's likelihood of reporting the freezing problem.

Using a chi-square distribution with 2 degrees of freedom, the tail area above $X^2 = 40.13$ is approximately 0.000000002.

Using a significance level of $\alpha = 0.05$, the null hypothesis is rejected since the p-value is much smaller. The data provide convincing evidence that the question asked did affect a seller's likelihood to tell the truth about problems with the iPod.

16.2.4 Another example: Diabetes treatment

An experiment evaluated three treatments for Type 2 Diabetes in patients aged 10–17 who were being treated with metformin. The three treatments were: continued metformin alone (met), metformin combined with rosiglitazone (rosi), or a lifestyle intervention program. Each patient had a primary outcome of either failure (lacked glycemic control) or success (maintained glycemic control). The results are shown below.

Treatment	Failure	Success	Total
met	120	114	234

Treatment	Failure	Success	Total
rosi	90	143	233
lifestyle	109	123	232
Total	319	380	699

What are appropriate hypotheses for this test?

-
- H_0 : There is no difference in the effectiveness of the three treatments.
 - H_A : There is some difference in effectiveness between the three treatments, e.g., perhaps the rosiglitazone treatment performed better than lifestyle intervention.

Compute the expected counts for each of the six cells in the diabetes treatment table.

Show answer

Row 1, Col 1: $\frac{234 \times 319}{699} = 106.8$. Row 1, Col 2: $\frac{234 \times 380}{699} = 127.2$. Row 2, Col 1: $\frac{233 \times 319}{699} = 106.3$. Row 2, Col 2: $\frac{233 \times 380}{699} = 126.7$. Row 3, Col 1: $\frac{232 \times 319}{699} = 105.9$. Row 3, Col 2: $\frac{232 \times 380}{699} = 126.1$.

Note: when analyzing 2-by-2 contingency tables (where both variables have only two categories), it is generally preferable to use the two-proportion methods from earlier chapters.

16.3 Comparing the simulation and chi-square approaches

Both the randomization test and the mathematical chi-square test address the same question: are two categorical variables independent? The two approaches generally give very similar results, especially when the sample size is large and the expected counts are well above 5.

The **randomization test** is more flexible because:

- It does not require any minimum expected count.
- It works well even for small samples.
- It requires no distributional assumptions beyond independence.

The **mathematical chi-square test** is more convenient because:

- It does not require a computer to simulate thousands of randomizations.
- It provides a simple formula and distribution for computing p-values.
- It is widely implemented in statistical software.

In practice, the chi-square test is the standard approach when conditions are met, while the randomization test provides a useful check or alternative when conditions are questionable.

16.4 Chapter review

16.4.1 Summary

In this chapter, we extended the chi-square framework to two-way tables, testing whether two categorical variables are independent. The randomization approach shuffles the data under the null hypothesis to build a distribution of the chi-square statistic, while the mathematical approach uses the chi-square distribution with $(R - 1) \times (C - 1)$ degrees of freedom. Both approaches require that observations are independent and that expected counts are sufficiently large (at least 5 in each cell).

The chi-square test of independence tells us *whether* two variables are related, but does not tell us *how* they are related or *which* groups differ. To understand the nature of the relationship, we must examine the individual cells and their contributions to the chi-square statistic.

16.4.2 Key terms

- Independence
- Expected counts (two-way table)
- Chi-square statistic
- Chi-square distribution
- Degrees of freedom (two-way table)

Chapter 17

Analysis of Variance

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In earlier chapters, we developed methods to compare the mean of a numerical outcome across two groups. An important aspect of those analyses was to look at the difference in sample means as an estimate for the difference in population means. When comparing more than two groups, simple subtraction will not fully capture the variation across three or more groups. As with two groups, the research question focuses on whether group membership is independent of the numerical response variable. In this chapter, we focus on a new statistic that incorporates differences in means across more than two groups. Although the ideas are similar to the t -test, they have earned their own name: **ANalysis Of VArance**, or **ANOVA**.

Sometimes we want to compare means across many groups. We might initially think to do pairwise comparisons. For example, if there were three groups, we might compare the first mean with the second, then with the third, and finally compare the second and third means – a total of three comparisons. However, this strategy can be treacherous. If we have many groups and do many comparisons, it is likely that we will eventually find a difference just by chance, even if there is no difference in the populations. Instead, we should apply a holistic test to check whether there is evidence that at least one pair of groups is different, which is where ANOVA saves the day.

ANOVA uses a single hypothesis test to check whether the means across many groups are equal:

- H_0 : The mean outcome is the same across all groups: $\mu_1 = \mu_2 = \dots = \mu_k$ where μ_j represents the mean of the outcome for observations in category j .
- H_A : At least one mean is different.

Generally, we must check three conditions before performing ANOVA:

- The observations are independent within and between groups.
- The responses within each group are nearly normal.
- The variability across the groups is about equal.

When these conditions are met, we may perform ANOVA to determine whether the data provide convincing evidence against the null hypothesis.

College departments commonly run multiple sections of the same introductory course. Consider a statistics department that runs three sections of an intro stats course. We might like to determine

whether there are substantial differences in first exam scores across sections A, B, and C. Describe appropriate hypotheses.

-
- H_0 : The average score is identical in all sections, $\mu_A = \mu_B = \mu_C$. The observed differences are due to chance.
 - H_A : The average score varies by class. We would reject the null hypothesis if there were larger differences among the class averages than what we might expect from chance alone.

Strong evidence favoring the alternative hypothesis in ANOVA is described by unusually large differences among the group means. Assessing the variability of the group means relative to the variability among individual observations within each group is key to ANOVA's success.

Consider the figure below showing two sets of three groups.

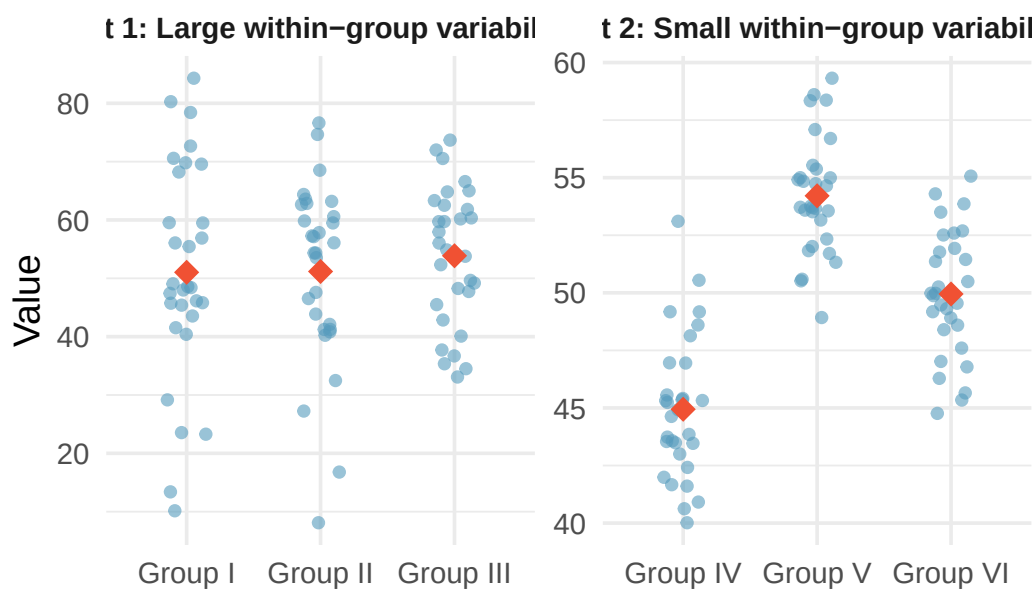


Figure 17.1: Side-by-side dot plots for two sets of three groups. In the first set (Groups I, II, III), the within-group variability is large relative to the between-group differences. In the second set (Groups IV, V, VI), the within-group variability is small and the between-group differences are clearly visible.

Compare Groups I, II, and III. Can you visually determine if the group differences are real? Now compare Groups IV, V, and VI.

Any real difference in Groups I, II, and III is difficult to discern because the data within each group are very volatile relative to differences in the group averages. On the other hand, Groups IV, V, and VI show noticeable differences because those differences are large *relative to the variability in individual observations within each group*.

17.1 Case study: MLB batting

We would like to determine whether there are real differences in batting performance among baseball players according to their position: outfielder (OF), infielder (IF), and catcher (C). We use data from 429 Major League Baseball players from the 2018 season who had at least 100 at-bats. The measure of batting performance is the on-base percentage (OBP), which roughly represents the fraction of the time a player successfully gets on base or hits a home run.

Summary statistics for each group:

Position	n	Mean	SD
OF	160	0.320	0.044
IF	205	0.318	0.037
C	64	0.302	0.038

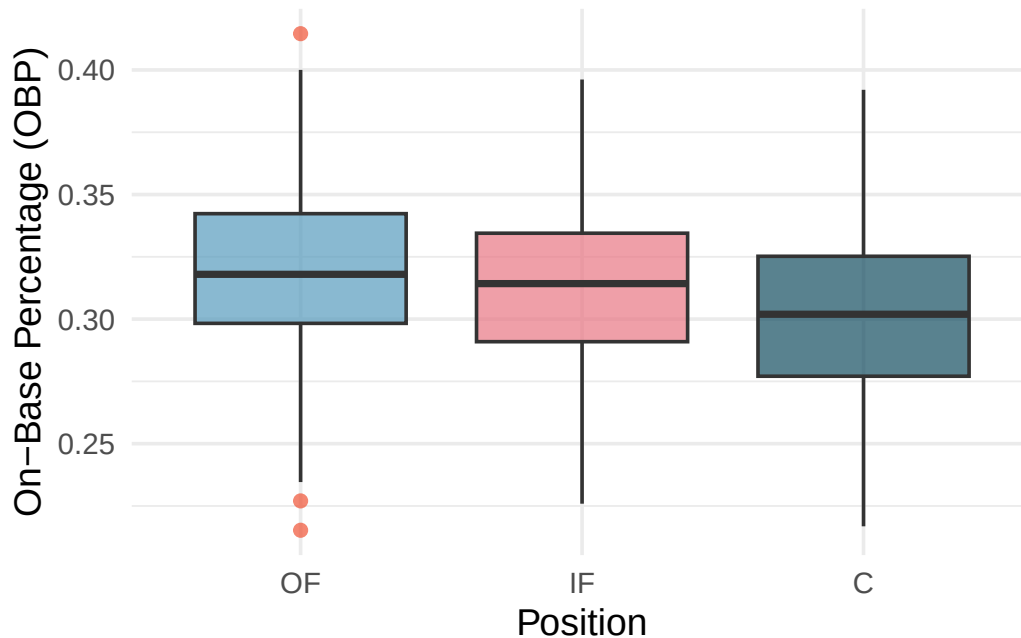


Figure 17.2: Side-by-side box plots of on-base percentage for 429 players across three positions. The variability appears approximately constant across groups. Catchers seem to have a slightly lower on-base percentage.

The null hypothesis under consideration is $\mu_{OF} = \mu_{IF} = \mu_C$. Write the null and alternative hypotheses in plain language.

Show answer

H_0 : The average on-base percentage is equal across the three positions. H_A : The average on-base percentage varies across some (or all) groups.

What would be an appropriate point estimate of the on-base percentage by outfielders, μ_{OF} ?

A good estimate of the on-base percentage by outfielders would be the sample average of OBP for just those players who are outfielders: $\bar{x}_{OF} = 0.320$.

17.1.1 Why not just compare the most different groups?

The largest difference between sample means is between catchers and outfielders. Why might it be inappropriate to simply test whether μ_C and μ_{OF} are different?

The primary issue is **data snooping** (also called **data fishing**). We are inspecting the data before picking the groups to compare. It is inappropriate to examine all data informally and only afterward decide which parts to formally test. Naturally, we would pick the groups with the largest differences, which inflates the Type I error rate.

For example, suppose we measure aptitude for students in 20 classrooms where all students are randomly assigned. With so many groups, we will probably observe a few that look different just by chance. If we select only those classes and perform a formal test, we will probably reach the wrong conclusion.

17.2 Randomization test for comparing many means

17.2.1 The F-statistic

The method of analysis of variance focuses on answering one question: is the variability in the sample means so large that it seems unlikely to be from chance alone? We quantify the between-group variability using the **mean square between groups (MSG)**, which has degrees of freedom $df_G = k - 1$ where k is the number of groups. The MSG can be thought of as a scaled variance formula for the group means.

We also need a benchmark for how much variability to expect if the null hypothesis is true. For this, we compute the **mean square error (MSE)**, which measures the variability *within* groups and has degrees of freedom $df_E = n - k$.

The **F-statistic** is the ratio of these two quantities:

$$F = \frac{MSG}{MSE}$$

The MSG represents between-group variability, and MSE measures within-group variability. When the null hypothesis is true, any differences among sample means are due to chance, and the MSG and MSE should be about equal, making F close to 1. Large values of F suggest that the between-group differences are larger than what we would expect from random variation.

The test statistic for three or more means is an F.

The F statistic is a ratio of how the groups differ (MSG) as compared to how the observations within a group vary (MSE).

$$F = \frac{MSG}{MSE}$$

When the null hypothesis is true and the conditions are met, F has an F-distribution with $df_1 = k - 1$ and $df_2 = n - k$.

Conditions:

- Independent observations, both within and across groups
- Large samples and no extreme outliers

17.2.2 Variability of the statistic

Suppose we have three exams (A, B, and C) and want to investigate whether they differ in difficulty:

- $H_0: \mu_A = \mu_B = \mu_C$. The exams are equally difficult.
- H_A : At least one exam is more (or less) difficult than the others.

Exam	n	Mean	SD	Min	Max
A	58	75.1	13.9	44	100
B	55	72.0	13.8	38	100
C	51	79.4	14.1	45	100

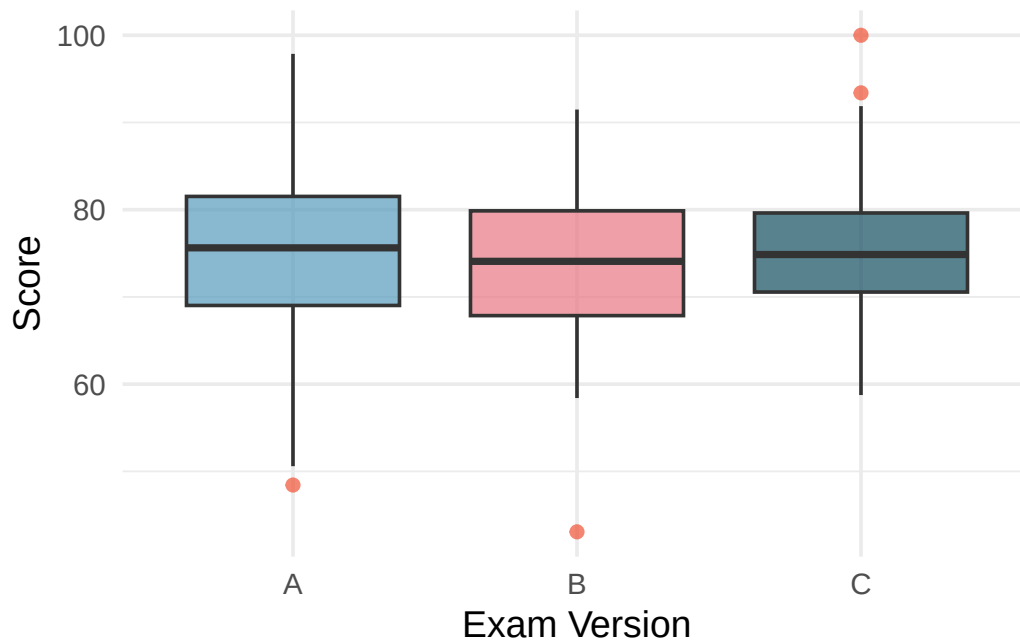


Figure 17.3: Box plots of exam scores for three exam versions. Exam C appears to have a higher median, but the distributions overlap.

To perform a randomization test, we randomly reassign the exam labels to the observed scores. If the null hypothesis is true, the exam version should not matter, so any score could belong to any exam version.

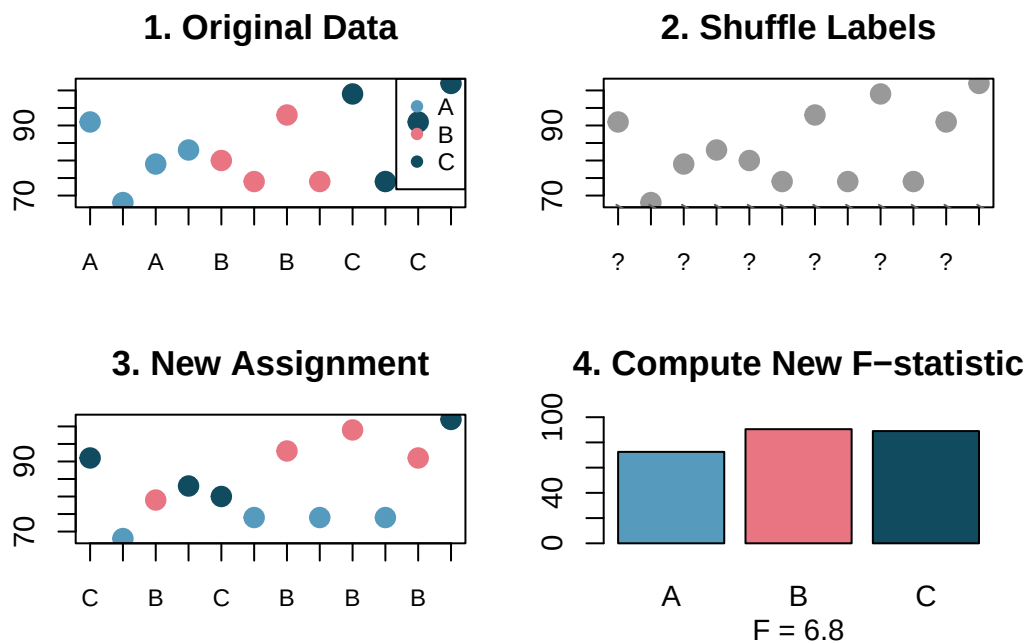


Figure 17.4: The randomization process: exam version labels are randomly shuffled across the observed scores to generate a new assignment, from which a new F-statistic is computed.

By repeating this process 1,000 times, we build a null distribution of F-statistics.

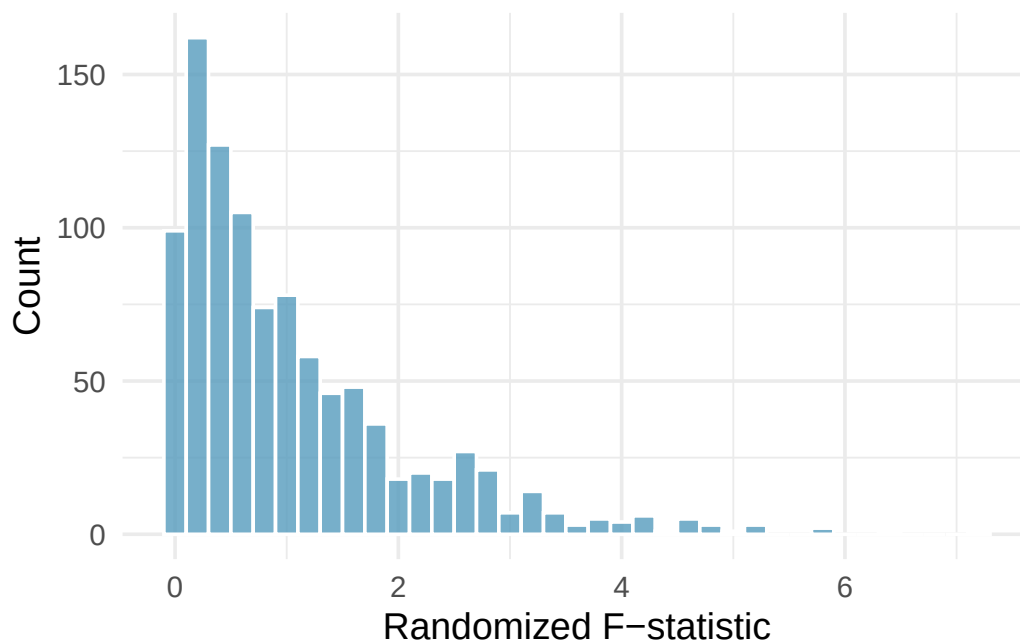


Figure 17.5: Histogram of F-statistics from 1,000 randomizations. The distribution is right-skewed with most values below 2. The tail extends to about 6.

17.2.3 Observed statistic vs. null statistics

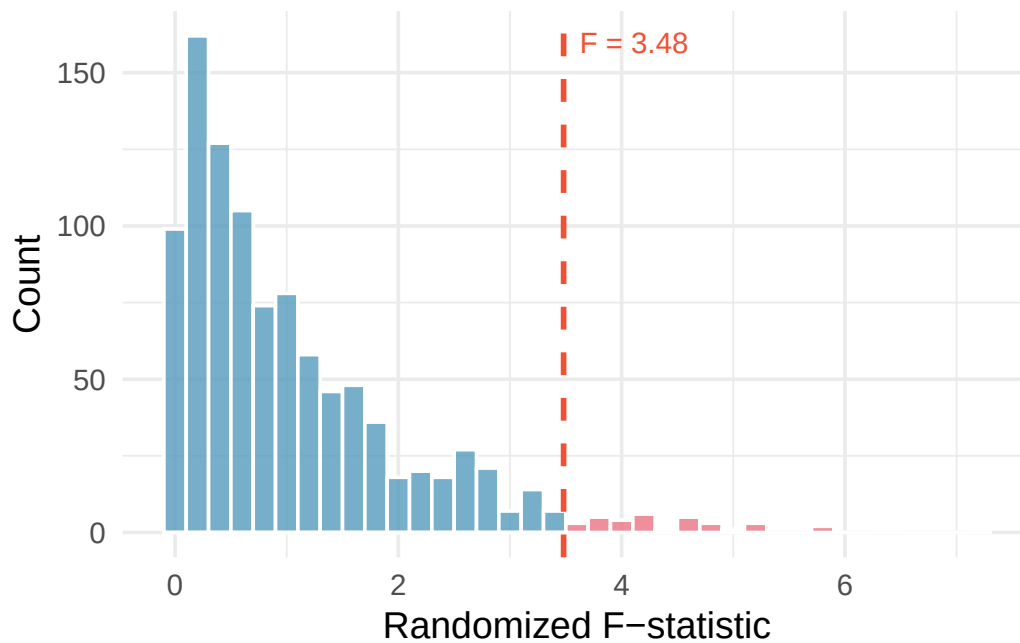


Figure 17.6: Histogram of F-statistics from 1,000 randomizations with the observed F-statistic of 3.48 marked by a red line. The area to the right represents the p-value.

Using statistical software, we find that 3.6% of the randomized F statistics were at or above the observed test statistic of $F = 3.48$. That is, the p-value is 0.036. Assuming a significance level of $\alpha = 0.05$, the p-value is smaller than the significance level, which leads us to reject the null hypothesis. We conclude that the difficulty level is different for at least one of the exams.

17.3 Mathematical model for ANOVA

The randomization test on the F-statistic has a corresponding mathematical theory: the **F-distribution**.

17.3.1 Variability of the statistic

The larger the observed variability in the sample means (MSG) relative to the within-group variability (MSE), the larger the F-statistic and the stronger the evidence against the null hypothesis. We use the upper tail of the F-distribution to compute the p-value.

The F-statistic and the F-test.

Analysis of variance (ANOVA) uses a test statistic, the F-statistic, which represents a standardized ratio of variability in the sample means relative to the variability within the groups. If H_0 is true and the model conditions are satisfied, the F-statistic follows an F-distribution with parameters $df_1 = k - 1$ and $df_2 = n - k$. The upper tail of the F-distribution is used to compute the p-value.

For the MLB data, $MSG = 0.00803$ and $MSE = 0.00158$. Identify the degrees of freedom associated with MSG and MSE and verify the F-statistic is approximately 5.077.

Show answer

There are $k = 3$ groups, so $df_G = k - 1 = 2$. There are $n = 429$ total observations, so $df_E = n - k = 426$. The F-statistic is $F = \frac{MSG}{MSE} = \frac{0.00803}{0.00158} = 5.08 \approx 5.077$.

17.3.2 The p-value from the F-distribution

A p-value can be computed from the F-statistic using an F-distribution with $df_1 = df_G$ and $df_2 = df_E$.

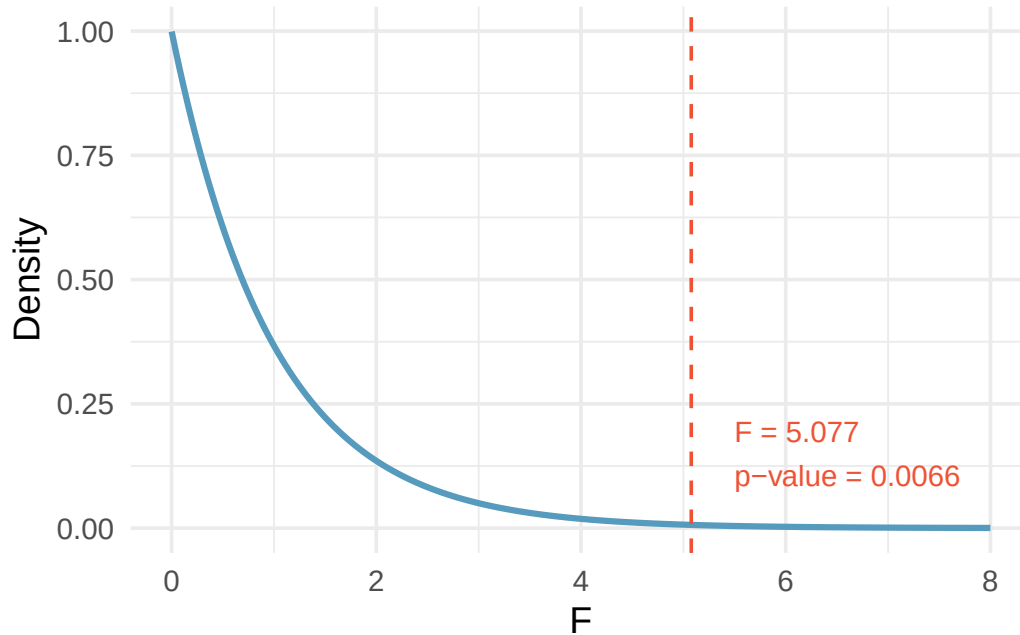


Figure 17.7: An F-distribution with $df_1 = 2$ and $df_2 = 426$. The area to the right of $F = 5.077$ is shaded, representing a p-value of approximately 0.0066.

The p-value corresponding to $F = 5.077$ with $df_1 = 2$ and $df_2 = 426$ is approximately 0.0066. Does this provide strong evidence against the null hypothesis?

The p-value is smaller than 0.05, indicating the evidence is strong enough to reject the null hypothesis at a significance level of 0.05. The data provide strong evidence that the average on-base percentage varies by player's primary field position.

Note that the small p-value tells us there is a notable difference somewhere, but ANOVA does **not** tell us *which* group is different. Follow-up analyses (discussed in the next chapter on multiple comparisons) are needed.

17.3.3 Reading an ANOVA table

ANOVA results are commonly presented in a table. Here is the ANOVA table for the MLB data:

Source	df	Sum Sq	Mean Sq	F value	p-value
position	2	0.0161	0.00803	5.077	0.0066
Residuals	426	0.6740	0.00158		

The key information is in the last two columns: the F-statistic (5.077) and the p-value (0.0066).

Components of an ANOVA table.

- **Sum of Squares Between Groups (SSG):** Measures the total variability of the group means around the overall mean. $SSG = \sum_{j=1}^k n_j(\bar{x}_j - \bar{x})^2$.
- **Sum of Squared Errors (SSE):** Measures the total variability of observations around their group means. $SSE = \sum_{j=1}^k (n_j - 1)s_j^2$.
- **Sum of Squares Total (SST):** The total variability in all observations. $SST = SSG + SSE$.
- **Mean Square Between Groups (MSG):** $MSG = SSG/(k - 1)$.
- **Mean Square Error (MSE):** $MSE = SSE/(n - k)$.
- **F-statistic:** $F = MSG/MSE$.

17.4 Conditions for ANOVA

There are three conditions to check before performing ANOVA:

- **Independence.** If the data are a simple random sample, this condition is generally satisfied. For experiments, carefully consider whether the data may be independent (e.g., no pairing).
- **Approximately normal.** As with one- and two-sample testing for means, the normality assumption is especially important when sample sizes are small. With large sample sizes within each group, we look mainly for extreme outliers.
- **Constant variance.** The variability in the groups should be about equal. This can be checked by examining side-by-side box plots or comparing the standard deviations across groups. A common rule of thumb is that the ratio of the largest to smallest group standard deviation should be less than 2.

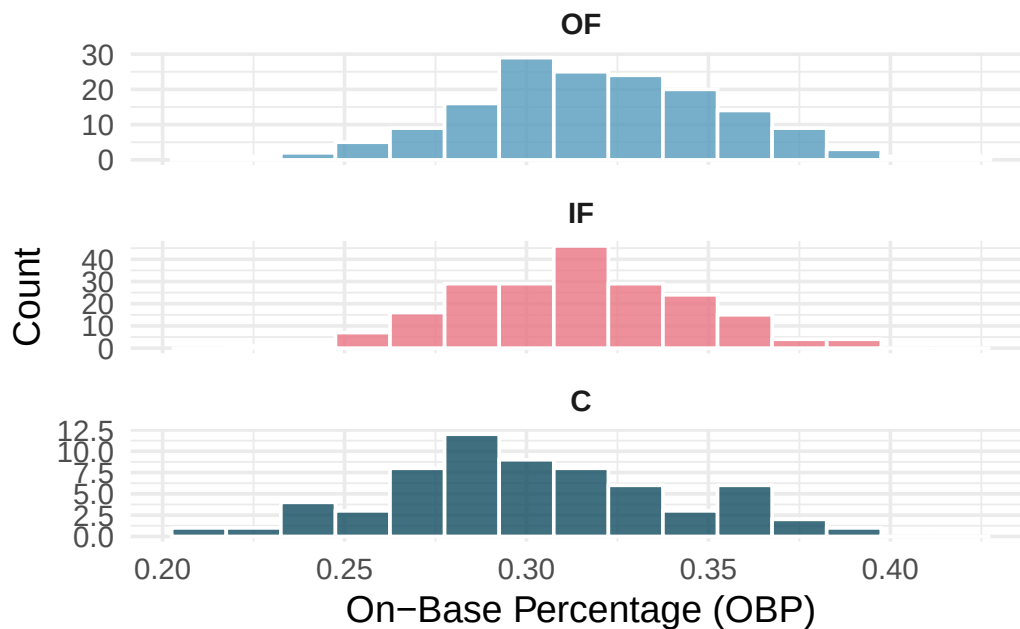


Figure 17.8: Histograms of OBP for each field position, showing that the distributions are reasonably bell-shaped and symmetric.

Diagnostics for an ANOVA analysis.

Independence is always important. The normality condition matters most when group sizes are small (with large groups, look only for extreme outliers). The constant variance condition is especially important when group sizes differ.

17.5 Chapter review

17.5.1 Summary

In this chapter, we introduced both the randomization test and the mathematical model for comparing means across two or more groups using ANOVA. The F-statistic measures the ratio of between-group variability (MSG) to within-group variability (MSE). When the null hypothesis is true and the conditions are satisfied, the F-statistic follows an F-distribution. A significant result tells us that at least one group mean differs from the others, but does not identify *which* groups differ. For that, we need multiple comparison procedures, which are covered in the next chapter.

17.5.2 Key terms

- ANOVA (analysis of variance)
- F-statistic
- F-distribution
- F-test
- Mean square between groups (MSG)
- Mean square error (MSE)
- Sum of squares between groups (SSG)
- Sum of squared errors (SSE)
- Sum of squares total (SST)
- Degrees of freedom (ANOVA)
- Data snooping / data fishing

Chapter 18

Multiple Comparisons

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

When an ANOVA test rejects the null hypothesis, we know that at least one group mean differs from the others – but we do not know *which* groups differ. This chapter addresses the natural follow-up question: which specific pairs of groups have different means? Attempting many pairwise comparisons inflates the overall Type I error rate, so we need specialized methods that control for the fact that we are conducting multiple tests simultaneously. We introduce the multiple testing problem, the Bonferroni correction, and Tukey's Honest Significant Difference (HSD) method.

18.1 The multiple testing problem

When we conduct a single hypothesis test at significance level $\alpha = 0.05$, there is a 5% chance of incorrectly rejecting a true null hypothesis (a Type I error). But what happens when we conduct many tests?

Suppose we compare $k = 4$ groups. How many pairwise comparisons are possible? If we conduct each test at $\alpha = 0.05$ and all null hypotheses are actually true, what is the probability that we make at least one Type I error?

With 4 groups, there are $\binom{4}{2} = 6$ pairwise comparisons. If we conduct each at $\alpha = 0.05$, the probability that any *single* test does not produce a Type I error is $1 - 0.05 = 0.95$. If the tests were independent, the probability that *none* of the 6 tests produces a Type I error would be:

$$P(\text{no errors}) = 0.95^6 = 0.735$$

So the probability of making **at least one** Type I error is:

$$P(\text{at least one error}) = 1 - 0.735 = 0.265$$

That is about a 27% chance of at least one false positive – much higher than the 5% we intended.

18.1.1 Family-wise error rate

The probability of making at least one Type I error across all tests in a family of comparisons is called the **family-wise error rate** (FWER). When conducting m independent tests, each at significance level α , the FWER is:

$$\text{FWER} = 1 - (1 - \alpha)^m$$

The table below shows how the FWER grows with the number of comparisons when $\alpha = 0.05$:

Number of groups (k)	Number of comparisons (m)	FWER
3	3	0.143
4	6	0.265
5	10	0.401
6	15	0.537
10	45	0.901

With 10 groups, there is a 90% chance of making at least one Type I error if we use unadjusted pairwise tests. This is the core of the multiple comparisons problem: we need procedures that control the FWER at an acceptable level (typically 0.05).

Family-wise error rate (FWER).

The **family-wise error rate** is the probability of making one or more Type I errors among all the hypothesis tests in a family of comparisons. Methods like the Bonferroni correction and Tukey's HSD are designed to keep the FWER at or below a specified level α .

18.2 The Bonferroni correction

The simplest method for controlling the family-wise error rate is the **Bonferroni correction**. The idea is straightforward: if we are conducting m tests and want the overall FWER to be at most α , we use a stricter significance level for each individual test.

Bonferroni correction.

When conducting m hypothesis tests and we want the family-wise error rate to be at most α , the Bonferroni correction uses an adjusted significance level for each individual test:

$$\alpha^* = \frac{\alpha}{m}$$

Equivalently, we can multiply each p-value by m (the adjusted p-value) and compare the result to the original α .

After an ANOVA with $k = 4$ groups rejects H_0 , we want to conduct all pairwise comparisons at an overall $\alpha = 0.05$. How many comparisons are there, and what adjusted significance level should we use for each?

There are $\binom{4}{2} = 6$ pairwise comparisons. The Bonferroni-adjusted significance level is:

$$\alpha^* = \frac{0.05}{6} = 0.0083$$

So each individual two-sample t -test must have a p -value below 0.0083 to be considered significant.

The Bonferroni correction is simple and broadly applicable – it can be used with any type of test, not just pairwise comparisons of means. However, it is **conservative**, meaning it may fail to detect real differences when they exist (higher Type II error). When the number of comparisons is very large, the Bonferroni correction can be overly strict.

18.2.1 Example: Exam scores

Suppose an ANOVA comparing three exam versions (A, B, C) gave $F = 3.48$ with $p = 0.036$. We rejected H_0 and now want to know which exams differ.

With $k = 3$ groups, there are $\binom{3}{2} = 3$ pairwise comparisons. The Bonferroni-adjusted significance level is $\alpha^* = 0.05/3 = 0.0167$.

Suppose the three pairwise t -test p -values are:

Comparison	p-value	Significant at $\alpha^* = 0.0167$?
A vs. B	0.245	No
A vs. C	0.068	No
B vs. C	0.011	Yes

Only the comparison between exams B and C is significant after the Bonferroni correction. We conclude that there is evidence of a difference in difficulty between exams B and C, but we cannot say the other pairs differ.

18.3 Tukey's Honest Significant Difference

A more powerful alternative to the Bonferroni correction for pairwise comparisons of means is **Tukey's Honest Significant Difference (HSD)** method. While the Bonferroni correction is general-purpose, Tukey's method is specifically designed for comparing all pairs of group means after ANOVA, and it is less conservative.

18.3.1 How Tukey's HSD works

Tukey's HSD computes a single critical difference – the minimum difference between two group means that would be considered statistically significant. The procedure uses the **Studentized range distribution** to determine this critical value.

For each pair of groups j and j' , the test statistic is:

$$q = \frac{\bar{x}_j - \bar{x}_{j'}}{\sqrt{MSE/n^*}}$$

where MSE is the mean square error from the ANOVA, and n^* is a measure of the group sample size (when group sizes are equal, $n^* = n_j$; when unequal, a harmonic mean is often used). The statistic q is compared to critical values from the Studentized range distribution.

Tukey's HSD.

Tukey's Honest Significant Difference method controls the family-wise error rate at α when making all pairwise comparisons of group means. It computes a confidence interval for each pairwise difference:

$$(\bar{x}_j - \bar{x}_{j'}) \pm q^* \cdot \sqrt{\frac{MSE}{n^*}}$$

where q^* is the critical value from the Studentized range distribution with k groups and $n - k$ degrees of freedom. If the confidence interval does not contain zero, the difference is significant.

18.3.2 Example: MLB batting positions

After the ANOVA for MLB batting data ($F = 5.077, p = 0.0066$), we rejected H_0 and want to identify which position pairs differ in on-base percentage. Tukey's HSD produces the following pairwise comparisons:

Comparison	Difference	95% CI	Significant?
IF - OF	-0.002	(-0.016, 0.012)	No
C - OF	-0.018	(-0.038, 0.002)	No
C - IF	-0.016	(-0.035, 0.003)	No

Interestingly, although the overall ANOVA was significant, none of the individual Tukey pairwise comparisons reaches significance at the 0.05 level. This can happen because the ANOVA considers all groups simultaneously, while the pairwise comparisons are more conservative. In this case, the overall pattern of catchers having lower OBP than the other two positions drives the ANOVA result, but no single pairwise difference is large enough to be declared significant on its own.

18.4 Interpreting post-hoc results

Post-hoc tests (like Bonferroni or Tukey's HSD) are conducted *after* a significant ANOVA result. Several important principles guide their interpretation:

18.4.1 When to use post-hoc tests

Post-hoc pairwise comparisons should only be conducted after ANOVA rejects the null hypothesis. If the ANOVA p-value is not significant, there is no justification for exploring pairwise differences – doing so would be data snooping.

18.4.2 Reporting results

When reporting results of multiple comparisons, it is common to use **compact letter displays**. Groups that share a letter are not significantly different from each other:

Group	Mean	Grouping
A	75.1	ab
B	72.0	a
C	79.4	b

In this notation, Groups A and B share the letter “a” (not significantly different), and Groups A and C share the letter “b” is not shared by B and C (they are significantly different). Group A, sharing a letter with both B and C, is not significantly different from either.

18.4.3 Confidence intervals for differences

An especially useful way to present multiple comparison results is through confidence intervals for pairwise differences. If a confidence interval contains zero, the two groups are not significantly different. If the interval is entirely positive or entirely negative, the groups differ.

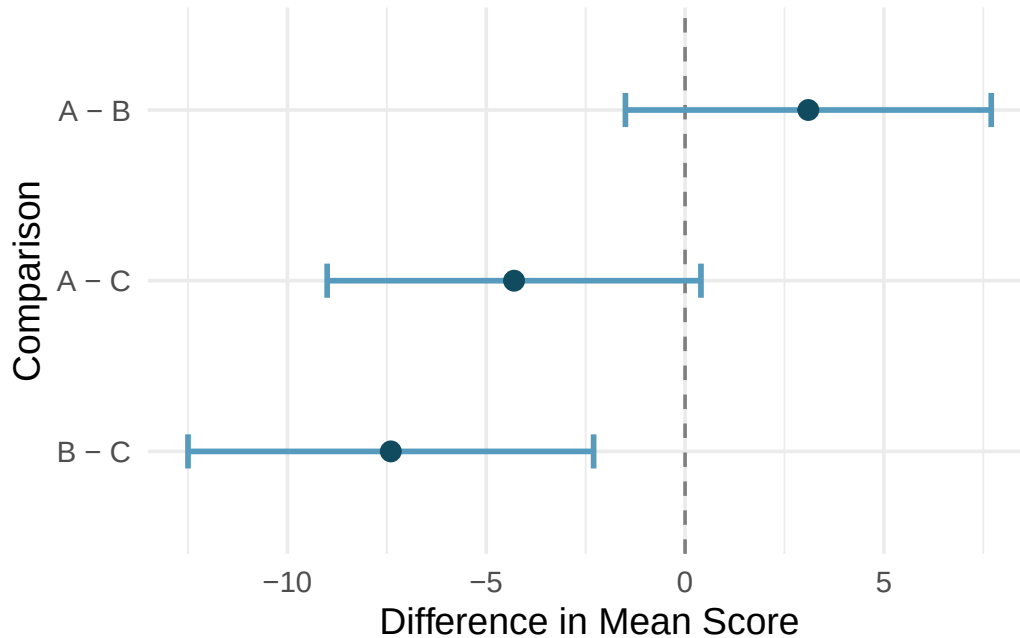


Figure 18.1: Confidence intervals for pairwise differences in mean exam scores. The interval for B vs. C does not contain zero, indicating a significant difference.

18.4.4 Common mistakes

1. **Running multiple pairwise tests without correction.** This inflates the Type I error rate. Always use a correction method when conducting multiple comparisons.
2. **Conducting post-hoc tests after a non-significant ANOVA.** If the overall F -test does not reject H_0 , there is no justification for exploring pairwise differences.
3. **Ignoring practical significance.** A statistically significant difference may be too small to matter in practice. Always consider the magnitude of the differences, not just the p-values.
4. **Over-interpreting non-significant pairwise comparisons.** A non-significant pairwise comparison after a significant ANOVA does not mean the groups are equal – it means we do not have enough evidence to declare them different individually.

18.5 Bonferroni vs. Tukey: When to use which

Both methods control the family-wise error rate, but they have different strengths:

Feature	Bonferroni	Tukey's HSD
Generality	Any set of tests	Pairwise comparisons of means only

Feature	Bonferroni	Tukey's HSD
Power	More conservative (less powerful)	Less conservative (more powerful)
Assumptions	None beyond the individual tests	Equal group variances (same as ANOVA)
Best for	Small number of planned comparisons	All pairwise comparisons after ANOVA

Rule of thumb: If you are comparing all pairs of group means after a significant ANOVA, use Tukey's HSD. If you have a small number of pre-planned comparisons (not necessarily all pairs), the Bonferroni correction is appropriate.

18.6 Chapter review

18.6.1 Summary

When ANOVA rejects the null hypothesis, the natural follow-up question is: which groups differ? Performing many pairwise tests without correction inflates the family-wise error rate, making false positives likely. The Bonferroni correction divides the significance level by the number of comparisons, which is simple but conservative. Tukey's HSD method is specifically designed for all pairwise comparisons of means after ANOVA and is typically more powerful. Both methods control the family-wise error rate at the desired level.

18.6.2 Key terms

- Multiple comparisons problem
- Family-wise error rate (FWER)
- Bonferroni correction
- Tukey's Honest Significant Difference (HSD)
- Post-hoc test
- Compact letter display

Chapter 19

Correlation

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

When we have two numerical variables, we often want to understand the relationship between them. Does one variable increase as the other increases? Does it decrease? Or is there no apparent pattern? In this chapter, we introduce tools for visualizing and quantifying the strength and direction of the linear relationship between two numerical variables. We begin with scatterplots, then introduce the correlation coefficient r , discuss its properties and limitations, and address the ever-important distinction between correlation and causation.

19.1 Scatterplots

A **scatterplot** displays the relationship between two numerical variables. Each point represents a single observation, with one variable on the horizontal axis and the other on the vertical axis. Scatterplots are the most useful tool for visualizing the relationship between two quantitative variables.

When examining a scatterplot, we look for four key features:

1. **Direction:** Is the relationship positive (upward trend), negative (downward trend), or neither?
2. **Form:** Is the relationship linear (points cluster around a line) or nonlinear (points follow a curve)?
3. **Strength:** How tightly do the points cluster around the trend? A strong relationship has little scatter; a weak relationship has lots of scatter.
4. **Unusual observations:** Are there any outliers that deviate from the overall pattern?

Researchers captured 104 brushtail possums in Australia and took body measurements before releasing them. Consider the relationship between total body length (cm) and head length (mm).

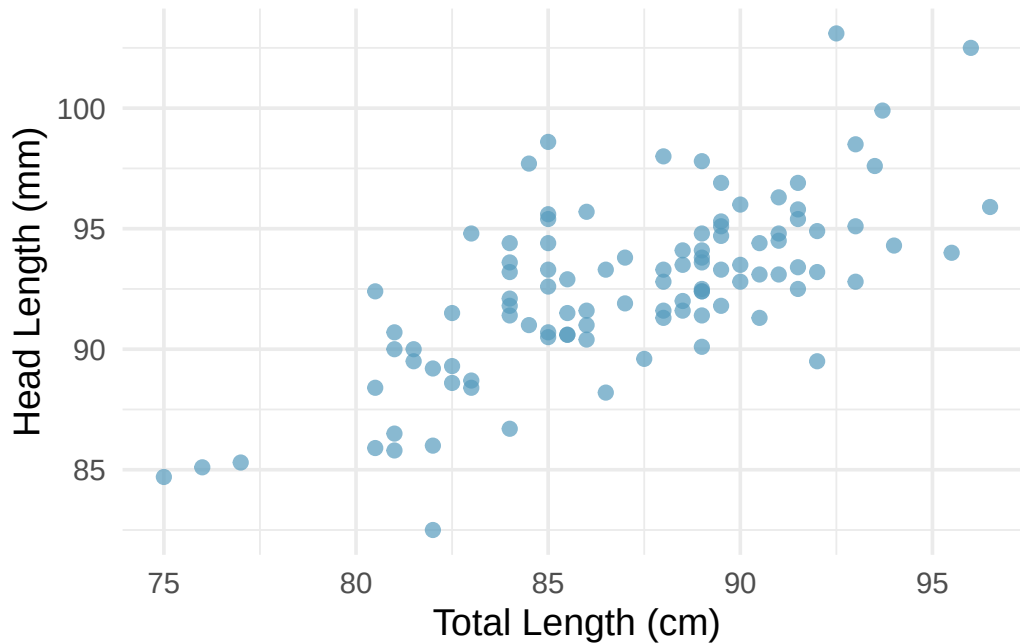


Figure 19.1: A scatterplot showing head length against total length for 104 brushtail possums. The relationship is moderately strong, positive, and linear.

Describe the relationship between total length and head length.

The relationship is **positive** (possums with longer total lengths tend to have longer heads), **linear** (the points cluster around a straight line, not a curve), and **moderate in strength** (the points are somewhat scattered around the trend, but the pattern is clear). There do not appear to be any extreme outliers.

19.1.1 When linear models are not appropriate

There are cases where fitting a straight line to the data is not helpful, even if there is a clear relationship between the variables.

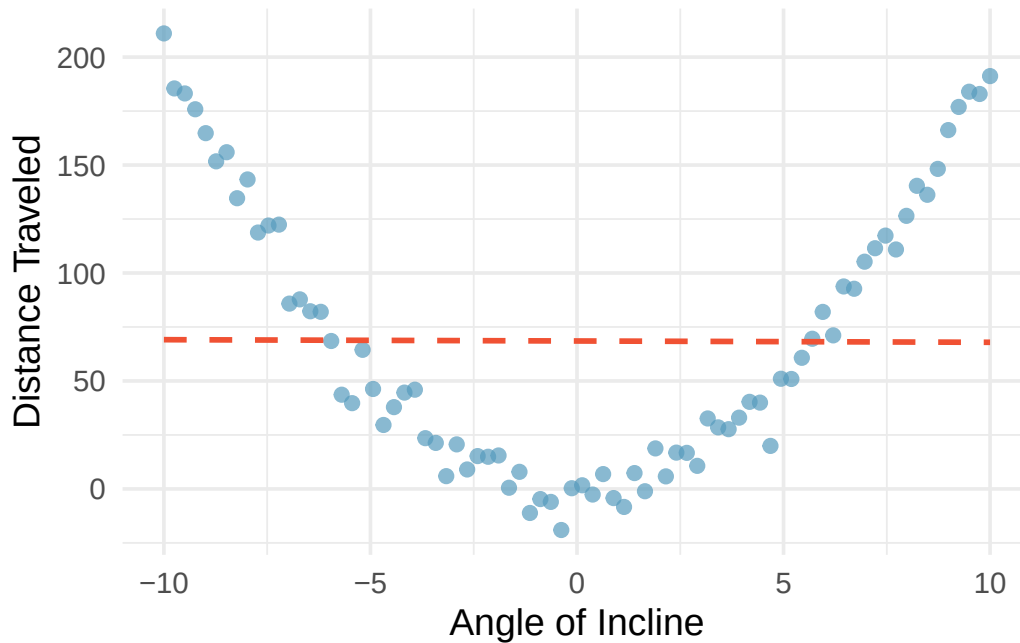


Figure 19.2: A scatterplot showing a clear quadratic (U-shaped) relationship between angle of incline and distance traveled. The best-fitting straight line is flat and does not capture the pattern at all.

When the relationship is nonlinear, fitting a straight line can be misleading. We must always examine scatterplots before computing correlations or fitting linear models.

19.2 The correlation coefficient

We need a way to quantify the strength and direction of a linear relationship. The **correlation coefficient** r does exactly this.

Correlation: strength of a linear relationship.

Correlation, denoted r , always takes values between -1 and 1 and describes the strength and direction of the linear relationship between two variables.

The correlation is unitless and is not affected by linear changes in the units of measurement (e.g., converting from inches to centimeters).

The formula for the correlation coefficient is:

$$r = \frac{1}{n-1} \sum_{i=1}^n \frac{x_i - \bar{x}}{s_x} \cdot \frac{y_i - \bar{y}}{s_y}$$

where $\bar{x}$ and $\bar{y}$ are the sample means, and s_x and s_y are the sample standard deviations for each variable. In practice, we always use a computer or calculator to compute r .

19.2.1 Interpreting correlation values

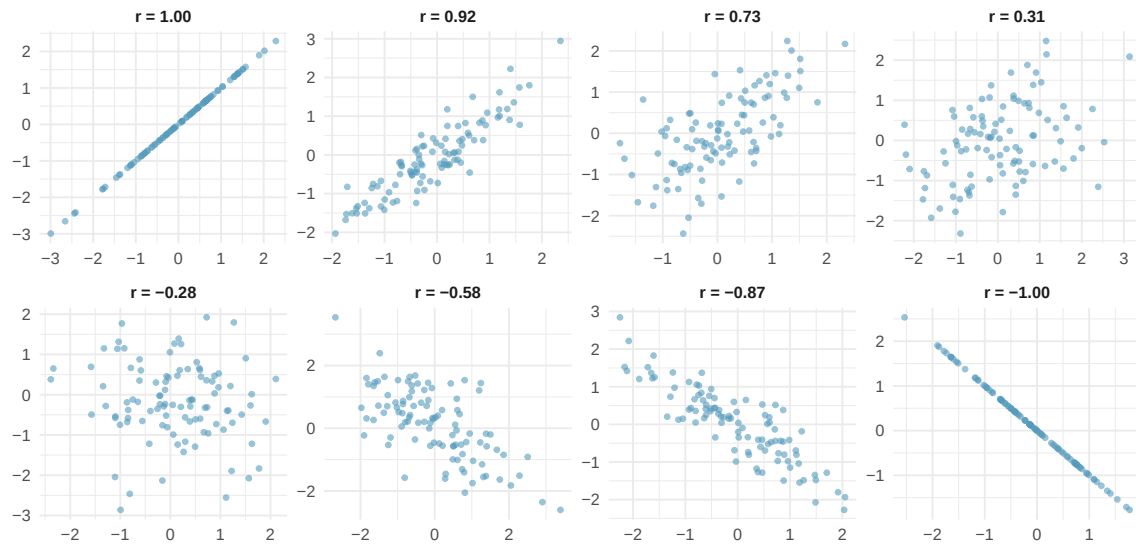


Figure 19.3: Eight scatterplots showing different correlation values. The first row shows positive relationships with r values of 1.00, 0.92, 0.73, and 0.31. The second row shows negative relationships with r values of -0.28, -0.58, -0.87, and -1.00.

Key patterns to notice:

- When $r = 1$ or $r = -1$, all points fall exactly on a straight line (a perfect linear relationship).
- When r is close to $+1$ or -1 , the relationship is strong and linear.
- When r is close to 0, there is no linear relationship (but there could still be a nonlinear relationship!).
- Positive r means both variables tend to increase together; negative r means one tends to decrease as the other increases.

19.2.2 Properties of correlation

The correlation coefficient has several important properties:

1. **Range:** $-1 \leq r \leq 1$.
2. **Unitless:** Correlation has no units. Changing the units of x or y (e.g., from cm to inches) does not change r .
3. **Symmetric:** The correlation of x with y is the same as the correlation of y with x .
4. **Not affected by linear transformations:** Adding a constant to or multiplying a variable by a positive constant does not change r .
5. **Only measures linear relationships:** Correlation can be misleading for nonlinear relationships.

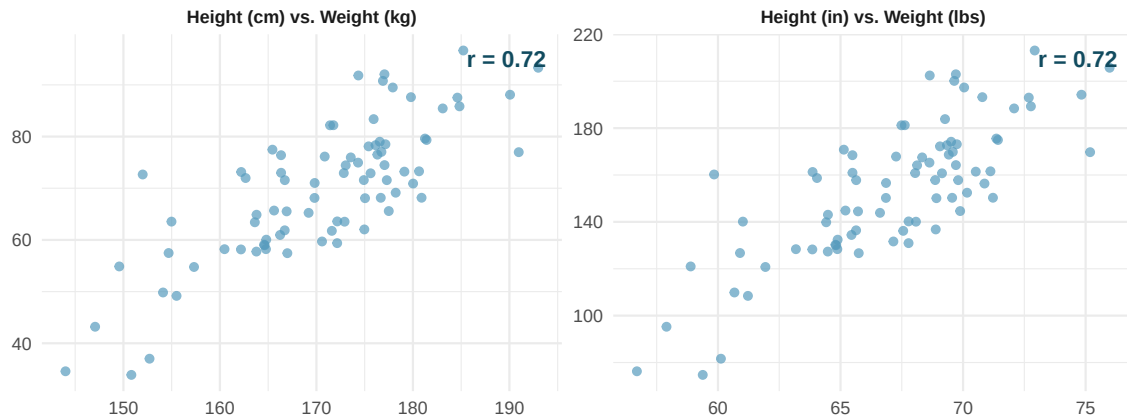


Figure 19.4: Two scatterplots showing the same relationship between weight and height, measured in different units (kg/cm vs. lbs/in). The correlation $r = 0.72$ is the same in both plots.

19.2.3 Correlation does not capture nonlinear relationships

The correlation coefficient is designed to measure the strength of *linear* trends. Nonlinear trends, even when strong, can produce misleading correlations.

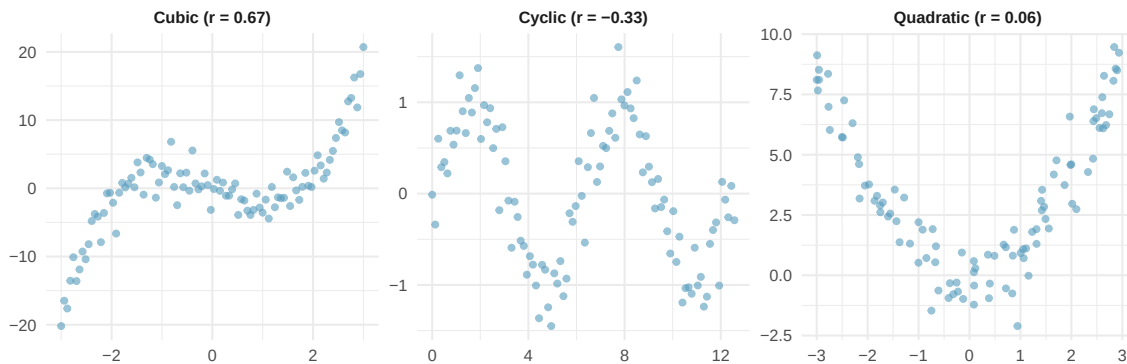


Figure 19.5: Three scatterplots showing strong nonlinear relationships (quadratic, cyclic, and a complex curve) that have weak correlations.

No straight line is a good fit for any of the datasets in the figure above. If you drew nonlinear curves on each plot, what general features would be important in your fit?

Show answer

The curves should be close to most points and reflect the overall nonlinear trend in the data.

Correlation only measures linear relationships. Always examine a scatterplot before computing or interpreting a correlation. A correlation near zero does not necessarily mean there is no relationship – it means there is no *linear* relationship. There may still be a strong nonlinear association.

Examine the scatterplots below showing the relationship between various crop yields across countries. Order the plots from strongest negative to strongest positive linear relationship.

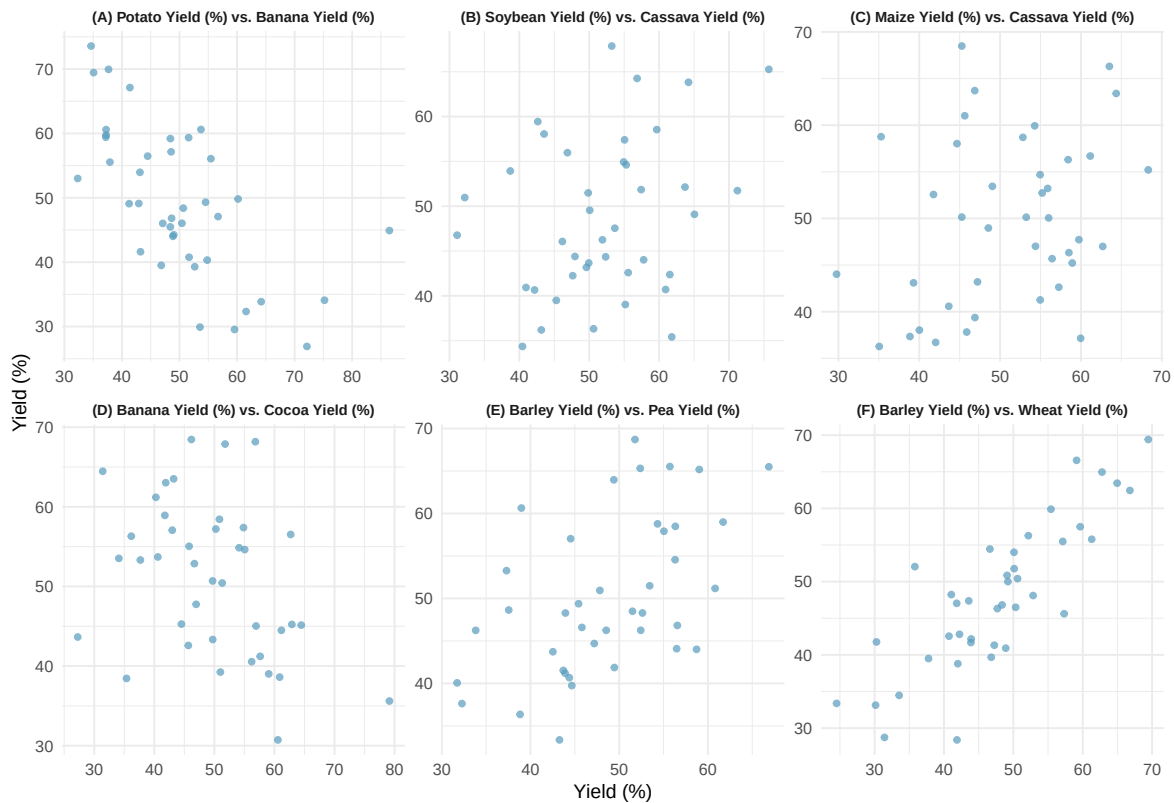


Figure 19.6: Six scatterplots of crop yield percentages across countries. Each shows a different pair of crop types with varying degrees of correlation.

Ordering from most negative to most positive correlation: the plots showing (A) bananas vs. potatoes, (D) cocoa vs. bananas, (B) cassava vs. soybeans, (C) cassava vs. maize, (E) peas vs. barley, and (F) wheat vs. barley.

19.3 Correlation and causation

Correlation does not imply causation.

Just because two variables are correlated does not mean that one causes the other. There may be a **confounding variable** (also called a lurking variable) that is associated with both x and y , creating a correlation between them even though there is no direct causal link.

Studies have found a positive correlation between the number of firefighters sent to a fire and the amount of damage the fire causes. Does this mean sending more firefighters causes more damage?

No. The confounding variable is the **severity of the fire**. Larger fires cause more damage *and* require more firefighters. The number of firefighters does not cause the damage; both are driven by the severity of the fire.

To establish causation, we need one of the following:

1. A **randomized experiment** where the explanatory variable is randomly assigned.

2. Strong theoretical justification combined with careful control of confounding variables.

Observational studies can reveal associations (correlations), but they cannot establish causation on their own.

19.4 Sampling variability of correlation

Like any sample statistic, the correlation coefficient r computed from a sample is just an estimate of the true population correlation ρ (the Greek letter “rho”). Different samples from the same population will produce different values of r .

When the sample size is small, r can vary substantially from sample to sample, even when the true population correlation is zero. This means we should be cautious about interpreting correlations from small samples.

Suppose the true population correlation between two variables is $\rho = 0$ (no linear relationship). If we draw random samples of size $n = 10$, the sample correlation r could reasonably fall anywhere between about -0.55 and $+0.55$ just by chance.

With larger samples ($n = 100$), the sampling variability shrinks, and r would typically fall between about -0.20 and $+0.20$.

This illustrates why statistical tests of significance are important: we need to determine whether an observed correlation is large enough to be unlikely under the null hypothesis that $\rho = 0$.

We will return to inference for correlation when we discuss inference for regression in a later chapter. In particular, testing whether the slope of a regression line is zero is equivalent to testing whether the population correlation is zero.

StatBench: Correlation Simulator. Use the Correlation Simulator to explore how the correlation coefficient changes as you add or move points in a scatterplot. Experiment with different sample sizes to see how sampling variability affects the observed correlation.

19.5 Chapter review

19.5.1 Summary

In this chapter, we introduced scatterplots as the primary tool for visualizing the relationship between two numerical variables. We then defined the correlation coefficient r as a measure of the strength and direction of the linear relationship. Key properties of r include: it ranges from -1 to $+1$, it is unitless, it is symmetric, and it measures only linear relationships. We emphasized that correlation does not imply causation – confounding variables can create spurious associations. Finally, we noted that the sample correlation is subject to sampling variability, especially in small samples.

19.5.2 Key terms

- Scatterplot
- Correlation (r)
- Positive association / negative association
- Nonlinear relationship
- Confounding variable
- Correlation does not imply causation
- Population correlation (ρ)

Chapter 20

Linear Regression

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Linear regression is a powerful statistical technique for describing the relationship between two numerical variables. Many people have some familiarity with regression just from reading the news, where straight lines are overlaid on scatterplots. In this chapter, we build on the ideas of correlation from the previous chapter to develop the least squares regression line. We learn how to interpret the slope and intercept, use residuals to assess model fit, quantify the strength of the model with R^2 , and perform inference on the slope to test whether a linear relationship exists in the population.

20.1 Fitting a line to data

Linear regression is the statistical method for fitting a line to data where the relationship between two variables, x and y , can be modeled by a straight line with some error:

$$y = b_0 + b_1x + e$$

The values b_0 and b_1 represent the model's **intercept** and **slope**, respectively, and the error is represented by e . These values are calculated from the data – they are sample statistics. If the observed data are a random sample from a target population, these values are point estimates for the population parameters β_0 and β_1 .

When we use x to predict y , we call x the **predictor** variable (or explanatory variable) and y the **outcome** variable (or response variable). We often write the model without the error term when focusing on the predicted average outcome:

$$\hat{y} = b_0 + b_1x$$

The “hat” on y signifies that this is a predicted (estimated) value.

It is rare for all data to fall perfectly on a straight line. Instead, data more commonly appear as a cloud of points, where the trend may be strong or weak.

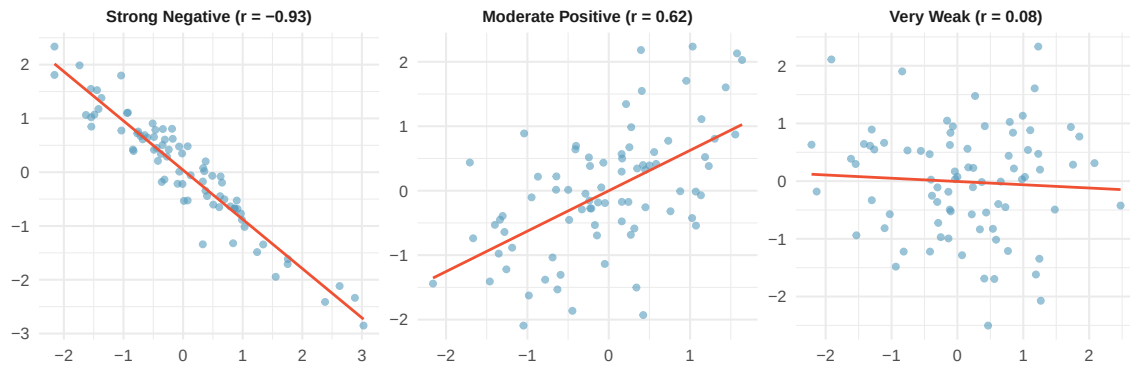


Figure 20.1: Three scatterplots with fabricated data showing a strong negative linear trend, a moderate positive linear trend, and a very weak trend.

There are also cases where fitting a straight line is inappropriate, even if there is a clear relationship between the variables. When the relationship is nonlinear, a linear model can be misleading.

20.1.1 Using linear regression to predict possum head lengths

Researchers captured 104 brushtail possums in Australia and took body measurements. We consider total body length (cm) as the predictor to predict head length (mm).

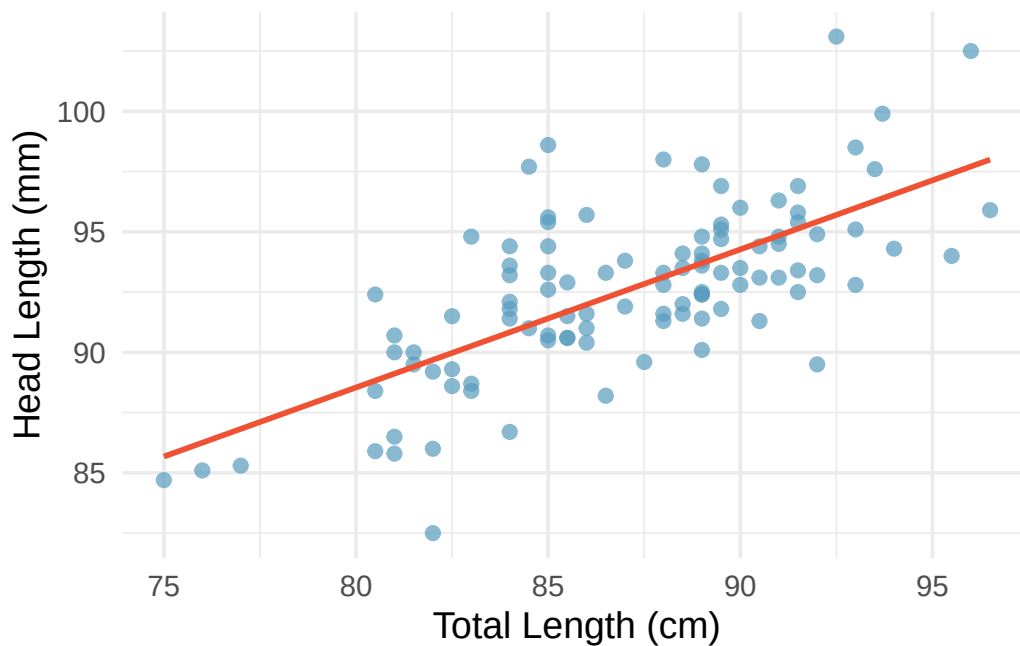


Figure 20.2: A scatterplot showing head length against total length for 104 brushtail possums, with a linear model fit to the data.

The equation for this line is:

$$\hat{y} = 41 + 0.59x$$

We can use this line to make predictions. For instance, the equation predicts a possum with a total length of 80 cm will have a head length of:

$$\hat{y} = 41 + 0.59 \times 80 = 88.2 \text{ mm}$$

This estimate may be viewed as an average: the equation predicts that possums with a total length of 80 cm will have an average head length of 88.2 mm.

20.2 Residuals

Residuals are the leftover variation in the data after accounting for the model fit:

$$\text{Data} = \text{Fit} + \text{Residual}$$

Each observation has a residual. If an observation is above the regression line, its residual is positive. If it is below the line, the residual is negative.

Residual: Difference between observed and expected.

The residual of the i^{th} observation (x_i, y_i) is the difference between the observed outcome and the predicted outcome:

$$e_i = y_i - \hat{y}_i$$

We identify $\hat{y}_i$ by plugging x_i into the model.

The linear fit is $\hat{y} = 41 + 0.59x$. Compute the residual for the observation (76.0, 85.1).

First compute the predicted value: $\hat{y} = 41 + 0.59 \times 76.0 = 85.84$. The residual is $e = 85.1 - 85.84 = -0.74$ mm. The negative residual indicates that the model overpredicted head length for this possum.

If a model underestimates an observation, will the residual be positive or negative? What about if it overestimates?

Show answer

If a model underestimates an observation, the residual is positive (the actual value exceeds the prediction). If it overestimates, the residual is negative.

20.2.1 Residual plots

Residuals are helpful in evaluating how well a linear model fits a dataset. We display them in a **residual plot**, where the horizontal axis shows the predicted values and the vertical axis shows the residuals.

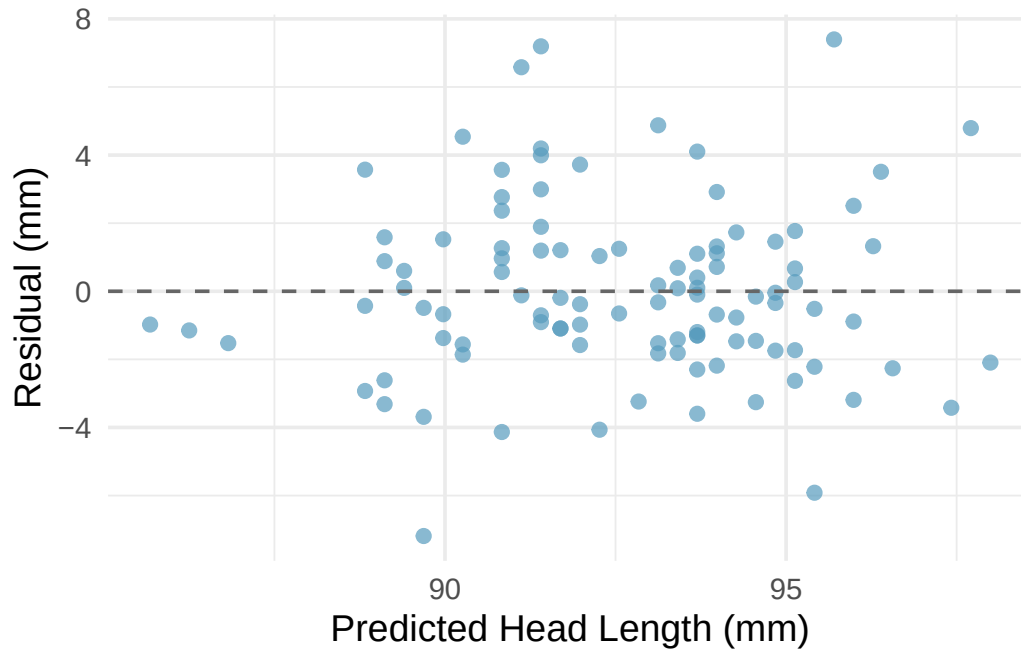


Figure 20.3: A residual plot for the possum model. The residuals appear randomly scattered around zero with no obvious pattern.

What patterns should we look for in a residual plot?

-
- **No pattern (good):** Residuals scattered randomly around zero suggest the linear model is appropriate.
 - **Curved pattern (bad):** A U-shape or other curve in the residuals indicates a nonlinear relationship that a straight line fails to capture.
 - **Fan shape (bad):** Increasing or decreasing spread in the residuals indicates non-constant variability.

20.3 Least squares regression

Fitting linear models by eye is subjective. **Least squares regression** provides an objective, rigorous approach.

20.3.1 The objective: minimize squared residuals

We want a line with small residuals. The **least squares line** minimizes the sum of the squared residuals:

$$e_1^2 + e_2^2 + \cdots + e_n^2$$

Why square the residuals rather than just sum their absolute values?

1. It is the most commonly used method.
2. It is widely supported in statistical software.
3. A residual twice as large is usually more than twice as bad. Squaring accounts for this.

4. The inference theory connecting the model to population parameters is most straightforward with least squares.

20.3.2 Interpreting the slope and intercept

For the Elmhurst College data, the least squares line predicting gift aid (in \$1,000s) from family income (in \$1,000s) is:

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times \text{family_income}$$

Term	Estimate	Std. Error	T statistic	p-value
(Intercept)	24.32	1.29	18.83	<0.0001
family_income	-0.0431	0.0108	-3.98	0.0002

What do the intercept and slope mean?

Slope ($b_1 = -0.0431$): For each additional \$1,000 of family income, we would expect a student to receive \$43.10 less in gift aid, on average. The negative coefficient means higher income is associated with less aid. We must be cautious: this is an observational study, so we cannot interpret a causal connection.

Intercept ($b_0 = 24.32$): The model predicts that a student whose family has no income would receive \$24,320 in gift aid, on average. This is meaningful because the family income for some students is near zero. In other applications, the intercept may have no practical meaning if $x = 0$ is outside the range of the data.

Interpreting parameters estimated by least squares.

The **slope** describes the estimated difference in the predicted average outcome of y if the predictor variable x were one unit larger.

The **intercept** describes the average outcome of y if $x = 0$, provided the linear model is valid at $x = 0$ (which may not be the case if $x = 0$ is far outside the observed data range).

20.3.3 Computing the slope and intercept from summary statistics

The slope can be computed using the correlation and the standard deviations of the two variables:

$$b_1 = \frac{s_y}{s_x} \cdot r$$

The least squares line always passes through the point $(\bar{x}, \bar{y})$. Using the point-slope form:

$$b_0 = \bar{y} - b_1 \cdot \bar{x}$$

Identifying the least squares line from summary statistics.

- Estimate the slope: $b_1 = (s_y/s_x) \cdot r$.
- Find the intercept using the fact that $(\bar{x}, \bar{y})$ is on the line: $b_0 = \bar{y} - b_1 \bar{x}$.

20.4 R-squared

We evaluated the strength of a linear relationship using r . More commonly, we describe the strength of a linear fit using R^2 , called **R-squared** or the **coefficient of determination**.

Coefficient of determination: proportion of variability explained by the model.

R^2 measures the proportion of variation in the outcome variable y that is explained by the linear model with predictor x . Since r is always between -1 and 1 , R^2 is always between 0 and 1 .

$$R^2 = 1 - \frac{SSE}{SST}$$

where:

- $SST = \sum (y_i - \bar{y})^2$ is the **total sum of squares**, measuring total variability in y .
- $SSE = \sum (y_i - \hat{y}_i)^2 = \sum e_i^2$ is the **sum of squared errors**, measuring leftover variability after using the model.

For simple linear regression, $R^2 = r^2$.

For the Elmhurst data, the correlation is $r = -0.499$. What proportion of the variability in gift aid is explained by family income?

$R^2 = (-0.499)^2 = 0.249$, or about 25%. Family income explains about 25% of the variability in gift aid among these students.

If a linear model has a very strong negative relationship with $r = -0.97$, how much of the variation in the outcome is explained by the predictor?

Show answer

$R^2 = (-0.97)^2 = 0.94$, or about 94% of the variation is explained.

20.5 Inference for the slope

We have been computing sample statistics b_0 and b_1 , but these are estimates of population parameters β_0 and β_1 . We now turn to formal inference: testing whether a linear relationship exists in the population.

20.5.1 Hypotheses

- $H_0: \beta_1 = 0$. There is no linear relationship between x and y .
- $H_A: \beta_1 \neq 0$. There is a linear relationship between x and y .

20.5.2 Randomization test for the slope

If the null hypothesis is true ($\beta_1 = 0$), then x and y are not linearly related. We can simulate this by **permuting** the response variable: randomly shuffling the y values while keeping the x values fixed. This breaks any relationship between x and y .

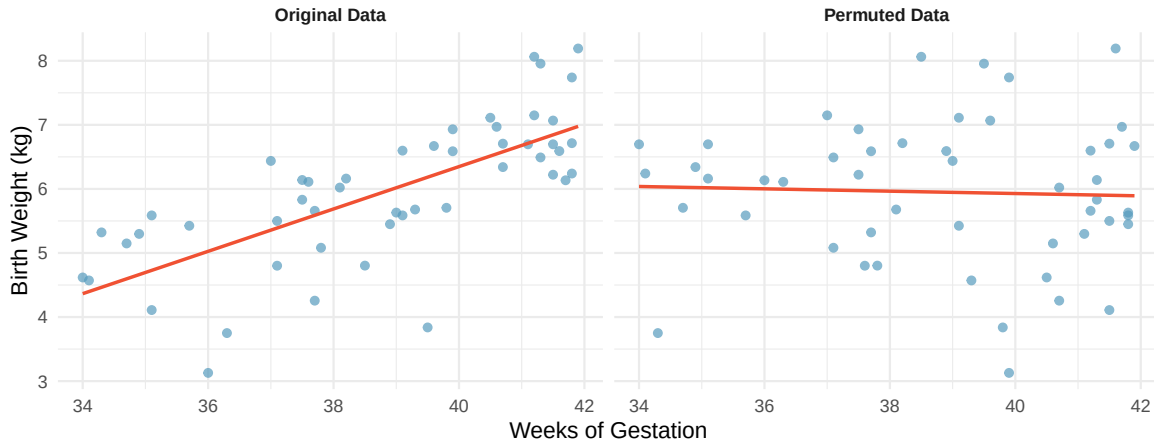


Figure 20.4: Original data showing a positive linear relationship between weeks of gestation and birth weight, alongside the same data after permuting the weight variable, which destroys the relationship.

By repeating the permutation many times and computing the slope each time, we build a null distribution of slopes.

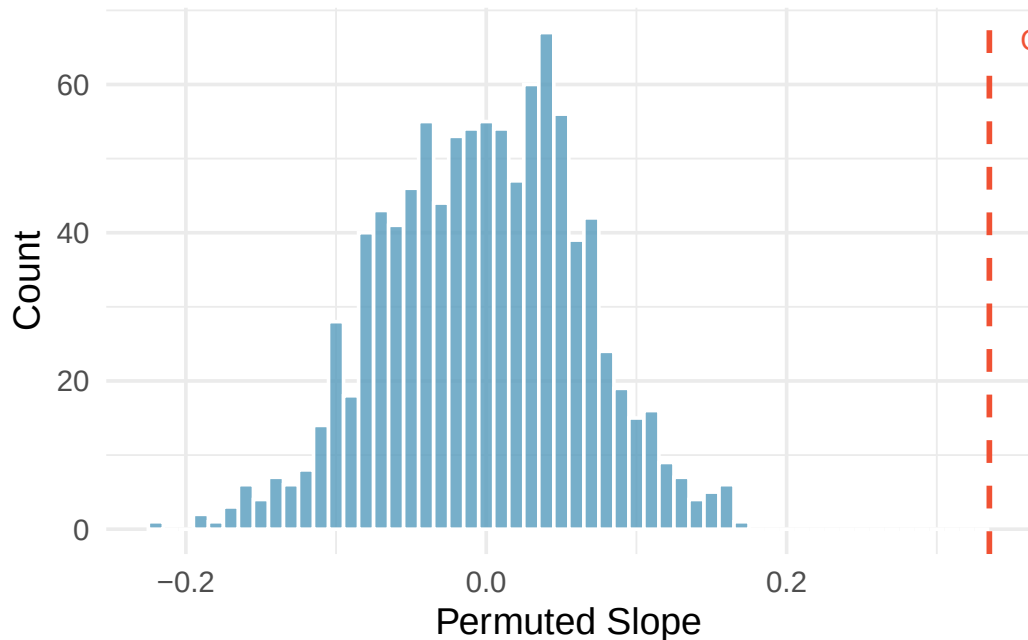


Figure 20.5: Histogram of slopes from 1,000 permutations of birth weight. The permuted slopes range from about -0.15 to $+0.15$, centered at zero. The observed slope of 0.335 is far from this distribution.

The observed slope of 0.335 is far from any value in the null distribution. We reject H_0 and conclude there is a linear relationship between weeks of gestation and birth weight.

20.5.3 Mathematical model: the t -test for slope

When the technical conditions are met, we can use the t -distribution to test the slope:

$$T = \frac{b_1 - 0}{SE_{b_1}}$$

This T -statistic follows a t -distribution with $df = n - 2$.

The regression output for predicting the change in House seats for the President's party from the unemployment rate is:

Term	Estimate	Std. Error	T statistic	p-value
(Intercept)	-7.36	5.16	-1.43	0.1649
unemp	-0.89	0.835	-1.07	0.2961

The p-value of 0.2961 is not significant. The data do not provide convincing evidence that the unemployment rate is a useful predictor of midterm election outcomes.

20.5.4 Confidence intervals for the slope

Confidence intervals for model coefficients.

Confidence intervals for model coefficients can be computed using the t -distribution:

$$b_1 \pm t_{df}^* \times SE_{b_1}$$

where t_{df}^* is the critical value corresponding to the confidence level, with $df = n - 2$.

Using the Elmhurst College regression output ($b_1 = -0.0431$, $SE = 0.0108$, $df = 48$), compute a 95% confidence interval for the slope.

The critical value is $t_{48}^* = 2.01$. The confidence interval is:

$$-0.0431 \pm 2.01 \times 0.0108 = (-0.0648, -0.0214)$$

We are 95% confident that for each additional \$1,000 in family income, the university's gift aid decreases by between \$21.40 and \$64.80 on average.

20.6 Conditions for linear regression

For the mathematical inference to be valid, we check four conditions, often remembered by the **LINE** mnemonic:

- **L – Linear model:** The data should show a linear trend. Check the scatterplot and residual plot.
- **I – Independent observations:** Be cautious with time series data or clustered observations.
- **N – Nearly normal residuals:** Residuals should be approximately normally distributed. This is less critical for large samples (Central Limit Theorem), but outliers are always a concern.
- **E – Equal variability:** The variability of points around the line should be roughly constant across all values of x . A “fan shape” in the residual plot indicates a violation.

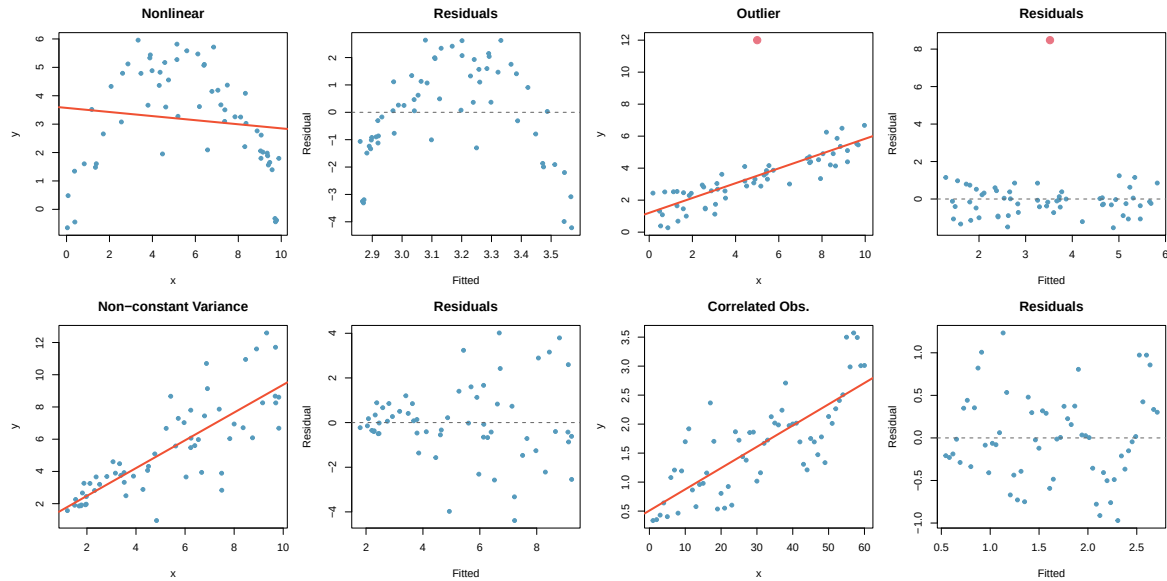


Figure 20.6: Four scatterplots and their residual plots showing common violations: nonlinearity, an outlier, increasing variability, and correlated observations.

Diagnostics for linear regression.

Independence is always important. The normality condition matters most for small samples. The constant variance condition is especially important for inference. The linearity condition is the most fundamental – if the true relationship is not linear, the slope estimate has no meaningful interpretation.

20.7 Chapter review

20.7.1 Summary

In this chapter, we developed the least squares regression line as a tool for modeling the linear relationship between two numerical variables. The slope describes the predicted change in y for a one-unit increase in x , and the intercept describes the predicted value of y when $x = 0$. Residuals measure the difference between observed and predicted values, and residual plots help assess whether the linear model is appropriate. The coefficient of determination R^2 quantifies the proportion of variability in y explained by the model. For inference, we use either a randomization test or the t -distribution to test whether the slope is significantly different from zero, and we can construct confidence intervals for the slope. The LINE conditions (linearity, independence, normality, equal variance) must be checked for the mathematical model to be valid.

20.7.2 Key terms

- Predictor / outcome variable
- Least squares line
- Slope (b_1) and intercept (b_0)
- Residual
- Residual plot
- R-squared (R^2) / coefficient of determination
- Total sum of squares (SST) / sum of squared errors (SSE)

- T-test for slope
- LINE conditions
- Extrapolation

Chapter 21

Prediction and Model Fit

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

In the previous chapter, we developed the least squares regression line, interpreted its slope and intercept, and performed inference to test whether a linear relationship exists. In this chapter, we focus on using the regression model for prediction and assessing how well it fits the data. We discuss the dangers of extrapolation, introduce residual diagnostics for identifying problems with the model, examine leverage and influential points, and briefly introduce confidence intervals and prediction intervals for forecasts.

21.1 Making predictions

One of the primary uses of a regression model is to predict the value of the response variable y for a given value of the explanatory variable x . To make a prediction, we simply plug the value of x into the regression equation.

The least squares line for predicting gift aid from family income at Elmhurst College is:

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times \text{family_income}$$

Predict the gift aid for a student whose family income is \$80,000 (i.e., $x = 80$ in \$1,000s).

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times 80 = 24.3 - 3.45 = 20.85$$

The model predicts this student will receive approximately \$20,850 in gift aid.

Suppose a high school senior is considering Elmhurst College. Can they simply use the linear equation to calculate their financial aid?

They may use it as an estimate, but with important qualifiers. First, the data come from a single year, and aid policies may change. Second, the equation provides an imperfect estimate – while the linear model captures the overall trend, individual students' aid will vary around the predicted value.

21.2 Extrapolation

Use the Elmhurst model to estimate the aid for a student whose family has an income of \$1 million (i.e., $x = 1000$).

$$\widehat{\text{aid}} = 24.3 - 0.0431 \times 1000 = -18.8$$

The model predicts $-\$18,800$ in aid! Elmhurst College does not charge extra tuition to wealthy students. This nonsensical prediction illustrates the danger of **extrapolation**.

Extrapolation is treacherous.

Extrapolation means applying a model to make predictions outside the range of the observed data. A linear model is only an approximation of the true relationship, and that approximation may break down badly outside the range of the data used to build it.

Generally, predictions should only be made within the range of the observed predictor values. Predictions far outside this range are unreliable and should be treated with great skepticism.

Before making a prediction, always check whether the value of x falls within the range of the data used to build the model. If it does not, acknowledge that you are extrapolating and that the prediction may be unreliable.

21.3 Residual diagnostics

In the previous chapter, we introduced residual plots as a way to assess whether a linear model is appropriate. In this section, we examine residual diagnostics in more detail.

21.3.1 Patterns in residual plots

A well-fitting linear model produces residuals that appear randomly scattered around zero. Systematic patterns in the residuals indicate problems:

Consider the three pairs of scatterplots and residual plots below.

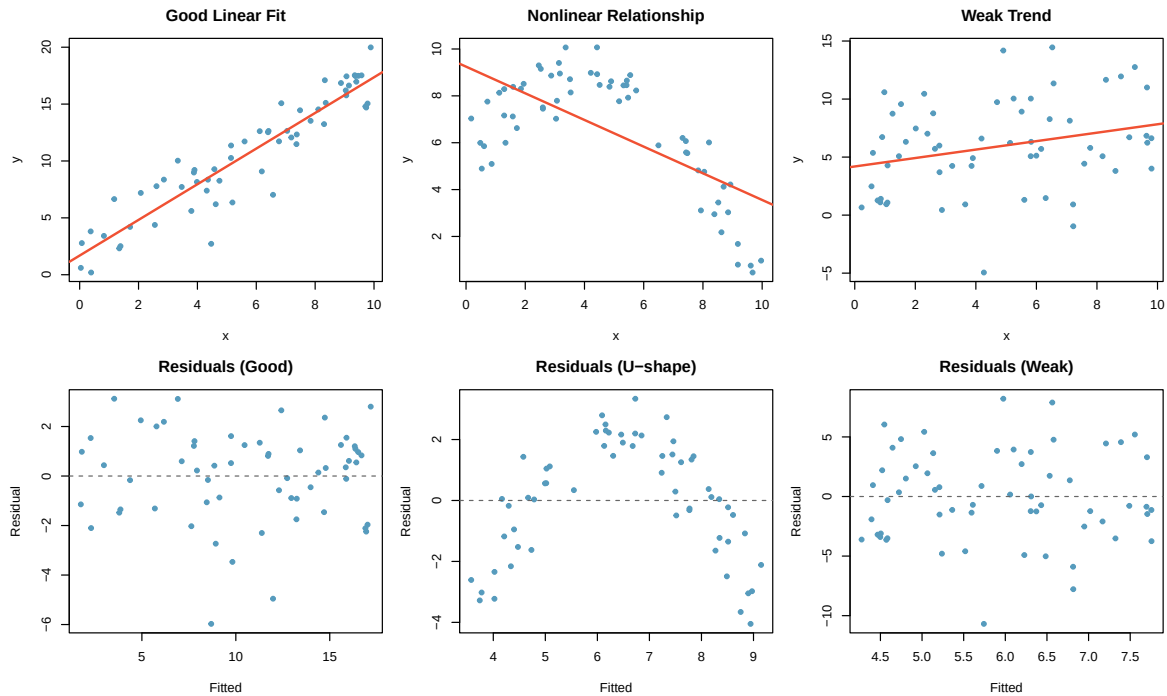


Figure 21.1: Three scatterplots with linear models in the top row and their corresponding residual plots in the bottom row. The first shows a good linear fit with random residuals. The second shows a curved relationship with a U-shaped residual pattern. The third shows a weak trend with random residuals.

Describe any patterns in the residuals.

Dataset 1: The residuals show no obvious patterns. They are scattered randomly around zero. The linear model is appropriate.

Dataset 2: There is a curved pattern in the residuals – they are negative at the edges and positive in the middle. This indicates a nonlinear relationship. A straight line is not a good model for these data.

Dataset 3: The residuals show no patterns, but the trend is very weak. The linear model is technically appropriate, but the relationship may not be practically meaningful.

21.3.2 Non-constant variability (heteroscedasticity)

When the spread of the residuals changes as the predicted value increases, we have **non-constant variability** (also called **heteroscedasticity**). This often appears as a “fan shape” in the residual plot – residuals spread out (or narrow) as the predicted values increase.

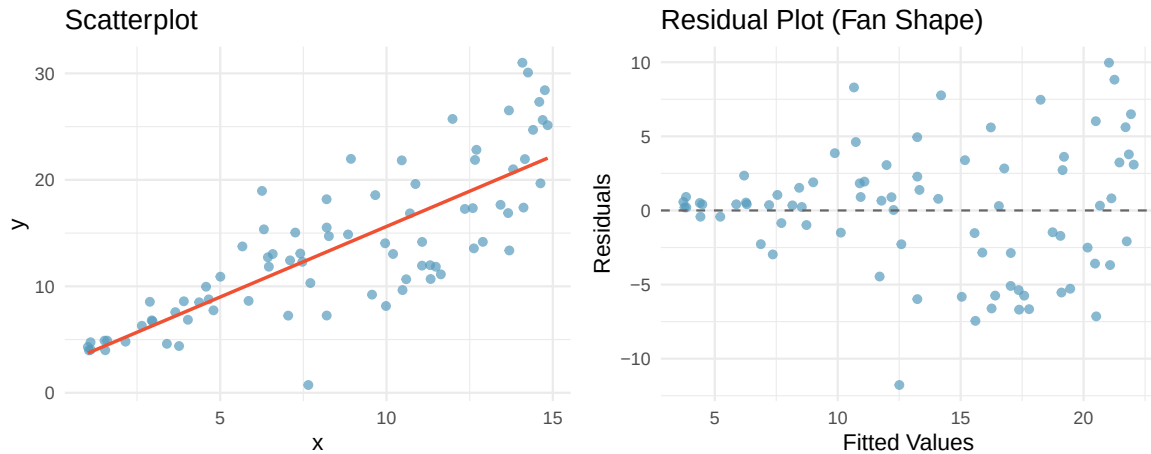


Figure 21.2: A scatterplot and residual plot showing increasing variability as x increases. The residual plot shows a fan shape.

Non-constant variability does not bias the regression line itself, but it does affect inference: standard errors, confidence intervals, and p -values may be unreliable.

21.4 Leverage and influential points

Not all observations have the same impact on the regression line. Some points, by virtue of their position, can exert a disproportionate influence on the model.

21.4.1 Leverage

Leverage.

A point has **high leverage** if its x -value is far from $\bar{x}$ (the mean of the predictor variable). High-leverage points have the *potential* to strongly influence the slope and intercept of the regression line.

High leverage does not automatically mean the point is problematic. If a high-leverage point falls close to the trend established by the other data, it may actually strengthen the model. A high-leverage point is only problematic if its y -value does not follow the pattern of the rest of the data.

21.4.2 Influential points

Influential point.

A point is **influential** if removing it would substantially change the regression line (i.e., the estimated slope or intercept). Influential points are typically high-leverage points that do not follow the pattern of the rest of the data.

Consider three scenarios:

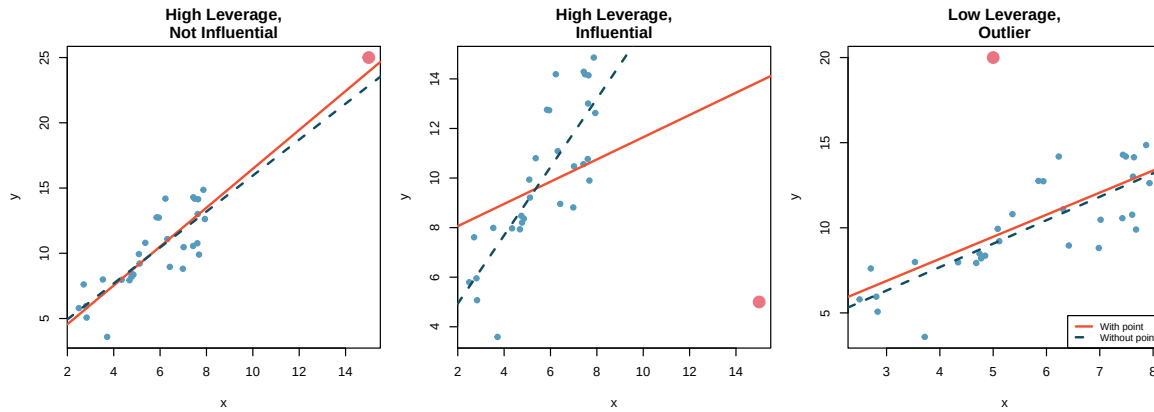


Figure 21.3: Three scatterplots with regression lines. In the first, a point in the corner has high leverage but follows the trend (not influential). In the second, a point far from the trend has high leverage and is influential – removing it changes the slope substantially. In the third, a point near the mean of x is an outlier in the y -direction but has low leverage and is not very influential.

Scenario 1: The point in the far right corner has high leverage (it is far from $\bar{x}$), but it falls along the trend of the other data. Removing it would barely change the regression line. It is *not* influential.

Scenario 2: The point in the far right corner has high leverage *and* falls far from the trend. Removing it would substantially change the slope. It *is* influential.

Scenario 3: The point near the center of the x -values is an outlier in the y -direction. It has low leverage because its x -value is near $\bar{x}$. Removing it would change the intercept slightly but not the slope very much. It has limited influence.

Dealing with influential points.

When you identify an influential point:

1. Verify that it is not a data entry error.
2. Run the analysis with and without the point and report both results.
3. Do not automatically remove it – if it is a legitimate observation, it contains real information.
4. Be transparent about its effect on the conclusions.

21.5 Confidence and prediction intervals

When we use a regression model to make predictions, there are two types of uncertainty to consider:

21.5.1 Confidence intervals for the mean response

A **confidence interval for the mean response** estimates the average value of y for all individuals with a particular value of x . This interval captures the uncertainty in estimating the population regression line.

For example, we might ask: “What is the average gift aid for all Elmhurst students with a family income of \$80,000?” The confidence interval for the mean response at $x = 80$ would give a range of plausible values for this population average.

The formula for a $(1 - \alpha) \times 100\%$ confidence interval for the mean response at $x = x^*$ is:

$$\hat{y} \pm t_{n-2}^* \cdot SE_{\hat{y}}$$

where the standard error depends on the distance of x^* from $\bar{x}$:

$$SE_{\hat{y}} = s_e \sqrt{\frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

Here $s_e = \sqrt{MSE}$ is the residual standard error.

21.5.2 Prediction intervals for individual observations

A **prediction interval** estimates the value of y for a *single* new individual with a particular value of x . This interval is always wider than the confidence interval for the mean response because it must account for both:

1. The uncertainty in estimating the regression line (same as the confidence interval).
2. The natural variability of individual observations around the line.

The prediction interval at $x = x^*$ is:

$$\hat{y} \pm t_{n-2}^* \cdot s_e \sqrt{1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

Notice the extra “1+” under the square root compared to the confidence interval formula. This is what makes the prediction interval wider.

21.5.3 Comparing the two intervals

For the possum data, suppose we want to predict the head length for possums with a total length of 85 cm.

The point prediction is $\hat{y} = 41 + 0.59 \times 85 = 91.15$ mm.

- A 95% **confidence interval for the mean** head length of all possums with total length 85 cm might be (89.8, 92.5) mm.
- A 95% **prediction interval** for the head length of a single possum with total length 85 cm might be (83.2, 99.1) mm.

The confidence interval is relatively narrow because it estimates an average, which is estimated with moderate precision. The prediction interval is much wider because it must also account for the natural variation among individual possums – even possums with the same total length have different head lengths.

Confidence interval vs. prediction interval.

- A **confidence interval for the mean response** estimates the average y for a given x . It accounts for uncertainty in the regression line.
- A **prediction interval** estimates the y for a single new observation at a given x . It accounts for both the uncertainty in the regression line and the variability of individual observations.
- Prediction intervals are always wider than confidence intervals.
- Both intervals are narrowest at $x = \bar{x}$ and widen as x moves away from $\bar{x}$.

The fact that both intervals widen as x moves away from $\bar{x}$ connects back to the danger of extrapolation: predictions become increasingly uncertain farther from the center of the data.

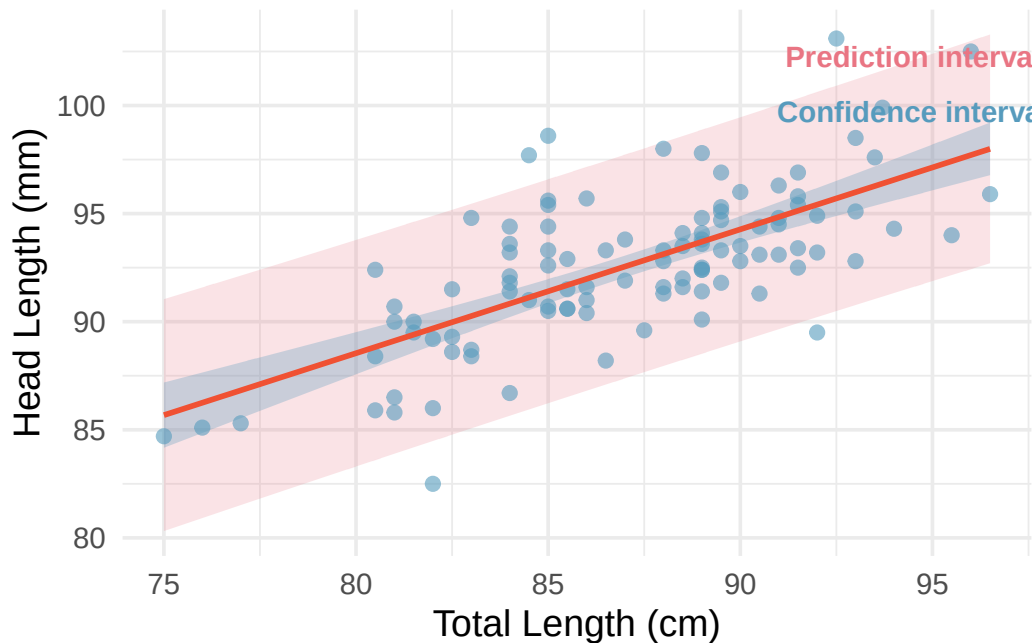


Figure 21.4: A scatterplot with a regression line and both 95% confidence bands (narrow) and 95% prediction bands (wide). Both bands are narrowest near the mean of x and flare outward.

21.6 Chapter review

21.6.1 Summary

In this chapter, we focused on using regression models for prediction and evaluating their fit. We learned that extrapolation – predicting outside the range of the observed data – is dangerous and can lead to nonsensical predictions. Residual diagnostics help us identify violations of the model assumptions, including nonlinearity, non-constant variance, and the influence of unusual points. Leverage measures how far a point's x -value is from the center of the data; influential points are high-leverage observations that do not follow the overall trend and substantially affect the regression line when removed. Finally, we distinguished between confidence intervals for the mean response and prediction intervals for individual observations, noting that prediction intervals are always wider because they account for individual variability.

21.6.2 Key terms

- Extrapolation
- Residual diagnostics
- Non-constant variability (heteroscedasticity)
- Leverage
- Influential point
- Confidence interval for the mean response
- Prediction interval

PART V

FOUNDATIONS

You have already been *doing* probability — every simulation you ran was an experiment in randomness. This part formalizes the rules governing random events, introduces random variables, and develops the binomial and normal distributions as mathematical models. These foundations explain *why* the methods from earlier parts work.

Chapter 22

Probability Rules

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

Probability forms the foundation of statistics, and you are probably already aware of many of the ideas we will discuss in this chapter. However, formalizing these concepts is likely new. While this chapter provides the theoretical foundation for the inference methods you have already been using, it also offers practical tools for reasoning about uncertainty. We will define probability, develop rules for combining probabilities, and explore conditional probability and independence — all concepts that underpin the inference methods from Parts II and III.

StatBench tools for this chapter: The probability concepts in this chapter provide the theoretical foundation for the distribution calculators and inference tools you used in earlier parts of the book. Try the **Binomial Distribution Calculator** to see probability in action — set $n = 1$ for a single Bernoulli trial.

[Launch Binomial Distribution Calculator](#)

22.1 Defining probability

Statistics is built on probability, and probability gives us the language to describe uncertainty. We begin with a few simple examples that may feel familiar.

A “die,” the singular of “dice,” is a cube with six faces numbered 1, 2, 3, 4, 5, and 6. What is the chance of getting a 1 when rolling a fair die?

If the die is fair, then the chance of a 1 is as good as the chance of any other number. Since there are six equally likely outcomes, the chance must be 1-in-6 or, equivalently, $1/6$.

What is the chance of getting a 1 or 2 on the next roll?

The outcomes 1 and 2 constitute two of the six equally likely outcomes, so the chance of getting one of these two is $2/6 = 1/3$.

What is the chance of not rolling a 2?

Since the chance of rolling a 2 is $1/6$, the chance of not rolling a 2 must be $1 - 1/6 = 5/6$. Alternatively, not rolling a 2 means getting a 1, 3, 4, 5, or 6, which makes up five of the six equally likely outcomes.

We use probability to describe and understand apparent randomness. We frame probability in terms of a **random process** giving rise to an **outcome**.

Random process	Possible outcomes
Roll a die	1, 2, 3, 4, 5, or 6
Flip a coin	H or T

The **probability** of an outcome is the proportion of times the outcome would occur if we observed the random process an infinite number of times. Probability always takes values between 0 and 1 (inclusive) and may also be expressed as a percentage between 0% and 100%.

Probability can be illustrated by rolling a die many times. Let $\hat{p}_n$ be the proportion of outcomes that are 1 after the first n rolls. As the number of rolls increases, $\hat{p}_n$ converges to the probability $p = 1/6$.

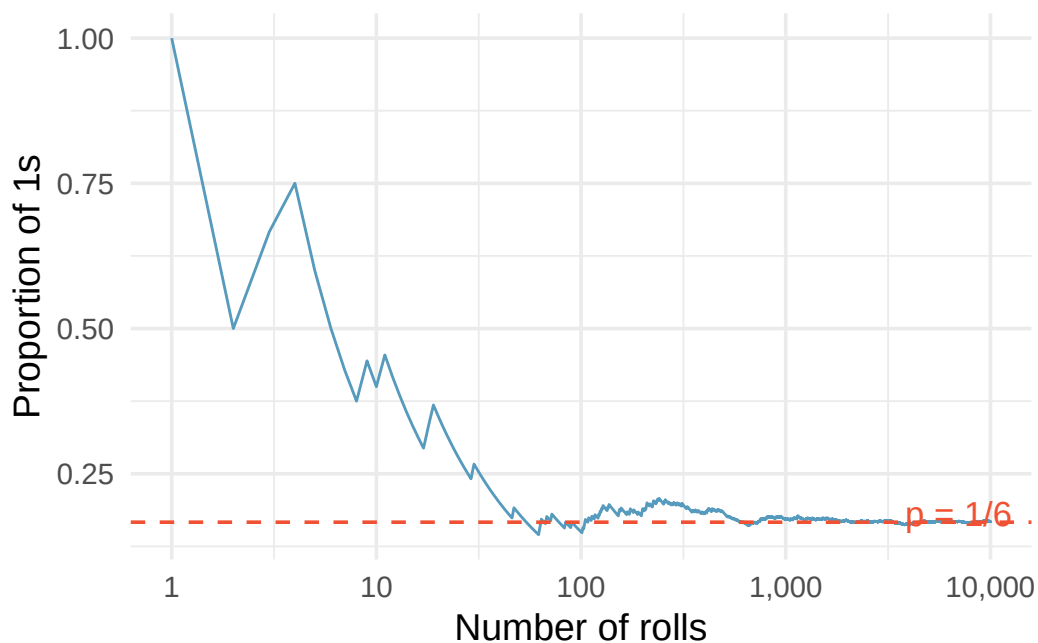


Figure 22.1: The fraction of die rolls that are 1 at each stage in a simulation. The proportion tends to get closer to $1/6$ as the number of rolls increases.

Law of Large Numbers. As more observations are collected, the proportion $\hat{p}_n$ of occurrences with a particular outcome converges to the probability p of that outcome.

Occasionally the proportion will veer away from the true probability, but these deviations become smaller as the number of trials increases.

We write the probability of rolling a 1 as $P(\text{rolling a 1})$. When the context is clear, we abbreviate this as $P(1)$.

Random processes include rolling a die and flipping a coin. (a) Think of another random process. (b) Describe all the possible outcomes of that process.

Show answer

- (a) and (b) Here are several examples. (i) Whether someone gets sick next month: outcomes are *sick* and *not sick*. (ii) Randomly selecting a person and measuring their height: the outcome is a positive number. (iii) Whether the stock market goes up or down next week: outcomes are *up*, *down*, and *no change*. (iv) The number of emails you receive tomorrow: the outcome is a non-negative integer.

22.1.1 Simulation: The law of large numbers in action

One of the most powerful ways to build intuition about probability is through simulation. Imagine writing a computer program that simulates flipping a fair coin 10,000 times and tracking the running proportion of heads after each flip.

- After 10 flips, the proportion of heads might be 0.30 or 0.70 — quite far from 0.50.
- After 100 flips, the proportion is likely between 0.40 and 0.60.
- After 1,000 flips, the proportion is usually between 0.47 and 0.53.
- After 10,000 flips, it is almost certainly within 0.01 of 0.50.

This is the Law of Large Numbers at work. The key insight is that probability is a *long-run* concept. In the short run, anything can happen. In the long run, patterns emerge with remarkable stability.

This simulation perspective will recur throughout our course. In Parts II and III, we used simulation to build sampling distributions and test hypotheses. Here in Part IV, we develop the *mathematical* rules that explain *why* those simulations work.

22.2 Sample spaces and events

The **sample space** S of a random process is the set of all possible outcomes. An **event** is a subset of the sample space — that is, a collection of one or more outcomes.

For rolling a single die, the sample space is $S = \{1, 2, 3, 4, 5, 6\}$.

Let A represent the event that the die roll results in 1 or 2, so $A = \{1, 2\}$. Let B represent the event that the die roll is 4 or 6, so $B = \{4, 6\}$. These events are shown visually in the diagram below.

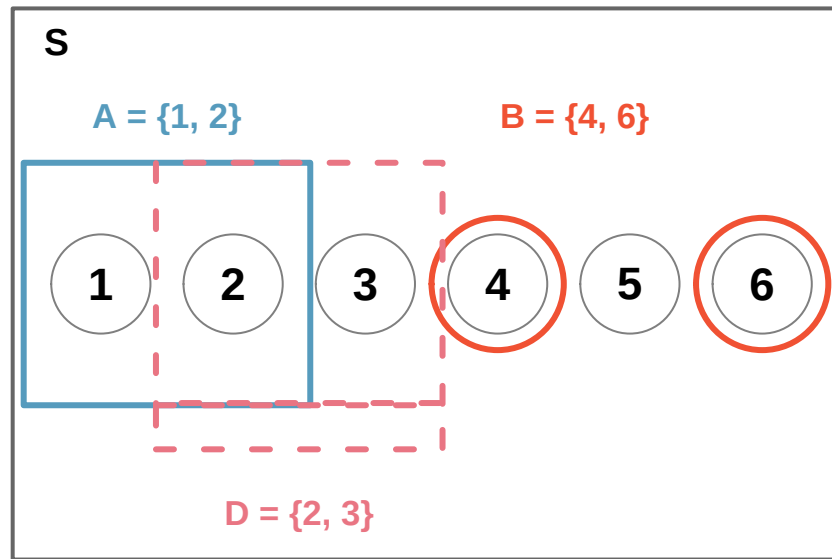


Figure 22.2: Three events for die outcomes. A and B are disjoint since they share no outcomes.

22.3 The complement rule

Let $D = \{2, 3\}$ represent the event that the outcome of a die roll is 2 or 3. Then the **complement** of D , denoted D^c , represents all outcomes in the sample space that are *not* in D :

$$D^c = \{1, 4, 5, 6\}$$

- (a) Compute $P(D^c) = P(\text{rolling a } 1, 4, 5, \text{ or } 6)$. (b) What is $P(D) + P(D^c)$?

Show answer

- (a) The outcomes are disjoint and each has probability $1/6$, so $P(D^c) = 4/6 = 2/3$. (b) $P(D) = 1/6 + 1/6 = 1/3$. Since D and D^c are disjoint, $P(D) + P(D^c) = 1/3 + 2/3 = 1$.

A complement A^c is constructed so that (i) every possible outcome not in A is in A^c , and (ii) A and A^c are disjoint. This gives us:

Complement rule. The complement of event A is denoted A^c and represents all outcomes not in A . The probabilities of A and A^c are related by:

$$P(A) + P(A^c) = 1 \quad \text{equivalently,} \quad P(A) = 1 - P(A^c)$$

The complement rule is especially useful when computing a probability directly would be tedious but the complement is simple.

Let A represent the event where the sum of two dice is less than 12. What is $P(A)$?

The complement A^c is the event that the sum equals 12. The only way to roll a sum of 12 is to get two 6s, so $P(A^c) = 1/36$. Using the complement rule:

$$P(A) = 1 - P(A^c) = 1 - \frac{1}{36} = \frac{35}{36}$$

Find the following probabilities for rolling two dice: (a) The sum is *not* 6. (b) The sum is at least 4. (c) The sum is no more than 10.

Show answer

- (a) $P(\text{sum} = 6) = 5/36$, so $P(\text{not } 6) = 1 - 5/36 = 31/36$. (b) $P(\text{sum} = 2 \text{ or } 3) = 1/36 + 2/36 = 1/12$, so $P(\text{at least } 4) = 1 - 1/12 = 11/12$. (c) $P(\text{sum} = 11 \text{ or } 12) = 2/36 + 1/36 = 1/12$, so $P(\text{no more than } 10) = 1 - 1/12 = 11/12$.

22.4 The addition rule

22.4.1 Disjoint (mutually exclusive) events

Two outcomes or events are called **disjoint** (or **mutually exclusive**) if they cannot both happen at the same time. For instance, when rolling a die, the outcomes 1 and 2 are disjoint since they cannot both occur on a single roll.

Addition rule for disjoint events. If A_1 and A_2 are disjoint events, then:

$$P(A_1 \text{ or } A_2) = P(A_1) + P(A_2)$$

More generally, if $A_1, A_2, \dots, A_k$ are all mutually disjoint, then:

$$P(A_1 \text{ or } A_2 \text{ or } \dots \text{ or } A_k) = P(A_1) + P(A_2) + \dots + P(A_k)$$

We are interested in the probability of rolling a 1, 4, or 5. (a) Explain why the outcomes 1, 4, and 5 are disjoint. (b) Apply the addition rule to determine $P(1 \text{ or } 4 \text{ or } 5)$.

Show answer

- (a) At most one of these outcomes can occur on a single die roll, so they are disjoint. (b) $P(1 \text{ or } 4 \text{ or } 5) = P(1) + P(4) + P(5) = 1/6 + 1/6 + 1/6 = 3/6 = 1/2$.

22.4.2 Probability distributions

A **probability distribution** is a table of all disjoint outcomes and their associated probabilities. For example, the distribution for the sum of two dice:

Dice sum	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Rules for probability distributions. A valid probability distribution must satisfy three rules:

1. The outcomes listed must be disjoint.
2. Each probability must be between 0 and 1.
3. The probabilities must total 1.

22.4.3 The general addition rule

When events are *not* disjoint, we cannot simply add their probabilities — doing so would double-count the outcomes they share.

Consider a standard deck of 52 cards. Let A be the event “the card is a diamond” and B be the event “the card is a face card.” These events overlap: the jack, queen, and king of diamonds are both diamonds and face cards.

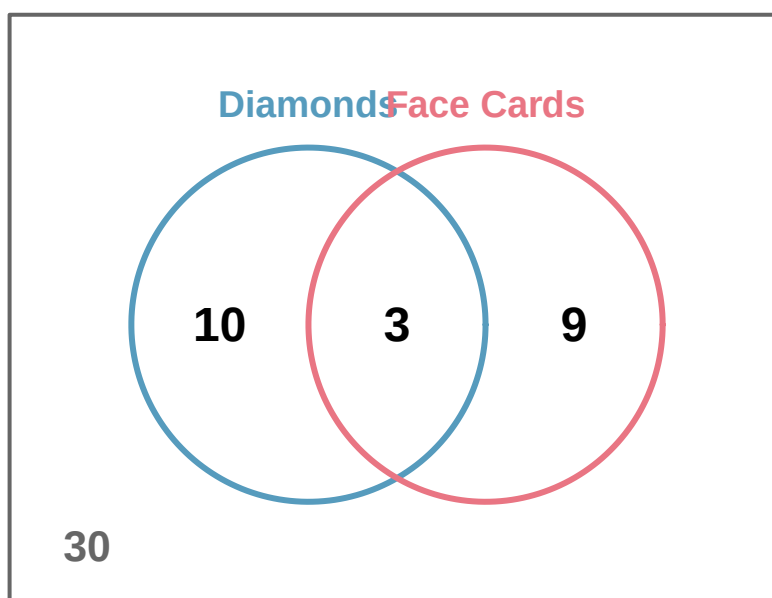


Figure 22.3: A Venn diagram for diamonds and face cards showing 10 diamonds that are not face cards, 9 face cards that are not diamonds, and 3 cards that are both.

If we compute $P(A) + P(B) = 13/52 + 12/52 = 25/52$, we have counted the 3 cards in the overlap twice. We correct this:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52} = \frac{11}{26}$$

General addition rule. For any two events A and B (disjoint or not):

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

“Or” is inclusive in statistics. When we write “ A or B ,” we mean “ A , B , or both” unless explicitly stated otherwise.

- (a) If A and B are disjoint, explain why $P(A \text{ and } B) = 0$. (b) Show that the general addition rule simplifies to the simpler addition rule when A and B are disjoint.

Show answer

- (a) If A and B are disjoint, they can never occur simultaneously, so $P(A \text{ and } B) = 0$. (b) Substituting 0 for $P(A \text{ and } B)$ in the general addition rule gives $P(A \text{ or } B) = P(A) + P(B) - 0 = P(A) + P(B)$, which is the simpler addition rule.

22.5 Conditional probability

22.5.1 Exploring conditional probability with data

There can be rich relationships between two or more variables, and conditional probability provides the framework for exploring them. Suppose we have data from a study classifying 1,822 photos using both a machine learning (ML) algorithm and human judgment:

	Truth: fashion	Truth: not	Total
ML: pred_fashion	197	22	219
ML: pred_not	112	1491	1603
Total	309	1513	1822

If a photo is actually about fashion, what is the chance the ML classifier correctly identified it?

Of the 309 fashion photos, the ML algorithm correctly classified 197:

$$P(\text{ML predicts fashion} \mid \text{truth is fashion}) = \frac{197}{309} = 0.638$$

We call this a **conditional probability** because we computed the probability *under a condition* — that the photo was actually about fashion.

22.5.2 Marginal and joint probabilities

A **marginal probability** is a probability based on a single variable, without regard to other variables. A **joint probability** is the probability of two or more outcomes occurring together.

From the photo classification data: $P(\text{ML predicts fashion}) = 219/1822 = 0.120$ is a marginal probability, while $P(\text{ML predicts fashion and truth is fashion}) = 197/1822 = 0.108$ is a joint probability.

22.5.3 The conditional probability formula

Conditional probability. The conditional probability of outcome A given condition B is:

$$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$$

The vertical bar “|” is read as “given.”

Using the photo classification data, we can verify our earlier calculation using proportions alone:

$$P(\text{truth is fashion} \mid \text{ML predicts fashion}) = \frac{P(\text{truth is fashion and ML predicts fashion})}{P(\text{ML predicts fashion})} = \frac{0.108}{0.120} = 0.900$$

If the ML classifier says a photo is about fashion, there is a 90% probability it actually is.

The smallpox dataset provides data on 6,224 individuals in Boston in 1721. Of these, 3.92% were inoculated. Among those not inoculated, 14.11% died. Among those inoculated, only 2.46% died. Write out, in formal notation, the probability a randomly selected person who was not inoculated died from smallpox.

Show answer

$$P(\text{result} = \text{died} \mid \text{inoculated} = \text{no}) = \frac{P(\text{died and not inoculated})}{P(\text{not inoculated})} = \frac{0.1356}{0.9608} = 0.1411.$$

22.6 The multiplication rule

22.6.1 Multiplication rule for independent processes

Consider rolling two dice. If $1/6$ of the time the first die is a 1, and $1/6$ of *those* times the second die is also a 1, then:

$$P(\text{both are 1}) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Multiplication rule for independent processes. If A and B represent events from two different and independent processes, then:

$$P(A \text{ and } B) = P(A) \times P(B)$$

If there are k events $A_1, \dots, A_k$ from k independent processes, then:

$$P(A_1 \text{ and } A_2 \text{ and } \dots \text{ and } A_k) = P(A_1) \times P(A_2) \times \dots \times P(A_k)$$

About 9% of people are left-handed. Suppose 2 people are selected at random from the US population. (a) What is the probability that both are left-handed? (b) What is the probability that both are right-handed?

Show answer

$$(a) 0.09 \times 0.09 = 0.0081. (b) P(\text{right-handed}) = 1 - 0.09 = 0.91, \text{ so } 0.91 \times 0.91 = 0.8281.$$

Suppose 5 people are selected at random. (a) What is the probability that all are right-handed? (b) What is the probability that not all are right-handed?

Show answer

$$(a) 0.91^5 = 0.624. (b) \text{ Using the complement: } P(\text{not all right-handed}) = 1 - 0.624 = 0.376.$$

22.6.2 The general multiplication rule

The general multiplication rule handles events that may not be independent:

General multiplication rule. If A and B represent two outcomes or events, then:

$$P(A \text{ and } B) = P(A \mid B) \times P(B)$$

This is simply a rearrangement of the conditional probability formula.

In the smallpox data, 96.08% of residents were not inoculated, and 85.88% of those who were not inoculated survived. What is the probability that a resident was not inoculated and lived?

$$P(\text{lived and not inoculated}) = P(\text{lived} \mid \text{not inoculated}) \times P(\text{not inoculated}) = 0.8588 \times 0.9608 = 0.8251$$

22.6.3 Tree diagrams

Tree diagrams are a tool to organize outcomes and probabilities when two or more processes occur in sequence and each process is conditioned on its predecessors.

Suppose 13% of students earned an A on the midterm. Of those who earned an A on the midterm, 47% received an A on the final. Of those who earned lower than an A on the midterm, 11% received an A on the final. You randomly pick up a final exam and notice the student received an A. What is the probability the student also earned an A on the midterm?

We organize the information into a tree diagram:

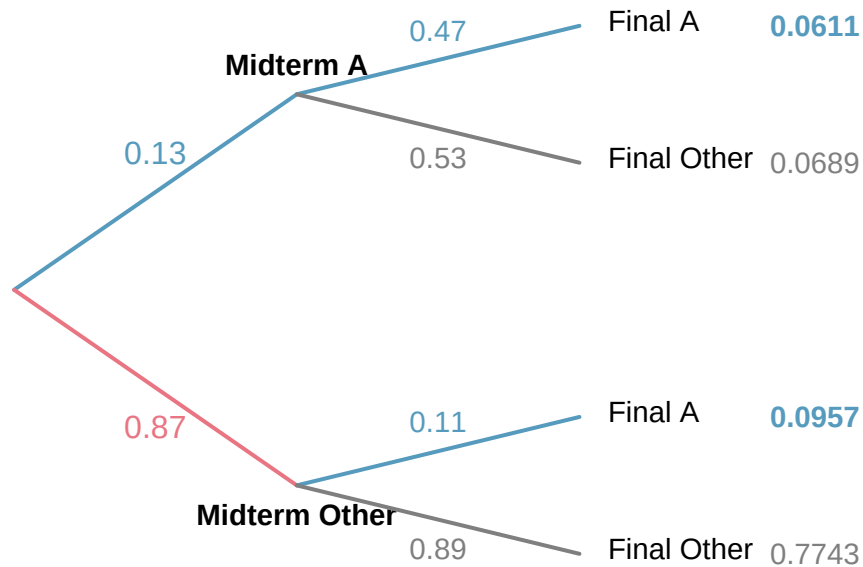


Figure 22.4: Tree diagram for midterm and final exam grades.

From the tree:

- $P(\text{midterm A and final A}) = 0.13 \times 0.47 = 0.0611$
- $P(\text{final A}) = 0.0611 + 0.0957 = 0.1568$
- $P(\text{midterm A} \mid \text{final A}) = \frac{0.0611}{0.1568} = 0.390$

The probability the student also earned an A on the midterm is about 0.39.

Sum of conditional probabilities. If $A_1, \dots, A_k$ represent all disjoint outcomes of a variable, and B is an event, then:

$$P(A_1 \mid B) + P(A_2 \mid B) + \dots + P(A_k \mid B) = 1$$

The complement rule also holds for conditional probabilities: $P(A \mid B) = 1 - P(A^c \mid B)$.

22.7 Independence

Two processes are **independent** if knowing the outcome of one provides no useful information about the outcome of the other. Mathematically, events A and B are independent if:

$$P(A \text{ and } B) = P(A) \times P(B)$$

Equivalently, A and B are independent if $P(A | B) = P(A)$.

If we draw one card from a shuffled deck, is the event that the card is a heart independent of the event that the card is an ace?

$P(\heartsuit) = 1/4$, $P(\text{ace}) = 1/13$, and $P(\heartsuit \text{ and ace}) = 1/52$. We check:

$$P(\heartsuit) \times P(\text{ace}) = \frac{1}{4} \times \frac{1}{13} = \frac{1}{52} = P(\heartsuit \text{ and ace})$$

Because the equation holds, “heart” and “ace” are independent events.

Let X and Y represent the outcomes of rolling two dice. (a) What is $P(X = 1)$? (b) What is $P(Y = 1 \text{ and } X = 1)$? (c) Compute $P(Y = 1 | X = 1)$. (d) Compare $P(Y = 1)$ with the result from (c).

Show answer

(a) $1/6$. (b) $1/36$. (c) $\frac{1/36}{1/6} = 1/6$. (d) $P(Y = 1) = 1/6$, the same as part (c). Knowledge about X does not change the probability for Y , confirming the dice are independent.

The gambler's fallacy. Ron is watching a roulette table and notices the last five spins were all black. He bets on red, reasoning that “black six times in a row” is unlikely. This reasoning is flawed: each spin is independent of the previous spins. Past outcomes provide no information about future ones. Casinos often post recent outcomes precisely to encourage this fallacy.

22.8 Bayes' theorem (optional)

In many situations, we are given $P(B | A)$ but need $P(A | B)$ — the “inverted” conditional probability. Bayes' theorem provides a systematic way to perform this inversion.

In Canada, about 0.35% of women over 40 develop breast cancer in any given year. Mammograms produce false negatives 11% of the time (indicating no cancer when cancer is present) and false positives 7% of the time (indicating cancer when none is present). If a randomly selected woman over 40 tests positive, what is the probability she actually has breast cancer?

We organize the information into a tree diagram:

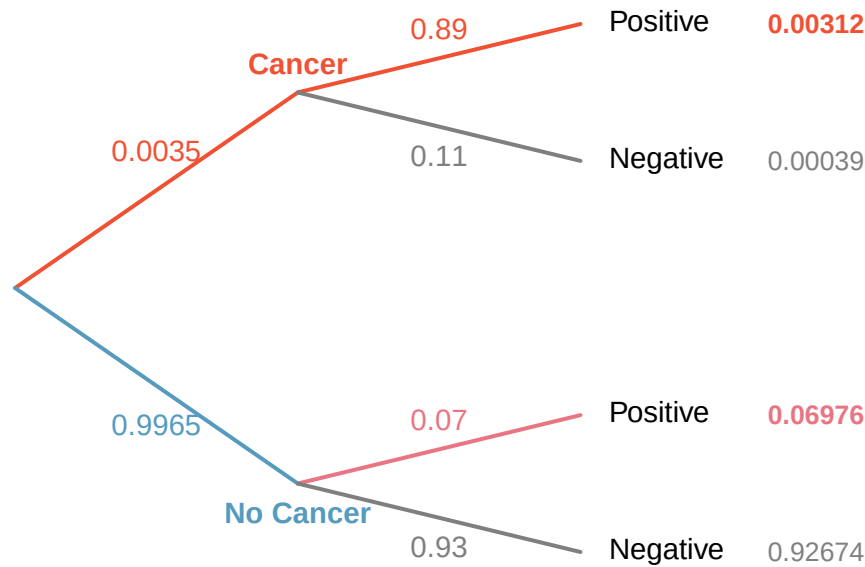


Figure 22.5: Tree diagram for breast cancer screening.

From the tree:

- $P(\text{cancer and positive}) = 0.0035 \times 0.89 = 0.00312$
- $P(\text{positive}) = 0.00312 + 0.06976 = 0.07288$
- $P(\text{cancer} \mid \text{positive}) = \frac{0.00312}{0.07288} \approx 0.043$

Even with a positive mammogram, there is only about a 4% chance the patient has breast cancer. This is why doctors often run additional tests.

Bayes' theorem. If $A_1, A_2, \dots, A_k$ represent all possible disjoint outcomes of a first variable, and B is an observed outcome of a second variable, then:

$$P(A_1 \mid B) = \frac{P(B \mid A_1) P(A_1)}{P(B \mid A_1) P(A_1) + P(B \mid A_2) P(A_2) + \dots + P(B \mid A_k) P(A_k)}$$

The numerator is the probability of getting both A_1 and B . The denominator is the total (marginal) probability of B .

To apply Bayes' theorem:

1. Identify the marginal probabilities: $P(A_1), P(A_2), \dots, P(A_k)$.
2. Identify the conditional probabilities: $P(B \mid A_1), P(B \mid A_2), \dots, P(B \mid A_k)$.
3. Substitute into the formula.

Jose visits campus on Thursday evenings. Academic events occur 35% of evenings, sporting events 20%, and no events 45%. The parking garage fills up 25% of the time during academic events, 70% during sporting events, and 5% when there are no events. If Jose finds the garage full, what is the probability there is a sporting event?

Show answer

Using Bayes' theorem with $A_1 = \text{sporting}$, $A_2 = \text{academic}$, $A_3 = \text{none}$, and $B = \text{garage full}$:

$$P(\text{sporting} \mid \text{full}) = \frac{0.7 \times 0.2}{0.7 \times 0.2 + 0.25 \times 0.35 + 0.05 \times 0.45} = \frac{0.14}{0.25} = 0.56.$$
 There is a 56% probability of a sporting event.

The strategy of updating beliefs based on new evidence is the foundation of **Bayesian statistics**, an important branch of statistics that we will not have time to explore further in this course.

22.9 Chapter review

22.9.1 Summary

In this chapter, we developed the mathematical rules of probability:

- **Probability** is the long-run proportion of times an outcome occurs, governed by the **Law of Large Numbers**.
- The **sample space** S is the set of all possible outcomes; an **event** is a subset of S .
- **Complement rule**: $P(A) = 1 - P(A^c)$.
- **Addition rule**: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. For disjoint events, $P(A \text{ and } B) = 0$.
- **Conditional probability**: $P(A \mid B) = P(A \text{ and } B) / P(B)$.
- **Multiplication rule**: $P(A \text{ and } B) = P(A \mid B) \times P(B)$. For independent events, $P(A \text{ and } B) = P(A) \times P(B)$.
- Events are **independent** if knowing one occurred does not change the probability of the other.
- **Bayes' theorem** inverts conditional probabilities.

22.9.2 Exercises

Exercises for this chapter are under development.

Chapter 23

Random Variables

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In the previous chapter, we developed rules for computing probabilities of events. Now we shift our focus to **random variables** — numerical summaries of random processes. A random variable assigns a number to each outcome of a random process, which allows us to apply mathematical tools for prediction and analysis. We distinguish between discrete and continuous random variables, introduce probability distributions, and build the foundation for the specific distributions (binomial and normal) that follow in later chapters.

23.1 Random variables

It is often useful to model a random process by assigning a number to each outcome. This numerical model is called a **random variable**.

A **random variable** is a numerical outcome of a random process. We typically denote random variables with capital letters such as X , Y , or Z , and their observed values with the corresponding lowercase letters x , y , or z .

Two books are assigned for a statistics class: a textbook (\$137) and a study guide (\$33). The university bookstore has determined that 20% of enrolled students buy neither book, 55% buy the textbook only, and 25% buy both books. If X represents the amount a single student spends on books, then X is a random variable with possible values \$0, \$137, and \$170.

Random variables come in two types — discrete and continuous — and the distinction matters because they require different mathematical tools.

23.2 Discrete random variables

A **discrete random variable** takes on a finite or countably infinite number of distinct values. These values can be listed (even if the list is infinitely long).

Examples of discrete random variables:

- The number showing when you roll a die: $X \in \{1, 2, 3, 4, 5, 6\}$

- The number of heads in 10 coin flips: $X \in \{0, 1, 2, \dots, 10\}$
- The number of customers arriving at a store in an hour: $X \in \{0, 1, 2, 3, \dots\}$

23.2.1 Why the distinction matters

In Parts II and III, we worked extensively with sample statistics like the sample mean $\bar{x}$ and the sample proportion $\hat{p}$. These are actually *observed values* of random variables. The sampling distribution of $\bar{x}$ that we studied in Chapter 9 is really the probability distribution of a random variable. Formalizing the language of random variables helps us state results more precisely and connect our earlier simulation-based reasoning to mathematical theory.

23.2.2 Probability mass functions

The complete description of a discrete random variable is given by its **probability mass function** (PMF), which specifies the probability of each possible value.

The **probability mass function** (PMF) of a discrete random variable X is the function $P(X = x)$ that gives the probability that X takes on each possible value x . A valid PMF must satisfy:

1. $P(X = x) \geq 0$ for every value x .
2. The sum of all probabilities equals 1: $\sum_{\text{all } x} P(X = x) = 1$.

The bookstore example from the previous section has the following PMF:

x_i	\$0	\$137	\$170
$P(X = x_i)$	0.20	0.55	0.25

We can verify this is a valid PMF: all probabilities are between 0 and 1, and $0.20 + 0.55 + 0.25 = 1.00$.

Suppose a campus shuttle arrives every 10 minutes. The number of students who board the shuttle at a particular stop can be modeled with the following distribution:

Number of students (x)	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10

- (a) Verify this is a valid probability distribution. (b) What is the probability that 2 or fewer students board? (c) What is the probability that more than 2 students board?

Show answer

- (a) All probabilities are between 0 and 1, and $0.10 + 0.25 + 0.35 + 0.20 + 0.10 = 1.00$. (b) $P(X \leq 2) = 0.10 + 0.25 + 0.35 = 0.70$. (c) $P(X > 2) = 1 - P(X \leq 2) = 1 - 0.70 = 0.30$, or equivalently $0.20 + 0.10 = 0.30$.

23.2.3 Visualizing discrete distributions

Discrete probability distributions are often displayed as bar plots where the height of each bar represents the probability of that outcome. When the outcomes are numerical, we place the bars at their numerical locations, creating a plot that resembles a histogram.

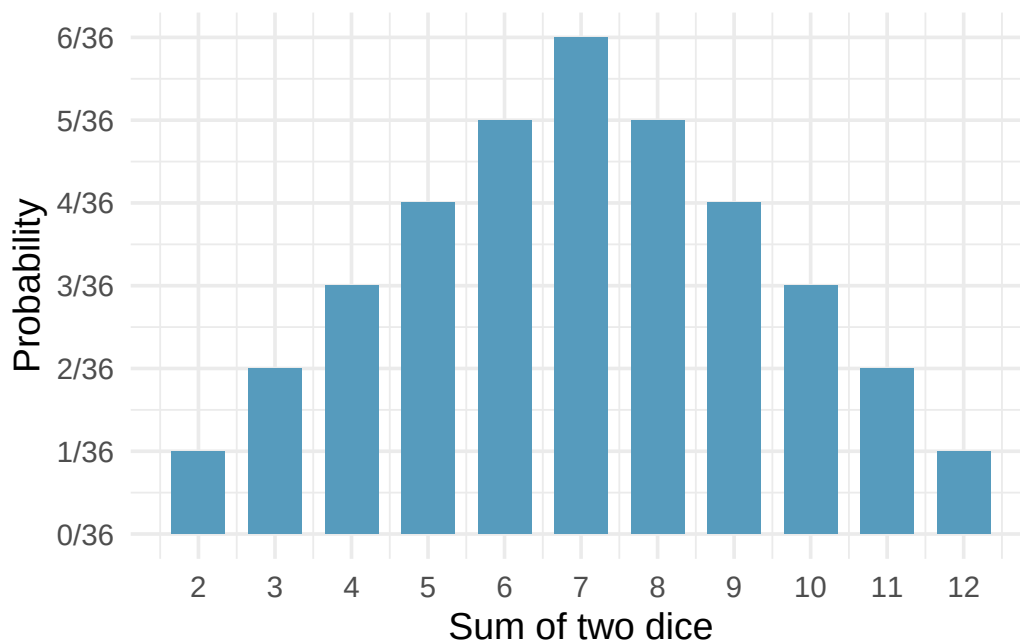


Figure 23.1: Probability distribution for the sum of two dice displayed as a bar plot.

23.2.4 Building intuition with simulation

We can also *estimate* a probability distribution by simulating the random process many times and recording the relative frequencies of each outcome. For example, if we simulate rolling two dice 10,000 times and record the sum each time, the histogram of those 10,000 sums will closely approximate the theoretical probability distribution. As the number of simulations increases, the approximation improves — this is the Law of Large Numbers in action.

This connection between simulation and theory is one of the most important ideas in the course. Every time we used simulation in Parts II and III to build a null distribution or a bootstrap distribution, we were approximating a theoretical probability distribution.

23.3 Cumulative distribution functions

The **cumulative distribution function** (CDF) of a random variable X is the function:

$$F(x) = P(X \leq x)$$

The CDF gives the probability that X takes a value less than or equal to x .

The CDF is a useful alternative to the PMF for answering “at most” or “at least” types of questions. For a discrete random variable, the CDF is a step function that jumps at each possible value of X .

Consider the sum of two dice. The CDF at a few key values:

x	2	3	4	5	6	7	8	9	10	11	12
$F(x) =$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{15}{36}$	$\frac{21}{36}$	$\frac{26}{36}$	$\frac{30}{36}$	$\frac{33}{36}$	$\frac{35}{36}$	$\frac{36}{36}$
$P(X \leq x)$											

Key properties of any CDF:

- $F(x)$ is non-decreasing: if $a < b$, then $F(a) \leq F(b)$.
- $F(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F(x) \rightarrow 1$ as $x \rightarrow +\infty$.
- $P(a < X \leq b) = F(b) - F(a)$.

The CDF becomes especially important for continuous random variables, where individual values have zero probability and we must always work with intervals.

Using the CDF for the sum of two dice above, find: (a) $P(X \leq 5)$. (b) $P(X > 9)$. (c) $P(4 < X \leq 8)$.

Show answer

- (a) $F(5) = 10/36$. (b) $P(X > 9) = 1 - F(9) = 1 - 30/36 = 6/36 = 1/6$. (c) $P(4 < X \leq 8) = F(8) - F(4) = 26/36 - 6/36 = 20/36 = 5/9$.

23.4 Continuous random variables

A **continuous random variable** can take any value in some interval of real numbers. Its probability distribution is described by a smooth curve called a **probability density function** (PDF).

Examples of continuous random variables:

- The height of a randomly selected adult
- The time until the next bus arrives
- The temperature at noon tomorrow

23.4.1 From histograms to density curves

When we have a very large sample and use very narrow bins in a histogram, the outline of the histogram starts to look like a smooth curve. This smooth curve is a probability density function.

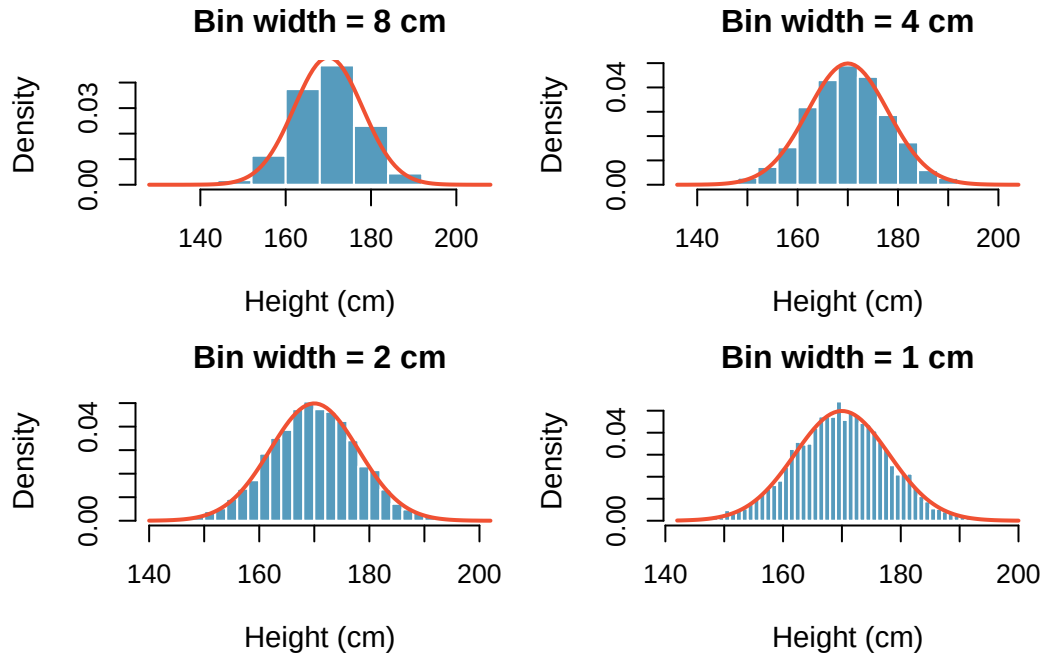


Figure 23.2: The transition from a histogram to a smooth density curve as the bin width decreases and sample size increases.

Properties of a probability density function (PDF):

1. The density curve is always on or above the horizontal axis: $f(x) \geq 0$ for all x .
2. The total area under the curve equals 1.
3. The probability that X falls between a and b equals the area under the curve between a and b : $P(a \leq X \leq b) = \text{area between } a \text{ and } b$.

23.4.2 Probabilities from continuous distributions

For a continuous random variable, probabilities are *areas* under the density curve.

The heights of US adults follow a continuous distribution. What proportion of the population has heights between 180 cm and 185 cm?

Using the density curve, we find the area under the curve between 180 and 185:

$$P(180 \leq X \leq 185) = \text{area between 180 and 185} \approx 0.1157$$

About 11.6% of US adults are between 180 and 185 cm tall.

A key difference: the probability of an exact value is zero for continuous random variables.

For a continuous random variable, $P(X = c) = 0$ for any specific value c . There is no area under the curve at a single point. This means:

$$P(X \leq a) = P(X < a)$$

This is *not* true for discrete random variables, where individual outcomes carry positive probability.

If $P(180 \leq \text{height} \leq 185) = 0.1157$ for a randomly selected US adult, and three adults are selected at random: (a) What is the probability all three have heights between 180 and 185 cm? (b) What is the probability none of them do?

Show answer

(a) Since the selections are independent: $0.1157^3 \approx 0.0015$. (b) $(1 - 0.1157)^3 = 0.8843^3 \approx 0.692$.

23.4.3 Discrete vs. continuous: a summary

Feature	Discrete	Continuous
Values	Finite or countable list	Any value in an interval
Described by	Probability mass function (PMF)	Probability density function (PDF)
$P(X = c)$	Can be positive	Always equals 0
Probabilities	Sum probabilities of individual values	Find areas under the curve
Visualization	Bar plot	Smooth curve
Example	Number of coin flips until heads	Height of a randomly selected person

In this course, we will work primarily with two specific distributions:

- The **binomial distribution** (Chapter 24) — a discrete distribution
- The **normal distribution** (Chapter 25) — a continuous distribution

Understanding the general concepts from this chapter will make those specific distributions much easier to work with.

23.5 Chapter review

23.5.1 Summary

- A **random variable** assigns a numerical value to each outcome of a random process.
- **Discrete random variables** take on a finite or countable set of values. Their distributions are described by probability mass functions (PMFs).
- **Continuous random variables** can take any value in an interval. Their distributions are described by probability density functions (PDFs), and probabilities are calculated as areas under the curve.
- The **cumulative distribution function** (CDF) gives $P(X \leq x)$ and works for both discrete and continuous random variables.
- For continuous random variables, $P(X = c) = 0$ for any specific value c .
- A valid probability distribution has non-negative probabilities that sum (or integrate) to 1.

23.5.2 Exercises

Exercises for this chapter are under development.

Chapter 24

Expected Value and Variance

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

How much revenue should a bookstore expect per student? How much variability should a stock investor anticipate in their portfolio? These questions require us to summarize a random variable's distribution with just two numbers: its **expected value** (center) and **variance** (spread). In this chapter, we define these quantities, develop useful computational properties, and extend them to linear combinations of random variables. These tools will be essential when we study the binomial and normal distributions in the chapters that follow.

24.1 Expected value of a random variable

We introduced random variables in Chapter 22. Now we develop the tools to summarize them numerically, starting with the expected value — the long-run average of a random variable.

Two books are assigned for a statistics class: a textbook (\$137) and a study guide (\$33). From past experience, the bookstore knows that 20% of enrolled students buy neither book, 55% buy the textbook only, and 25% buy both books. If there are 100 students enrolled, how many books should the bookstore expect to sell?

About 20 students buy 0 books, 55 buy 1 book (55 books total), and 25 buy 2 books (50 books total). The bookstore should expect to sell about $0 + 55 + 50 = 105$ books.

What is the average revenue per student?

The expected total revenue is $\$0 \times 20 + \$137 \times 55 + \$170 \times 25 = \$11,785$, and there are 100 students. The expected revenue per student is $\$11,785/100 = \117.85 .

Notice that the “average revenue per student” was computed by multiplying each outcome by its probability and summing:

$$0 \times 0.20 + 137 \times 0.55 + 170 \times 0.25 = 117.85$$

This is the **expected value** of the random variable X .

Expected value of a discrete random variable. If X takes outcomes $x_1, x_2, \dots, x_k$ with probabilities $P(X = x_1), P(X = x_2), \dots, P(X = x_k)$, the **expected value** of X is:

$$E(X) = x_1 \cdot P(X = x_1) + x_2 \cdot P(X = x_2) + \dots + x_k \cdot P(X = x_k) = \sum_{i=1}^k x_i P(X = x_i)$$

The Greek letter μ may be used in place of $E(X)$.

The expected value represents the **long-run average** of the random variable. If we were to observe the random process many, many times, the average of all observed values would converge to $E(X)$.

24.1.1 The expected value as a center of gravity

In physics, the expected value corresponds to the **center of gravity** (or balance point) of a distribution. Imagine cutting out the shape of a probability distribution from a piece of cardboard. The expected value is the point where you could balance the cutout on the tip of your finger. This is why we sometimes call $E(X)$ the “mean” — it is exactly analogous to the sample mean $\bar{x}$, but for a probability distribution rather than a data set.

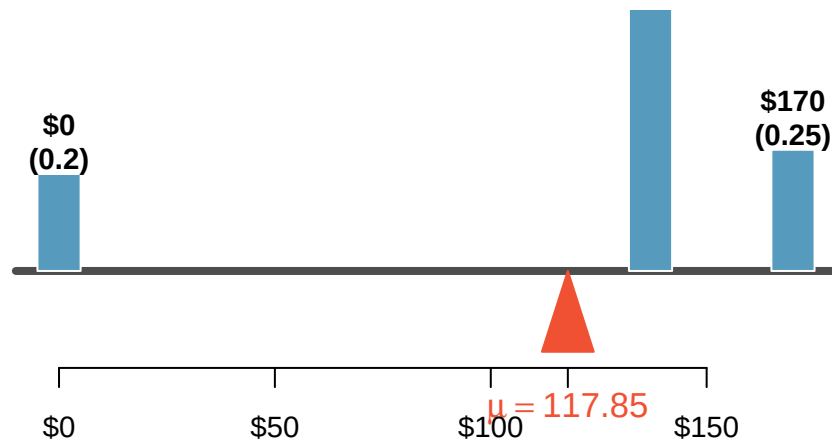


Figure 24.1: A weight system representing the bookstore distribution balanced at the mean of \$117.85.

A campus shuttle arrives every 10 minutes. The number of students boarding follows this distribution:

x	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10

What is the expected number of students who board?

Show answer

$E(X) = 0(0.10) + 1(0.25) + 2(0.35) + 3(0.20) + 4(0.10) = 0 + 0.25 + 0.70 + 0.60 + 0.40 = 1.95$. On average, about 1.95 students board the shuttle at this stop.

The expected value does not have to be a value that the random variable can actually take. In the shuttle example, $E(X) = 1.95$, but you can never have 1.95 students board. The expected value is a long-run average, not a prediction of any single outcome.

24.2 Variance of a random variable

The expected value tells us where the center of a distribution lies, but it says nothing about the *spread*. The bookstore might want to know not just the average revenue per student, but how much that revenue varies from student to student.

Variance of a discrete random variable. If X takes outcomes $x_1, \dots, x_k$ with probabilities $P(X = x_1), \dots, P(X = x_k)$ and expected value $\mu = E(X)$, then the **variance** of X is:

$$\text{Var}(X) = \sigma^2 = \sum_{i=1}^k (x_i - \mu)^2 P(X = x_i)$$

The **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.

The variance is the average squared deviation from the mean, weighted by the probabilities. The standard deviation has the same units as X , which makes it more interpretable.

Compute the variance and standard deviation of the bookstore revenue per student.

We organize the calculation in a table with $\mu = 117.85$:

	$x_1 = 0$	$x_2 = 137$	$x_3 = 170$	Total
$P(X = x_i)$	0.20	0.55	0.25	
$x_i - \mu$	-117.85	19.15	52.15	
$(x_i - \mu)^2$	13,888.6	366.7	2,719.6	
$(x_i - \mu)^2 \cdot P(X = x_i)$	2,777.7	201.7	679.9	3,659.3

The variance is $\sigma^2 = 3,659.3$ and the standard deviation is $\sigma = \sqrt{3,659.3} = \60.49 .

The bookstore also sells a chemistry textbook (\$159) and supplement (\$41). From past experience, 15% of chemistry students buy neither, 25% buy the textbook only, and 60% buy both. Let Y be the revenue from a single chemistry student. (a) Write out the probability distribution of Y . (b) Compute $E(Y)$. (c) Compute $\text{SD}(Y)$.

Show answer

- (a) Y takes values \$0, \$159, \$200 with probabilities 0.15, 0.25, 0.60. (b) $E(Y) = 0(0.15) + 159(0.25) + 200(0.60) = 0 + 39.75 + 120 = \159.75 . (c) $\text{Var}(Y) = (0 - 159.75)^2(0.15) + (159 - 159.75)^2(0.25) + (200 - 159.75)^2(0.60) = 3828.0 + 0.1 + 972.0 = 4800.1$, so $\text{SD}(Y) = \sqrt{4800.1} \approx \69.28 .

24.3 Properties of expected value

The expected value has several important algebraic properties that make it easy to work with. These properties follow from the definition but are worth stating explicitly because we will use them frequently.

Properties of expected value. For a random variable X and constants a and b :

1. **Adding a constant shifts the mean:** $E(X + b) = E(X) + b$
2. **Multiplying by a constant scales the mean:** $E(aX) = a \cdot E(X)$
3. **General linear transformation:** $E(aX + b) = a \cdot E(X) + b$

Why these make sense: If every student's textbook cost increases by \$10 (adding a constant $b = 10$), then the average cost also increases by \$10. If the bookstore marks up all prices by 20% (multiplying by $a = 1.20$), then the average revenue also increases by 20%.

Suppose the bookstore offers a 10% discount to students who buy both books. The discounted price is $0.90 \times X$, where X is the original price. What is the expected revenue per student under the discount?

Using the property $E(aX) = aE(X)$:

$$E(0.90X) = 0.90 \times E(X) = 0.90 \times 117.85 = \$106.07$$

The expected revenue per student drops from \$117.85 to \$106.07.

The bookstore charges a flat \$5 processing fee in addition to book costs. If X is the cost of books and the total charge is $Y = X + 5$, what is $E(Y)$?

Show answer

$$E(Y) = E(X + 5) = E(X) + 5 = 117.85 + 5 = \$122.85.$$

24.4 Properties of variance

Variance has different algebraic properties than expected value. The most important distinction: adding a constant does not change the variance.

Properties of variance. For a random variable X and constants a and b :

1. **Adding a constant does not change variance:** $\text{Var}(X + b) = \text{Var}(X)$
2. **Multiplying by a constant scales variance by the square:** $\text{Var}(aX) = a^2 \cdot \text{Var}(X)$
3. **General linear transformation:** $\text{Var}(aX + b) = a^2 \cdot \text{Var}(X)$

For standard deviations:

$$\text{SD}(aX + b) = |a| \cdot \text{SD}(X)$$

Why adding a constant does not change variance: If every student pays \$5 more, the *spread* of payments does not change — everyone's payment shifts by the same amount. The center moves, but the variability stays the same.

Why multiplying scales by the square: If every price is doubled ($a = 2$), then deviations from the mean are also doubled, so *squared* deviations are multiplied by $2^2 = 4$.

For the bookstore revenue with $\text{Var}(X) = 3,659.3$ and $\sigma_X = \$60.49$:

- If a \$5 fee is added: $\text{Var}(X + 5) = \text{Var}(X) = 3,659.3$ and $\text{SD}(X + 5) = \$60.49$.

- If a 10% discount is applied: $\text{Var}(0.90X) = 0.90^2 \times 3,659.3 = 0.81 \times 3,659.3 = 2,964.0$ and $\text{SD}(0.90X) = 0.90 \times 60.49 = \54.44 .

Suppose temperatures in Celsius are converted to Fahrenheit using $F = 1.8C + 32$. If the standard deviation of daily high temperatures is $\sigma_C = 5^\circ\text{C}$, what is the standard deviation in Fahrenheit?

Show answer

$\text{SD}(1.8C + 32) = 1.8 \times \text{SD}(C) = 1.8 \times 5 = 9^\circ\text{F}$. The additive constant 32 does not affect the spread.

24.5 Linear combinations of random variables

Sometimes the total outcome is best described as a sum of several random variables. For instance, a person's total weekly commute time is the sum of five daily commute times.

A **linear combination** of random variables X and Y is an expression of the form $aX + bY$, where a and b are fixed constants.

John travels to work five days a week. Let X_i represent his travel time on day i . His total weekly commute time is:

$$W = X_1 + X_2 + X_3 + X_4 + X_5$$

If the average daily commute is 18 minutes, the expected weekly total is:

$$E(W) = E(X_1) + E(X_2) + \cdots + E(X_5) = 18 + 18 + 18 + 18 + 18 = 90 \text{ minutes}$$

24.5.1 Expected value of a linear combination

Expected value of a linear combination. For random variables X and Y with constants a and b :

$$E(aX + bY) = a \cdot E(X) + b \cdot E(Y)$$

This generalizes to any number of random variables:

$$E(a_1X_1 + a_2X_2 + \cdots + a_kX_k) = a_1E(X_1) + a_2E(X_2) + \cdots + a_kE(X_k)$$

This property holds whether or not the random variables are independent.

Elena is selling a TV at auction (expected profit X : \$175) and buying a toaster oven (expected cost Y : \$23). Her net gain is $X - Y$. What is her expected net gain?

Show answer

$$E(X - Y) = E(X) - E(Y) = 175 - 23 = \$152.$$

Leonard has invested \$6,000 in Caterpillar stock (CAT) and \$2,000 in Exxon Mobil stock (XOM). If X represents CAT's monthly percentage return (in decimal form) and Y represents XOM's return, his portfolio change is $6000X + 2000Y$. If $E(X) = 0.020$ and $E(Y) = 0.002$:

$$E(6000X + 2000Y) = 6000 \times 0.020 + 2000 \times 0.002 = \$124$$

Leonard expects to gain \$124 per month on average.

24.5.2 Variance of a linear combination

Variance of a linear combination (independent case). If X and Y are **independent** random variables, then:

$$\text{Var}(aX + bY) = a^2 \cdot \text{Var}(X) + b^2 \cdot \text{Var}(Y)$$

The standard deviation is: $\text{SD}(aX + bY) = \sqrt{a^2 \text{Var}(X) + b^2 \text{Var}(Y)}$.

Independence is required for the variance formula. The formula $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$ is only valid when X and Y are independent. If the variables are not independent, an additional covariance term is needed.

However, the expected value formula $E(aX + bY) = aE(X) + bE(Y)$ always holds, regardless of independence.

John's daily commute has a standard deviation of 4 minutes, and the commute times for different days are independent. What is the standard deviation of his weekly commute time $W = X_1 + X_2 + X_3 + X_4 + X_5$?

Each coefficient is 1, and $\text{Var}(X_i) = 4^2 = 16$:

$$\text{Var}(W) = 1^2(16) + 1^2(16) + 1^2(16) + 1^2(16) + 1^2(16) = 5 \times 16 = 80$$

$$\text{SD}(W) = \sqrt{80} \approx 8.94 \text{ minutes}$$

Notice that the standard deviation of the sum (8.94 minutes) is *not* five times the daily standard deviation (which would be 20 minutes). Variabilities do not add up as simply as means do — we must add *variances*, not standard deviations.

Leonard's portfolio has $\text{Var}(X) = 0.0057$ and $\text{Var}(Y) = 0.0021$ (where the stocks are approximately independent). What is the standard deviation of his monthly return?

$$\text{Var}(6000X + 2000Y) = 6000^2 \times 0.0057 + 2000^2 \times 0.0021 = 205,200 + 8,400 = 213,600$$

$$\text{SD} = \sqrt{213,600} \approx \$462$$

While an average monthly return of \$124 sounds nice, a standard deviation of \$462 means the returns are highly volatile.

Elena's TV sale has $\text{SD}(X) = \$25$ and her toaster oven purchase has $\text{SD}(Y) = \$8$. Assuming the auctions are independent, compute the standard deviation of her net gain $X - Y$.

Show answer

Note that $X - Y = 1 \cdot X + (-1) \cdot Y$. So $\text{Var}(X - Y) = 1^2 \times 25^2 + (-1)^2 \times 8^2 = 625 + 64 = 689$. Thus $\text{SD}(X - Y) = \sqrt{689} \approx \26.25 . The negative coefficient is eliminated when we square it — subtracting a random variable introduces *more* variability, not less.

24.5.3 Why variances add even when we subtract

A common source of confusion: the variance of a difference is the *sum* of the variances (not the difference). This makes sense intuitively. If you buy a bag of apples and a bag of oranges, the uncertainty in the total weight is a combination of both uncertainties — regardless of whether you are adding or subtracting the weights. Two sources of randomness always create more total uncertainty.

24.6 Chapter review

24.6.1 Summary

- The **expected value** $E(X) = \sum x_i P(X = x_i)$ is the long-run average of a random variable.
- The **variance** $\text{Var}(X) = \sum (x_i - \mu)^2 P(X = x_i)$ measures spread; the **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.
- **Expected value properties:** $E(aX + b) = aE(X) + b$ (always holds).
- **Variance properties:** $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (adding a constant does not change variance).
- **Linear combinations:** $E(aX + bY) = aE(X) + bE(Y)$ (always). $\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y)$ (requires independence).
- Variances always add, even when random variables are subtracted.

24.6.2 Exercises

Exercises for this chapter are under development.

Chapter 25

Binomial Distribution

Draft Chapter. This chapter is part of UWL's STAT 145 curriculum and is under active development. Content is substantially complete but may undergo revisions. Feedback welcome.

How many of 400 randomly sampled adults are left-handed? How many of 20 randomly selected parts from an assembly line are defective? These “count of successes” questions arise constantly in statistics and are answered by the **binomial distribution**. In this chapter, we identify the four conditions that define a binomial setting, develop the formula for computing binomial probabilities, derive the mean and standard deviation, and show how the normal distribution can approximate binomial probabilities when the sample size is large.

StatBench tools for this chapter: Use the **Binomial Distribution Calculator** to compute binomial probabilities, visualize the distribution, and explore how changing n and p affects the shape.

[Launch Binomial Distribution Calculator](#)

25.1 The binomial setting

The binomial distribution describes the number of successes in a fixed number of independent trials, where each trial has only two possible outcomes.

The four binomial conditions. A random variable X follows a binomial distribution if:

1. **Fixed number of trials:** The number of trials n is fixed in advance.
2. **Binary outcomes:** Each trial results in one of two outcomes, which we label “success” and “failure.”
3. **Independence:** The trials are independent — the outcome of one trial does not affect the outcome of any other trial.
4. **Constant probability:** The probability of success p is the same for every trial.

If all four conditions are met, we say X follows a binomial distribution with parameters n and p , written $X \sim \text{Binomial}(n, p)$.

A health insurance company found that 70% of the people they insure stay below their deductible in any given year. Consider a random sample of four individuals. Does the number who stay below their deductible follow a binomial distribution?

1. **Fixed n :** Yes, $n = 4$ individuals.
2. **Binary:** Yes, each person either stays below the deductible (success) or exceeds it (failure).

3. **Independence:** Yes, because the sample is random from a large population.
4. **Constant p :** Yes, each person has a 70% chance of staying below the deductible.

All four conditions are met, so $X \sim \text{Binomial}(n = 4, p = 0.70)$.

The labels “success” and “failure” are just conventions — a “success” does not have to be a good outcome. If we are counting defective parts, a “success” might be finding a defective part. The mathematical framework does not depend on which outcome we call a success, as long as we are consistent.

The probability that a random smoker will develop a severe lung condition is about 0.3. If you have 4 friends who smoke, are the conditions for the binomial model satisfied?

Show answer

This is questionable. If the friends know each other, the independence condition may be violated. Acquaintances may have similar smoking habits or might influence each other. If, however, we could treat them as a random sample of smokers, the conditions would be approximately met.

25.1.1 The Bernoulli distribution: a single trial

A single trial with two outcomes is called a **Bernoulli trial**, and the resulting random variable is a **Bernoulli random variable**.

If X is a random variable that takes value 1 with probability p (success) and 0 with probability $1 - p$ (failure), then X is a **Bernoulli random variable** with:

$$\mu = p \quad \sigma = \sqrt{p(1 - p)}$$

A binomial random variable is simply the sum of n independent Bernoulli random variables, each with the same probability p .

25.2 Computing binomial probabilities

25.2.1 Building the formula

Let's find the probability that exactly 1 of 4 randomly selected insured individuals exceeds the deductible (so 3 “successes” — staying below — and 1 “failure”).

Consider the specific scenario where person A exceeds and persons B, C, D do not:

$$P(A = \text{exceed}, B = \text{not}, C = \text{not}, D = \text{not}) = (0.3)(0.7)(0.7)(0.7) = (0.7)^3(0.3)^1 = 0.103$$

But there are four such scenarios — any one of the four people could be the one who exceeds. Each scenario has the same probability. So:

$$P(\text{exactly 3 successes in 4 trials}) = 4 \times (0.7)^3(0.3)^1 = 0.412$$

The general pattern is:

$$P(X = k) = [\text{number of arrangements}] \times [\text{probability of one arrangement}]$$

25.2.2 The binomial coefficient

The number of ways to arrange k successes in n trials is given by the **binomial coefficient**:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

read as “ n choose k ,” where $n!$ (read “ n factorial”) means $n \times (n-1) \times \cdots \times 2 \times 1$, with the special case $0! = 1$.

For our example: $\binom{4}{3} = \frac{4!}{3! \cdot 1!} = \frac{24}{6 \times 1} = 4$, confirming the four arrangements we identified.

25.2.3 The binomial probability formula

Binomial probability formula. If $X \sim \text{Binomial}(n, p)$, the probability of exactly k successes is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

for $k = 0, 1, 2, \dots, n$.

What is the probability that 5 of 8 randomly selected insured individuals will not exceed the deductible?

With $n = 8$, $k = 5$, and $p = 0.70$:

$$P(X = 5) = \binom{8}{5} (0.70)^5 (0.30)^3$$

Computing the binomial coefficient:

$$\binom{8}{5} = \frac{8!}{5! \cdot 3!} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

And the probability of a single arrangement: $(0.70)^5 (0.30)^3 \approx 0.00454$.

$$P(X = 5) = 56 \times 0.00454 \approx 0.254$$

Suppose 4 friends who smoke can be treated as a random sample, with $P(\text{severe lung condition}) = 0.3$ for each. (a) What is the probability that none develop a severe lung condition? (b) What is the probability that exactly one does? (c) What is the probability that at most one does?

Show answer

(a) $P(X = 0) = \binom{4}{0} (0.3)^0 (0.7)^4 = 1 \times 1 \times 0.2401 = 0.2401$. (b) $P(X = 1) = \binom{4}{1} (0.3)^1 (0.7)^3 = 4 \times 0.3 \times 0.343 = 0.4116$. (c) $P(X \leq 1) = P(X = 0) + P(X = 1) = 0.2401 + 0.4116 = 0.6517$.

What is the probability that at least 2 of the 4 smoking friends will develop a severe lung condition?

Show answer

Using the complement: $P(X \geq 2) = 1 - P(X \leq 1) = 1 - 0.6517 = 0.3483$.

25.2.4 Computing binomial probabilities in practice

For small values of n and k , the binomial formula can be computed by hand. For larger problems, use technology:

- **R:** `dbinom(k, n, p)` for $P(X = k)$; `pbinom(k, n, p)` for $P(X \leq k)$
- **Calculator:** Most graphing calculators have a `binompdf` and `binomcdf` function
- **StatBench:** The Binomial Distribution Explorer computes probabilities and visualizes the distribution

For example, to find $P(X \leq 42)$ when $X \sim \text{Binomial}(400, 0.15)$, computing 43 individual probabilities by hand would be impractical. In R:

```
pbinom(42, 400, 0.15)
```

returns approximately 0.0054.

25.3 Mean and standard deviation

Mean and standard deviation of a binomial distribution. If $X \sim \text{Binomial}(n, p)$, then:

$$\mu = np \quad \sigma = \sqrt{np(1-p)}$$

These formulas follow from the fact that a binomial random variable is the sum of n independent Bernoulli random variables, each with mean p and variance $p(1-p)$.

If 40 individuals are randomly sampled and each has a 70% chance of staying below their insurance deductible, how many would you expect to stay below? What is the standard deviation?

$$\mu = np = 40 \times 0.70 = 28$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{40 \times 0.70 \times 0.30} = \sqrt{8.4} \approx 2.9$$

We expect about 28 individuals to stay below their deductible, give or take about 2.9. Using the rough “95% within 2 standard deviations” guideline, we would expect between about 22 and 34 individuals in most samples.

Suppose 7 friends who smoke can be treated as a random sample, with $p = 0.3$ for developing a severe lung condition. How many would you expect to develop the condition? What is the standard deviation?

Show answer

$\mu = np = 7 \times 0.3 = 2.1$. $\sigma = \sqrt{7 \times 0.3 \times 0.7} = \sqrt{1.47} \approx 1.21$. You would expect about 2 of the 7 friends to develop a severe lung condition.

25.3.1 Simulation perspective: Does the formula match reality?

We can verify the binomial formulas using simulation. Consider $X \sim \text{Binomial}(n = 20, p = 0.3)$:

- **Theory:** $\mu = 20 \times 0.3 = 6$ and $\sigma = \sqrt{20 \times 0.3 \times 0.7} = \sqrt{4.2} \approx 2.05$
- **Simulation:** Generate 10,000 binomial random samples, each with $n = 20$ and $p = 0.3$. The sample mean of the 10,000 counts will be very close to 6, and the sample standard deviation will be very close to 2.05.

This is another instance of the Law of Large Numbers: the simulated distribution converges to the theoretical distribution as we increase the number of simulations.

Try it in StatBench: Use the Binomial Distribution Explorer to set $n = 20$ and $p = 0.3$. Simulate 1,000 binomial experiments and compare the histogram of results to the theoretical formula. As you increase the number of simulations, the histogram converges to the theoretical distribution.

25.4 Normal approximation to the binomial

The binomial formula can be cumbersome for large n , especially when computing cumulative probabilities like $P(X \leq 42)$ for $n = 400$. Fortunately, when n is large enough, the binomial distribution is well approximated by the normal distribution.

Approximately 15% of the US population smokes cigarettes. A local government surveyed 400 randomly selected individuals and found only 42 smokers. If the true proportion is 0.15, what is the probability of observing 42 or fewer smokers?

The exact answer requires summing 43 binomial probabilities, which gives $P(X \leq 42) = 0.0054$.

As the sample size increases, the binomial distribution becomes increasingly bell-shaped and symmetric. This is illustrated in the figure below.

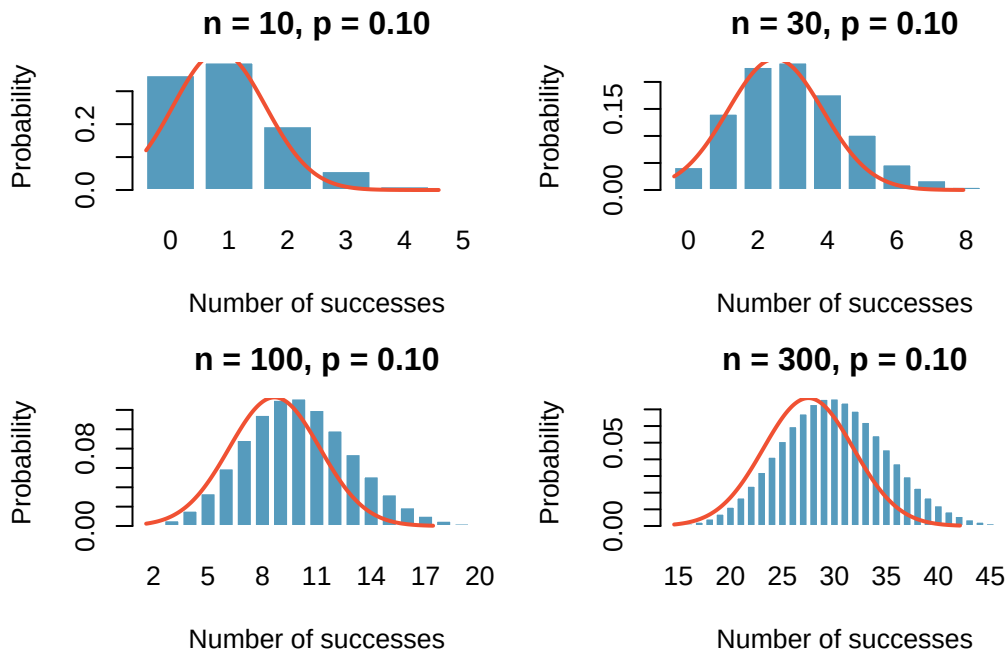


Figure 25.1: Binomial distributions with $p = 0.10$ and sample sizes $n = 10, 30, 100$, and 300 . As n increases, the distribution approaches a normal curve.

Normal approximation to the binomial. The binomial distribution with parameters n and p is approximately normal when both $np \geq 10$ and $n(1 - p) \geq 10$. The approximating normal distribution has:

$$\mu = np \quad \sigma = \sqrt{np(1 - p)}$$

Use the normal approximation to estimate $P(X \leq 42)$ for $X \sim \text{Binomial}(400, 0.15)$.

First, check the conditions: $np = 400 \times 0.15 = 60 \geq 10$ and $n(1 - p) = 400 \times 0.85 = 340 \geq 10$. The normal approximation is appropriate.

The approximating distribution is $N(\mu = 60, \sigma = 7.14)$, where $\sigma = \sqrt{400 \times 0.15 \times 0.85} = 7.14$.

Compute the Z-score:

$$Z = \frac{42 - 60}{7.14} = \frac{-18}{7.14} = -2.52$$

Using the standard normal distribution: $P(Z \leq -2.52) = 0.0059$.

This is very close to the exact binomial probability of 0.0054.

A coin is flipped 100 times. Use the normal approximation to find the probability of getting 60 or more heads.

Show answer

Check conditions: $np = 100 \times 0.5 = 50 \geq 10$ and $n(1 - p) = 50 \geq 10$. The approximation is appropriate. $\mu = 50, \sigma = \sqrt{100 \times 0.5 \times 0.5} = 5$. $Z = (60 - 50)/5 = 2$. $P(Z \geq 2) = 1 - 0.9772 = 0.0228$. There is about a 2.3% chance of getting 60 or more heads in 100 flips of a fair coin.

25.4.1 Connecting to inference

The normal approximation to the binomial is not just a computational shortcut — it is the theoretical basis for the inference methods we developed in Parts II and III. When we studied the sampling distribution of the sample proportion $\hat{p}$ in Chapter 9, we were implicitly using the fact that the count of successes X in a sample of size n follows a binomial distribution, and that this binomial distribution is approximately normal for large n .

Specifically, if $X \sim \text{Binomial}(n, p)$, then $\hat{p} = X/n$, and the normal approximation tells us:

$$\hat{p} \approx N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

This is exactly the result we used when constructing confidence intervals and performing hypothesis tests for proportions in Chapter 13. The theory developed in this chapter *explains* why those methods work.

25.5 Chapter review

25.5.1 Summary

- The **binomial distribution** counts the number of successes in n independent trials, each with the same probability of success p .
- **Four conditions:** fixed n , binary outcomes, independence, constant p .
- **Binomial probability formula:** $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
- **Mean and standard deviation:** $\mu = np$ and $\sigma = \sqrt{np(1 - p)}$.
- When $np \geq 10$ and $n(1 - p) \geq 10$, the binomial distribution is well approximated by the normal distribution $N(np, \sqrt{np(1 - p)})$.
- The normal approximation to the binomial provides the theoretical justification for the inference methods on proportions from Chapter 13.

25.5.2 Exercises

Exercises for this chapter are under development.

Chapter 26

Normal Distribution

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

The normal distribution is the most important probability model in statistics. Its familiar bell shape appears in an extraordinary range of settings — from SAT scores and adult heights to measurement errors and stock returns. In this chapter, we study the normal distribution as a **probability model** for individual observations: we learn to compute probabilities, find percentiles, and assess whether data plausibly come from a normal distribution. This complements Chapter 10, which uses the normal distribution as an approximation for **sampling distributions** in inference. Here, the focus is squarely on probability.

StatBench tools for this chapter: Use the **Normal Distribution Calculator** to visualize normal curves, shade areas, compute probabilities, and find percentiles. Adjust μ and σ to see how the distribution changes.

[Launch Normal Distribution Calculator](#)

26.1 The normal distribution as a probability model

Among all distributions we encounter in practice, one is overwhelmingly the most common. The symmetric, unimodal, bell-shaped curve known as the **normal distribution** (also called the **Gaussian distribution**) arises in a remarkable variety of settings.

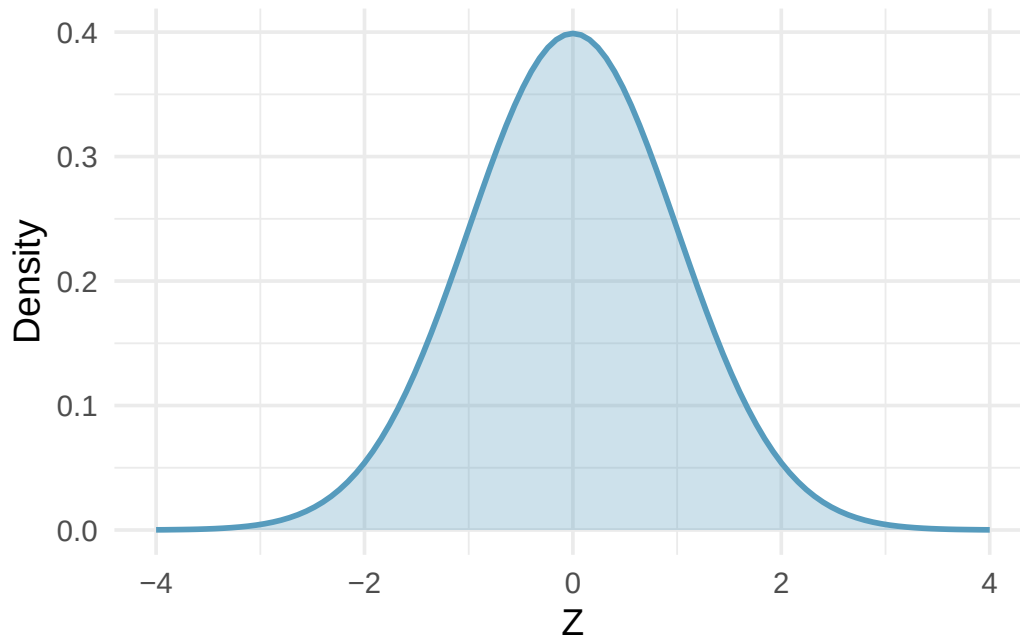


Figure 26.1: A standard normal curve — symmetric, unimodal, and bell-shaped.

Many variables are *nearly* normal, but none are *exactly* normal. The normal distribution, while not a perfect description of any single real-world variable, is an extraordinarily useful model for a wide variety of problems.

26.1.1 Parameters: mean and standard deviation

The normal distribution is completely specified by two **parameters**:

- The **mean** μ determines the center (location) of the bell curve.
- The **standard deviation** σ determines the spread (width) of the bell curve.

If a random variable X follows a normal distribution with mean μ and standard deviation σ , we write:

$$X \sim N(\mu, \sigma)$$

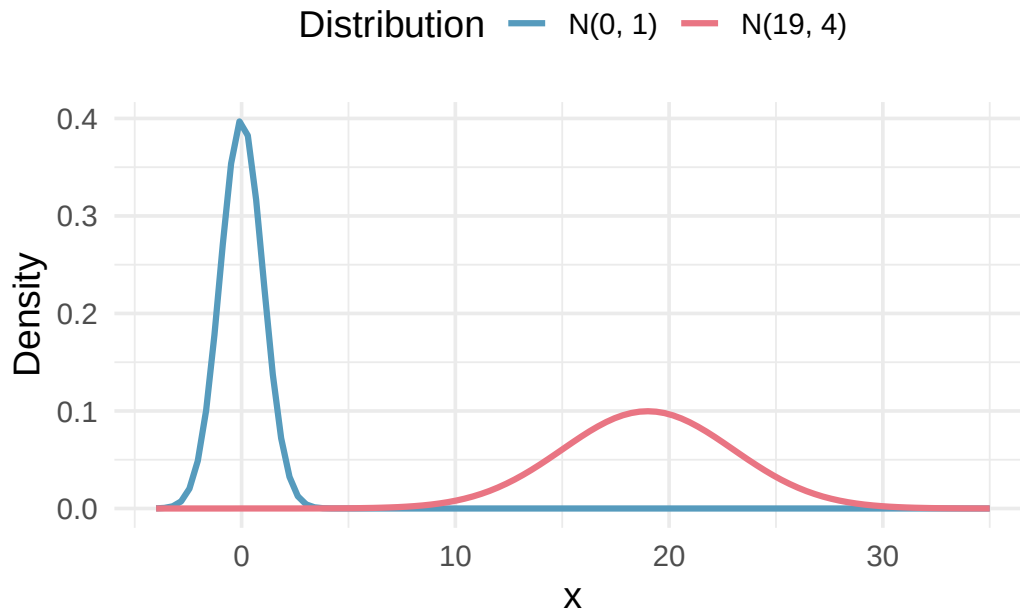


Figure 26.2: Two normal distributions: $N(0, 1)$ and $N(19, 4)$. Changing μ shifts the curve; changing σ stretches or compresses it.

The normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$ is called the **standard normal distribution**. It is often denoted $Z \sim N(0, 1)$.

Write the shorthand notation for normal distributions with: (a) mean 5 and standard deviation 3, (b) mean -100 and standard deviation 10, (c) mean 2 and standard deviation 9.

Show answer

(a) $N(\mu = 5, \sigma = 3)$. (b) $N(\mu = -100, \sigma = 10)$. (c) $N(\mu = 2, \sigma = 9)$.

26.1.2 The 68-95-99.7 rule

A useful rule of thumb describes how probability is distributed within a normal distribution:

The 68-95-99.7 rule. For a normal distribution:

- About **68%** of observations fall within 1 standard deviation of the mean ($\mu \pm \sigma$).
- About **95%** of observations fall within 2 standard deviations of the mean ($\mu \pm 2\sigma$).
- About **99.7%** of observations fall within 3 standard deviations of the mean ($\mu \pm 3\sigma$).

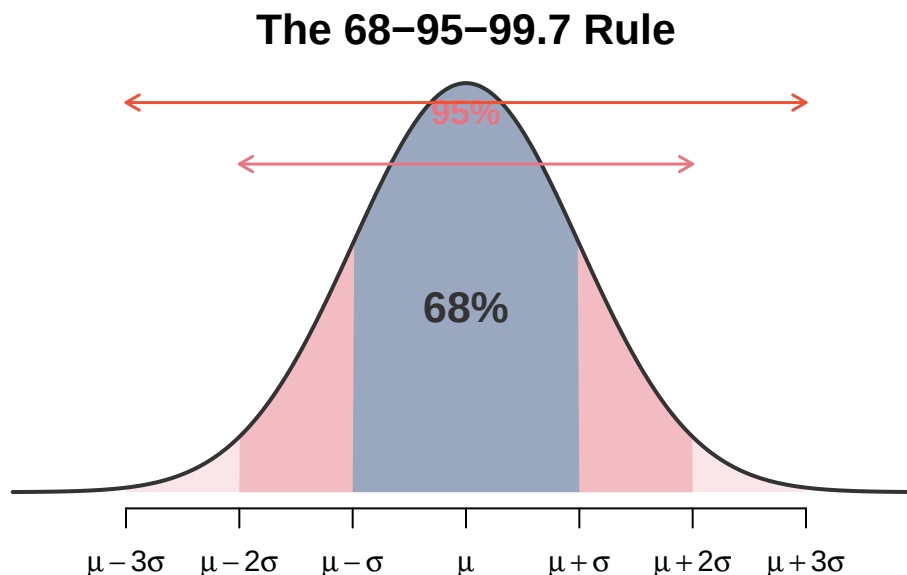


Figure 26.3: The 68-95-99.7 rule illustrated on a normal distribution.

It is possible for a normal random variable to fall 4, 5, or even more standard deviations from the mean, but such events are extremely rare. The probability of being more than 4 standard deviations from the mean is about 1 in 15,000.

SAT scores closely follow $N(\mu = 1100, \sigma = 200)$. (a) About what percent of test takers score between 700 and 1500? (b) What percent score between 1100 and 1500?

Show answer

- (a) 700 and 1500 are each 2 standard deviations from the mean, so about 95% of test takers score in this range. (b) By symmetry, half of the 95% falls above the mean: $95\%/2 = 47.5\%$.

26.1.3 Scope note: Chapter 25 vs. Chapter 10

In Chapter 10, you used the normal distribution as an approximation for **sampling distributions** — describing the behavior of statistics like $\bar{x}$ and $\hat{p}$ across many samples. Here, we study the normal distribution in a different role: as a **probability model for individual observations**. You use it to answer questions like:

- What proportion of adults are taller than 6 feet?
- What SAT score puts a student in the 95th percentile?
- Is it plausible that this data set came from a normal distribution?

The same mathematical object — the normal distribution — serves both purposes, but the *interpretation* is different. In Chapter 10, we modeled the behavior of sample statistics; here, we are modeling data.

26.2 Finding probabilities: $P(X < a)$

26.2.1 Z-scores: standardizing observations

To find probabilities from a normal distribution, we first convert the observation to a **Z-score**, which measures how many standard deviations it falls above or below the mean.

The **Z-score** of an observation x from a distribution with mean μ and standard deviation σ is:

$$Z = \frac{x - \mu}{\sigma}$$

Observations above the mean have positive Z-scores; observations below the mean have negative Z-scores; an observation equal to the mean has a Z-score of 0.

SAT scores follow $N(1100, 200)$ and ACT scores follow $N(21, 6)$. Ann scored 1300 on the SAT and Tom scored 24 on the ACT. Who performed better relative to their peers?

$$Z_{\text{Ann}} = \frac{1300 - 1100}{200} = 1.0 \quad Z_{\text{Tom}} = \frac{24 - 21}{6} = 0.5$$

Ann is 1 standard deviation above the SAT mean, while Tom is only 0.5 standard deviations above the ACT mean. Ann performed relatively better.

Head lengths of brushtail possums follow $N(92.6, 3.6)$ mm. Compute the Z-scores for possums with head lengths of 95.4 mm and 85.8 mm. Which is more unusual?

Show answer

$Z_1 = (95.4 - 92.6)/3.6 = 0.78$ and $Z_2 = (85.8 - 92.6)/3.6 = -1.89$. Since $|-1.89| > |0.78|$, the 85.8 mm head length is more unusual.

26.2.2 Computing tail areas

Once we have a Z-score, we can find the probability of observing a value at least that extreme. This is called a **tail area** or **tail probability**.

Always draw a picture first. For any normal probability problem, draw and label the normal curve, shade the area of interest, and then find the Z-score. The picture provides a visual estimate and helps avoid errors.

There are several ways to find the area under the normal curve:

1. **Statistical software** (most common in practice): In R, `pnorm(z)` returns the area to the left of z . For example, `pnorm(1)` returns 0.8413.
2. **Calculator**: Most graphing calculators have a `normalcdf` function.
3. **Probability tables**: Tables of standard normal areas are provided in many textbooks (see Appendix).

Shannon is a randomly selected SAT taker. What is the probability she scores at least 1190?

Step 1: Draw the picture — shade the area above 1190 on $N(1100, 200)$.

Step 2: Compute the Z-score.

$$Z = \frac{1190 - 1100}{200} = \frac{90}{200} = 0.45$$

Step 3: Find the area. Using software, the area to the *left* of $Z = 0.45$ is 0.6736. Since we want the area to the *right*:

$$P(X \geq 1190) = 1 - 0.6736 = 0.3264$$

The probability Shannon scores at least 1190 is about 32.6%.

Finding areas to the right. Most software returns the area to the *left* of a Z-score. To find the area to the *right*, subtract from 1:

$$P(X > a) = 1 - P(X \leq a)$$

Edward scored 1030 on his SAT. What is his percentile?

The **percentile** is the proportion of scores *below* his. Compute the Z-score:

$$Z = \frac{1030 - 1100}{200} = -0.35$$

Using software: $P(Z \leq -0.35) = 0.3632$. Edward is at the 36th percentile.

Stuart scored 1500 on the SAT. (a) What is his percentile? (b) What percent of test takers scored higher?

Show answer

- (a) $Z = (1500 - 1100)/200 = 2.0$. $P(Z \leq 2) = 0.9772$. Stuart is at the 97.7th percentile. (b) $1 - 0.9772 = 0.0228$, or about 2.3%.

26.2.3 Finding probabilities between two values

To find the probability between two values, compute the area between them by subtracting tail areas.

Adult male heights in the US follow $N(70.0'', 3.3'')$. What is the probability a randomly selected adult male is between 5'9'' (69'') and 6'2'' (74'')?

Z-scores:

$$Z_{69} = \frac{69 - 70}{3.3} = -0.30 \quad Z_{74} = \frac{74 - 70}{3.3} = 1.21$$

Areas:

- $P(Z \leq -0.30) = 0.3821$ (area to the left of 69'')
- $P(Z \leq 1.21) = 0.8869$ (area to the left of 74'')

Area between:

$$P(69 \leq X \leq 74) = 0.8869 - 0.3821 = 0.5048$$

About 50.5% of adult males are between 5'9'' and 6'2''.

SAT scores follow $N(1100, 200)$. What percent of test takers score between 1100 and 1400?

Show answer

$Z_{1100} = 0 \rightarrow P(Z \leq 0) = 0.5000$. $Z_{1400} = 1.5 \rightarrow P(Z \leq 1.5) = 0.9332$. Area between: $0.9332 - 0.5000 = 0.4332$, or about 43.3%.

26.3 Finding percentiles (inverse normal)

Sometimes we know a tail probability (or percentile) and need to find the corresponding observation value. This is the *inverse* of the normal probability problem.

Strategy:

1. Draw the picture and identify the tail area.
2. Find the Z-score corresponding to that tail area (using software, a calculator, or a table).
3. Use the Z-score formula to solve for x : $x = \mu + Z \cdot \sigma$.

Erik's height is at the 40th percentile of the US adult male height distribution $N(70.0, 3.3)$. How tall is he?

Step 1: The lower tail has area 0.40.

Step 2: Using software, the Z-score corresponding to the 40th percentile is $Z = -0.25$.

Step 3: Solve for Erik's height:

$$x = \mu + Z \cdot \sigma = 70.0 + (-0.25)(3.3) = 70.0 - 0.825 = 69.18 \text{ inches}$$

Erik is about 5'9''.

What is the height at the 82nd percentile for adult males?

Using software, the Z-score at the 82nd percentile is $Z = 0.92$. Then:

$$x = 70.0 + 0.92 \times 3.3 = 70.0 + 3.04 = 73.04 \text{ inches}$$

The 82nd percentile corresponds to about 6'1''.

SAT scores follow $N(1100, 200)$. (a) What is the 95th percentile for SAT scores? (b) What is the 97.5th percentile?

Show answer

- (a) $Z_{0.95} = 1.645$, so $x = 1100 + 1.645 \times 200 = 1429$. (b) $Z_{0.975} = 1.96$, so $x = 1100 + 1.96 \times 200 = 1492$.

26.3.1 Using StatBench for normal probabilities and percentiles

The StatBench Normal Distribution Explorer provides an interactive way to work with the normal distribution. You can:

- Set μ and σ to define the distribution
- Shade regions to find probabilities (left tail, right tail, or between two values)
- Enter a probability to find the corresponding cutoff value (inverse normal)
- Toggle between showing the Z-score and the original scale

Try it in StatBench: Open the Normal Distribution Explorer and set $\mu = 1100$, $\sigma = 200$ for SAT scores. Shade the region above 1300 to find the probability of scoring above 1300. Then switch to "find value" mode and enter 0.95 to find the 95th percentile.

26.4 QQ plots and checking normality

Before using the normal distribution as a model, we should check whether the assumption of normality is reasonable. Two visual methods are commonly used.

26.4.1 Method 1: Histogram with normal curve overlay

The simplest approach is to plot a histogram of the data and overlay the best-fitting normal curve (using $\bar{x}$ for μ and s for σ). If the histogram closely matches the curve, the normal model is reasonable.

26.4.2 Method 2: Normal probability plots (QQ plots)

A **normal probability plot** (also called a **QQ plot**, short for “quantile-quantile plot”) plots the observed data values against the values we would expect if the data were perfectly normal. If the data are approximately normal, the points will fall close to a straight line.

How to read a QQ plot:

- **Points on or near the line:** The data are consistent with a normal distribution.
- **Systematic curve (S-shape or other):** The data deviate from normality in a specific way.
- **Points curving up at both ends:** The distribution has heavier tails than normal (more extreme values).
- **Points curving down at both ends:** The distribution has lighter tails than normal.
- **Points curving up at the right:** The distribution is right-skewed.
- **Points curving down at the left:** The distribution is left-skewed.

26.4.3 Interpreting QQ plots: Examples

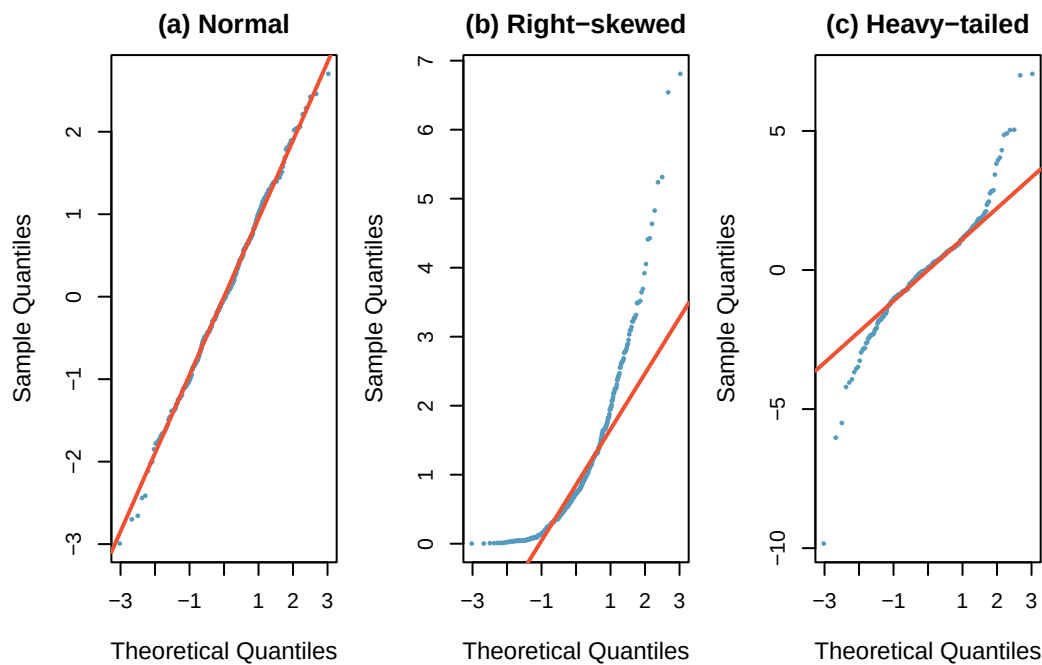


Figure 26.4: Three QQ plots: (a) approximately normal data with $n = 400$ — points fall close to the line; (b) right-skewed data — points curve upward at the right; (c) heavy-tailed data — points curve away from the line at both ends.

Sample size matters. With small sample sizes ($n < 30$), we should expect more variability in the QQ plot even when the data truly come from a normal distribution. Deviations must be more severe to be considered evidence against normality when n is small.

A researcher collects the following data and creates a QQ plot. The points follow the diagonal line closely in the middle of the distribution but curve upward at the right end. What does this suggest about the shape of the data?

Show answer

The QQ plot suggests the data are right-skewed. The bulk of the data are approximately normal, but there are larger values in the upper tail than the normal distribution would predict.

26.4.4 When is the normal model appropriate?

The normal distribution is a good model for data that are:

- **Symmetric:** The histogram is roughly mirror-image around the center.
- **Unimodal:** There is a single peak.
- **Bell-shaped:** The histogram has the characteristic tapered shape of the normal curve.
- **No extreme outliers:** Outliers can distort both the mean and standard deviation, making the normal model a poor fit.

Many variables are approximately normal:

- Standardized test scores (SAT, ACT, GRE)
- Heights of adults (within a sex)
- Measurement errors in scientific instruments
- Many biological measurements (blood pressure, cholesterol levels)

Many variables are **not** normal:

- Income distributions (right-skewed)
- Waiting times (often right-skewed)
- Count data (often Poisson or binomial)
- Variables with natural boundaries (proportions, ages)

When the normal model is not appropriate, other distributions may be used, or nonparametric methods may be applied.

26.5 Chapter review

26.5.1 Summary

- The **normal distribution** $N(\mu, \sigma)$ is a continuous probability model defined by its mean μ (center) and standard deviation σ (spread).
- The **68-95-99.7 rule** provides quick estimates: 68% of values fall within 1 SD, 95% within 2 SDs, and 99.7% within 3 SDs of the mean.
- The **Z-score** $Z = (x - \mu)/\sigma$ standardizes an observation by measuring how many standard deviations it falls from the mean.
- **Finding probabilities:** Convert to a Z-score, then use software, a calculator, or a table to find the area under the standard normal curve.
- **Finding percentiles** (inverse normal): Look up the Z-score corresponding to the desired area, then solve $x = \mu + Z\sigma$.
- **QQ plots** assess whether data are approximately normal by plotting observed values against expected normal values. Points near a straight line indicate normality.

- Recall that Chapter 10 used the normal distribution for **sampling distributions** in inference. This chapter treats it as a **probability model** for individual observations — the theoretical foundation for that earlier work.

26.5.2 Exercises

Exercises for this chapter are under development.

PART VI

DESIGN

A hypothesis test can fail to detect a real effect — a Type II error. This part explores when and why this happens, introduces statistical power, and shows how to plan studies that are large enough to detect meaningful effects. These topics are essential for designing good research studies and critically evaluating published results.

Chapter 27

Type II Error

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In Chapter 7, we introduced the hypothesis testing framework and saw how data can provide evidence against a null hypothesis. We focused on the risk of incorrectly rejecting a true null hypothesis — a Type I error. But there is a second, equally important kind of mistake: failing to detect a real effect. This chapter examines **Type II error**, explores the tradeoff between the two error types, and develops intuition for why some studies miss real effects.

StatBench tools for this chapter: Use the randomization test simulators from Chapter 7 to explore Type II error. Run tests with small sample sizes or small effect sizes and notice how often the test fails to reject H_0 — those are Type II errors in action.

[Launch Randomization Test: Difference in Means](#)

27.1 Review: hypothesis testing and Type I error

Recall the basic structure of a hypothesis test. We begin with two competing claims:

- H_0 (the **null hypothesis**): There is no effect, no difference, or no relationship.
- H_A (the **alternative hypothesis**): There *is* an effect, difference, or relationship.

We collect data, compute a test statistic, and calculate a p-value — the probability of observing results at least as extreme as ours if H_0 were true. If the p-value is small enough (below our chosen significance level α), we reject H_0 .

But rejecting H_0 when it is actually true is a mistake. This mistake has a name.

A **Type I error** occurs when we reject the null hypothesis H_0 even though H_0 is actually true. In other words, we conclude there is an effect when in reality there is none.

The probability of a Type I error is controlled by the significance level α . When we set $\alpha = 0.05$, we are accepting a 5% chance of making this kind of mistake — declaring a “discovery” that is really just noise.

Type I error is the kind of mistake that gets the most attention in introductory statistics, and for good reason: it is the error rate we directly control when we choose α . But controlling one kind of error does

not mean we have eliminated all risk. There is a second kind of mistake lurking in every hypothesis test, and it can be just as costly — sometimes more so.

27.2 Type II error: missing a real effect

Suppose a new medication truly does lower blood pressure compared to a placebo. A clinical trial is conducted, data are collected, and the researchers perform a hypothesis test. But the p-value comes back at 0.14 — not small enough to reject H_0 at the $\alpha = 0.05$ level. The researchers conclude: “We did not find sufficient evidence that the medication works.”

But the medication *does* work. The study simply failed to detect the real effect. This is a **Type II error**.

A **Type II error** occurs when we fail to reject the null hypothesis H_0 even though the alternative hypothesis H_A is actually true. In other words, there is a real effect, but our test misses it.

Type II errors are sometimes called “false negatives” — the test comes back negative (fail to reject H_0) when the truth is positive (the effect exists). In medical screening, this is analogous to a test that says “no disease” when the patient actually has the disease. In a court system, it is analogous to acquitting a guilty person.

27.3 The decision table

Every hypothesis test ends with one of two decisions: reject H_0 or fail to reject H_0 . And the truth is one of two realities: H_0 is true or H_A is true. Combining these gives four scenarios:

Table 27.1: Four possible outcomes of a hypothesis test.

	H_0 is true	H_A is true
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

Two of the four outcomes are correct decisions:

- Rejecting H_0 when H_A is true — we correctly detected the real effect.
- Failing to reject H_0 when H_0 is true — we correctly avoided a false alarm.

The other two outcomes are errors:

- **Type I error** (upper-left): We rejected H_0 but it was actually true. False alarm.
- **Type II error** (lower-right): We failed to reject H_0 but H_A was actually true. Missed detection.

In a US court, the defendant is either innocent (H_0) or guilty (H_A). What does a Type I error represent in this context? What does a Type II error represent? Table 27.1 may be useful.

If the court makes a Type I error, this means the defendant is innocent (H_0 true) but wrongly convicted. A Type II error means the court failed to reject H_0 (i.e., failed to convict the person) when they were in fact guilty (H_A true).

A pharmaceutical company tests whether a new drug reduces cholesterol more than a placebo. They set up H_0 : the drug has no effect, and H_A : the drug reduces cholesterol. Describe what a Type I error and a Type II error would mean in this context.

Show answer

A Type I error would mean concluding the drug reduces cholesterol when it actually does not — the company might invest millions in producing an ineffective drug. A Type II error would mean failing to detect a real cholesterol-lowering effect — the company might shelve a drug that actually works.

A quality control inspector tests whether a batch of light bulbs has a defect rate above 2%. She sets up H_0 : defect rate ≤ 0.02 and H_A : defect rate > 0.02 . Which error is worse in this context: Type I or Type II? Why?

Show answer

A Type II error is arguably worse here: the inspector would fail to detect that the defect rate is too high, and defective bulbs would be shipped to customers. A Type I error means rejecting a good batch, which wastes product but doesn't harm customers. The relative costs depend on the specific situation, but in quality control, missing a real problem (Type II) often has more serious consequences.

27.4 Error rates: α and β

We give the probabilities of the two error types their own Greek-letter names:

The **Type I error rate**, denoted α , is the probability of rejecting H_0 when H_0 is true:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true})$$

The **Type II error rate**, denoted β , is the probability of failing to reject H_0 when H_A is true:

$$\beta = P(\text{fail to reject } H_0 \mid H_A \text{ is true})$$

The significance level α is something we choose before the test — it is under our direct control. Common choices are $\alpha = 0.05$, $\alpha = 0.01$, and $\alpha = 0.10$.

The Type II error rate β , on the other hand, is **not** something we directly choose. It depends on several factors that we will explore shortly: the true effect size, the sample size, the variability in the data, and the chosen significance level α . Unlike α , β is usually unknown because we don't know the true state of the world — if we knew H_A were true and knew the exact effect size, we wouldn't need to run the test.

This asymmetry between α and β is a fundamental feature of the hypothesis testing framework. We control one error rate directly (α) and try to manage the other (β) through careful study design.

27.5 The tradeoff between α and β

How could we reduce the Type I error rate in US courts? What influence would this have on the Type II error rate?

To lower the Type I error rate, we might raise our standard for conviction from “beyond a reasonable doubt” to “beyond a conceivable doubt” so fewer people would be wrongly convicted. However, this would also make it more difficult to convict the people who are actually guilty, so we would make more Type II errors.

How could we reduce the Type II error rate in US courts? What influence would this have on the Type I error rate?

Show answer

To lower the Type II error rate, we want to convict more guilty people. We could lower the standards for conviction from “beyond a reasonable doubt” to “beyond a little doubt”. Lowering the bar for guilt will also result in more wrongful convictions, raising the Type I error rate.

The example and guided practice above provide an important lesson: if we reduce how often we make one type of error, we generally make more of the other type.

This tradeoff is not just a courtroom analogy — it is baked into the mathematics of hypothesis testing. Here is the core intuition:

- **Lowering α** means we require stronger evidence (a smaller p-value) before rejecting H_0 . This makes us less likely to reject H_0 when it is true (good), but also less likely to reject H_0 when it is false (bad). So β increases.
- **Raising α** means we accept weaker evidence for rejecting H_0 . This makes us more likely to detect real effects (β decreases), but also more likely to make false discoveries (α increases).

The α - β tradeoff. For a fixed sample size and effect size, decreasing α increases β , and vice versa. You cannot reduce both error rates simultaneously without changing something else — typically the sample size.

To see why this makes sense, imagine a number line representing all possible values of a test statistic. Under H_0 , the test statistic is centered at some null value. The rejection region is the set of values far enough from the null to trigger rejection. Making α smaller means shrinking the rejection region — fewer values lead to rejection. But if H_A is true, the test statistic is centered somewhere else. A smaller rejection region means less of the alternative distribution falls inside it, so we are less likely to reject — β goes up.

The only way to reduce *both* α and β at the same time is to collect more data. More data makes the test statistic more precise (smaller standard error), which makes it easier to distinguish between H_0 and H_A — allowing us to keep α small while also keeping β small. This is why sample size planning (Chapter 28) is so important.

27.6 Choosing a significance level

The **significance level** (also called the **discernibility level**) provides the cutoff for the p-value which will lead to a decision of “reject the null hypothesis.” The traditional level is 0.05. However, it is sometimes helpful to adjust the significance level based on the application. We may select a level that is smaller or larger than 0.05 depending on the consequences of any conclusions reached from the test.

If making a Type I error is dangerous or especially costly, we should choose a small significance level (e.g., 0.01 or 0.001). If we want to be very cautious about rejecting the null hypothesis, we demand very strong evidence favoring the alternative H_A before we would reject H_0 .

If a Type II error is relatively more dangerous or much more costly than a Type I error, then we should choose a higher significance level (e.g., 0.10). Here we want to be cautious about failing to reject H_0 when the null is actually false.

Significance levels should reflect consequences of errors. The significance level selected for a test should reflect the real-world consequences associated with making a Type I or Type II error.

A city water department tests whether the lead content in drinking water exceeds the safe limit. They set H_0 : lead level $\leq$ safe limit, and H_A : lead level $>$ safe limit. Should they use $\alpha = 0.01$ or $\alpha = 0.10$?

A Type II error here — failing to detect dangerous lead levels — could have serious public health consequences. A Type I error — declaring the water unsafe when it is actually fine — would cause inconvenience and expense but is far less harmful than allowing people to drink contaminated water. Therefore, the water department should use a larger α (like 0.10) to make it easier to detect real contamination, accepting a slightly higher chance of false alarms.

A social media company wants to test whether a new recommendation algorithm increases average time spent on the platform. They set H_0 : no difference, H_A : the new algorithm increases engagement. Should they use $\alpha = 0.01$ or $\alpha = 0.10$?

The stakes here are relatively low. A Type I error means rolling out an algorithm that doesn't actually help — some wasted development effort. A Type II error means keeping the old algorithm when the new one would have been slightly better — a missed opportunity but not a disaster. A standard $\alpha = 0.05$ or even a conservative $\alpha = 0.01$ is reasonable, since the company can easily run another test with more data.

27.7 Visualizing Type II error

The most effective way to build intuition about Type II error is to think about what happens when there *is* a real effect but our test fails to detect it. Simulation makes this concrete.

27.7.1 A simulation thought experiment

Imagine the following scenario. A tutoring program truly raises students' exam scores by an average of 5 points (on a 100-point scale). The standard deviation of exam scores in the population is 15 points. A researcher recruits $n = 20$ students, has them go through the tutoring program, and compares their scores to the known population mean using a one-sample t -test at $\alpha = 0.05$.

Here is what might happen:

- **The truth:** $\mu = 75$ (population mean with tutoring), while $\mu_0 = 70$ (the null hypothesis value without tutoring). The real effect is $\mu - \mu_0 = 5$ points.
- **One simulated study:** The researcher collects 20 scores. Due to natural variability, the sample mean might be $\bar{x} = 73.2$ with $s = 14.8$. The test statistic is:

$$t = \frac{73.2 - 70}{14.8/\sqrt{20}} = \frac{3.2}{3.31} = 0.97$$

The p-value for this test is about 0.17 — not significant at $\alpha = 0.05$. The researcher fails to reject H_0 , even though the tutoring program really does work. This is a Type II error.

- **Another simulated study:** A different random sample of 20 students happens to include several who benefited greatly from tutoring. The sample mean is $\bar{x} = 76.8$ with $s = 13.5$. Now:

$$t = \frac{76.8 - 70}{13.5/\sqrt{20}} = \frac{6.8}{3.02} = 2.25$$

The p-value is about 0.018 — significant at $\alpha = 0.05$. This time the researcher correctly rejects H_0 . This is a correct decision.

27.7.2 What happens across many studies?

If we repeated this experiment 1,000 times — each time drawing a fresh sample of 20 students from a population where the true effect is 5 points — we would find that roughly 340 of those experiments reject H_0 and roughly 660 do not. That means:

- About 34% of the time, we correctly detect the real effect.
- About 66% of the time, we commit a Type II error.

The Type II error rate is $\beta \approx 0.66$ — alarmingly high! Nearly two-thirds of studies with this design would miss the real effect. This is not because the effect doesn't exist; it's because the sample size is too small relative to the effect size and variability.

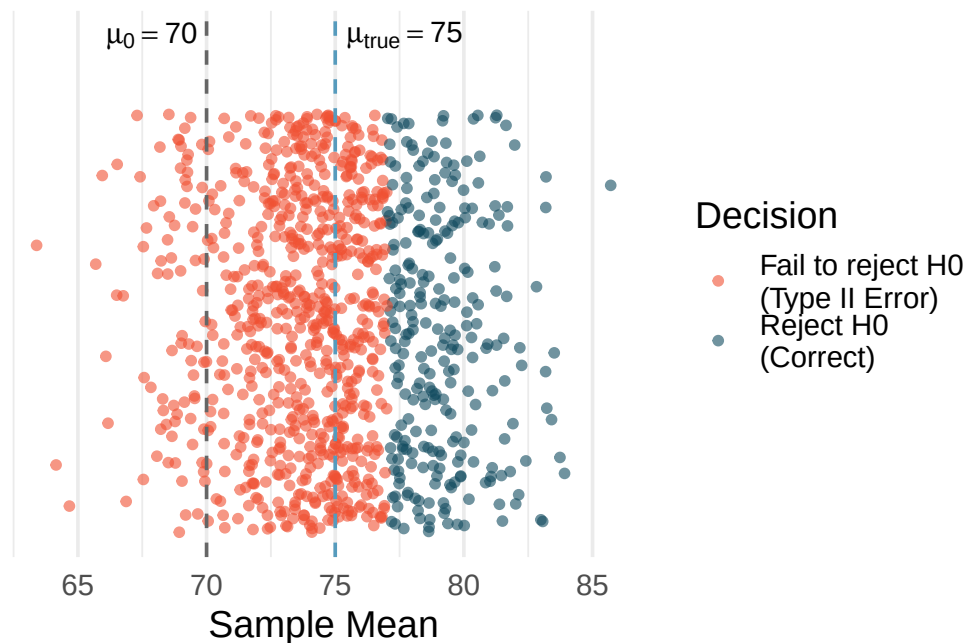


Figure 27.1: Visualization of 1,000 simulated hypothesis tests when the true effect is 5 points. Each dot represents one simulated study. The green dots are studies that correctly rejected H_0 (correct decisions), and the red dots are studies that failed to reject H_0 (Type II errors).

In the tutoring program simulation above, what would happen to the Type II error rate if the researcher increased the sample size from 20 to 100 students? Would β increase, decrease, or stay the same?

Show answer

The Type II error rate would decrease substantially. With 100 students instead of 20, the standard error drops from $15/\sqrt{20} = 3.35$ to $15/\sqrt{100} = 1.50$. The test becomes much more precise, making it easier to distinguish the true mean of 75 from the null value of 70. In fact, with $n = 100$, the power would be about 94%, meaning $\beta \approx 0.06$ — the test would miss the real effect only about 6% of the time.

StatBench: Power Simulation Tool. Use the Power Simulation Tool to explore Type II error visually. Set the true effect size, sample size, significance level, and population variability, then run 1,000 simulated experiments. Watch how the proportion of studies that reject H_0 changes as you adjust each parameter. The simulations that fail to reject H_0 represent Type II errors.

27.8 Factors that affect the Type II error rate

Four key factors determine how likely a study is to commit a Type II error:

27.8.1 1. Effect size

The **effect size** is the magnitude of the difference between the true parameter value and the null hypothesis value. Larger effects are easier to detect.

- If a tutoring program raises scores by 20 points, almost any reasonable study will detect it.
- If a tutoring program raises scores by 2 points, you need a very large or very precise study to detect it.

Intuitively, a large effect “sticks out” from the noise, while a small effect can easily be hidden by random variability.

27.8.2 2. Sample size

Larger samples provide more precise estimates and smaller standard errors. The standard error of the sample mean is $SE = \sigma/\sqrt{n}$, which decreases as n increases. A smaller standard error means the sampling distribution is narrower, making it easier to distinguish the true effect from zero.

More data, fewer missed effects. Increasing the sample size is the most common way to reduce the Type II error rate. This is why large clinical trials with thousands of participants are more likely to detect treatment effects than small pilot studies.

27.8.3 3. Population variability (σ)

When individual observations are highly variable, it is harder to estimate the population mean precisely. The standard error depends directly on σ : high variability means a wide sampling distribution, which makes it harder to distinguish H_A from H_0 .

For example, if exam scores range from 20 to 100 with $\sigma = 25$, it is much harder to detect a 5-point effect than if scores range from 60 to 80 with $\sigma = 5$.

A researcher is studying whether a dietary supplement improves reaction time. She can either recruit participants from the general population (where reaction times are highly variable) or from a group of trained athletes (where reaction times are more consistent). Which group would give her study a lower Type II error rate, assuming the supplement has the same true effect in both populations?

Show answer

The group of trained athletes would give a lower Type II error rate because their reaction times are less variable (smaller σ). A smaller σ leads to a smaller standard error, making it easier to detect the true effect of the supplement.

27.8.4 4. Significance level (α)

As we discussed in Section 27.5, a smaller α makes the test more conservative — it demands stronger evidence before rejecting H_0 . This directly increases β because the rejection region shrinks.

The table below summarizes these four factors:

Table 27.2: How each factor affects the Type II error rate.

Factor	Change	Effect on β
Effect size	Larger effect	β decreases
Sample size (n)	Larger n	β decreases
Population variability (σ)	Larger σ	β increases
Significance level (α)	Smaller α	β increases

A researcher plans a two-sample t -test to compare a new therapy to a standard therapy. She knows from pilot data that the standard deviation of patient outcomes is about $\sigma = 12$. She is hoping to detect a difference of at least 4 points between the two therapies. Her initial plan calls for 30 patients per group with $\alpha = 0.05$.

The standard error for the difference in means is approximately:

$$SE = \sqrt{\frac{12^2}{30} + \frac{12^2}{30}} = \sqrt{\frac{144}{30} + \frac{144}{30}} = \sqrt{9.6} = 3.10$$

The rejection boundary for a two-sided test is approximately $1.96 \times SE = 1.96 \times 3.10 = 6.07$. Since the true effect is only 4 points — smaller than the rejection boundary of 6.07 — the study will more often than not fail to detect the effect. The Type II error rate will be high.

What could she change to reduce the Type II error rate?

She has several options: (1) increase the sample size to reduce the standard error, (2) try to reduce variability through better measurement or a more homogeneous study population, or (3) increase α to 0.10 (though this increases the Type I error rate). The most practical option is usually to increase the sample size.

27.9 Practical vs. statistical significance revisited

The concepts of Type I and Type II error are closely connected to the distinction between **statistical significance** and **practical significance** that we first encountered in earlier chapters.

Statistical significance means the data provide enough evidence to reject H_0 at our chosen α level. It says: “The effect is probably not zero.”

Practical significance means the effect is large enough to matter in the real world. It says: “The effect is big enough to be useful.”

These are different questions, and they can give different answers:

Scenario	Statistically significant?	Practically significant?	What happened?
Large effect, large n	Yes	Yes	Ideal outcome
Large effect, small n	Possibly not	Yes	Type II error risk — effect exists but may be missed

Scenario	Statistically significant?	Practically significant?	What happened?
Small effect, large n	Yes	No	“Significant but trivial” — very large studies can detect tiny, unimportant effects
Small effect, small n	No	No	No news

The lower-left cell is the Type II error danger zone: a real, important effect exists, but the study is too small to detect it. The upper-right cell represents a different problem: a large study that detects a trivially small effect that has no practical importance.

A study with 50,000 participants finds that a new teaching method raises test scores by 0.3 points on a 100-point scale, with $p = 0.002$. Is this result practically significant?

The result is statistically significant ($p = 0.002 < 0.05$), meaning we have strong evidence that the effect is not zero. However, a 0.3-point improvement on a 100-point scale is almost certainly too small to matter in practice. No school would adopt a new teaching method for a gain of one-third of a point. This is a case of statistical significance without practical significance — the study was so large that it detected a real but trivially small effect.

Statistical significance is not the same as practical significance. A statistically significant result tells you the effect is probably real, but it does not tell you the effect is important. Always consider the *size* of the effect, not just whether it is statistically significant.

A small pilot study with 12 participants finds that a new exercise program reduces resting heart rate by 8 beats per minute, but the result is not statistically significant ($p = 0.09$). Does this mean the exercise program doesn’t work?

Show answer

Not necessarily. An 8 bpm reduction in resting heart rate is a clinically meaningful effect. The study may simply have been too small to detect it — this is a potential Type II error. The non-significant result means we don’t have enough evidence to be confident the effect is real, but it doesn’t prove the program is ineffective. A larger study with more participants would be warranted.

27.10 Chapter review

27.10.1 Summary

A hypothesis test can go wrong in two ways. A **Type I error** occurs when we reject H_0 even though it is true — a false alarm. A **Type II error** occurs when we fail to reject H_0 even though H_A is true — a missed detection. The probability of a Type I error is α (the significance level), and the probability of a Type II error is β .

There is a fundamental tradeoff: for a fixed sample size and effect size, decreasing α increases β , and vice versa. The only way to reduce both error rates simultaneously is to increase the sample size, which makes the test more precise.

Four factors affect the Type II error rate: the effect size (larger effects are easier to detect), the sample size (more data reduces β), the population variability (more variability increases β), and the significance level (smaller α increases β).

Finally, statistical significance and practical significance are not the same thing. A large study can detect trivially small effects, and a small study can miss important effects. Researchers should always consider both the statistical evidence and the practical importance of their findings.

27.10.2 Exercises

Exercises for this chapter are under development.

Chapter 28

Statistical Power

Extended Content. This chapter extends beyond UWL’s core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

In Chapter 26, we saw that a hypothesis test can miss a real effect — a Type II error. The flip side of this coin is **power**: the probability that a test *correctly* detects a real effect. Power is not just a theoretical concept; it is the foundation of good study design. Before collecting data, researchers need to ask: “Is my study large enough to find the effect I’m looking for?” This chapter defines power, explores the factors that influence it, and shows how simulation can estimate power when formulas are unavailable.

StatBench tools for this chapter: Use the randomization test simulators to explore power empirically. Try running tests with increasing sample sizes and observe how the rejection rate changes — that’s power. A dedicated Power Simulation Tool is planned for a future release.

[Launch Randomization Test: Difference in Means](#)

28.1 What is power?

In Chapter 26, we defined the Type II error rate β as the probability of failing to reject H_0 when H_A is true. Power is simply the complement of β .

The **power** of a hypothesis test is the probability of rejecting the null hypothesis when the alternative hypothesis is true:

$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_A \text{ is true})$$

Power answers the question: “If there is a real effect, how likely is my study to detect it?”

The four possible outcomes of a hypothesis test, with their probabilities, are:

Table 28.1: Probabilities of each outcome in a hypothesis test.

	H_0 is true	H_A is true
Reject H_0	α (Type I error)	$1 - \beta$ (Power)
Fail to reject H_0	$1 - \alpha$ (Correct)	β (Type II error)

H_0 is true	H_A is true
---------------	---------------

Notice the two columns must each sum to 1:

- When H_0 is true: $\alpha + (1 - \alpha) = 1$
- When H_A is true: $(1 - \beta) + \beta = 1$

Power = $1 - \beta$. A test with high power (close to 1) is very likely to detect a real effect. A test with low power (close to 0) will often miss real effects, even when they exist.

28.2 Why power matters

Power matters for both scientific and ethical reasons.

28.2.1 Designing studies that can actually detect effects

Imagine spending a year and \$500,000 on a clinical trial, only to discover at the end that the study was too small to detect the treatment effect. The trial produces an inconclusive result — a non-significant p-value — even though the treatment actually works. The money is gone, the patients' time was wasted, and the scientific question remains unanswered.

This scenario is unfortunately common. A study with low power is essentially a gamble: you might detect the effect (if you're lucky with your sample), or you might not (if natural variability obscures the signal). Low-powered studies are wasteful because they frequently produce null results that leave everyone uncertain.

28.2.2 Ethical considerations

In medical research, every patient enrolled in a clinical trial is exposed to some degree of risk — side effects, time away from proven treatments, or the inconvenience of participation. If a study is underpowered and unlikely to produce a clear answer, then patients are being exposed to risk with little prospect of scientific benefit. Many institutional review boards (IRBs) now require a power analysis before approving research involving human subjects.

28.2.3 The scientific literature

Low-powered studies distort the scientific literature. When only a small fraction of underpowered studies happen to produce significant results, the published literature overestimates effect sizes (because only the “lucky” studies with unusually large observed effects made it past the significance threshold). This phenomenon is sometimes called the **winner's curse** — the studies that “win” (reject H_0) in a low-power setting tend to have inflated effect sizes.

A researcher wants to study whether a new fertilizer increases crop yield. Based on previous research, she expects the fertilizer to increase yield by about 8%. She plans to compare treated and untreated fields. If her study has power of 0.50, what does this mean in practical terms?

Power of 0.50 means that even if the fertilizer truly increases yield by 8%, there is only a 50% chance that her study will detect it — essentially a coin flip. She has a 50% chance of committing a Type II error and concluding there is no evidence the fertilizer works. This is generally considered unacceptable. She should increase her sample size or consider other design improvements before proceeding.

28.3 Factors affecting power

The same four factors that affect β (Chapter 26) determine power, since $\text{power} = 1 - \beta$. Here we frame each factor in terms of power rather than Type II error rate.

28.3.1 Sample size: $\uparrow n \rightarrow \uparrow \text{power}$

This is the most important lever researchers have. Larger samples produce smaller standard errors, which make the sampling distribution narrower. A narrower distribution makes it easier to distinguish the alternative from the null.

Power.

The power of the test is the probability of rejecting the null claim when the alternative claim is true.

How easy it is to detect the effect depends on both how big the effect is (e.g., how good the medical treatment is) as well as the sample size.

Consider the blood pressure medication example from Chapter 26. A new medication is expected to lower blood pressure by 3 mmHg compared to a standard medication. The standard deviation of blood pressure change is $\sigma = 12$ mmHg, and we use a two-sided test with $\alpha = 0.05$.

What is the approximate power with $n = 100$ patients per group? With $n = 500$ per group?

With $n = 100$ per group:

The standard error for the difference in means is:

$$SE = \sqrt{\frac{12^2}{100} + \frac{12^2}{100}} = \sqrt{2.88} \approx 1.70$$

The rejection boundary for a two-sided test at $\alpha = 0.05$ is approximately $\pm 1.96 \times 1.70 = \pm 3.33$ mmHg.

If the true effect is -3 mmHg, then the alternative distribution is centered at -3 with standard deviation 1.70. We need to find the probability that a value from this distribution falls below -3.33 :

$$Z = \frac{-3.33 - (-3)}{1.70} = \frac{-0.33}{1.70} = -0.20 \rightarrow P(Z < -0.20) \approx 0.42$$

The power is approximately **42%** — far too low.

With $n = 500$ per group:

$$SE = \sqrt{\frac{12^2}{500} + \frac{12^2}{500}} = \sqrt{0.576} \approx 0.76$$

The rejection boundary is $\pm 1.96 \times 0.76 = \pm 1.49$ mmHg.

$$Z = \frac{-1.49 - (-3)}{0.76} = \frac{1.51}{0.76} = 1.99 \rightarrow P(Z < 1.99) \approx 0.977$$

The power is approximately **97.7%** — excellent. With 500 patients per group, we would almost certainly detect a 3 mmHg effect.

28.3.2 Effect size: $\uparrow$ effect $\rightarrow \uparrow$ power

A larger effect is easier to detect. If a drug lowers blood pressure by 10 mmHg instead of 3 mmHg, the signal is much stronger relative to the noise, and even a modest sample size will detect it.

Using the same setup as the worked example above ($\sigma = 12$, $\alpha = 0.05$, $n = 100$ per group), estimate the power to detect a true difference of -8 mmHg instead of -3 mmHg.

Show answer

With $SE = 1.70$ and rejection boundary at -3.33 , the alternative is centered at -8 . We compute $Z = (-3.33 - (-8))/1.70 = 4.67/1.70 = 2.75$, giving $P(Z < 2.75) \approx 0.997$. The power is about 99.7%. An 8 mmHg effect is easy to detect even with only 100 patients per group.

28.3.3 Significance level: $\uparrow \alpha \rightarrow \uparrow$ power

A larger α means a wider rejection region. With more “room” to reject H_0 , the test is more sensitive — but also more prone to false alarms (Type I errors).

Using the blood pressure setup ($\sigma = 12$, $n = 100$, true effect = -3 mmHg), compare the power at $\alpha = 0.05$ versus $\alpha = 0.10$.

Show answer

At $\alpha = 0.05$: rejection boundary = $\pm 1.96 \times 1.70 = \pm 3.33$. $Z = (-3.33 - (-3))/1.70 = -0.20$, power ≈ 0.42 . At $\alpha = 0.10$: rejection boundary = $\pm 1.645 \times 1.70 = \pm 2.80$. $Z = (-2.80 - (-3))/1.70 = 0.12$, power ≈ 0.55 . Power increases from 42% to 55% when we relax α from 0.05 to 0.10, but at the cost of a higher false alarm rate.

28.3.4 Variability: $\uparrow \sigma \rightarrow \downarrow$ power

Greater variability in the data means a larger standard error, which spreads out the sampling distribution and makes it harder to distinguish the alternative from the null.

Table 28.2: Factors that affect power. All relationships assume the other factors are held constant.

Factor	Change	Effect on Power
Sample size (n)	Increase n	Power increases
Effect size	Larger effect	Power increases
Significance level (α)	Increase α	Power increases
Population variability (σ)	Decrease σ	Power increases

The researcher’s primary tool for increasing power is sample size. Effect size is determined by nature (we don’t control how effective a treatment is). Variability can sometimes be reduced through better measurement or study design, but it is largely a feature of the population. The significance level α is usually fixed by convention. That leaves sample size as the main adjustable dial.

28.4 Simulation-based power estimation

When exact power formulas are available (as for t -tests and z -tests), we can compute power mathematically. But many real-world situations involve test procedures where exact formulas are complicated or unavailable — for example, bootstrap tests, permutation tests, or tests involving non-normal data. In these cases, **simulation** offers a straightforward alternative.

28.4.1 The simulation recipe

The idea is simple: if we know (or hypothesize) the true state of the world under H_A , we can simulate the data-collection process many times, run the hypothesis test each time, and count how often we reject H_0 .

Here is the step-by-step recipe:

1. **Specify the alternative hypothesis:** Choose the effect size you want to detect. For example: “The true mean difference is 5 points.”
2. **Specify the data-generating process:** Define the population(s) from which data will be drawn. For example: “Each observation comes from a normal distribution with mean 75 and standard deviation 15.”
3. **Choose the sample size and significance level.**
4. **Simulate one experiment:** Draw a random sample of size n from the specified population, compute the test statistic, and determine whether you reject H_0 .
5. **Repeat many times:** Run step 4 a total of B times (e.g., $B = 1,000$ or $B = 10,000$).
6. **Estimate power:** Count the proportion of simulations that rejected H_0 . This proportion is the estimated power.

$$\widehat{\text{Power}} = \frac{\text{Number of simulations that reject } H_0}{B}$$

A biologist plans to test whether a new habitat restoration technique increases the average number of bird species observed at a site. She expects the technique to increase the count from a baseline of 12 species to about 15 species, with a standard deviation of 6 species. She will survey $n = 25$ sites using a one-sample t -test at $\alpha = 0.05$.

Describe how she could use simulation to estimate the power of her study.

-
1. **Alternative:** true mean = 15, null value = 12.
 2. **Data-generating process:** draw 25 observations from a normal distribution with mean 15 and standard deviation 6.
 3. **Sample size** $n = 25$, **significance level** $\alpha = 0.05$.
 4. **One simulated experiment:** generate 25 random values from $N(15, 6)$, compute $\bar{x}$ and s , calculate $t = (\bar{x} - 12)/(s/\sqrt{25})$, find the p-value, and check if $p < 0.05$.
 5. **Repeat** 10,000 times.
 6. **Estimate power:** suppose 8,247 of the 10,000 simulations rejected H_0 . Then $\widehat{\text{Power}} = 8,247/10,000 = 0.825$.

She would estimate the power at about 82.5%, which exceeds the conventional 80% target.

The beauty of simulation-based power analysis is its flexibility. It works for any test procedure — not just t -tests — and can accommodate complex scenarios like non-normal data, unequal group sizes, missing data, or multi-stage designs.

StatBench: Power Simulation Tool. The Power Simulation Tool implements exactly this recipe. Set the true effect size, sample size, variability, and significance level. Click “Run Simulation” to generate 1,000 simulated experiments. The tool displays the proportion that reject H_0 (the estimated power) and shows a histogram of simulated test statistics with the rejection region shaded.

28.5 Power curves

A single power calculation tells us the power for one specific combination of sample size, effect size, α , and σ . But researchers often want to see how power changes across a range of values. **Power curves** show this relationship graphically.

28.5.1 Power vs. sample size

The most common power curve plots power on the y-axis against sample size on the x-axis, for a fixed effect size, α , and σ . This curve answers: “How many observations do I need to achieve a desired level of power?”

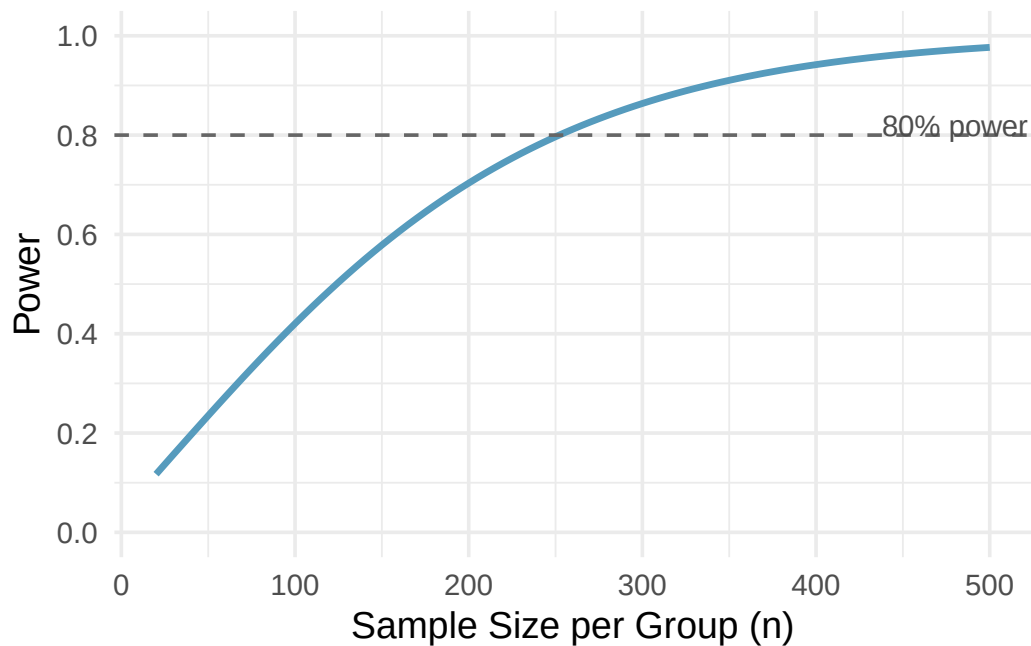


Figure 28.1: A power curve showing power (y-axis) versus sample size per group (x-axis) for the blood pressure example with true effect = 3 mmHg, $\sigma = 12$, $\alpha = 0.05$. The dashed horizontal line marks 80% power; the curve crosses it at approximately $n = 250$ per group.

Key features of power curves:

- **Power starts low** for small sample sizes and **increases toward 1** as n grows.
- The curve has a characteristic **S-shape**: power increases slowly at first, then more rapidly in the middle range, then levels off as it approaches 1.
- The curve **never reaches exactly 1** (there is always some chance of a Type II error, though it becomes negligibly small for very large n).
- The steepest part of the curve — where adding more observations gives the biggest payoff in power — is typically in the middle range.

28.5.2 Power vs. effect size

We can also plot power against effect size, holding n , α , and σ fixed. This curve shows how the detectable effect depends on its magnitude.

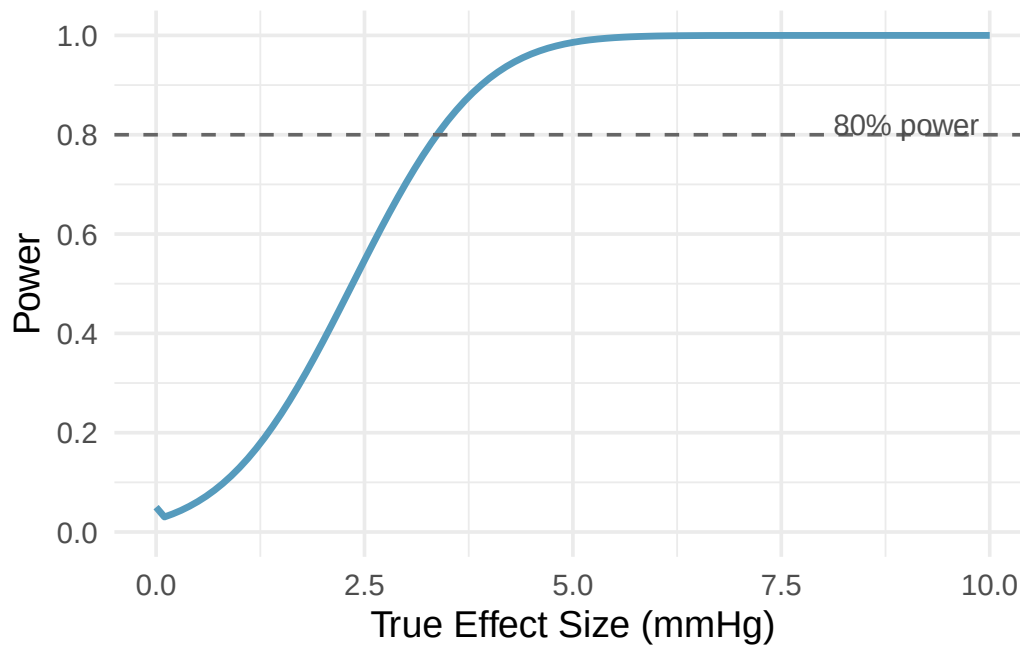


Figure 28.2: A power curve showing power (y-axis) versus true effect size (x-axis) for the blood pressure example with $n = 200$ per group, $\sigma = 12$, $\alpha = 0.05$. Larger effects are easier to detect.

At an effect size of 0, the “power” is just α — this is the Type I error rate, the probability of rejecting H_0 when there is no effect. As the effect size grows, power increases.

Look at the power-vs-sample-size curve description above. If a researcher can only afford to recruit 100 patients per group, approximately what power would she have? Is this adequate?

Show answer

From the curve, at $n = 100$ per group the power is approximately 42%. This is far below the conventional minimum of 80%. The researcher should either find a way to increase the sample size, or reconsider whether the study is worth conducting at this scale.

28.6 Typical power targets

How much power is “enough”? There is no universal answer, but strong conventions have emerged in statistical practice.

The 80% power convention. The most widely used target is **power = 0.80** (80%). This means the study has an 80% chance of detecting the specified effect. Equivalently, $\beta = 0.20$ — a 20% chance of a Type II error.

Some fields or funding agencies require higher power:

- **80% power** is the conventional minimum for most research.
- **90% power** is sometimes required for clinical trials, especially when the consequences of a Type II error are severe.
- **95% power** is occasionally used in high-stakes contexts but requires substantially larger sample sizes.

Why is 80% a reasonable target? Why not 95% or 99%?

The choice of 80% reflects a practical tradeoff. Increasing power beyond 80% requires increasingly large sample sizes, and the gains become smaller. Going from 80% to 90% power might require 30–50% more observations. Going from 90% to 95% might require another 30% or more. At some point, the cost and logistical difficulty of recruiting additional participants outweigh the marginal reduction in Type II error risk.

The 80% target also reflects the observation that in most settings, a Type II error (missing a real effect) is considered less serious than a Type I error (claiming a false effect), so we are comfortable with $\beta = 0.20$ even when $\alpha = 0.05$. If the consequences of missing an effect are severe — for example, in a drug safety trial — a higher power target is appropriate.

Power analysis must be done before data collection. A power analysis conducted *after* data have been collected and analyzed is called **post hoc power analysis** (or “observed power”), and it is widely regarded as misleading. Post hoc power is mathematically determined by the p-value and adds no new information. The time to think about power is during the study design phase.

28.7 Computing power for a two-sample test

While a full derivation of power formulas is beyond the scope of this course, it is useful to see the mechanics of a power calculation for the common case of comparing two group means. This section walks through the logic step by step using the blood pressure medication example.

28.7.1 Setting up the null and alternative distributions

Suppose a pharmaceutical company has developed a new drug to lower blood pressure. They plan a randomized experiment comparing the new drug to a standard medication. The hypotheses are:

- $H_0: \mu_{\text{treatment}} - \mu_{\text{control}} = 0$ (no difference)
- $H_A: \mu_{\text{treatment}} - \mu_{\text{control}} \neq 0$ (there is a difference)

From pilot studies, the standard deviation of blood pressure measurements within each group is about $\sigma = 12$ mmHg. The test will use $\alpha = 0.05$.

Step 1: Determine the null distribution. Under H_0 , the difference $\bar{x}_{\text{treatment}} - \bar{x}_{\text{control}}$ has mean 0 and standard error:

$$SE = \sqrt{\frac{\sigma^2}{n} + \frac{\sigma^2}{n}} = \sigma \sqrt{\frac{2}{n}}$$

With $n = 100$ per group: $SE = 12\sqrt{2/100} = 12 \times 0.1414 = 1.70$.

So under H_0 , the test statistic approximately follows a normal distribution centered at 0 with standard deviation 1.70.

Step 2: Find the rejection regions. At $\alpha = 0.05$ (two-sided), we reject H_0 when the observed difference falls more than $1.96 \times SE = 1.96 \times 1.70 = 3.33$ units from zero. That is, we reject when the observed difference is below -3.33 or above $+3.33$.

Step 3: Specify the alternative. Suppose the new drug truly lowers blood pressure by 3 mmHg more than the standard medication, so the true difference is $\mu_{\text{treatment}} - \mu_{\text{control}} = -3$.

Under this alternative, the observed difference follows a normal distribution centered at -3 with standard deviation 1.70 (the standard error doesn’t change — it depends on σ and n , not the true mean).

Step 4: Compute the power. Power is the probability that the observed difference falls in the rejection region when the alternative is true. The rejection boundary closest to the true effect of -3 is at -3.33 . So we need:

$$P(\bar{x}_{\text{treatment}} - \bar{x}_{\text{control}} < -3.33 \mid \text{true difference} = -3)$$

Standardizing:

$$Z = \frac{-3.33 - (-3)}{1.70} = \frac{-0.33}{1.70} = -0.20$$

$$P(Z < -0.20) \approx 0.42$$

The power is approximately **42%**. This is quite low — the study has less than a coin-flip chance of detecting the 3 mmHg effect.

Using the same setup ($\sigma = 12$, $\alpha = 0.05$, true effect = -3 mmHg), compute the power when $n = 250$ per group.

Show answer

With $n = 250$: $SE = 12\sqrt{2/250} = 12 \times 0.0894 = 1.073$. Rejection boundary: $1.96 \times 1.073 = 2.10$. Under H_A centered at -3 : $Z = (-2.10 - (-3))/1.073 = 0.90/1.073 = 0.84$. $P(Z < 0.84) \approx 0.80$. The power is approximately 80% — right at the conventional target. So about 250 patients per group would be needed to achieve 80% power for this effect size.

28.8 Chapter review

28.8.1 Summary

Power is the probability of correctly rejecting a false null hypothesis: $\text{Power} = 1 - \beta$. A well-designed study should have power of at least 0.80 (80%), meaning it has an 80% chance of detecting the effect of interest.

Power depends on four factors: sample size (more data gives more power), effect size (larger effects are easier to detect), significance level (higher α gives more power but more false alarms), and population variability (less variability gives more power). Of these, sample size is the factor most under the researcher's control.

When exact power formulas are unavailable, simulation provides a flexible alternative: generate many datasets under the assumed alternative, run the test on each, and count the proportion that reject H_0 . Power curves visualize how power changes with sample size or effect size, helping researchers identify the “sweet spot” where power is adequate without wasting resources.

Power analysis should always be done *before* data collection. Post hoc power analysis — computing power after the study is over — is misleading and widely discouraged.

28.8.2 Exercises

Exercises for this chapter are under development.

Chapter 29

Sample Size Planning

Extended Content. This chapter extends beyond UWL's core STAT 145 curriculum. It is included for completeness and for instructors who wish to cover additional topics. Content is in draft form.

How many observations do you need? This deceptively simple question is one of the most important in all of applied statistics. Too few observations and the study lacks the power to detect real effects (Chapter 27). Too many and the study wastes time, money, and — in medical contexts — exposes unnecessary participants to risk. This chapter develops formulas and strategies for determining the right sample size before a study begins.

StatBench tools for this chapter: Power Simulation Tool (sample size slider)

29.1 Why plan sample size?

Sample size planning — also called **sample size determination** or **power analysis** — is the process of deciding how many observations to collect before beginning a study. There are three main reasons to plan carefully.

29.1.1 Statistical reasons

As we saw in Chapter 27, the power of a hypothesis test depends critically on the sample size. A study that is too small has low power and a high probability of committing a Type II error — missing a real effect. A study that is too large may have power close to 100%, but the excess observations provide diminishing returns.

Similarly, for estimation, the width of a confidence interval depends on the sample size through the standard error. More observations produce narrower intervals and more precise estimates. Sample size planning for confidence intervals asks: “How many observations do I need to achieve a desired margin of error?”

29.1.2 Ethical reasons

In medical research, participants are exposed to potential risks: side effects, discomfort, time away from proven treatments. If a study is underpowered and unlikely to produce useful results, participants bear risk without commensurate benefit. Many institutional review boards (IRBs) and research ethics committees require a sample size justification as part of the study approval process.

At the other extreme, an unnecessarily large study exposes more people to risk than needed. If a treatment has a large, harmful side effect, ethical principles demand that we detect it with as few participants as possible.

29.1.3 Practical reasons

Research costs money. Each additional participant means more time, more supplies, more lab work, more data entry. A well-planned sample size ensures that resources are used efficiently — enough to answer the research question, but not so many that the budget is strained or the timeline is unrealistic.

Plan your sample size before collecting data. Determining the sample size after data have been collected defeats the purpose. The goal is to ensure that the study *will have* adequate power or precision, not to verify that it *did*.

29.2 Sample size for estimating a mean

One of the most common sample size questions is: “How many observations do I need to estimate a population mean with a given margin of error?”

29.2.1 The margin of error formula

Recall that a confidence interval for a population mean μ takes the form:

$$\bar{x} \pm z^* \cdot \frac{\sigma}{\sqrt{n}}$$

where z^* is the critical value for the desired confidence level (e.g., $z^* = 1.96$ for 95% confidence) and σ is the population standard deviation. The **margin of error** (ME) is the “plus-or-minus” part:

$$ME = z^* \cdot \frac{\sigma}{\sqrt{n}}$$

If we want the margin of error to be no larger than some target value, we can solve for n :

$$ME = z^* \cdot \frac{\sigma}{\sqrt{n}} \implies \sqrt{n} = \frac{z^* \cdot \sigma}{ME} \implies n = \left(\frac{z^* \cdot \sigma}{ME} \right)^2$$

The **sample size for estimating a mean** with margin of error ME at confidence level C is:

$$n = \left(\frac{z^* \cdot \sigma}{ME} \right)^2$$

where z^* is the critical value corresponding to confidence level C (e.g., $z^* = 1.96$ for 95% confidence) and σ is the population standard deviation.

Always round up. When the formula gives a non-integer answer, always round *up* to the next whole number. Rounding down would give a margin of error slightly larger than desired.

A university wants to estimate the average amount of student loan debt among its graduates to within \$1,500 with 95% confidence. Previous studies suggest the standard deviation of student loan debt is about \$12,000. How many graduates should they survey?

We have:

- $ME = 1,500$
- $z^* = 1.96$ (for 95% confidence)
- $\sigma = 12,000$

$$n = \left(\frac{1.96 \times 12,000}{1,500} \right)^2 = \left(\frac{23,520}{1,500} \right)^2 = (15.68)^2 = 245.86$$

Rounding up, the university should survey $n = 246$ **graduates**.

A health researcher wants to estimate the average daily calorie intake of adults in a city to within 50 calories with 99% confidence. From national data, the standard deviation of daily calorie intake is approximately 500 calories. How large a sample does she need?

Show answer

With $ME = 50$, $z^* = 2.576$ (for 99% confidence), and $\sigma = 500$: $n = (2.576 \times 500/50)^2 = (25.76)^2 = 663.6$. Rounding up, she needs $n = 664$ participants.

29.2.2 Where does σ come from?

There is a practical catch in the formula: we need to know the population standard deviation σ before we collect data. But if we already knew σ , we would know more about the population than we actually do. In practice, σ is estimated from:

- **Previous studies:** Published research on similar populations often reports standard deviations that can be used for planning.
- **Pilot studies:** A small preliminary study can provide a rough estimate of σ .
- **Expert judgment:** Subject-matter experts may have reasonable estimates based on experience.
- **Conservative estimates:** When in doubt, use a larger estimate of σ . This produces a larger (more conservative) sample size, ensuring the margin of error target is met even if the true variability is somewhat less than assumed.

Using s from pilot data. If you estimate σ from a pilot study with a small sample, the estimate may be unreliable. When using pilot data, it is wise to inflate the estimate somewhat — for example, using the upper end of a confidence interval for σ — to provide a safety margin.

Referring to the student loan debt example, suppose the \$12,000 standard deviation came from a pilot survey of 30 graduates. The sample standard deviation was $s = 12,000$. Should the researcher use exactly $\sigma = 12,000$ in the sample size formula?

With only 30 observations, the estimate of σ is itself uncertain. A 95% confidence interval for σ would be roughly \$9,600 to \$16,000. To be safe, the researcher might use $\sigma = 14,000$ or even $\sigma = 16,000$ in the formula, which would give $n = (1.96 \times 14,000/1,500)^2 = 335$ or $n = (1.96 \times 16,000/1,500)^2 = 437$. Using the larger value ensures the margin of error target is likely met.

29.3 Sample size for estimating a proportion

When estimating a population proportion p , the confidence interval takes the form:

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

The margin of error is:

$$ME = z^* \sqrt{\frac{p^*(1-p^*)}{n}}$$

where p^* is a planning value for p (since we don't know p yet — that's what we're trying to estimate). Solving for n :

$$n = \left(\frac{z^*}{ME} \right)^2 \cdot p^*(1-p^*)$$

The **sample size for estimating a proportion** with margin of error ME at confidence level C is:

$$n = \left(\frac{z^*}{ME} \right)^2 \cdot p^*(1-p^*)$$

where z^* is the critical value for confidence level C and p^* is a planning estimate of the proportion.

29.3.1 Choosing p^*

The value $p^*(1-p^*)$ is largest when $p^* = 0.5$ and smallest when p^* is near 0 or 1. This means:

- If you have a reasonable estimate of p from prior research, use it as p^* .
- If you have no idea what p might be, **use** $p^* = 0.5$. This gives the largest (most conservative) sample size, ensuring the margin of error target is met regardless of the true proportion.

A political polling firm wants to estimate the proportion of voters who support a ballot measure to within 3 percentage points with 95% confidence. They have no prior information about voter support. How many voters should they survey?

With no prior information, we use $p^* = 0.5$ for the most conservative estimate:

- $ME = 0.03$ (3 percentage points)
- $z^* = 1.96$
- $p^* = 0.5$

$$n = \left(\frac{1.96}{0.03} \right)^2 \times 0.5 \times 0.5 = (65.33)^2 \times 0.25 = 4,268.4 \times 0.25 = 1,067.1$$

Rounding up, they need $n = 1,068$ **voters**.

This is why political polls typically survey about 1,000 people — it gives roughly a $\pm 3\%$ margin of error at 95% confidence.

A quality control manager wants to estimate the proportion of defective items in a production run to within 1 percentage point with 95% confidence. Based on historical data, the defect rate is usually around 4%. How large a sample should she inspect?

Show answer

With $ME = 0.01$, $z^* = 1.96$, and $p^* = 0.04$: $n = (1.96/0.01)^2 \times 0.04 \times 0.96 = (196)^2 \times 0.0384 = 38,416 \times 0.0384 = 1,475.2$. Rounding up, she needs $n = 1,476$ items. Note: if she had used $p^* = 0.5$

instead, she would have gotten $n = 9,604$ — a much larger and unnecessarily conservative sample. Using prior information about p can save a lot of resources.

Compare the required sample sizes when $p^* = 0.5$ versus $p^* = 0.10$ for a 95% confidence interval with $ME = 0.04$.

With $p^* = 0.5$:

$$n = \left(\frac{1.96}{0.04} \right)^2 \times 0.5 \times 0.5 = (49)^2 \times 0.25 = 2,401 \times 0.25 = 600.25 \rightarrow n = 601$$

With $p^* = 0.10$:

$$n = \left(\frac{1.96}{0.04} \right)^2 \times 0.10 \times 0.90 = 2,401 \times 0.09 = 216.09 \rightarrow n = 217$$

Using prior knowledge that p is near 0.10 reduces the required sample from 601 to 217 — a 64% reduction. This illustrates why it is valuable to use a good planning estimate when one is available.

29.4 Sample size for comparing two groups

When comparing two group means, the sample size calculation involves the power framework from Chapter 27. The goal is to find n such that the test has at least 80% power (or another target) to detect an effect of a specified size.

29.4.1 Conceptual approach

The idea is the same as in the one-sample case, but now we need to account for two sources of variability (one from each group). For a two-sample t -test comparing means with equal group sizes (n per group), equal standard deviations (σ), and $\alpha = 0.05$ (two-sided):

$$SE = \sigma \sqrt{\frac{2}{n}}$$

To achieve 80% power, we need the true effect size to be about $2.8 \times SE$ away from the null value.¹ Setting the minimum detectable effect δ equal to $2.8 \times SE$ and solving for n :

$$\delta = 2.8 \times \sigma \sqrt{\frac{2}{n}} \implies n = 2 \left(\frac{2.8 \cdot \sigma}{\delta} \right)^2$$

Returning to the blood pressure example from Chapter 27: a new drug is expected to lower blood pressure by at least 3 mmHg compared to a standard medication. The standard deviation is $\sigma = 12$ mmHg. How many patients per group are needed for 80% power with $\alpha = 0.05$?

¹The 2.8 comes from $z_{0.025} + z_{0.20} = 1.96 + 0.84 = 2.80$, where $z_{0.025}$ defines the rejection boundary and $z_{0.20}$ defines the 80% power point on the alternative distribution.

Using the formula:

$$n = 2 \left(\frac{2.8 \times 12}{3} \right)^2 = 2 \left(\frac{33.6}{3} \right)^2 = 2 \times (11.2)^2 = 2 \times 125.44 = 250.88$$

Rounding up, $n = 251$ **patients per group** — a total of 502 patients.

This matches the result we found in Chapter 27 when we computed the power for various sample sizes and found that $n \approx 250$ per group gave approximately 80% power.

A psychologist plans a study comparing reaction times under two conditions. She expects a difference of 25 milliseconds, with a standard deviation of 60 ms in each group. How many participants does she need per group for 80% power at $\alpha = 0.05$?

Show answer

Using the formula: $n = 2(2.8 \times 60/25)^2 = 2(6.72)^2 = 2 \times 45.16 = 90.32$. Rounding up, she needs $n = 91$ per group, or 182 total participants.

29.4.2 Different power targets

The constant 2.8 in the formula above is specific to 80% power and $\alpha = 0.05$ (two-sided). For other targets:

Table 29.1: Multiplier values ($z_{\alpha/2} + z_{\beta}$) for common power and significance level combinations.

Power	$\alpha = 0.05$ (two-sided)	$\alpha = 0.01$ (two-sided)
80%	2.80	3.36
90%	3.24	3.86
95%	3.60	4.20

Using Table 29.1, how many patients per group would be needed in the blood pressure example ($\sigma = 12$, $\delta = 3$) for **90% power** at $\alpha = 0.01$?

Show answer

Using the multiplier 3.86: $n = 2(3.86 \times 12/3)^2 = 2(15.44)^2 = 2 \times 238.4 = 476.8$. Rounding up, $n = 477$ per group, or 954 total — nearly double the 502 needed for 80% power at $\alpha = 0.05$. Higher power and stricter α both demand more observations.

29.5 Practical considerations

Formulas give a starting point, but real-world sample size planning involves several additional considerations.

29.5.1 Budget and resources

The formula might say you need 500 participants, but your budget only supports 200. In this case, you have several options:

- **Accept lower power** and acknowledge the study may not detect small effects.
- **Narrow the research question** to focus on larger effects that require fewer observations to detect.
- **Reduce variability** through more precise measurements, a more homogeneous study population, or a paired design (if applicable).

- **Seek additional funding** if the research question is important enough.
- **Don't run the study** if the available sample size gives unacceptably low power. A study that is almost certain to produce null results wastes everyone's time and resources.

Sometimes the best decision is not to run the study. If a power analysis shows that the achievable sample size gives power below 50%, the study is more likely to miss a real effect than to detect one. Researchers should seriously consider whether conducting such a study is worthwhile.

29.5.2 Recruitment and attrition

Not everyone you recruit will complete the study. Participants drop out, miss appointments, provide unusable data, or are lost to follow-up. The sample size formula tells you how many *completed* observations you need. You should plan to recruit more participants to account for expected attrition.

A longitudinal health study requires $n = 300$ completed observations. Based on similar studies, the researchers expect 15% attrition. How many participants should they recruit initially?

If 15% will be lost, then 85% of the recruited participants will complete the study. Let n_{recruit} be the number to recruit:

$$0.85 \times n_{\text{recruit}} = 300 \implies n_{\text{recruit}} = \frac{300}{0.85} = 352.9$$

They should recruit **353 participants** to end up with approximately 300 completed observations.

29.5.3 Pilot studies

A **pilot study** is a small preliminary study conducted before the main study. Pilot studies serve several purposes relevant to sample size planning:

- **Estimate σ (or p)** for use in the sample size formula.
- **Test procedures:** Identify problems with the survey instrument, experimental protocol, or data collection process.
- **Estimate attrition rates:** How many participants drop out or provide unusable data?
- **Check feasibility:** Is it realistic to recruit the number of participants the formula requires?

Pilot studies typically involve 10-30 participants — enough to get rough estimates but not so many that significant resources are consumed before the main study begins.

Pilot study data should not be combined with main study data (at least not without careful statistical adjustment). The pilot study is used to *design* the main study. Combining the two can bias the results because the main study design was chosen based on the pilot data.

29.5.4 When formulas don't apply: simulation

The sample size formulas in this chapter assume specific test procedures (z-tests and t-tests) and specific distributional assumptions (normality). When these conditions don't hold — for example, when using a bootstrap or permutation test, or when the data are highly skewed — simulation provides an alternative.

The simulation approach for sample size planning is an extension of the simulation-based power estimation from Chapter 27:

1. **Choose a candidate sample size n .**
2. **Simulate** many datasets of size n under the assumed alternative.

3. **Run the analysis** on each simulated dataset.
4. **Compute the proportion** that reject H_0 — this is the estimated power for that n .
5. **Repeat** for different values of n until you find the smallest n that achieves the desired power.

This approach is computationally intensive but extremely flexible. It can handle any test procedure, any data distribution, and any study design.

A researcher wants to use a permutation test (Chapter 7) to compare two groups, but the data are expected to be heavily right-skewed. Formulas for the t -test may not apply. How could she determine the sample size?

She would use simulation:

1. Specify the alternative: for example, Group A has median 50 with a right-skewed distribution, and Group B has median 60 with the same shape.
2. For a candidate n (say, $n = 40$ per group), simulate 10,000 experiments: draw 40 values from each group's assumed distribution, run a permutation test, and record whether H_0 is rejected.
3. Compute the proportion of rejections — this is the estimated power at $n = 40$.
4. Repeat for $n = 50, 60, 80, 100, \dots$ and find the smallest n where power ≥ 0.80 .

This avoids any assumption about normality and accounts for the specific properties of the permutation test.

StatBench: Power Simulation Tool (Sample Size Slider). Use the Power Simulation Tool's sample size slider to explore the relationship between sample size and power interactively. Set the true effect size and variability, then slide the sample size up and down to see how power changes. The tool will highlight when you cross the 80% power threshold.

29.6 Summary of sample size formulas

The table below collects the key formulas from this chapter for quick reference.

Table 29.2: Summary of sample size formulas. In all cases, round the result up to the next whole number.

Goal	Formula	Key inputs
Estimate a mean to within ME	$n = \left(\frac{z^* \cdot \sigma}{ME}\right)^2$	z^*, σ, ME
Estimate a proportion to within ME	$n = \left(\frac{z^*}{ME}\right)^2 \cdot p^*(1 - p^*)$	z^*, p^*, ME
Compare two means (80% power, $\alpha = 0.05$)	$n = 2 \left(\frac{2.8 \cdot \sigma}{\delta}\right)^2$ per group	σ, δ (min. detectable effect)

These formulas are starting points, not final answers. Always adjust for attrition, consider practical constraints, and verify assumptions. When in doubt, simulation provides a robust alternative.

29.7 Chapter review

29.7.1 Summary

Sample size planning is a critical step in study design. For estimating a population mean, the required sample size is $n = (z^* \cdot \sigma / ME)^2$, where ME is the desired margin of error. For estimating a proportion, $n = (z^* / ME)^2 \cdot p^*(1 - p^*)$, where p^* is a planning value for the proportion (use $p^* = 0.5$ if unknown). For comparing two group means with 80% power, $n = 2(2.8\sigma/\delta)^2$ per group, where δ is the minimum detectable effect.

In all cases, the formula requires prior information about variability (σ or p), which may come from previous studies, pilot data, or conservative estimates. Results should always be rounded up and adjusted for expected attrition.

When standard formulas do not apply — for example, with non-normal data, complex study designs, or non-standard test procedures — simulation-based power analysis offers a flexible alternative. The researcher simulates many datasets under the assumed alternative, runs the planned analysis on each, and identifies the sample size that achieves the desired power.

Practical considerations — budget, recruitment feasibility, attrition, and ethical constraints — always play a role in the final sample size decision. Sometimes the most honest conclusion of a power analysis is that the study cannot be conducted at an adequate scale, and resources should be directed elsewhere.

29.7.2 Exercises

Exercises for this chapter are under development.

Chapter A

Datasets

This appendix will catalog all datasets used in the textbook with descriptions, sources, and download links.

Datasets used in this textbook are drawn primarily from:

- The [openintro](#) R package (400+ datasets)
- The IMS dataset collection
- Public sources (Data.gov, UCI Machine Learning Repository)
- Local and regional datasets from the University of Wisconsin-La Crosse

All datasets are available in CSV format for use with any statistical software.

Chapter B

StatBench Quick Reference

[StatBench](#) is a free, browser-based statistics toolkit used throughout this textbook. This appendix provides a quick reference for all available tools.

B.1 Distribution Calculators

Tool	URL	Use For
Normal	Normal Calculator	Probabilities and percentiles for normal distributions
t	t Calculator	t-distribution probabilities (Ch 11-12)
Chi-Square	Chi-Square Calculator	Chi-square probabilities (Ch 14-15)
F	F Calculator	F-distribution probabilities (Ch 16)
Binomial	Binomial Calculator	Binomial probabilities (Ch 24)

B.2 Simulation Tools

Tool	URL	Use For
Bootstrap CI: One Sample	Bootstrap Mean	Confidence intervals for a mean (Ch 8)
Bootstrap CI: One Proportion	Bootstrap Proportion	CIs for a proportion (Ch 8, 13)
Bootstrap CI: Paired	Bootstrap Paired	CIs for paired differences (Ch 8)
Bootstrap CI: Two Samples	Bootstrap Two Means	CIs for difference in means (Ch 8)
Randomization: Diff in Means	Randomization Means	Hypothesis test for two means (Ch 7)
Randomization: One Proportion	Randomization Prop	Hypothesis test for one proportion (Ch 7)
Randomization: Diff in Proportions	Randomization Props	Hypothesis test for two proportions (Ch 7)
Randomization: Chi-Square	Randomization Chi-Sq	Randomization test for independence (Ch 15)
Bootstrap CI: Slope	Bootstrap Slope	CI for regression slope (Ch 19)

B.3 Conceptual Demos

Tool	URL	Use For
Sampling Distributions	Sampling Dist	Visualizing sampling variability (Ch 6, 9)
CI Coverage	CI Coverage	Understanding confidence level (Ch 8)

B.4 Explore Data

Tool	URL	Use For
Descriptive Statistics	Descriptive	Summary stats, histograms, boxplots (Ch 3-4)
Regression	Regression	Scatterplots, regression lines (Ch 18-20)
Categorical Data	Categorical	Bar charts, contingency tables (Ch 3, 14-15)

B.5 Keyboard Shortcuts

All StatBench pages support keyboard navigation:

Key	Action
1	Generate +1 sample/resample
2	Generate +10
3	Generate +100
4	Generate +1000
0	Reset simulation
Space	Play/pause animation
?	Show keyboard help
Esc	Close dialog

B.6 URL Parameters

StatBench supports URL parameters for reproducible demonstrations:

?seed=42	Seeded random number generator (deterministic results)
?n=30	Sample size
?data=1,2,3,4,5	Inline data (comma-separated)
?ci=95	Confidence level (%)

Example: [Bootstrap CI with seed](#)

Chapter C

Statistical Tables

This appendix will include standard normal (z), t -distribution, and chi-square tables for reference. With StatBench's distribution calculators available, tables are provided as a convenience for exams and offline use.